Twelve Angry Models:
Using Cognitive Models in JASP

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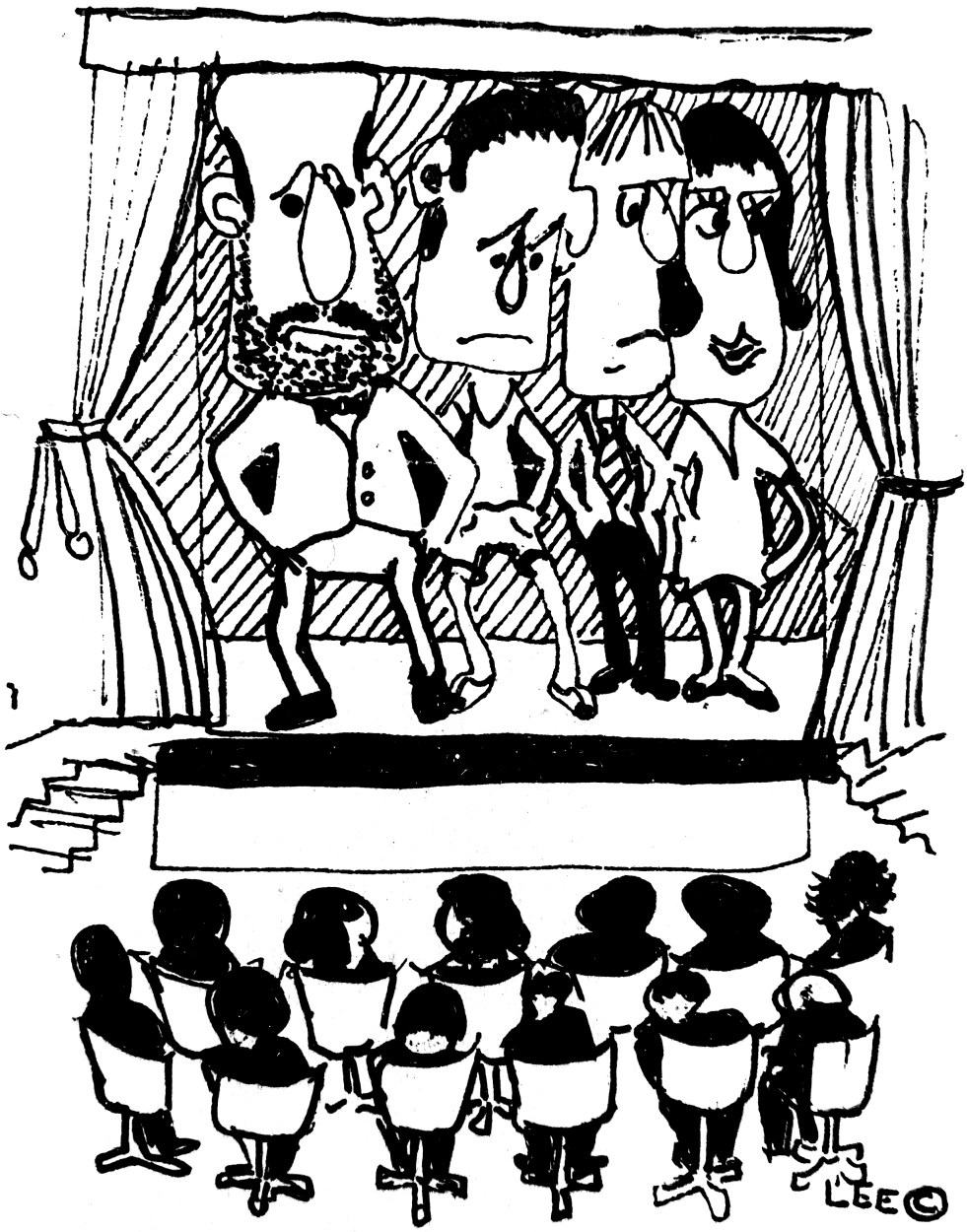

PLAYING TO A SMALL AUDIENCE

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Goals of this Book

This book is designed as an introduction to cognitive modeling using JASP. From conception, the book has been motivated by three complementary goals.

The first goal is to *promote data analysis using cognitive models*. Current practice in experimental psychology relies heavily on the statistical analysis of data through on regression and analysis of variance methods based on the generalized linear model. There are at least two potential advantages to using cognitive as well as statistical models for analysis. The first advantage is that cognitive model parameters have inherent psychological meaning: they represent interpretable quantities—such as learning rates, memory capacities, or risk propensities—rather than the slopes and intercepts of lines, or the means and variances of Gaussians. The second advantage is that cognitive models make stronger theoretical assumptions about the behavioral data being analyzed. To the extent these assumptions are good ones, this inductive bias allows more information to be extracted from the behavioral data. This can be especially useful if behavioral data are expensive or difficult to gather in large numbers, which often happens in clinical, developmental, and other settings.

The second goal is to *emphasize naturally occurring data*. One major change in the behavioral sciences over the last few decades has been the increasing availability of naturally occurring data. Companies maintain digital records of everyday consumer behavior, smartphones track the details of decision making in navigation, and social media crowdsources preferences and opinions related to almost every human endevour. Psychology needs to make progressively greater use of these data sources in its quest to understand people. There will always be a role for controlled experimentation, which allows precise measurement, systematic manipulation of key factors, and the possibility of critical tests of theories. Naturally occurring data have the complementary strengths of measuring people as they really behave outside the laboratory, in the complicated real world, and doing so at scale.

The third goal is to *encourage the use of Bayesian inference*. Although the tide is turning, psychology continues to rely too heavily on frequentist statistical methods. This has proved to be very limiting. Frequentist methods have all sorts of gaps, weaknesses, and pathologies in their application to the core statistical challenges of the empirical sciences. Bayesian statistical methods are far better suited, because of the founding commitments of represent uncertainty and adhering to the logic of probability theory. There are many practical advantages of the theoretical completeness, coherence, and consistency of Bayesian inference. I have always been less interested in upgrading existing statistical approaches to use Bayesian rather than frequentist methods, and more interested in the new modeling possibilities, such as the use of cognitive models, that Bayesian inference affords.

Combining these three points of emphasis, this book is about using cognitive models and Bayesian methods to analyze both experimental and naturally occurring data that measure human behavior.

Structure and Use of this Book

After an introduction to scientific modeling and Bayesiani nference in Chapter 0, each of the model chapters has the same format. The first section covers the background psychological theory of the model in a non-technical way. A number of worked examples then follow, applying the model to various data sets and research problems. For most modeling chapters, a hierarchical extension of the basic model is also presented. Once again, a section covers the background psychological theory before a number of worked examples are presented. At the end of each chapter is a section covering the formal details of the models, including their statistical definitions and their implementation in JAGS.

Whether you read the formal details of the models depends on whether it is important to you. It is possible to use the models in your own research based on a conceptual understanding of the model and seeing a few worked examples. Deeper understanding is gained by delving into the statistical details of the models, but it is not strictly necessary. The formal details *are* important for those who want to develop new models or tweak the existing ones, but can be skipped if the goal is simply to apply the existing models and interpret the results in your own research.

Which modeling chapters you cover is also up to you. They are roughly ordered from the simplest to the most complicated ones, but this is very approximate, because there really is not a one-dimensional ordering. The explanation of key concepts, like how to interpret joint, posterior, and representative posterior distributions, or what a hierarchical model involves, is more detailed for earlier chapters and is gradually assumed knowledge for later ones. So, it is probably a good idea to cover at least the first few chapters before focusing on later ones of special interest.

All of the applications have been implemented in JASP files that are referenced and available on the OSF. This gives you access to the data and models. The natural way to analyze your own data is to find the application that matches yours, and sync your own data to the relevant JASP file.

About the Doodles

In the formal details sections at the end of each chapter, and just occasionally elsewhere, you will notice doodles in the margins. I have long delighted in these sorts of drawings, in contexts as diverse as the Spy vs Spy battles in MAD magazines of the 1970s and 80s to displays of medieval marginalia at the Getty Museum in Los Angeles. As it happens, my father, Emeritus Professor David Lee, was an avid doodler during an academic career that involved senior administrative positions and the associated hours of meetings. The doodles in the margins of this book come from his collection, extracted from the edges of meetings agenda, draft policy documents, and whatever else was at hand when it was no longer possible just to sit and listen. You should feel free to add your own in the remaining margin space, motivated by your own frustrations and need for distraction.

0 Introduction

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0.1 Scientific Models

0.1.1 Goals of Scientific Models

Figure 1 shows two models of a plane. Neither are real planes. Models are not meant to be copies or emulations of the real thing. A model is, by definition, an abstraction of the real thing. It is something that simplifies or characterizes the real thing by creating an abstraction of it.

The reason for having models is that this simplification or abstraction can be useful. The flight simulator on the left in Figure 1 is useful for training pilots and designing cockpits. The wing in a wind tunnel on the right is useful for engineering the aerodynamics of the plane. Neither model plane is "right," but both are useful.

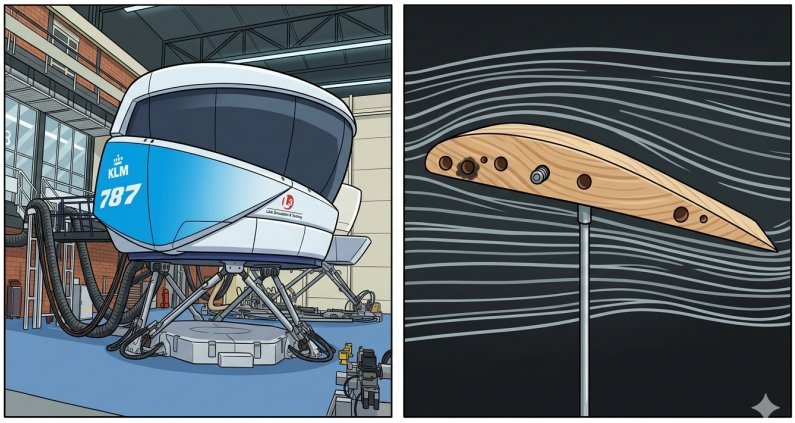


Figure 1: Two models of a plane. The model on the left is useful for training pilots. The model on the right is useful for engineering aerodynamics.

There are three goals a model should aim for to be useful: description, explanation, and prediction. For a model plane these goals try to answer the questions:

- **description**: What is the plane like, in terms of it shape, parts, and so on?
- **explanation**: How does the plane work, in terms of how it moves, flies, and so on?
- **prediction**: How would the plane operate under new or different circumstances?

Models in cognitive science have the same description, explanation, and prediction goals as models in other sciences. Other sciences focus on the phenomena that are important for them: physics cares about things like the aerodynamics of flight, astronomy cares about things like the movement of the planets, biology cares about things like the growth of plants, and so on. In cognitive science, the focus is on cognitive phenomena like people's ability to perceive and recognize stimuli, remember and recall information, learn associations and rules, and so on.

0.1.2 Elements of Scientific Models

One way to think about models is that they are processes with settings that produce observable behavior. As a starting analogy, consider a faucet with taps that produces water. This system has *settings*, in the form of the tap(s) that control the temperature and amount of flow of water, *processes*, in the forms of the inner workings of the faucet that produces the water temperature and flow according to the settings of the tap(s), and *behavior*, the resultant water temperature and flow that is observed and can be measured.

Figure 2 shows different possible "one tap" and "two tap" systems for producing water behavior.

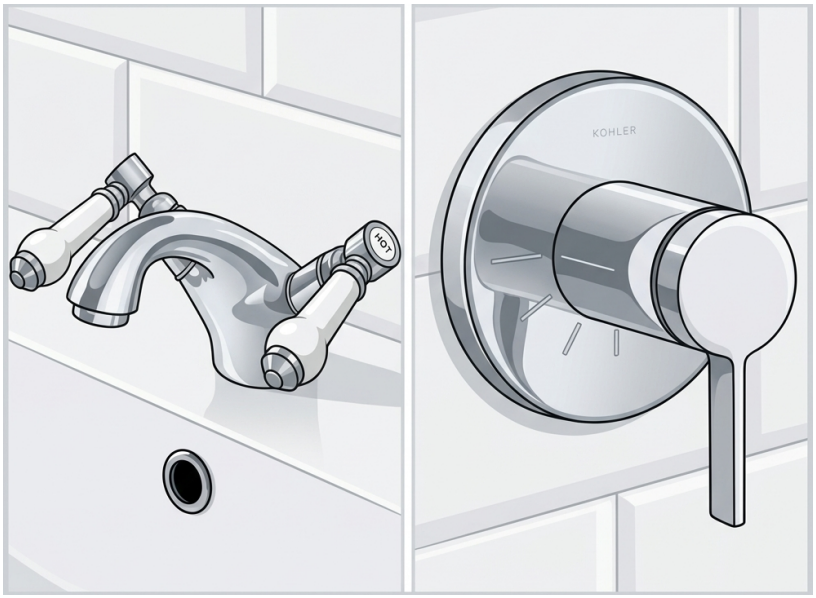


Figure 2: Two faucet systems for producing water with controllable pressure and temperature.

The settings, processes, and behaviors for faucets systems provide an analogy to the elements of scientific models of cognition. Models in cognitive science have

- **parameters**: Latent (unobservable, hidden) psychological variables, which have settings that control psychological processes (e.g., working memory capacity, learning rate, IQ, . . .)
- **processes**: The latent psychological mechanisms that detect and comprehend stimuli, encode, process, and retrieve information, and make decisions and produce actions
- **data**: The observable and measurable behavior resulting from the processes (e.g., button presses, response time, verbal answers, . . .)

Just as there are many ways to build a faucet, there are many ways to build a cognitive model. Different models make different assumptions about psychological processes, and have potentially different numbers and different types of parameters.

0.1.3 Scientific Inferences

A key difference between working with faucets and working with scientific models is that the parameters and processes in a scientific model are

latent. Imagine you could not see the settings on the tap(s), and could not open up the faucet to see its internal construction.

In this situation, you could collect data measuring different behaviors of the water, including its temperature and level of flow. From these data, it would be possible to make inferences about the setting of the tap(s) and the operation of the faucet system. For example, suppose the temperature could range from very hot to very cold, and either extreme could either be a light flow or a heavy flow of water. Then you could infer that it is likely that there are two taps independently controlling hot and cold water levels, rather than a single tap. This is an example of the modeling goal of *description*. You could also infer that when the water is flowing lightly at an extremely hot temperature, it is likely because just the hot tap is fully turned on. This is an example of the modeling goal of *explanation*. And you could infer that if both taps are turned on as far as they will go—even though this was never observed in data collection—a very high level of water at a moderate temperature will be produced. This is an example of the modeling goal of *prediction*.

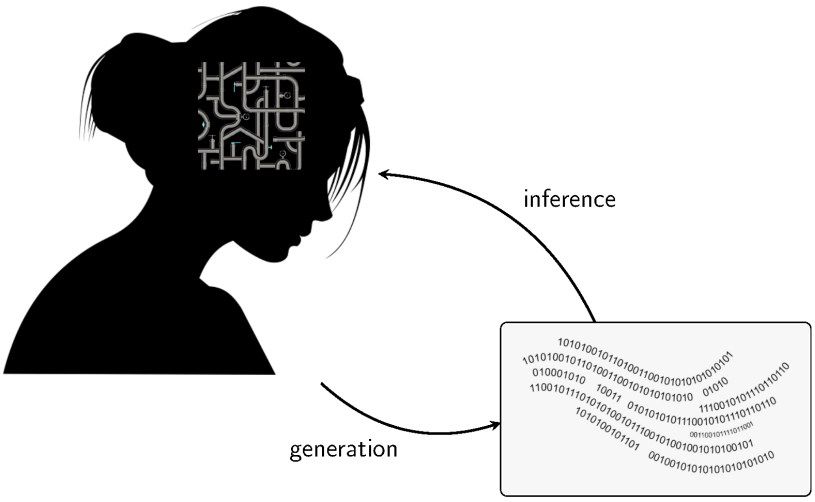


Figure 3: Making scientific inferences based on models that generate behavior.

Figure 3 shows schematically latent psychological variables and cognitive processes, inside the head, generating observed data, out in the world. These modeling assumptions can be based on theory, previous observation, and whatever insights scientists have. Constructing the model is an inherently creative process. The key outcome of the model-building process, however, is that it must generate data, and hence make predictions about what will be observed.

What statistical methods do is use the modeling assumptions and the observed data to make inferences about cognition. These inferences come in at least four forms:

- **inference about model adequacy**: statistical methods can examine whether the model is capable of generating the data that were observed
- **model comparisons**: given two or more competing models, statistical methods can provide a relative measure of which models are more likely to have generated the data
- **inferences about parameters**: for an adequate model, statistical methods can make inferences about what values of the parameters are likely to have generated the observed data

- **predictions about data**: given an adequate model, statistical methods can generate data that make predictions about observed behavior

0.2 Bayesian Methods

Throughout this book we use *Bayesian* statistical methods. This is what the JAGS language that implements the cognitive models inside JASP uses. Nothing in science (or life) is without the potential for controversy and debate, but there are good arguments that Bayesian statistics are an excellent way to analyze theories, models, and data in the empirical sciences. The special issue introduced by Vandekerckhove et al. (2018) presents many arguments in favor of using Bayesian methods in psychology, and McElreath (2020) is an accessible introduction to Bayesian statistical ideas and practices.

0.2.1 Probability Distributions

At the heart of Bayesian statistics is the idea that uncertainty should be fully represented in scientific inferences and predictions. Rarely in the empirical sciences is it possible to be certain what the value of a parameter is, or what future data will be. But it also the case that through theory and observation we do know something. Probability distributions provide a formal way to represent our uncertainty: that is, what we do and do not know. The results provided by Bayesian analysis are based on probability distributions. It is important to understand exactly what they are and how they can be used.

Discrete Distributions

Figure 4 shows three examples of *discrete* probability distributions. The top-left panel shows a distribution that represents uncertainty about working memory capacity. The bar graph shows the probability for possible values from 1 to 15. Following classic memory theory (Miller, 1956), the most likely values are 5, 6, 7, 8, and 9, all of which have probability 0.15 (or 15%). The values 3, 4, 10, and 11 have probability 0.015. The values 2 and 12 have probability 0.025. The sum of all of these probabilities is 1, meaning that these are all the possible values.

Formally, the working memory distribution is called a *probability mass function*. There is a finite number of possible discrete values, every possible value has a probability between 0 and 1 (with the extremes of 0 and 1 being allowed), and the sum of the probabilities is 1. Probability mass functions simply represent the probability of each possible value. These probabilities can be combined to express the total probability of a range of outcomes. For example, the probability the working memory capacity is somewhere from 5 to 9 is 0.90 (or 90%). Probabilities can also be compared in a relative way, in terms of their ratios. For example, it is twice as likely that working memory capacity is 7 rather than 6.

The top-right panel of Figure 4 represents uncertainty about the total baking time of a cake. This distribution is based on the baking times

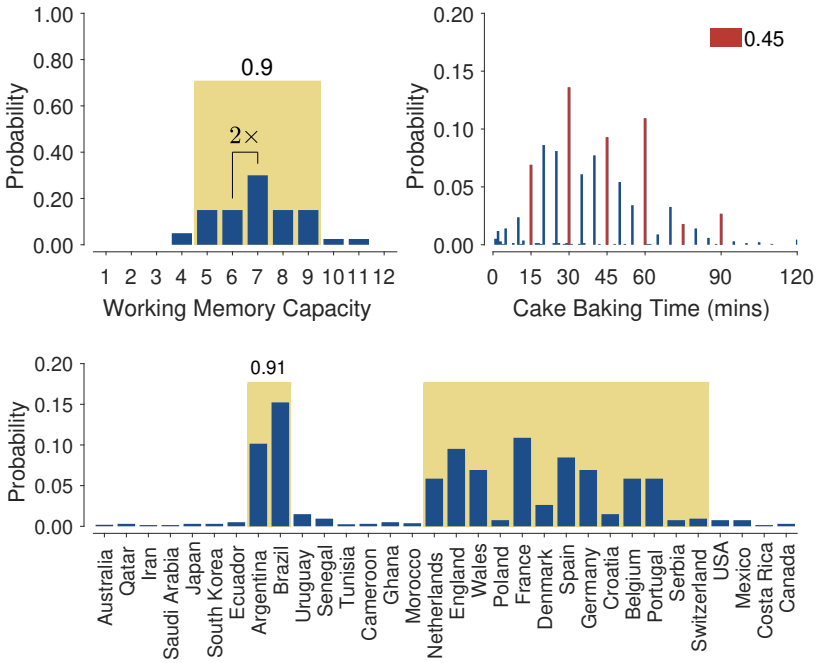


Figure 4: Three examples of discrete probability distributions, representing information about (top left) working memory capacity, (top right) the total baking time of a cake, and (bottom) the probability of various football teams winning the World Cup.

listed for 1346 cakes on recipes.com. Even though time is inherently continuous, the baking times can be represented as a probability mass function because each time is measured as a whole number of minutes. The most likely baking time is 30 minutes, but slightly different baking times like 29 minutes or 31 minutes have little or no probability. Instead, the baking times are mostly in multiples of 5 minutes, and multiples of 15 or 30 minutes are especially likely. For example, the probability that the baking time is either 15, 30, 60, or 90 minutes is 0.45.

These patterns are highly interpretable and reasonable in the context of baking. But, they are not standard in the sense they do not follow anything that looks like a Normal (Gaussian) curve or other familiar statistical distribution. The cake baking time distribution demonstrates that probability distributions have the flexibility to represent any pattern of uncertainty. It also highlights the limitations of using summary measures to communicate uncertainty. The working memory capacity distribution is reasonably well summarized by its mean and standard deviation, or by giving a range like "seven plus or minus two." But these sorts of summaries would be very misleading for the cake baking times. Probably the only way to convey the uncertainty about baking time is to present the entire probability distribution visually or in a table.

The bottom panel of Figure 4 represents uncertainty about the winner of the 2022 World Cup based on betting markets before the tournament started. Every team is given some probability of winning, but many of these probabilities are small. Brazil were the favorites, with about a 0.15 probability of winning. Once again, probabilities can be combined. For example, summing the winning probabilities of Argentina and Brazil with the European teams listed from the Netherlands to Switzerland gives a combined probability of 0.91 of one of these teams winning.

It is important to understand that the distributions in Figure 4 represent *uncertainty* rather than *variability*. The winner of the World Cup is not changing over time according to the distribution. There will be only one

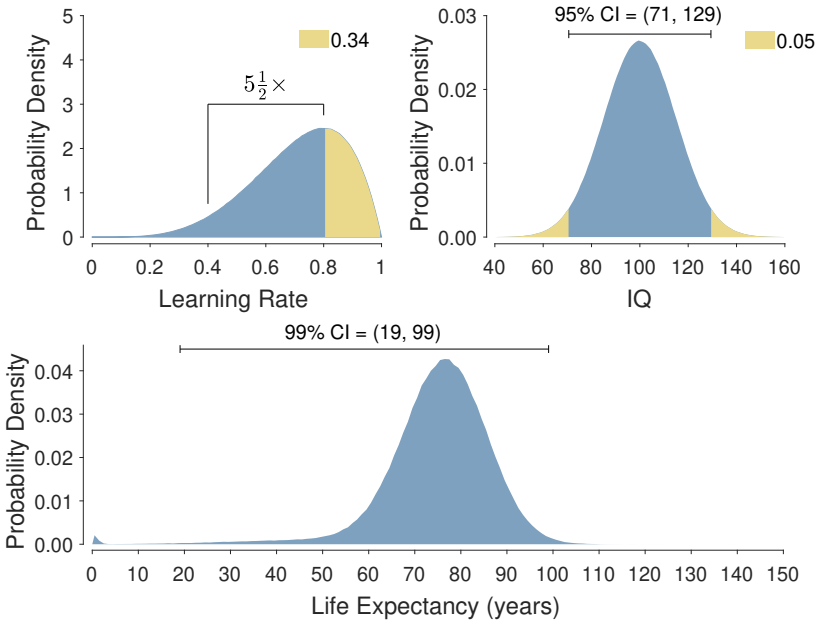


Figure 5: Three examples of continuous probability distributions, representing information about (top left) a learning rate, (top right) the IQ of a person, and (bottom) the life expectancy of a person.

winner. The distribution shows the uncertainty we have about which team will be the winner. In the same way, there is assumed to be only one working memory capacity and one total baking time, and the distributions show how probable each possibility is.

Continuous Distributions

Figure 5 shows three examples of *continuous* probability distributions. The top-left panel shows a distribution that represents uncertainty about a latent learning rate that can take values from 0 to 1. The y-axis values no longer represents probabilities: this is clear, since the values go well above 1. Instead, the y-axis values represent what are called probability densities, and the distribution is now called a *probability density function*. Probability densities are *not* the same as probabilities and are not interpretable in isolation. But the other properties of discrete distributions still apply. Just as the sum of the bar graphs for a discrete probability distribution is 1, the area under the curve of a continuous probability distribution is 1. This means that ranges of values have interpretable probabilities. For example, the probability that the learning rate is greater than 0.8 is about 0.34. This is the shaded yellow area under the curve. Similarly, ratios of densities continue to quantify relative probabilities. For example, it is about $5\frac{1}{2}$ times more likely the learning rate is 0.8 rather than 0.4, since the probability density is about $5\frac{1}{2}$ times higher at 0.8.

The top-right panel shows a probability density function representing uncertainty about a person's IQ. The shaded regions show where a person either has a very high IQ above 129 or a very low IQ below 71. The total area under these shaded regions is 0.05, which means there is a 5% probability the person's IQ falls in one of these extremes. This information can also be expressed in terms of a *credible interval* which identifies the bounds between which values are likely to lie. The 95% credible interval for IQ from 71 to 129 indicates that it is 95% likely that IQ lies between these two bounds.

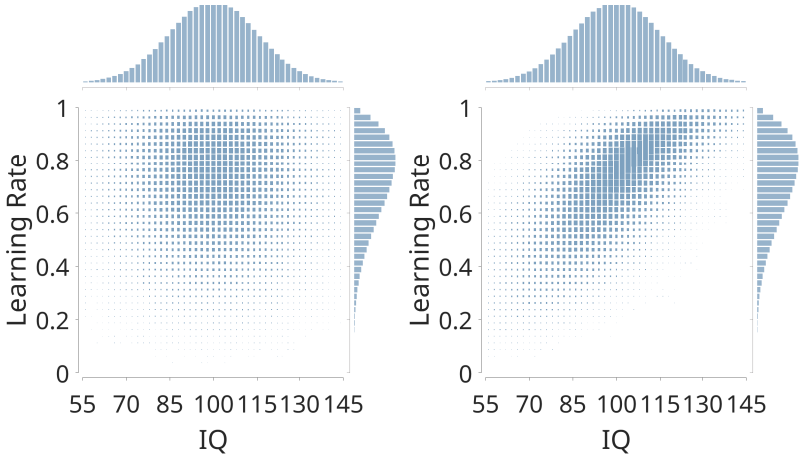


Figure 6: Two examples of joint and marginal distributions for IQ and learning rate, showing cases in which the joint distribution (left) does not show a correlation between IQ and learning rate, and (right) shows a positive correlation between IQ and learning rate.

The bottom panel shows a probability density function representing uncertainty about how long a person will live, based on the statistical theory developed by Zanotto et al. (2021). The bulk of the distribution is approximately Normal, but skews towards younger ages. For example there is more probability between 30 and 50 years than there is above 100 years. There is also some probability for very young ages, corresponding to infant mortality.

The life expectancy distribution again emphasizes the ability of probability distribution to represent complicated patterns and the limitations of summary measures such as credible intervals. The 99% credible interval for life expectancy ranges from 19 to 99 years, which fails to include the important infant mortality case. Once again, probably the only effective way to communicate the information in the life expectancy distribution is to present it in full. Any attempt at summarizing it seems likely to lose important information.

Joint and Marginal Distributions

All of the probability distributions shown in Figures 4 and 5 relate to a single quantity. The memory capacity, baking time, learning rate, IQ, and life span are just a single number, and the World Cup winner is just a single team. This means that the probability distributions representing uncertainty about which number or team is the true one are all one dimensional.

Often, however, scientific models involve multiple latent parameters. Consider a model for a learning task that assumes people's overall intelligence and their ability to learn both contribute to their performance. This model would include both an IQ parameter and a learning rate parameter. The goal of scientific inference is to identify the *combination* of IQ and learning rate that underlies a person's performance. Uncertainty about this combination takes the form of a *joint* distribution over both of the parameters simultaneously, and is inherently two dimensional.

Figure 6 shows two examples of joint distributions for this learning task model. The x-axis corresponds to the IQ parameter and the y-axis corresponds to the learning rate parameter. In the example on the left the uncertainties about IQ and learning rate are independent. The shading in the central panel shows the probability density of each combination

of IQ and learning rate. The heavier the shading, the more likely that combination of IQ and learning rate is the one belonging to the person. The histograms above and to the right of the central panel show the two one-dimensional *marginal* distributions. Metaphorically, think of the joint distribution as being on the page, and the marginal distribution as being in the upper and right margins of the page. The marginal distribution for the x-axis shows the uncertainty about IQ and the maginal distribution y-axis shows the uncertainty about learning rate.

The independence of uncertainty in the joint distribution on the left of Figure 6 is clear because the probability density of a combination of IQ and learning rate parameter values is determined by multiplying the marginal densities. For example, the probability density that a person's IQ is 100 and their learning rate is 0.8—as showing by the size of the square at that combination of values in the central panel—can be found by multiplying the marginal density for an IQ of 100 and the marginal density of a learning rate of 0.8. This "the joint is the product of the marginals" property means that the joint distribution does not contain any more information than the two marginal distributions.

This independence between parameters does not, however, always hold. The example on the right of Figure 6 shows a correlation that signifies dependency. The correlation is that higher values of IQ have more probability density in combination with higher values of learning rate, and lower values of IQ have more probability density in combination with lower values of learning rate. A situation like this seems very plausible. The representation of uncertainty provided by the joint posterior distribution can be interpreted as showing that the person's learning rate and IQ track each either: if one is high, the other is likely to be high; if one is low, the other is likely to be low, and so on. The joint distribution now has more information than the two marginal distributions. As a concrete example, an IQ of 115 is plausible under the marginal distribution at the top, as is a learning rate of 0.6 under the marginal distribution to the right. But their combination is very unlikely. The joint is no longer the product of the marginals.

The two examples in Figure 6 are constructed to have exactly the same marginal distributions. There is no way of knowing about the correlation in the uncertainty between IQ and learning rate from the marginal distributions alone. This again highlights the limitations of relying on summaries of the complete representation of uncertainty provided by the Bayesian approach. Just as means, medians, standard deviations, credible intervals may or may not be adequate summaries of a complete probability distribution, marginal distributions may or may not be adequate summaries of complete joint distributions. Preparing a summary, by definition, always involves losing information. The key question is whether it loses important information. Good scientific practice is always to examine the complete representation of uncertainty, and only to use summaries of these representations when they preserve the relevant information for the goals of the inference. For models with two or more parameters, this requires examining the joint distributions to assess whether marginal distributions are reasonable summaries.

0.2.2 Bayesian Inference

Bayesian statistics uses discrete and continuous distributions to represent latent psychological parameters and potentially observable data. Working memory capacity, learning rates, and IQ are latent parameters in psychological models that cannot directly be observed. The baking time of a cake, the winner of a World Cup, or the lifespan of a person are all observable, and models can make predictions about their outcomes.

What most often changes our uncertainty about parameters and future data is observing new data. This is the basis of the scientific method used by the empirical sciences (Feynman, 1994). It is the basic motivation for running experiments or collecting naturally occurring data. Observed outcomes provide information about the plausible values of latent parameters and future observations. For example, if someone performs well on a digit span test, that provides evidence that their working memory capacity is high. Or, if a cake is observed to have been baking for 30 minutes, that eliminates shorter baking times as possibilities.

Bayes Rule

Bayes Rule is a statistical theorem that formalizes how the evidence provided by data should be used to update probability distributions to incorporate the new information. The distribution before the data are observed, $p\left(\text{parameters}\right)$, is known as the *prior distribution* and the distribution that includes the information provided by the data, $p\left(\text{parameters} \mid \text{data}\right)$, is called the *posterior distribution*. They are linked by a predictive updating factor, shown in Equation 0.1.

$$\underbrace{p\left(\text{parameters} \mid \text{data}\right)}_{\text{posterior distribution}} = \underbrace{p\left(\text{parameters}\right)}_{\text{prior distribution}} \times \underbrace{\frac{p\left(\text{data} \mid \text{parameters}\right)}{p\left(\text{data}\right)}}_{\substack{\text{predictive} \\ \text{updating factor}}} \tag{0.1}$$

The predictive updating factor is made up of two parts. The numerator, $p\left(\text{data} \mid \text{parameters}\right)$, is known as the *likelihood*, and is the important part. The denominator, $p\left(\text{data}\right)$ acts as a normalizing constant to ensure that the total mass or density of the posterior is 1.

The likelihood measures the predictive accuracy of each possible parameter value, or combination of parameter values for joint distributions. It quantifies the relative probability of the data arising for any given values of the parameters. Intuitively, parameter values that predicted the data are made more likely as the prior is updated to the posterior, while parameter values that did not predict the data are made less likely. Bayes Rule formalizes and quantifies that intuition.

Prior, Posterior, Parameter, and Predictive Distributions

Prior and posterior distributions are always relative to specific data. If multiple data arrive in a sequence, the posterior for the first set of data becomes the prior for the second set, and so on. At any point in this

iterative process it is possible to make predictions about data, such as the outcome of the next experiment. Each parameter value makes its own predictions. Weighting those predictions according to the prior, so that the predictions of more likely parameters are given more emphasis, forms an overall prediction.

Just as uncertainty about parameters is prior or posterior to specific data, uncertainty about outcomes are prior or posterior to specific data. If the prior is used for weighting to form a predictive distribution, the result is called the *prior predictive distribution*. Rather than existing in latent parameter space, it exists in observable data space. If the posterior is used for weighting, the result is called the *posterior predictive distribution*

The prior predictive distributions make genuine predictions about the outcomes of experiments. Posterior predictive distributions are unfortunately named. It has already incorporated the data, so it does not make genuine predictions. Instead, it provides re-description of the previously observed data. Prior predictive distributions are especially useful for model comparison, since they can test how well competing models fare in predicting data. Posterior predictive distributions are useful for checking the descriptive adequacy (often called ''fit") of a model. Because this book focuses on mature cognitive models that have already been tested and are widely accepted, we will generally focus on posterior predictive distributions, which provide a way of checking that the model is well suited to specific data sets.

A Working Memory Capacity Example

A concrete example probably helps make the general ideas of priors and posteriors for parameters and predictions clear. Consider an experiment designed to measure a person's working memory capacity. The experiment involves the person taking both a forward and a reverse digit span test. In both of these tests they have to recall 10 sequences of digits and their score is the number they get correct. In the forward test they must recall the digits in the presented order, and in the reverse test they must recall them in reverse order. A classic finding is that there is lower test-retest reliability for the reverse order test (Blackburn & Benton, 1957), but it does provide another way to assess working memory capacity.

A model of the person's behavior involves specifying a prior and a likelihood. We assume the prior probabilities of different possible capacities follows the distribution in Figure 4. The possibilities are that their working memory capacity is $4, 5, \ldots, 11$. The likelihood specifies how many correct responses are expected on both tests for a given working memory capacity. Figure 7 presents these predictions for a simple account of how different capacities lead to different probabilities of recall.*

Each panel in Figure 7 corresponds to the predictions made by a specific working memory capacity from 4 to 11. The joint distribution in each panel shows the probability of the person answering some combination of questions correctly on the forward and reverse tests, given that working memory capacity. For example, if their working memory capacity is 4, it is most likely they will recall 0, 1, or 2 of the digit sequences in the

* We do not specify the model formally here. But, roughly, it is a logistic choice model of the type covered in Chapter **??**.

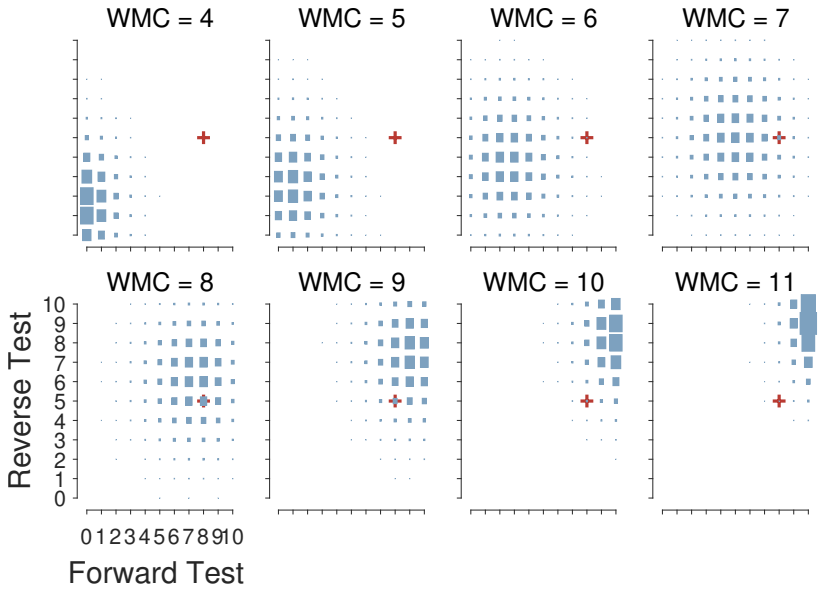


Figure 7: Predictions about the outcome of a forward and reverse digit span test. Each panel corresponds to a different possible working memory capacity (WMC), and the joint probability distribution shows how likely different numbers of correct recalls from 0 to 10 are for the two tests.

forward test, and somewhere up to about 5 in the reverse test. There is more uncertainty about the reverse test outcome, consistent with its lower reliability. At the other extreme, if their working memory capacity is 11, it is most likely they will recall all the digit sequences for the forward test, and most of the digit sequences for the reverse test. Intermediate working memory capacities lead to predictions of intermediate test performance.

The red crosses in all of the panels of Figure 7 shows the observed performance of the person once the tests are conducted. The person recalled 8 digit sequences in the forward test and 5 in the reverse test. Working memory capacities of 7, 8, and 9 predicted this outcome the best, and the observed data are most likely for a working memory capacity of 8.

The top-left panel of Figure 8 shows the prior for the person's working memory capacity and the top-right panel shows its update to the posterior. This transition can be understood in terms of the predictions in Figure 7. The working memory capacities of 7, 8, and 9 are the only ones with significant probability in the posterior, because they are the only ones that predicted the observed data. The working memory capacity of 8 has become especially probable, because it made the best prediction. The prior probabilities continue to have a residual impact on the posterior. Both working memory capacities 7 and 9 predicted the data about equally well, but 7 is more probable in the posterior because it was more probable in the prior.

The bottom-left and bottom-right panels of Figure 8 show, respectively, the corresponding prior and posterior predictive distributions. The prior predictive distribution is the combination of all of the value-specific predictions in Figure 7, weighted by the prior. This means that the prediction for a working memory capacity of 7 is given the most weight, but all of the values between 2 and 11 make a contribution. The posterior predictive distribution weights the individual predictions by the posterior, so only the predictions for working memory capacities of 7, 8, and 9 make any meaningful contribution. The difference is clear: the prior predictive distribution is much more uncertain, giving probability to most possible

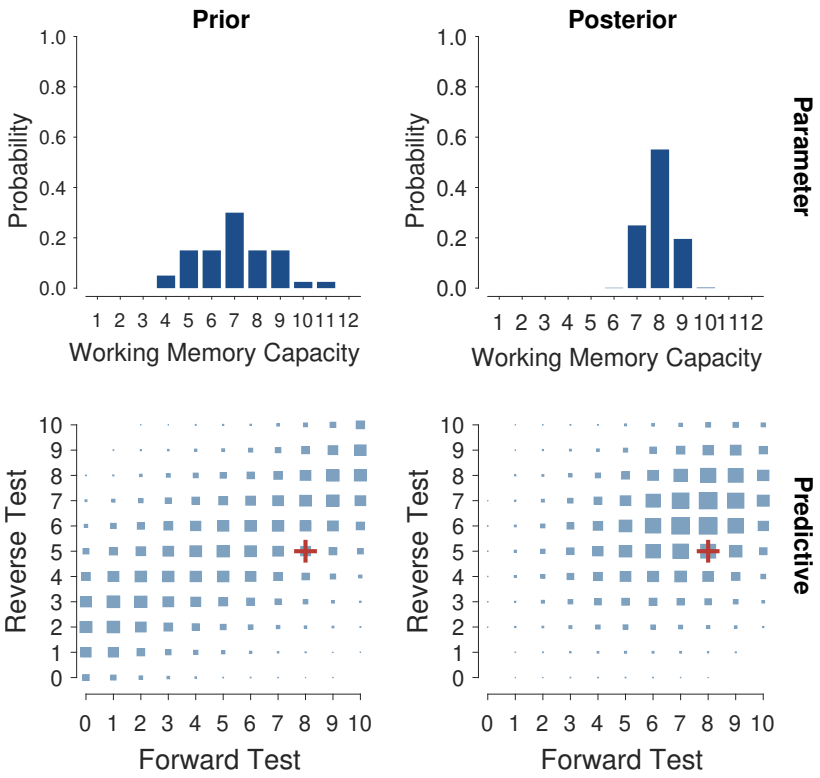
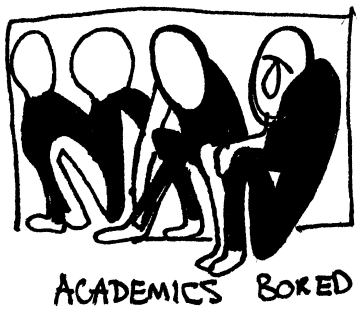


Figure 8: The (top left) prior and (top right) posterior distributions of the working memory capacity parameters, and the corresponding (bottom left) prior predictive and (bottom right) posterior predictive distributions over the possible observable outcomes of the digit span tests.

testing outcomes, whereas the posterior predictive distribution generally expects higher scores in both tests.

One way to think of the posterior predictive distribution is that it makes a prediction for the outcome of a replication of the experiment. If the same experiment was conducted again and more data became available, Bayesian inference would follow exactly the same process. The posterior distribution in Figure 8 would become the prior, and Bayes Rule would update it again to incorporate the new data. This, in turn, would lead to updated posterior predictions representing what is expected from future experiments.

Conceptually, that is all there is to Bayesian inference. As data are collected, probability distributions are updated, making more informed inferences about latent psychological parameters, and more informed predictions about future observable data.



Success Rate 1

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1.1 The Success Rate model

1.1.1 Motivation

A common form of behavioral data in the cognitive sciences is a set or sequence of binary outcomes, with one outcome corresponding to a "success" and the other to a "failure." Examples include students providing the right or wrong response to each of a set of true-or-false questions, participants completing a set of visual identification trials producing a series or correct and incorrect responses, or participants in a memory task either succeeding or failing to recall each of a sequence of stimuli.

A standard model for these sorts of data makes the simplifying assumption that the probability of a success is the same throughout. It assumes, for example, that the probability of success does not improve or worsen over the course of producing the behavior. It also assumes there is no dependency between the observations, so that the probability of the next outcome will be a success is not affected by the previous sequence of successes and failures. Given these simplifications, the key inference for the success rate model is "what is the underlying rate of success?"

1.1.2 Model

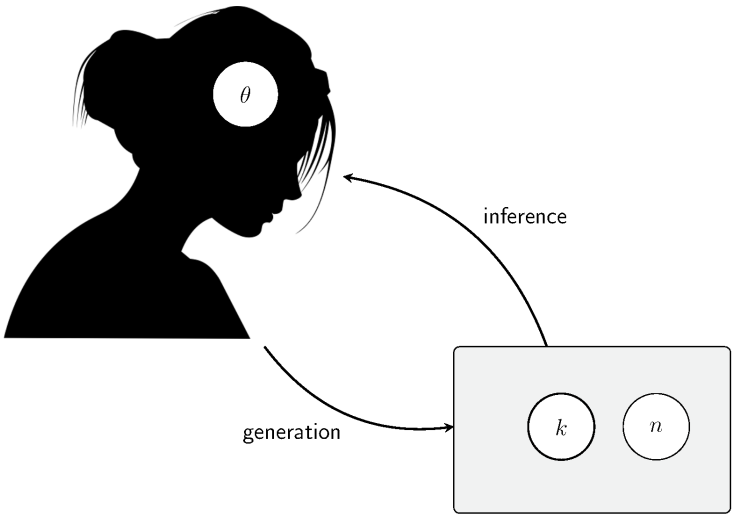


Figure 1.1: The structure of the success rate model.

Figure 1.1 summarizes the structure of the success rate model, showing how it relates the success rate parameter to the observed data. The success rate is denoted by the Greek letter θ, and is shown inside the person's head, to indicate it is a latent unobservable parameter. The observed data are represented by the count k of the number of successes out of n observations, and are shown in a box representing the external observable environment. The border of the k parameter is bold, to indicate that it

is a measure of behavior. This is not the case for n, which is usually part of the design for data collection, such as the number of trials in an experiment.

1.1.3 Mechanics

JASP does not use Greek letters like θ that are commonly used in describing cognitive models. Table 1.1 details the meaning of the parameters and variables in the success rate model, their name in the model as shown in Figure 1.1, and their name in JASP.

Name	Model	JASP
success rate	θ	`theta`
observed success count	k	`k`
number of observations	n	`n`
predicted success count	$\hat{k}$	`kPrd`
number of observations for prediction	$\hat{n}$	`nPrd`
comparison probability	π	`pi`

Table 1.1: The meaning of the parameters and variables in the success rate model, with their model and JASP names.

The first three variables in Table 1.1 are the core parameters and data included in Figure 1.1. These are the success rates θ, success counts k, and number of observations n for each of an individual. Note that the term "individual" is deliberately vague and general. In many situations, the different individuals are different people. But they could also be different individual experimental conditions, or different individual stimulus types, and so on.

The final three variables in Table 1.1, below the dividing line, are not included in Figure 1.1. They are not fundamental to the generative process assumed by the model, but they are useful additional parts of making inferences using the model, and so are provided in the JASP implementation.

Two of the additional variables relate to prediction. The model can use what is inferred about an individual success rate θ to make predictions about the count $\hat{k}$ of the number of successes that would occur out of $\hat{n}$ observations. The "hat" notation for $\hat{k}$ and $\hat{n}$ indicate that the variables are related to prediction.

The other additional variable in Table 1.1 is the comparison probability π. The JASP implementation of the success rate model can analyze multiple independent individuals simultaneously. When there are multiple individuals, each with their own success rate, $\boldsymbol{\pi}$ is a matrix with the inferred probabilities each individual has a higher success rate than another, for all possible pairs of individuals. The boldface typesetting of $\boldsymbol{\pi}$ indicates that it contains more than one value.

Input data

Table 1.2 demonstrates the general JASP data format using the data from the *Online Sellers* example.

The three columns correspond to the three required data variables

k	n	nPrd
10	10	200
48	50	200
186	200	200

Table 1.2: JASP data format for success rate model.

- the observed success count `k`
- the number of observations `n`
- the number of observations `nPrd` for posterior prediction

Each row in the data corresponds to an individual. There is no requirement that different individuals have the same number of observations. Table 1.2 demonstrates this, showing individuals with 10, 50, and 200 trials. There is also no requirement to make predictions about the same number of hypothetical observations. Table 1.2 uses 200 trials for all three individuals, because of the logic of the example, but these numbers could be different.

Output analyses

The JASP implementation provides three analyses:

- **success rate posteriors**: the posterior distribution of the success rate `theta` for each individual
- **success count posterior predictions**: the posterior predictive distribution of the success count `kPrd` for each `nPrd` observations for each individual
- **success rate comparisons**: the comparison probability that the success rate of an individual is greater than another, provided in a matrix `pi` that considers all possible pairs of individuals

The analysis of the success rate posterior is the central one for the success rate model. It provides the inference about the underlying success rate for each individual, based on their observed data. The posterior predictive analysis helps understand the implications of the success rate inference, especially in terms of making predictions. The comparison probabilities are easy to interpret measures that help in evaluating the relationships between the success rates of multiple individuals.

1.2 *Coin Tossing*

`coinTossing.jasp`

Bartoš et al. (2023) tested the hypothesis that a coin toss is slightly more likely to land on the side of the coin that was facing upwards when it was tossed. They found evidence for this hypothesis based on 350,757 tosses, and gave themselves something to talk about at parties. It also provides a source of experimental data in the classic domain of coin tossing for a first example of the success rate model.

Data

The data repository for the study is here, and suggests that the lead author contributed about 8,000—a little over 2%—of the tosses. Their

first 102 tosses were for coins with tails facing upwards, and resulted in 52 tails. The first $n = 10$ of these tosses resulted in $k = 7$ tails.

The success rate model can be used to make an inference about the success rate of tails from the first 10 tosses, and then make a posterior prediction about the outcome after 102 tosses.

Results

Success rate posterior

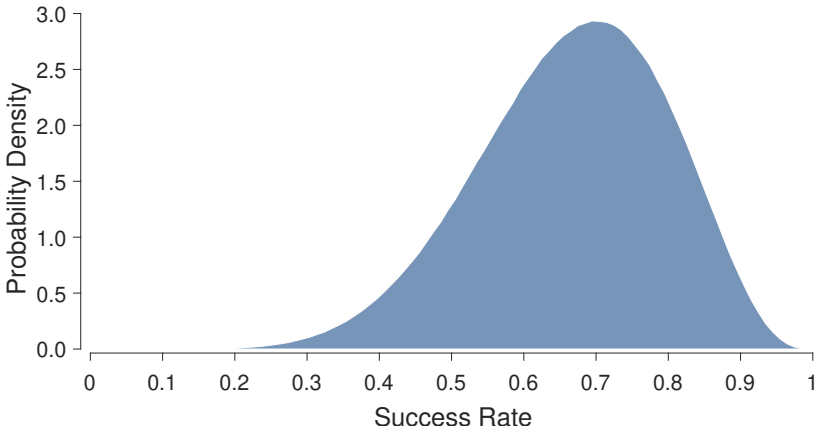


Figure 1.2: The posterior distributions of the success rate θ for the coin tossing data.

Figure 1.2 shows the posterior distribution for the success rate θ. Because it is based on only 10 tosses there is significant uncertainty. The most likely rates are around 0.7, but rates as low as about 0.3 and as high as near 1 remain possible. The 95% credible interval for θ is $(0.39, 0.89)$.

Success count posterior predictions

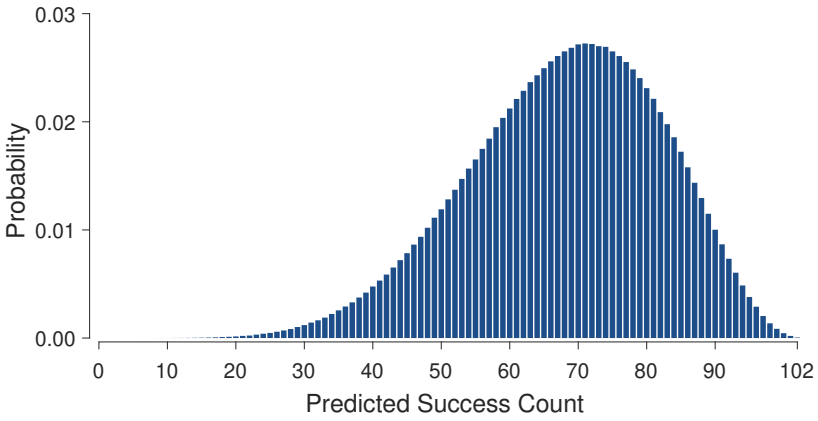


Figure 1.3: The posterior distributions of the success rate θ for the coin tossing data.

Figure 1.3 shows the posterior predictive distribution for the number of successes $\hat{k}$ from $\hat{n} = 102$ tosses. This distribution also shows significant uncertainty. Counts around 70 are most likely, but it is possible the outcome will be fewer than 30 or greater than 100 successes. The 95% credible interval for $\hat{k}$ is $(38, 92)$.

The posterior in Figure 1.2 and the posterior in Figure 1.3 are visually similar, and they are closely related. But it is important to understand that

they represent two different types of uncertainty. Figure 1.1 emphasizes the conceptual difference, showing the success rate as a latent parameter, and the success counts as observed or observable data.

The posterior represents uncertainty about a latent property of the coin-tossing data-generating mechanism, measuring the probability a coin will land on the side that was facing upwards when it was tossed. The posterior predictive represents uncertainty about the observable outcome of coin tosses. The posterior predictive distribution, for example, can be compared to the eventual outcome that there were 52 successes in the first 102 tosses. Whether this is genuinely a predictive test or not is subtle, since information about the first 10 tosses is included in the posterior distribution that generates the posterior predictive distribution, but information about the subsequent 92 tosses is not.

1.3 *Online Sellers*

`onlineSellers.jasp`

An example involving success rate inferences for multiple decision sources comes from the youtube channel 3Blue1Brown. It compares the positive ratings of three online sellers. The sellers have different numbers of positive ratings from different numbers of total customer evaluations. The research question is to infer from these data which seller has the highest rate of successful sales.

Data

Table 1.3: The online seller data.

Seller	Positive	Sales	Percentage
A	10	10	100%
B	48	50	96%
C	186	200	93%

The online sellers data are shown in Table 1.3. Seller A has only been evaluated positively $k_1 = 10$ out of $n_1 = 10$ sales. Seller B has been evaluated positively $k_2 = 48$ out of $n_2 = 50$ sales. Seller C has been the most evaluated, with $k_3 = 186$ positive ratings out of $n_3 = 200$ sales.

Results

Success rate posteriors

Figure 1.4 shows the posterior distributions for the success rates of the three online sellers. The modes of the distributions (the peaks) reflect the percentage of positive evaluations listed in the final column of Table 1.3. The width of the distributions represents the degree of uncertainty, which is greatest for Seller A, because they have been evaluated least often, and smallest for Seller C, because they have been evaluated most often.

The posterior distributions could be summarized by their means and 95% credible intervals. For Seller A, the posterior mean is $\theta_1 = 0.91$ with 95% credible interval $(0.70, 1.00)$. For Seller B, the posterior mean

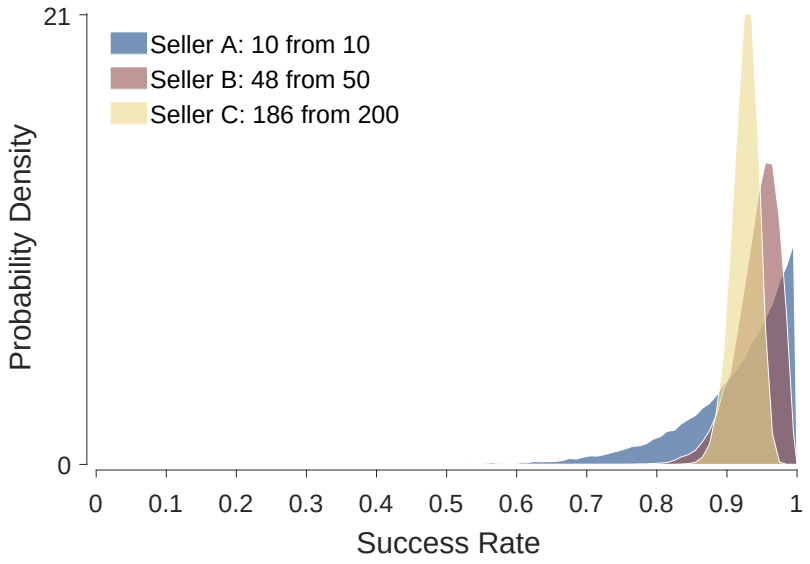


Figure 1.4: The success rate posterior distributions for the three online sellers.

is $\theta_2 = 0.94$ with 95% credible interval $(0.86, 0.99)$. For Seller C, the posterior mean is $\theta_3 = 0.93$ with 95% credible interval $(0.89, 0.96)$. These summaries are arguably effective for Sellers B and C, in the sense that they convey most of the information that is available visually in Figure 1.4. The mean and credible interval summary seems less effective for Seller A, because it does not communicate the clear asymmetry in the posterior distribution.

Success count posterior predictions

The posterior distributions in Figure 1.4 represent the relative probabilities of all of the possible underlying success rates of positive evaluations. But, answering "Which seller is better?" requires being clear on what it means to be better, and applying that criterion to the posterior distributions.

One approach is to use posterior predictive distributions to compare the number of successes Sellers A and B will have once they have the $\hat{n} = 200$ evaluations already reached by Seller C.

Figure 1.5 shows the results of this posterior predictive analysis. The top and bottom panels show, respectively, the posterior predictive probabilities for the success counts $\hat{k}_1$ and $\hat{k}_2$ for Sellers A and B after $\hat{n} = 200$ evaluations. Highlighted in both panels is the observed success count of $k_3 = 186$ for Seller C.

These analyses provide a way of thinking about which seller to prefer in terms of possible outcomes. While Seller A could easily have more than 190 positive ratings after 200 evaluations, they could also have fewer than 150. Seller B does not have the same downside risk, but their upside reward is more limited, with very high positive rating counts near 200 being less likely than for Seller A.

Success rate comparisons

Another approach to comparing the online sellers is provided by the pairwise comparisons. Table 1.4 shows the comparison probabilities for

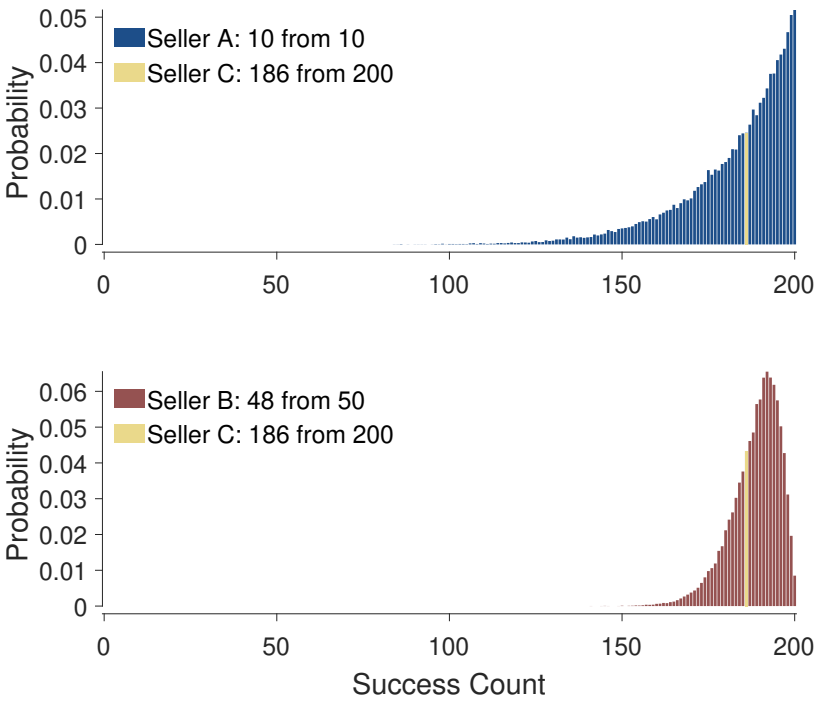


Figure 1.5: The posterior predictive distribution after $\hat{n} = 200$ evaluations for (top panel) Seller A and (bottom panel) Seller B.

Table 1.4: Comparison probabilities of the success rates for the online sellers data.

	Seller A	Seller B	Seller C
Seller A	–	0.45	0.56
Seller B	0.55	–	0.71
Seller C	0.44	0.29	–

all pairs of sellers. For example, because the entry in the second row and first column is 0.54, there is a 54% probability that Seller B has a higher success rate than Seller A. In the same way, there is a 70% probability that Seller B has a higher success rate than Seller C, and a 57% probability that Seller A has a higher success rate than Seller C. This analysis leads to the ordering Seller B, followed by Seller A, with Seller C last.

The conceptual differences between posterior predictive analyses and comparison probabilities for evaluation are unpacked in more detail in the following *Free Throws* example in Section 1.4.

1.4 *Free Throws*

`freeThrows.jasp`

In basketball, players often have to attempt free throws, which are (relatively) controlled attempts to shoot the ball through the basket. Basketball journalism usually measures free throw shooters in terms of the percentage of attempts a player scores. The success rate model provides an alternative approach based on inferring the underlying success rate they have for each free throw they attempt.

Data

Table 1.5 shows the number of free throws scored from the number attempted by four players, and the percentage of attempts scored for

Player	Scored	Attempted	Percentage
Scott Skiles	1548	1741	88.91%
Peja Stojakovic:	2237	2500	89.48%
Rick Barry	5713	6397	89.31%
Stephen Curry	3422	3766	90.87%

Table 1.5: The free throw data.

each, based on data found here.

Over the course of their (now completed) careers, Scott Skiles, Peja Stojakovic, and Rick Barry were regarded as excellent free throw shooters. At the time the data were collected, the fourth basket player listed, Stephen Curry, was a current player, and was also regarded as an excellent free throw shooter.

Results

Success rate posteriors

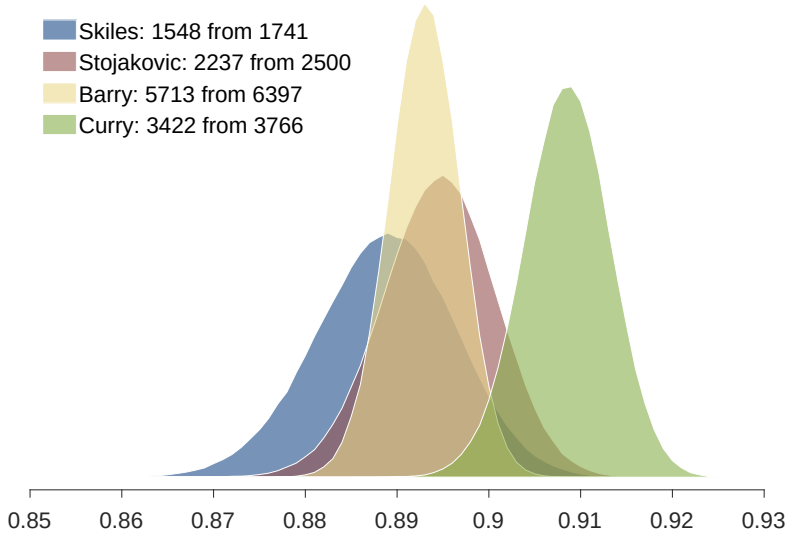


Figure 1.6: The success rate posterior distributions for the four basketball players.

Figure 1.6 shows the posterior distributions for the success rates of the four basketball players. The peaks of the posterior distributions are determined by the percentage of free throws each player has made, while the widths of the distributions correspond to the number they have attempted. For example, there is more uncertainty about the underlying success rate for Skiles than for Barry.

Success count posterior predictions

Figure 1.7 shows the results of the posterior predictive analysis. Each panel corresponds to a player, showing the distribution of how many free throws they are predicted to have scored from once they have attempted $\hat{n} = 10,000$ free throws. Using this analysis to compare the players suggests that Curry is likely to have scored the most free throws, but it is not certain. For example, it is possible Curry will have scored fewer

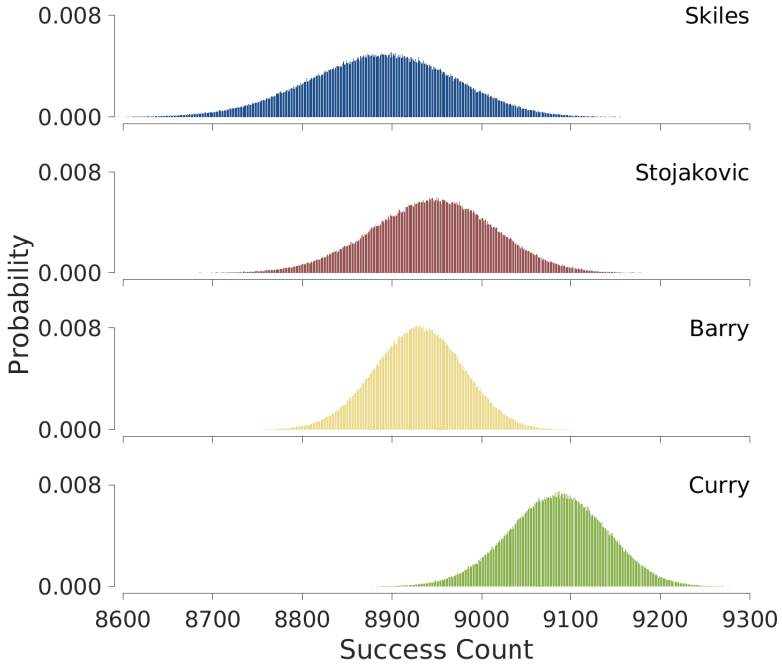


Figure 1.7: The posterior predictive distribution for each player after $\hat{n} = 10,000$ free throws have been attempted.

Table 1.6: Comparison probabilities of the success rates for the free throw data.

	Skiles	Stojakovic	Barry	Curry
Skiles	–	0.28	0.31	0.01
Stojakovic	0.72	–	0.59	0.04
Barry	0.69	0.41	–	0.01
Curry	0.99	0.96	0.99	–

than 9,000 free throws, and there is some probability that any or all of the other players will have scored more than 9,000.

Success rate comparisons

Table 1.6 shows the comparison probabilities. These results provide compelling evidence Curry is the best free throw shooter. His probability of having a better success rate is 99%, 97% and 99% in comparison to Skiles, Stojakovic, and Barry, respectively.

It is conceptually useful to understand why the posterior predictive analysis shows that it is unclear Curry will score the most free throws, but the comparison probabilities for the success rates show that Curry very likely has the best rate. The comparison probabilities are based on the underlying success rates shown in Figure 1.6. For example, the probability that Stojakovic's success rate is better than Curry's is the probability that Stojakovic's true success rate, with uncertainty described by the red distribution, falls above Curry's, with uncertainty described by the green distribution. This would only be the case if Stojakovic's success rate turned out to be at the upper end of what is possible and Curry's turned out to be at the low end of what is possible. Very roughly, Stojakovic's rate would need to be in the upper 10% and Curry's in the lower 10%. The probability of both of these independent outcomes

happening is $0.10 \times 0.10 = 0.01 = 1\%$, which is consistent with the 99% result in favor of Curry in Table 1.6.

The posterior predictive analysis, in contrast, is based on observable outcomes. This introduces additional uncertainty, in the form of what the outcomes will be. Even if it is known that a player's success rate is exactly 0.9, a wide range of outcomes could be observed after they have attempted 10,000 free throws. The posterior predictive distribution incorporates both the uncertainty about the underlying success rate in Figure 1.6 and the uncertainty about outcomes. The comparison of probabilities deals only with uncertainty about the underlying success rates, which is why it leads to a more confident conclusion.

There are technical terms for both of these sorts of uncertainty. The uncertainty about the underlying success rate is known as *epistemic* uncertainty. It is uncertainty that could be resolved with more information. If the players kept attempting free throws, the posterior distributions for their success rates would continue to narrow. Conceptually, after infinitely many free throw attempts, the posterior distribution would be a spike at the true success rate.

The uncertainty about outcomes is known as *aleatory* uncertainty. It is uncertainty that is inherent to the sampling process involved and cannot be eliminated. Even if the exact underlying success rate of a player is known, it is impossible to predict exactly how many free throws they will make from their sequence of attempts.

Given the fundamental conceptual distinction in the uncertainties involved, it is no surprise that there is no right or wrong analysis for comparing the free throw shooting abilities of the players. It depends what question is being asked. The question "who is inherently a better free throw shooter?" is about an underlying ability represented by the success rate, and the answer is very likely Curry. The question "who is likely to score more free throws?" incorporates additional uncertainty represented by the posterior predictive distribution. The answer is still probably Curry, but with less certainty.

1.5 *Student Grades*

`studentGrades.jasp`

Imagine a class of students taking a fifty-item exam. Three students have completed the exam, two are part way through the exam, and the rest have not started. This situation raises the new modeling issue of how to deal with students who have not started the exam and so have produced no observed data.

Data

Table 1.7 shows the observed performance of the students. The three students A, B, and C who have completed the exam got 39, 47, and 40 correct. The two students D and E who are part way through have both finished 10 questions, getting 8 and 5 correct. Student F is representative of all of the students who have not started the exam. They are listed

Student	Correct	Completed	Percentage
A	39	50	78%
B	47	50	94%
C	40	50	80%
D	8	10	80%
E	5	10	50%
F	0	0	–

Table 1.7: The student grades data.

as having 0 correct answers from 0 completed questions, and their percentage accuracy is undefined.

Results

Success rate posteriors

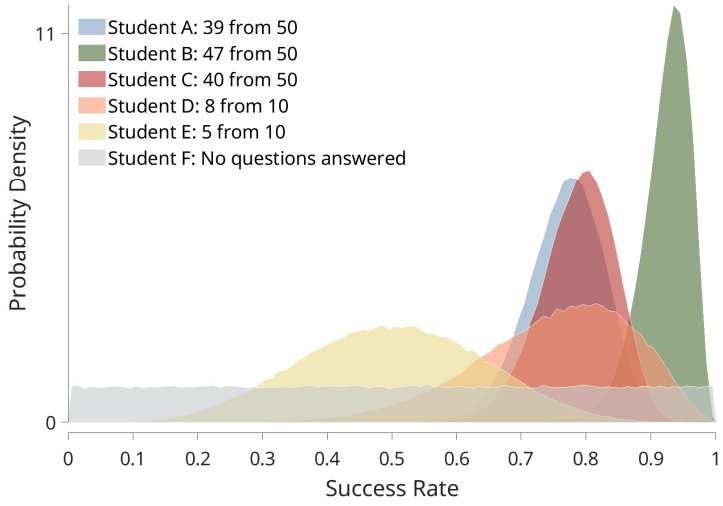


Figure 1.8: The success rate posterior distributions for the six students.

Figure 1.8 shows the posterior distributions for the success rates for the six students. The peaks of the posterior distributions are determined by the percentage of questions each student has answered correctly, while the widths of the distributions correspond to the number of questions they have answered. There is more uncertainty about the underlying success rate of students D and E who are part way through the exam than for students A, B, and C, who have completed it.

Posterior uncertainty is the greatest for student F. Because they have not answered any questions, there is no information that can be used to update their prior distribution to a posterior distribution. This means that the posterior distribution shown is the same as their prior distribution. It gives equal probability density to all possible success rates from 0 to 1.

Success count posterior predictions

Figure 1.9 shows the results of the posterior predictive analysis with $\hat{n} = 50$, corresponding to all of the students having completed the exam. Each panel corresponds to a student, showing the distribution of how many questions they are predicted to have answered correctly. For

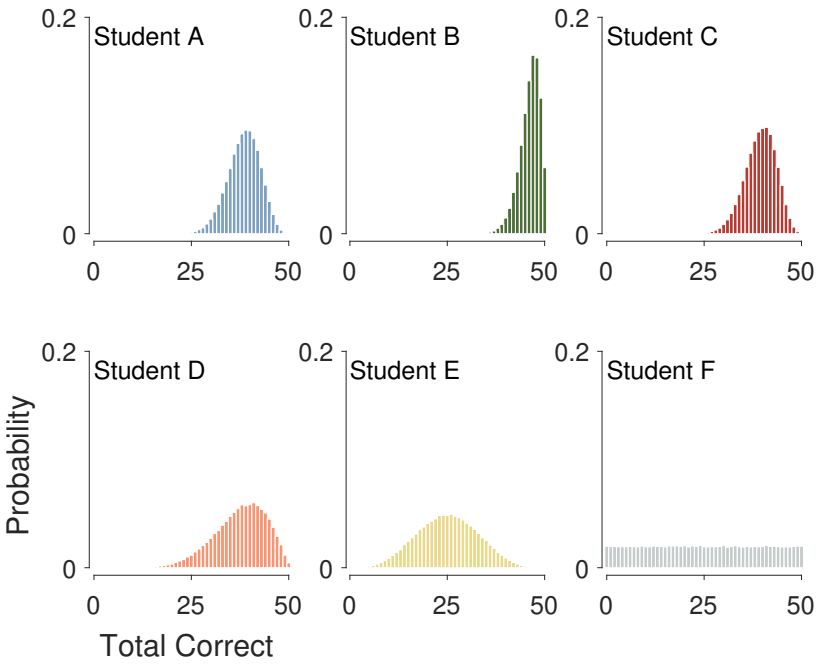


Figure 1.9: The posterior predictive distribution for each student after all 50 exam questions have been answered.

student F the prediction is that all possible totals from 0 to 50 correct answers are equally probable. For students D and E, who have answered only 10 questions, the most probable predicted totals reflect their current percentage accuracy. For example, because student E has answered 5 from 10 questions correctly, the peak of the posterior predictive distribution is 25 correct out of 50 questions.

Motivations for hierarchical models

The posterior and posterior predictive analyses for the student grades are useful for providing good concrete examples of the motivations for extending a model hierarchically. To understand these motivations, focus on the results for students E and F. Take a moment to think about the reasonableness of the inferences about their underlying rate of answering questions correctly and the predictions about their final scores. Stepping back from the details of modeling, just think about what you would intuitively expect students E and F to score once they have completed the exam.

♪ 𝅗𝅥 ♩ ♫ ♪ 𝅗𝅥 ♩ ♫

Some thinking music to help

A common intuition is that the inferences are not reasonable. It seems conceivable, even likely, that student E just got off to a bad start, and has an ability to answer questions correctly at a rate better than 0.5. It also seems likely that, even though student F has not answered any questions, that their success rate is more likely above 0.5 than below 0.5. If we were asked to predict student F's final score, it seems unlikely we would predict any total from 0 to 50 to be equally likely.

The disconnect between these intuitions and the results of the success rate model lie in the assumed independence of the students. If we learned that a lone student has 5 correct answers from 10 questions without any other context, the inferences for student E would seem reasonable. But the scores of the other students, especially those who have completed

the exam and scored 39, 47, and 40 out of 50, provides a relevant context. Their performance suggests that the exam is easy enough that students generally manage to perform well. As a consequence, a reasonable expectation is that student E got off to a bad start getting 5 questions wrong in the first 10, and that they are still likely to get more than 25 correct answers by the time they have completed the exam. The same ideas apply to student F, but even more strongly, since they have provided no observed data. All of our intuitions about their likely performance come from the observations that have been made of other students. The exam seems like one that students can do well in, so it is reasonable to expect student F to do well once they start answering questions.

By treating each student as independent of the others, the success rate model misses the context provided by other students. A hierarchical extension to the model introduces this context. The key idea is to connect the success rates of the students by introducing a group distribution that is effectively the class curve. This means that the inferred success rates take into consideration both the performance of the student and the plausible values allowed by the class curve. For example, the relatively poor initial performance of student E has to be reconciled with a class curve that, based on the performance of students A–D, expects relatively high success rates. The result is that the success rate for student E will be inferred to be higher than it would be using the standard success rate model. Similarly, the availability of the class curve allows a meaningful prediction to be made about the success rate of student F, and hence meaningful inferences about their future performance.

1.6 The Hierarchical Success Rate model

1.6.1 Motivation

The hierarchical extension of the success rate model provides an account of how multiple individuals are related to each other. The key idea is that the individuals are not independent of one another, but all belong to the same group distribution. The results from the *Student Grades* example provide a concrete example of the motivations for extending the success rate model hierarchically. These motivations are discussed in the previous Section 3.5.1. If you have not done so, the example is worth reading to understand the goals of hierarchical modeling.

The independence assumption of the original non-hierarchical success rate model is shown schematically in Figure 1.10. The total of N individuals have success rates $\theta_1, \theta_2, \ldots, \theta_N$ that generate their observed success counts $k_1, k_2, \ldots, k_N$. There is no interaction between the individuals. Applying the success rate model to multiple individuals—as in the online sellers, free throws, and student grades applications—effectively amounts to applying the model one at a time to each individual.

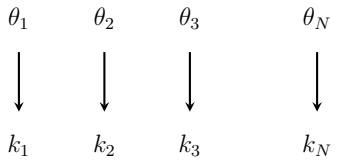


Figure 1.10: Conceptual representation of the independence assumption for modeling student grades.

A hierarchical extension of the success rate model is shown schematically in Figure 1.11. It adds the context that the individuals are related to one another by introducing a group distribution that links their success rates. The group distribution is represented by a Gaussian distribution with mean μ and standard deviation σ. There is still a success rate and success count for each of the individuals, but the introduction of the group distribution means that the success rates are no longer independent of one another. They are now connected by their common membership of the group.

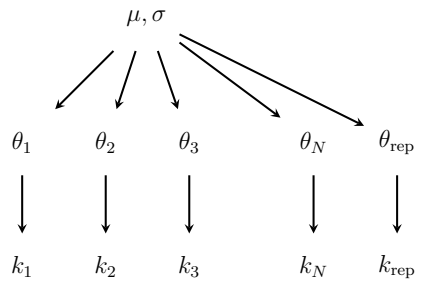


Figure 1.11: Conceptual representation of the hierarchical assumption for modeling student grades.

Figure 1.11 makes visually clear why the extension is called "hierarchical." The group distribution generates individual success rates that, in turn, generate observed success counts. There is no such psychological hierarchy in the independent structure shown in Figure 1.10 .

Because the hierarchical model makes different assumptions, it can provide new analyses, and produce different results.

- **Different inferences about success rates**: Hierarchical models make different inferences about success rates and different predictions about success counts. As Figure 1.11 shows, success rates are "sandwiched" between the group distribution from which they are generated, and the observed data they must explain. Inferences about success rates must simultaneously satisfy (or compromise) between these two requirements. A consequence of this property is that the behavior of each individual affects not only inferences about its own success rate, but also affects the inferences about the success rates of the other individuals.
- **New inferences about the group distribution**: The hierarchical model includes parameters that represent information above the level of individuals. The group mean provides information about the average success rate across all individuals, and the group standard deviation provides information about how much variability there is across individuals.
- **New inferences about a representative decision source**: The hierarchical model provides an account of how the success rates of individuals are generated. They are assumed to come from the group distribution. This allows inferences to be made about a *representative* individual for which no observed data are available. The success rate of this representative individual is shown in Figure 1.11 as θ_{rep}, and the prediction about its behavior is shown as k_{rep} successes from n_{rep} observations.

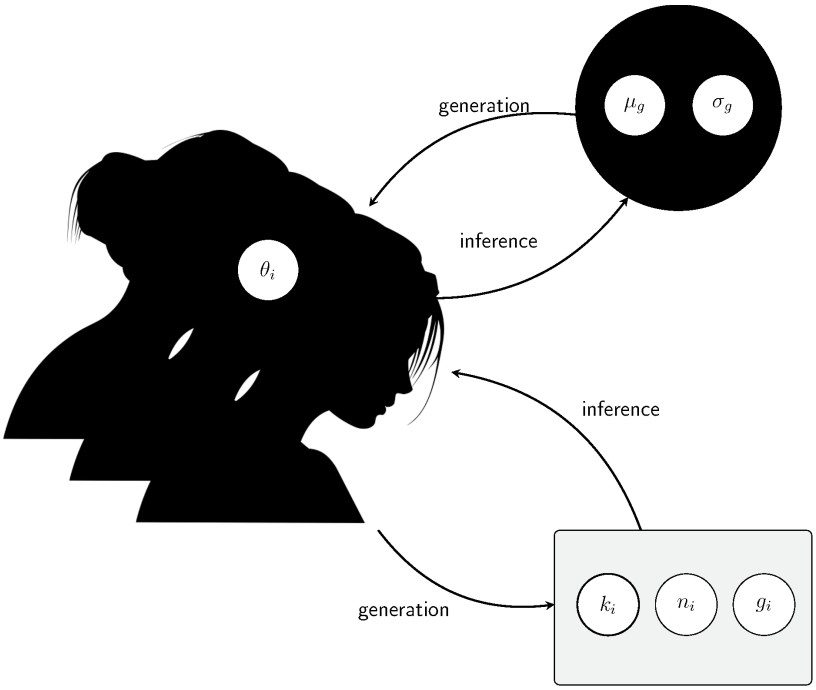


Figure 1.12: The structure of the hierarchical success rate model.

1.6.2 Model

Figure 1.12 summarizes the structure of the hierarchical success rate model. Multiple individuals are now shown. Once again, the word "individuals" is being used flexibly. Although the individuals are drawn as people in Figure 1.12, the individuals in applications could be individual experimental conditions, individual stimulus types, and so on.

Individual i has success rate θ_i. The success rates of multiple individuals are linked by an overarching group distribution with mean μ and standard deviation σ. Inference about the rates comes from the counts k_i of the number of successes out of n_i observations. Inferences about the success rates, in turn, allow inferences about the mean and standard deviation of the group distribution.

The hierarchical model allows for one additional level of structure. The individuals can be assigned to different groups by an observed variable g_i. All the individuals in the first group have $g_i = 1$, all the individuals in the second group have $g_i = 2$, and so on. If there is only one group, then every individual has $g_i = 1$. Each group has its own mean μ_g and standard deviation σ_g.

1.6.3 Mechanics

Table 1.8 lists the observed variables and latent parameters and variables for the hierarchical success model, giving both their statistical symbol and their JASP code name. The first five variables are the core parameters and data included in Figure 1.12. The core parameters are the success rates and the group means μ_g and standard deviations σ_g. The core observations are the success counts k_i and total n_i for each individual, and the group g_i to which they belong.

Name	Model	JASP
observed success count	k_i	`k[i]`
number of observations	n_i	`n[i]`
group	g_i	`group[i]`
success rate	θ_i	`theta[i]`
mean success rate for group	μ_g	`mu[g]`
standard deviation of success rate for group	σ_g	`sigma[g]`
predicted success count for individual	$\hat{k}_i$	`kPrd[i]`
representative success rate for group	$\tilde{\theta}_g$	`thetaRep[g]`
predicted success count for group	$\hat{\hat{k}}_g$	`kRepPrd[g]`
number of predicted observations	$\hat{n}$	`nPrd`

Table 1.8: The meaning of the parameters and variables in the hierarchical success rate model, with their model and JASP names.

Table 1.8 also lists additional parameters and variables. The inclusion of group distributions allow the success rate of a representative individual $\tilde{\theta}_g$ for each group to be inferred. This representative success rate is used, in turn, to make predictions about the number of successes $\hat{k}_g$ that occur out of $\hat{n}$ observations.

Input data

Table 1.9 demonstrates the general JASP data format, using the first rows of the data from the *US Elections* example. It includes the success count `k[i]`, the total number of observations `n[i]`, and the group member `group[i]` for each individual. The final column `nPrd` specifies the predicted success counts. This is a single number that only occupies the first row of the column.

k	n	group	nPrd
804,283	1,688,060	1	1000
690,080	1,688,060	2	
183,109	1,688,060	3	
102,000	258,506	1	
78,294	258,506	2	
73,481	258,506	3	
...	...	...	

Table 1.9: JASP data format for the hierarchical success rate model.

Output analyses

The JASP implementation of the hierarchical success rate model provides four analyses.

- **success rate posteriors**: The posterior distribution of the success rate `theta[i]` for each individual
- **success count posterior predictions**: The posterior predictive distribution of the success count `kPrd[i]` from `nPrd` observations for each individual
- **group posteriors**: The posterior distributions of the group means `mu[g]` and standard deviations `sigma[g]` for each group

- **representative success rates**: The posterior distribution of the success rate `thetaRep[g]` for the representative individual of each group
- **representative success counts**: The posterior predictive distribution of the success count `kRepPrd[g]` from `nPrd` observations for the representative individual of each group

1.7 *Hierarchical Grades*

`hierarchicalGrades.jasp`

The hierarchical grades application uses the same situation and data as the *Student Grades* example in Section 1.5. A class of students are taking a fifty-item exam. Students A–C have completed the exam, scoring 39, 47, and 40 correct. Students D and E are part way through the exam, with 8 and 5 correct from 10 questions. Student F has not started the exam.

Data

The data are provided in the *Student Grades* example in Table 1.7. Because all of the students belong to the same group, $g_i = 1$ for all of the students. The number of predictive observations remains set to $\hat{n} = 50$, corresponding to the completed exam.

Results

Success rate posteriors

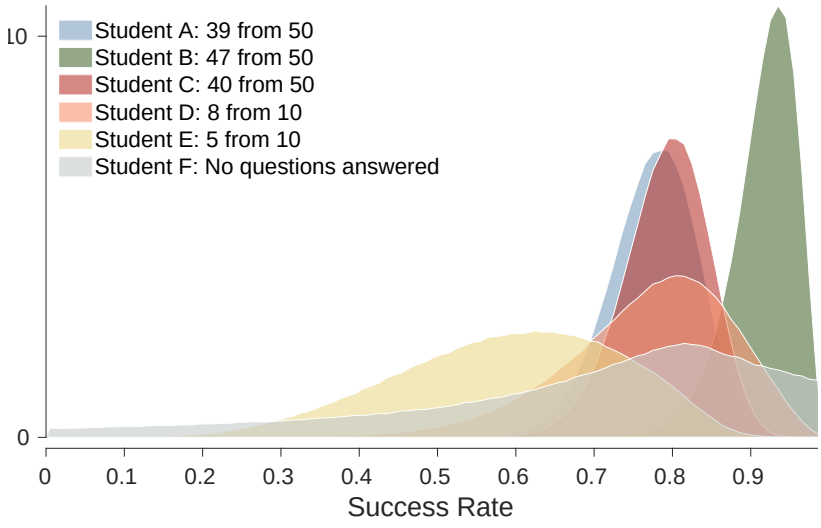


Figure 1.13: The success rate marginal posterior distributions for the six students, based on the hierarchical success rate model.

Figure 1.13 shows the marginal posterior distributions for the success rates θ_i for all of the students. For students A–D, the inferences are very similar to those obtained using the original non-hierarchical model in Figure 1.8.

The inference for student E, however, gives more probability density to higher rates than the non-hierarchical model. This is a good demonstration of the impact of hierarchical models on inference at the individual

level. The relatively poor initial performance of student E has to be reconciled with a class curve that, based on the performance of other students (especially students A–C, but also student D), expects relatively high success rates.

The success rate inference for student F is also very different from the uniform distribution inferred by the non-hierarchical model. Despite student F not having completed any questions, the hierarchical model has expectations about which success rates are more or less likely through their belonging to the group distribution. The posterior distribution for student F still shows large uncertainty, with all values between 0 and 1 being plausible, but there is more probability density for values above 0.6. As per the discussion motivating hierarchical models in Section 3.5.1, the inferences for students E and F seem more intuitively reasonable than those made by the non-hierarchical model.

One way to interpret the agreement between the non-hierarchical and hierarchical models for the other students is that, for students A–C, their completion of the exam provides a lot of information about their success rate. Since the observed data are the same for both the non-hierarchical and hierarchical models, the inferences about their rates are very similar. Student D does not provide as much information, but the information they do provide is consistent with the group distribution, and so there is no need for reconciliation. This means the hierarchical model makes a similar inference to the non-hierarchical model for these students, although the posterior distributions are a little narrower using the hierarchical model, because the additional consistency with the group distribution makes the inference a little more certain.

Success count posterior predictions

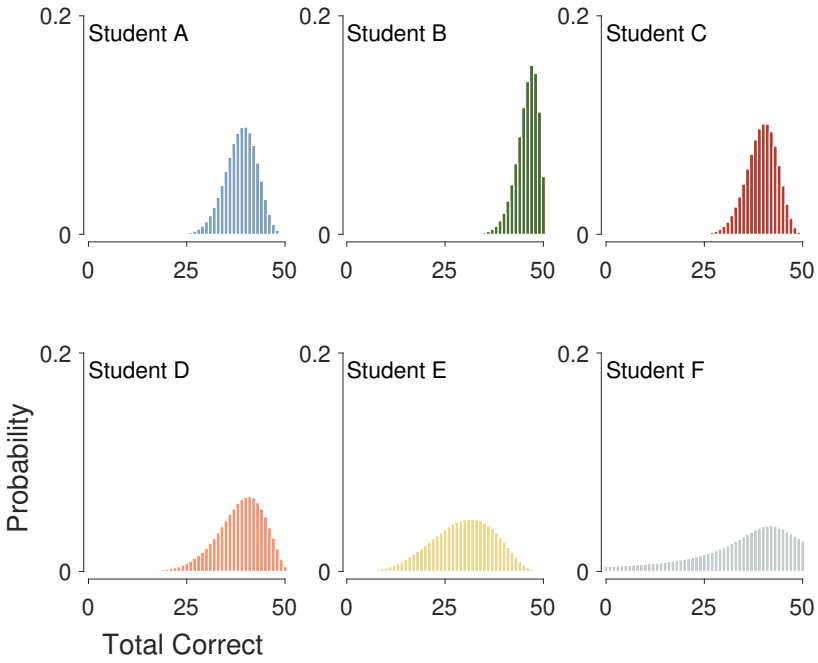


Figure 1.14: The posterior predicted success counts for the six students, based on the hierarchical success rate model.

Figure 1.14 shows the results of the posterior predictive analysis for 50 questions, corresponding to all of the students having completed the exam. Each panel corresponds to a student, showing the distribution

of how many questions they are predicted to have answered correctly. The distributions for students A, B, and C are very similar to those found using the non-hierarchical model. For students D and E, who have answered only 10 questions, the most probable predicted totals are now higher than those found using the non-hierarchical model. The predictions do not simply follow the current percentage accuracy of the students. For example, the peak of the distribution for student E, who has answered 5 from 10 questions correctly, is now above 30 rather than 25. This increase comes from the incorporation of the information that they are part of a class curve.

Group posteriors

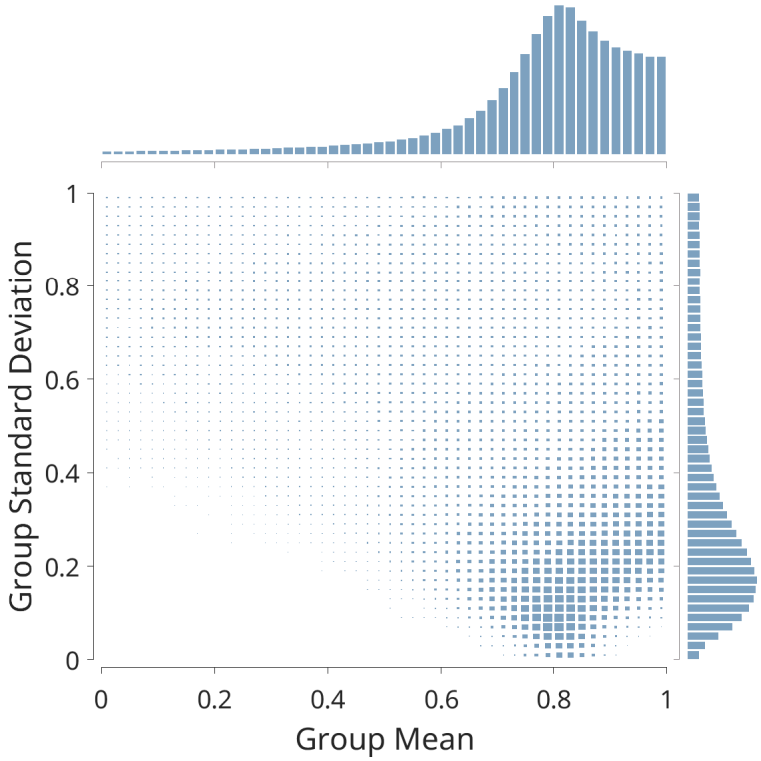


Figure 1.15: The joint and marginal posterior distributions of the group mean μ and group standard deviation σ for the student grades data using the hierarchical success rate model.

Figure 1.15 shows the joint and marginal posterior distribution for the group mean μ and standard deviation σ. The x-axis corresponds to the mean and the y-axis corresponds to the standard deviation. The shading in the central panel shows the probability density of each possible combination of a mean and standard deviation. In other words, it shows how likely it is that every possible (truncated) Gaussian distribution is to be the class curve. The heavier the shading, the more likely that combination is. Above and to the right of the joint distribution are the marginal distributions. The one above the x-axis shows the uncertainty about mean and the one to the right of the y-axis shows the uncertainty about standard deviation.

It is clear from the joint posterior distribution that the most likely class curve has a mean around 0.8 and a standard deviation around 0.1. It is also possible the mean could be a little higher, closer to 0.9, combined with a larger standard deviation around 0.2. The marginal distributions provide

simple summaries of the uncertainty about the mean and standard deviation, but miss some of the nuance provided by the joint distribution. In particular, the marginal distributions do not capture the information that if the mean is around 0.8 the standard deviation is likely to be smaller, but if the mean is higher the standard deviation is also likely to be higher. This is a concrete demonstration of the general point about the relationship between joint and marginal distributions discussed in Section 0.2.1.

Representative success rates and counts

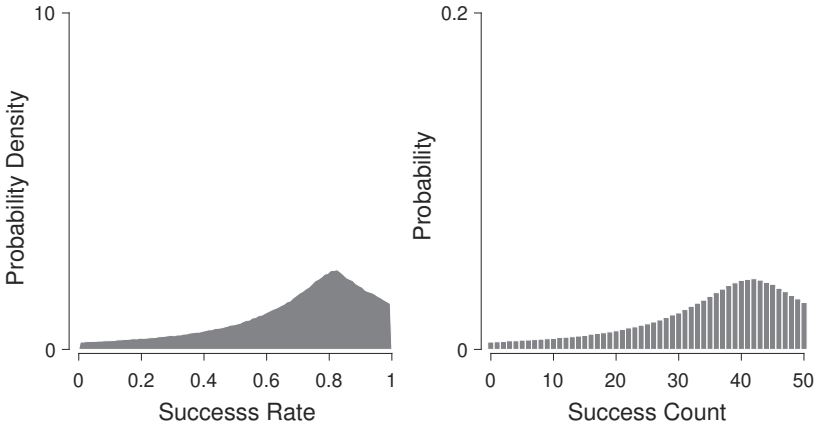


Figure 1.16: The (left panel) representative success rate and (right panel) predicted success count for the student grades.

The presence of a group distribution allows a representative individual to be modeled. Figure 1.16 focuses on the results for student F. Because student F has not started the exam, their posterior rate of success in Figure 1.13 and posterior predictive analysis of future success counts in Figure 1.14 corresponds to a representative analysis. The left panel shows the representative rate $\tilde{\theta}$ and the right panel shows the corresponding predicted success count $\hat{k}$ after 50 questions. This highlights the meaning of the representative analysis. It provides inferences and predictions about a representative student, for whom there are no data. It is natural to think of this representative student as a "random" student in the class, or perhaps as the next student to sit down and take the exam.

1.8 *Movie Ratings*

`movieRatings.jasp`

Rotten tomatoes is a review aggregation website for movies. Both general users and expert movie critics can rate movies as "fresh" or "rotten." This application involves a data set collected from the rotten tomatoes site, and focuses on one movie critic, Richard Mowe who provided 145 ratings from 2004 to 2012.

The research question is to understand the success rate with which Mowe provides positive ratings in each year. A hierarchical model is appropriate, treating success rates for each individual year as linked by a group distribution that corresponds to Mowe's overall disposition towards providing fresh ratings. Note that this is an application of the hierarchical success rate model that does not focus on individual differences between people. There is only person generating the data.

The hierarchical model captures the relationship across different years, modeling the shared context that the same person is generating the observed ratings for each year.

Data

Year	Fresh	Total	Percentage
2004	4	4	100%
2005	0	1	0%
2006	20	24	83%
2007	17	19	89%
2008	26	27	96%
2009	26	32	81%
2010	20	21	95%
2011	10	15	67%
2012	2	2	100%

Table 1.10: The Richard Mowe movie ratings data.

Table 1.10 lists the number of fresh ratings and the total number of movies by Mowe, for each year from 2004 to 2012. Mowe generally provides a high percentage of fresh ratings. His 86% fresh rate over the period contrasts with a 64% fresh percentage (720,210 fresh from 1,130,017 total ratings) for all of the critics in the data set.

Results

Success rate posteriors

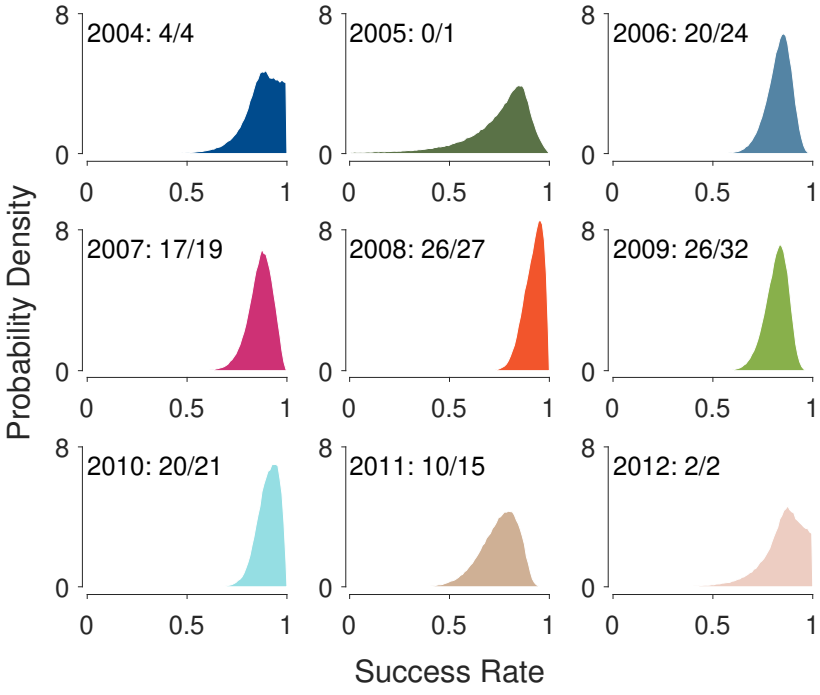


Figure 1.17: The success rate marginal posterior distributions for all years, based on the hierarchical success rate model.

Figure 1.17 shows the marginal posterior distributions for the success rates θ_i for every year. The striking result is that high success rates are very likely for all of the years, including ones like 2005 in which only one

movie was rated (and was evaluated as rotten). For years in which fewer movies were rated there is more uncertainty, but the overall inference is that high rates of fresh ratings are consistently expected.

Group posteriors

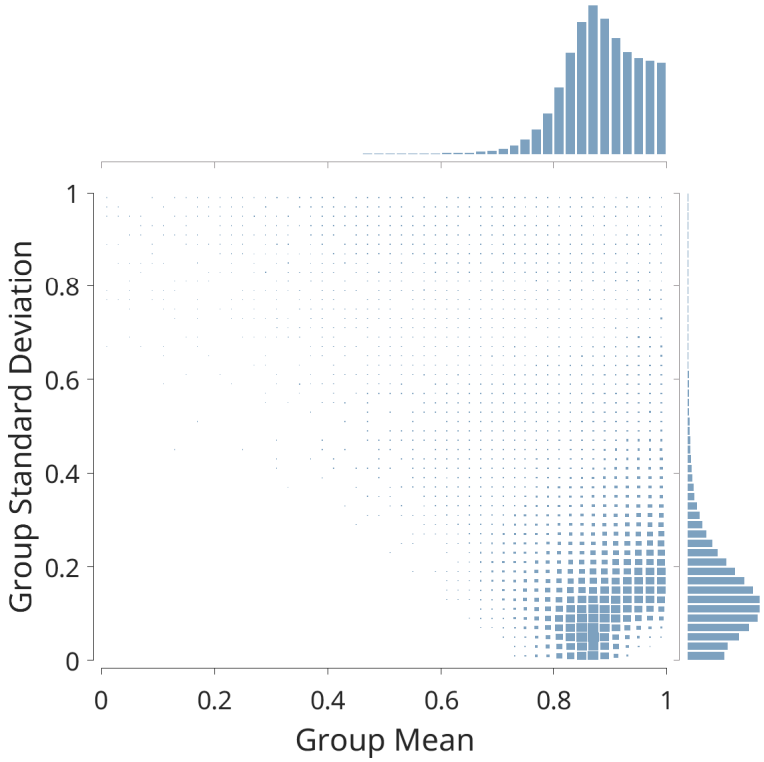


Figure 1.18: The joint and marginal posterior distributions of the group mean μ and group standard deviation σ for the movie ratings data using the hierarchical success rate model.

Figure 1.18 shows the joint and marginal posterior distribution for the group mean μ and standard deviation σ. The joint posterior gives high density to group distribution with high means and low standard deviations. The most likely possibilities are that Mowe has a mean rate of about 0.85 in providing fresh ratings, with almost no variability from year to year. It is also possible that the mean rating is even higher, but with a little variability.

These group inferences explain the pattern of inferences about individual years in Figure 1.17. The way to understand this pattern of result is that Mowe consistently provides fresh ratings. The hierarchical model captures the information Mowe consistently provides fresh ratings. The group distribution then "fills in the gaps" in years with few movie ratings, leading to the inference that, even in those years, the underlying success rate likely remains high.

Representative success rates and counts

Figure 1.19 shows the results of the representative analysis. The left panel shows the representative rate $\tilde{\theta}$. It is very likely Mowe's underlying rate of giving fresh ratings is at least 0.6 in a representative year, and it is likely the rate will be 0.8 or higher. The right panel shows the corresponding predicted success count $\hat{k}$ after $\hat{n} = 20$ movies are rated. One way to

summarize these predictions is that there is about a 77% probability of 15 or more fresh ratings from a year in which 20 movies are rated. This is the sum of the probabilities from 15 to 20 fresh ratings.

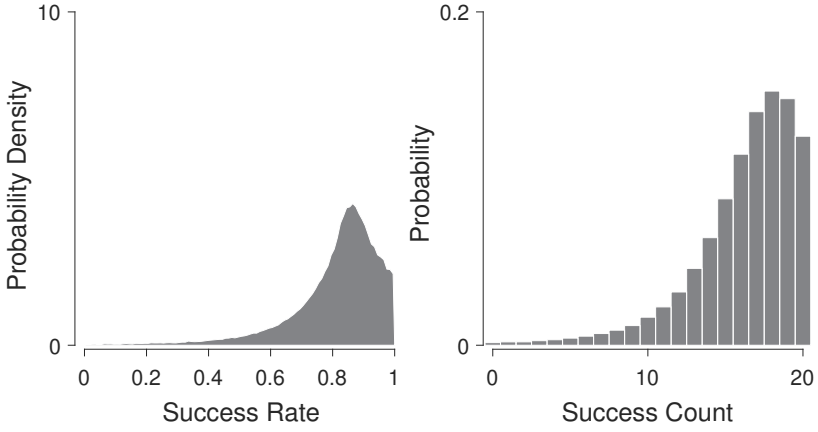


Figure 1.19: The (left panel) representative success rate and (right panel) predicted success count for the movie ratings using the hierarchical success rate model.

1.9 *US Election*

`usElection.jasp`

This example involves voting for four candidates in the 1992 US presidential election (MIT Election Data and Science Lab, 2017). The candidates are Bill Clinton, George H. W. Bush, Ross Perot, and Earl Farwell Dodge Jr. Clinton and Bush were the nominees of the Democratic and Republican party, respectively, and Perot was a prominent independent candidate. All three appeared on the ballots in all 50 US states and the District of Columbia. Dodge represented the prohibition party and, according to the data, appeared on only two state ballots.*

The research question is to understand the success rate with which each candidate attracted votes from the states and overall. This is non-trivial, because the states vary widely in terms of the number of voters, and Dodge is different from the other candidates by being involved in only two state votes.

Data

Table 1.11 shows some of the data. The first three rows correspond to the 1,6888,060 votes cast in Alabama, with Bush receiving 804,283, Clinton receiving 690,080, and Perot receiving 183,109. The next three rows correspond to the votes for the same three candidates in Alaska. The bottom four rows correspond to New Mexico and show that Dodge received 120 of the 569,986 votes cast. Overall, there are $3 \times 51 + 2 = 155$ rows in the data, each giving a success count of votes received out of a total number of votes cast for one of the four candidates.

Results

The hierarchical success rate model is applied to the data by treating each vote for a candidate in a state as an "individual," and assuming each

* This conflicts with other sources, in terms of both the number of states and which states were involved. But it is clear Dodge was on few ballots and received few votes.

Candidate	Votes	Total	State
Bush	804,283	1,688,060	AL
Clinton	690,080	1,688,060	AL
Perot	183,109	1,688,060	AL
Bush	102,000	258,506	AK
Clinton	78,294	258,506	AK
Perot	73,481	258,506	AK
...	...	...	...
Bush	261,617	569,986	NM
Clinton	212,824	569,986	NM
Perot	91,895	569,986	NM
Dodge	120	569,986	NM
...	...	...	...

Table 1.11: The structure of the US Elections data.

comes for a group distribution corresponding to the candidates. This means that there are four groups, with 51 individuals in the first three groups corresponding to Bush, Clinton, and Perot, and two individuals in the fourth group corresponding to Dodge.

Using the model in this way, the interesting level of analysis is not at the individual level of the 155 observations for specific candidates in specific states. It is at the group level of the candidates themselves, which integrated across all of the state-level information.

Group posteriors

Figure 1.20 shows the joint and marginal posterior distribution for the group means μ_g and standard deviations σ_g. The center panel shows the joint posterior distributions , the top panel shows the marginal posterior distributions of the means, and the right panel shows the marginal posterior distributions of the standard deviations.

It is likely that the mean success rate for Clinton is a little higher than for Bush, and it is clear that Perot is lower than both. In the context of the election, this can be interpreted as meaning Clinton is likely to win the national vote, with Bush the only other possibility. The inferences about the standard deviations show that Clinton, Bush, and Perot are similarly variable in their success rates across states. The marginal posterior distributions show some certainty that all have standard deviations around 0.1.

The group inferences for Dodge are different. Both the marginal posteriors for the mean and standard deviation peak at small values but have long tails across the entire range of possible values. This means that it is most likely that his mean success rate is small and has low variability across states. But, because of the lack of information, there is considerable uncertainty, and other values are possible.

Representative success rates

Figure 1.21 shows the representative success rate for each candidate. They can be interpreted as the distribution of possible success rates for the candidate in a random state. Or, if a new state was added to the US

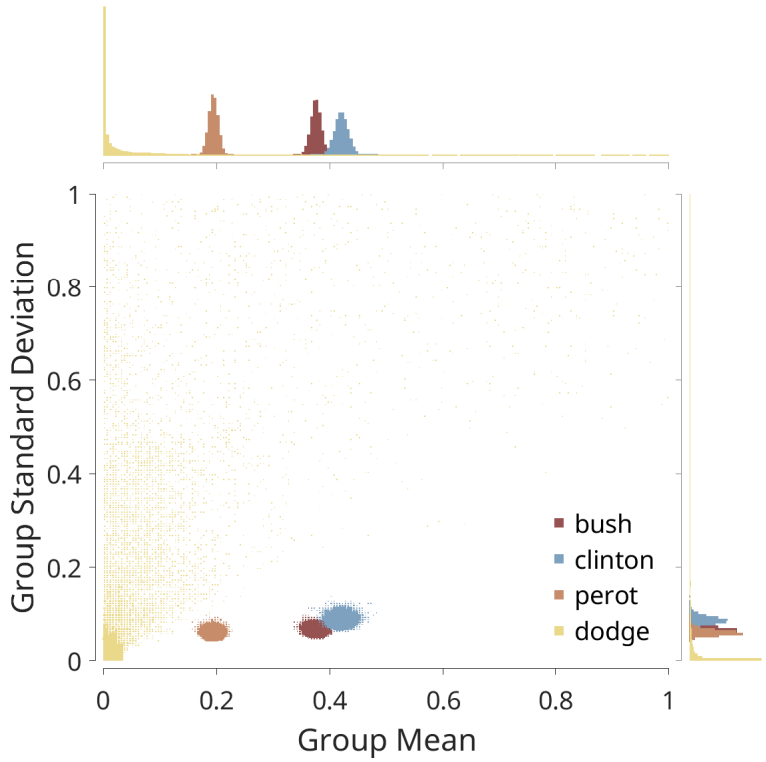


Figure 1.20: The joint and marginal posterior distributions of the group mean μ and group standard deviation σ for the hierarchical success rate model analysis of the US Election data.

election elections, the candidate's success rate in receiving votes in that state.

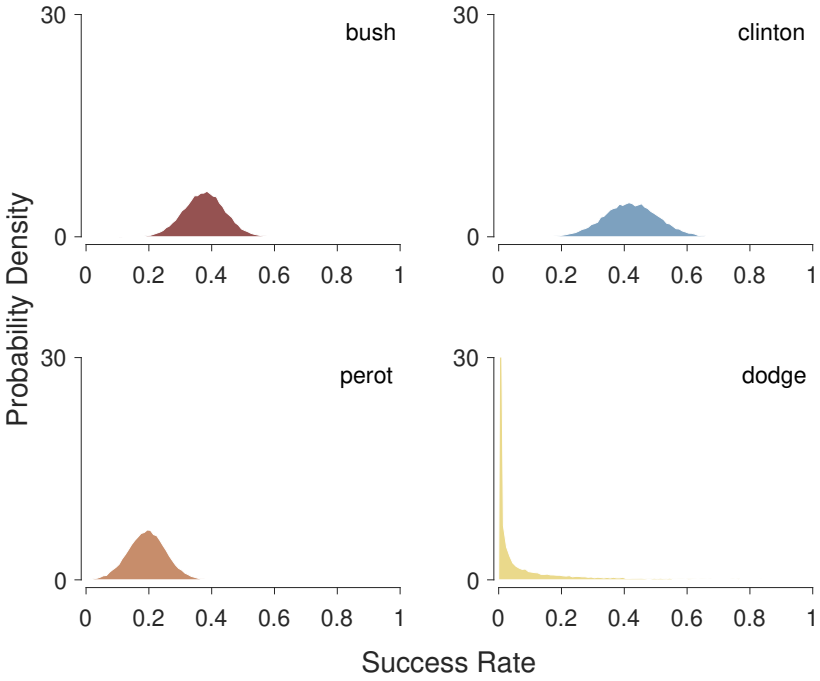


Figure 1.21: The representative success rate for the four presidential candidates, based on the hierarchical success rate model.

The results emphasize the variability across states in the success rates for Bush, Clinton, and Perot. For example, Clinton's average success rate is about 0.40, but it varies as low as about 0.20 and as high as about 0.60. This is consistent with the variability in the popularity of candidates across different states. The overlap in the posterior representative distributions of Bush and Clinton shows that either could receive the most votes in

different states, and there is also some overlap with Perot. For example, it is possible that there is a state for which Perot has a 0.30 success rate but both Clinton and Bush are closer to 0.20.

The posterior representative distribution of Dodge shows different results. His most likely success rate is very low, but there is some probability that it could be as high as 0.20. The low expectation comes from the small number of votes he received in the two states with available data. The uncertainty comes from the lack of additional information from other states.

It is interesting to compare the group inferences about success rates in Figure 1.20, which naturally relate to the overall popular vote, with the representative individual inferences about success rates, which naturally relate to individual states. For example, Figure 1.20 shows that Perot has a lower mean success rate than both Clinton and Bush, but the representative distribution in Figure 1.21 shows that he has a non-negligible probability of being able to win a state vote.

◇

1.10 Formal Details

1.10.1 The Success Rate model

Priors

The success rate model assumes the probability of success for individual i is equally likely to take any possible value, and so the uncertainty about its value is represented by the uniform distribution from 0 to 1. In statistical notation, this is written as

$$\theta_i \sim \text{uniform}\big(0, 1\big), \tag{1.1}$$

where the $\sim$ symbol can be read "is distributed as."

Likelihood

The model assumes that the process that generates the success count k_i out of n_i observations for individual i is the binomial distribution. Statistically, this is written as

$$k_i \sim \text{binomial}\big(\theta_i, n_i\big). \tag{1.2}$$

The likelihood acts as the bridge between latent parameters and observable data, and serves two important purposes. The first is that it allows inference, so that the information provided by observing k_i successes out of n_i observations can be incorporated into the uncertainty about θ_i. The second is that it allows prediction, so that what is known about θ_i can be translated into uncertain expectations about the probability of

different predicted success counts $\hat{k}_i$ from $\hat{n}_i$ observations. Quantifying these predictions simply uses the likelihood the same way, so that

$$\hat{k}_i \sim \text{binomial}\big(\theta_i, \hat{n}_i\big). \tag{1.3}$$

Pairwise comparison

The probability that the success rate for individual i is greater than the success rate for individual j is defined as

$$\pi_{ij} = \Pr\big(\theta_i > \theta_j\big). \tag{1.4}$$

The π_{ij} comparison probability is a deterministic function of the θ_i and θ_j success rate parameters.

JASP Implementation

The JAGS code for the success rate model is shown below.

```
# Success Rate
data{
  nIndividuals = length(k)
}
model{
  for (i in 1:nIndividuals){
    # Likelihood
    k[i] ~ dbin(theta[i], n[i])
    # Prior
    theta[i] ~ dunif(0, 1)
    # Predictive
    kPrd[i] ~ dbin(theta[i], nPrd[i])
    # Probability each rate is greater than each other
    for (j in 1:nIndividuals){
      pi[i, j] = step(theta[i] - theta[j])
    }
  }
}
```

The JAGS code starts with a data block to determine the number of individuals `nIndividuals`.

The main `model` block is organized in terms of the

- **likelihood** that connects the success counts `k[i]` to the success rate `theta[i]` and number of observations `n[i]` for each individual
- **prior distribution** for `theta[i]`
- **predictive distribution** for generating `kPred[i]` from `theta[i]` and `nPrd[i]`
- **pairwise comparison** of success rates `pi[i,j]`

1.10.2 The Hierarchical Success Rate model

Priors

The hierarchical success rate model has priors for the mean and standard deviation of the group distributions. Both are assumed to be uniform

from 0 to 1, giving prior density to every possible value

$$\begin{aligned} \mu_g &\sim \text{uniform}(0, 1) \\ \sigma_g &\sim \text{uniform}(0, 1). \end{aligned} \tag{1.5}$$

Likelihood

Because of the hierarchical extension, the success rates are generated from their group distribution. This means they are defined as part of the likelihood. Specifically, each θ_i is generated from the Gaussian group distribution determined by its group membership g_i. The value is truncated to lie in the range from 0 to 1 to ensure it is a valid success rate:

$$\theta_i \sim \text{Gaussian}_{(0,1)}\left(\mu_{g_i}, \frac{1}{\sigma^2_{g_i}}\right). \tag{1.6}$$

The success rate for the representative individual is generated in exactly the same way:

$$\tilde{\theta}_g \sim \text{Gaussian}_{(0,1)}\left(\mu_g, \frac{1}{\sigma^2_g}\right). \tag{1.7}$$

The posterior predictive success count $\hat{k}_i$ out of $\hat{n}$ observations for individual i is given by:

$$\hat{k}_i \sim \text{binomial}\left(\theta_i, \hat{n}\right). \tag{1.8}$$

The posterior predictive success count $\hat{k}_g$ also out of $\hat{n}$ observations for the representative individual from group g is based on the success rate $\tilde{\theta}_g$, and is generated the same way:

$$\hat{\tilde{k}}_g \sim \text{binomial}\left(\tilde{\theta}_g, \hat{n}\right). \tag{1.9}$$

JASP implementation

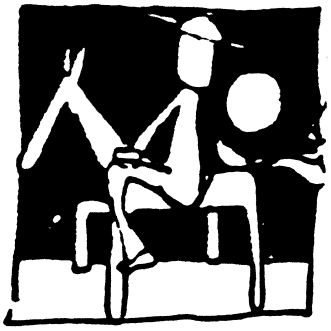
The JAGS code for the hierarchical success rate model is shown below.

```
# Hierarchical Success Rate
data{
  nIndividuals = length(k)
  nGroups = max(group)
}
model{
  # Individuals
  for (i in 1:nIndividuals){
    k[i] ~ dbin(theta[i], n[i])
    theta[i] ~ dnorm(mu[group[i]], 1/sigma[group[i]]^2)T(0, 1)
    kPrd[i] ~ dbin(theta[i], nPrd[1])
  }
  # Groups
  for (g in 1:nGroups){
    mu[g] ~ dunif(0, 1)
    sigma[g] ~ dunif(0, 1)
    # Representative
    thetaRep[g] ~ dnorm(mu[g], 1/sigma[g]^2)T(0, 1)
    kRepPrd[g] ~ dbin(thetaRep[g], nPrd[1])
  }
}
```

The JAGS code starts with a data block to determine the number of decision sources `nIndividuals` and the number of groups `nGroups`.

The main `model` block has a loop over the `nIndividuals` that defines the likelihood of the observed success counts `k[i]`, the generation of individual success rates `theta[i]` from the appropriate group `g[i]`, and the posterior predicted individual success counts `kPrd[i]` from `nPrd` observations.

There is also a loop over the `nGroups` that defines the group prior distributions for the group means `mu[g]` and standard deviations `sigma[g]`, the generation of the representative individual `thetaRep[g]` for each group, and the posterior predicted representative individual success counts `kRepPrd[g]` from `nPrd` observations.



Two-Choice Signal Detection 2

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2.1 The Two-Choice Signal Detection Model

2.1.1 Motivation

Signal detection theory (Green & Swets, 1966; Macmillan, 2001) is a widely used framework for data analysis in science and engineering. As a cognitive model, signal detection theory deals with the general situation in which people have to decide whether or not there is a "signal" or "target" in the environment. As one concrete example imagine you are a submarine sonar operator and your job is to monitor the outside surrounding environment trying to distinguish between a signal that indicates a missile is coming towards you, and just general noise caused by fish in the seabed or normal currents or something that does not require action. Or imagine that your job is to spot forged paintings. The signal in this case is the presence of a forgery among genuine paintings.

In both of these situations there are two possible states of the world. There is either really a "signal," a missile or a forgery, or there is only what is termed "noise," which is just the normal environment absent any signal. The decision to be made is whether a signal is present or not. This leads to the 2×2 table with four possibilities shown in Table 2.1.

	really signal	really noise
detect signal	hit	false alarm
detect no signal	miss	correct rejection

Table 2.1: The four outcomes in a signal detection task.

The four possibilities are:

- If the decision is that there is a signal when there really is one, the decision is called a *hit*. This corresponds to there really being a missile and you detecting it, or a painting really being a forgery and you spotting it.
- If the decision is that there is a signal when there really is not, the decision is called a *false alarm*. This corresponds to thinking you detected a missile when there really was not one, or claiming a painting is a forgery when it is genuine.
- If the decision is that there is no signal when there really is one, the decision is called a *miss*. This corresponds to failing to detect an incoming missile, or not noticing that a painting is a forgery.
- If the decision is that there is not a signal and there really is not, the decision is called a *correct rejection*. This corresponds to deciding there is no missile when there is not one, or correctly believing a painting to be authentic.

Signal detection theory provides a modeling framework for understanding and interpreting patterns of hits, false alarms, misses, and correct rejections in psychological terms. It does this by making assumptions about both stimulus representation and decision making.

The representational assumption leads to the concept of *discriminability*. Signal detection theory assumes there is a uni-dimensional latent continuum on which stimuli are experienced. The idea is that whatever perceptual and cognitive processes are involved in detecting missiles, spotting forgeries, or generally identifying target stimuli can be summarized in terms of a number that measures "strength" of evidence for a signal. That strength is called discriminability. It is a measure of how far the signal and noise distribution are separated from each other, and is often a property of the stimulus.

The decision-making assumption leads to the concept of *bias*. Signal detection theory assumes there is a criterion level of strength that determines whether or not a signal is present. Given that perfect discriminability is rarely possible, it is inevitable that errors will be made. A tendency to believe signals are present will lead to many false alarms but few misses. This would correspond to deciding that every hint of a missile was in fact a missile. A tendency to believe signals are not present will lead to many misses but few false alarms. This would correspond to discounting the possibility of forgery and deciding paintings are usually genuine. This decision-making tendency is called bias, and is usually under some level of conscious control.

2.1.2 Model

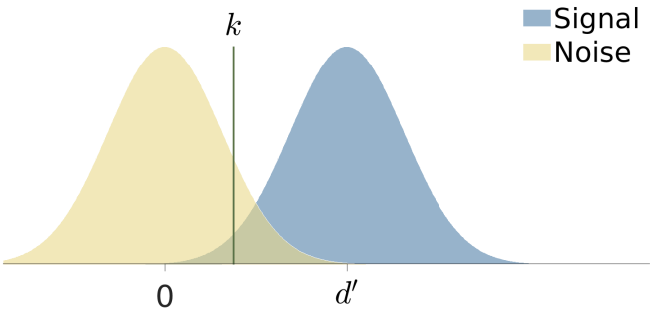


Figure 2.1: The assumptions of the two-choice signal detection model.

Figure 2.1 shows the representational and decision-making assumptions for a two-choice signal detection model. The signal and noise distributions have equal variance and are separated by a discriminability d' (pronounced "d prime"). A decision criterion k is also shown. On each trial, the model assumes a strength value for the current stimulus is sampled from either the signal or noise distribution, depending on whether or not the stimulus really does include a signal. If the sampled strength is above a decision criterion k, the decision is made that a signal is present.

Figure 2.2 shows how the probabilities of hits, false alarms, misses, and correct rejections are determined by the discriminability d' and the criterion k parameters. If there really is a signal, and the sample is greater than the criterion, then a hit will result, because there was a signal and the "detect signal" decision is made. That corresponds to the blue area. If the sample is smaller than the criterion, then a miss will result. That corresponds to the green area. Together the hit and miss probabilities have to sum to one, since they are the only two possibilities when a

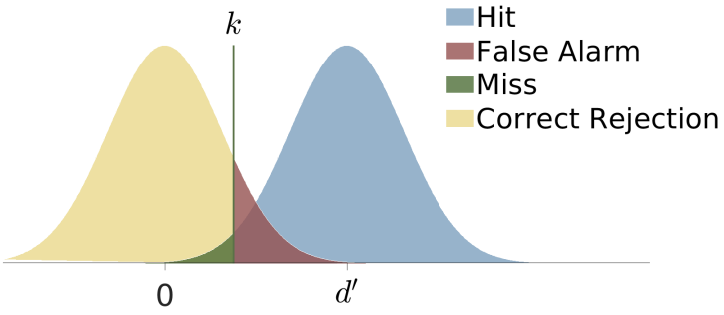


Figure 2.2: Hit, false alarm, miss, and correct rejection decision probabilities according to the two-choice signal detection model.

signal is present. Geometrically, the area under the signal distribution is one, and the blue and green areas partition it into the complementary hit and miss probabilities. The same logic applies if there is no signal, and the sample is drawn from the noise distribution. Samples greater than the criterion will lead to false alarms, and correspond to the red area. Samples smaller than the criterion will lead to correct rejections, and correspond to the yellow area. The sum of false alarms and correct rejections is also one.

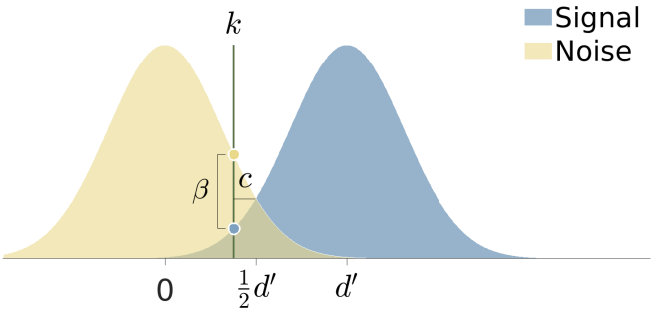


Figure 2.3: Measures of bias in the two-choice signal detection theory model.

Figure 2.3 shows how d' and k can be used to measure bias. One way is to compare the criterion value k to the value $\frac{1}{2}d'$. Exactly halfway between the means of of the signal and noise distributions, at $\frac{1}{2}d'$, is where the two distributions have equal probability. That makes the value $\frac{1}{2}d'$ an unbiased criterion, because it guarantees an equal probability for both false alarms and misses. Accordingly, the difference $c = k - \frac{1}{2}d'$ provides a measure of bias. The criterion is unbiased if $c = 0$. If the criterion is smaller than the unbiased value, the bias c becomes negative, it becomes more like to make "detect signal" responses, and false alarms become more likely than misses. If the criterion is greater than the unbiased value, the bias c becomes positive, it becomes more like to make "detect noise" responses, and misses become more likely than false alarms.

Another way to measure bias is in terms of the relative probability of signal versus noise at the criterion value. The relative heights of the signal and noise distributions provide this relative probability. In Figure 2.3 the noise distribution is about three times higher than the signal distribution at the criterion value k. This means that the probability is three times greater that a sample with strength k will correspond to noise rather than signal. Thus, setting the criterion here corresponds to a prior bias in favor of the "detect noise" response. This relative bias β compares the two heights shown in Figure 2.3.

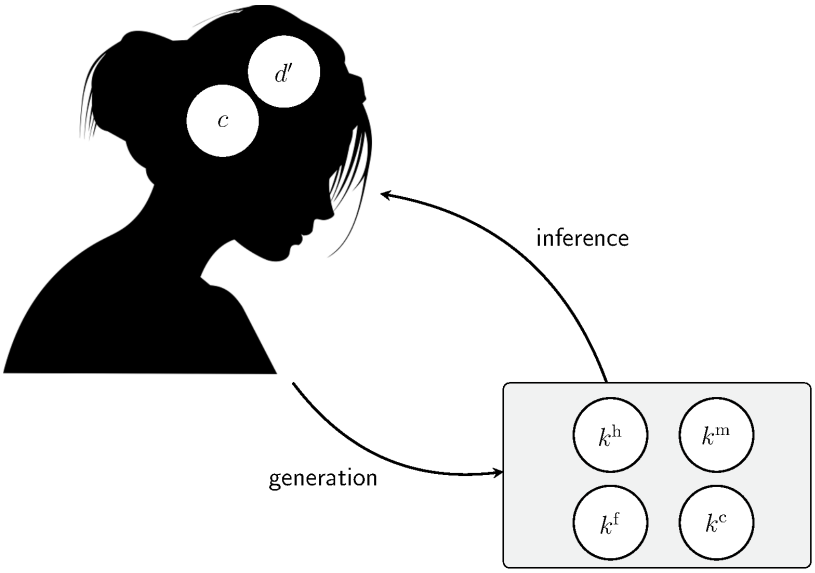


Figure 2.4: The structure of the two-choice signal detection model.

Figure 2.4 summarizes the structure of the two-choice signal detection model, showing how it relates the discriminability d' and the bias c parameters to the observed counts of hits k^{h}, false alarms k^{f}, misses k^{m}, and correct rejections k^{c}.* Inference about d' and c come from the counts. The model can use what is inferred about d' and cs to make predictions about how many hits $\hat{k}^{\mathrm{h}}$ will be observed in $\hat{n}^{\mathrm{s}}$ signal trials, and how many false alarms $\hat{k}^{\mathrm{f}}$ will be observed in $\hat{n}^{\mathrm{n}}$ noise trials. These predicted counts immediately also determine predictions about misses and correct rejections, since every signal trial must produce either a hit or a miss, and every noise trial must produce either a false alarm or a correct rejection.

The bias parameter c is shown in Figure 2.4 because it is the most commonly used, but it would also be reasonable to use either the criterion k or relative bias β parameters. Any of these three parameters paired with d' define the two-choice signal detection model. One reason to prefer the bias c or relative bias β over the criterion k is that they are often more independent of the discriminability d'. Whether bias c or relative bias β is more useful varies according to context, as the worked examples demonstrate.

2.1.3 Mechanics

Table 2.2 details the meaning of the parameters and variables in the signal detection theory model, their name in the model as shown in Figure 2.4, and their name in JASP.

The first set of variables in Table 2.2 are the core parameters and data included in Figure 2.4. These are the discriminability d' and the bias c parameters, and the four signal detection data counts. The second set of variables include the alternative criterion k and relative bias β measures of decision making, and the variables associated with posterior predictive analysis.

* It is unfortunate that the standard symbol the literature uses for the criterion is k, and that this clashes with the hit, false alarm, miss, and correct rejection counts k^{h}, k^{f}, k^{m}, and k^{c}. They can be distinguished by the labeling superscripts for the counts.

Name	Model	JASP
discriminability	d'	`d`
bias	c	`c`
hit count	k^{h}	`hit`
false alarm count	k^{f}	`fa`
miss count	k^{m}	`miss`
correct rejection count	k^{c}	`cr`
criterion	k	`k`
relative bias	β	`beta`
predicted hit count	$\hat{k}^{\text{h}}$	`hitPrd`
predicted false alarm count	$\hat{k}^{\text{f}}$	`faPrd`
number of signal observations for prediction	$\hat{n}^{\text{s}}$	`nPrdSig`
number of noise observations for prediction	$\hat{n}^{\text{n}}$	`nPrdNoi`

Table 2.2: The meaning of the observed variables and latent parameters in the two-choice signal detection model, with their model and JASP names.

Input data

Table 2.3 shows the general JASP data format using the data from the *Pilot Landing* example.

hit	fa	miss	cr	nPrdSig	nPrdNoi
3	0	0	27	1	99
2	1	2	25	1	99
3	0	3	24	1	99
2	0	4	23	1	99

Table 2.3: JASP data format for the two-choice signal detection model.

The first four columns correspond to the signal detection counts. The final two columns correspond to the number of signal and noise trials to be used for the posterior predictive analysis. Each row in the data corresponds to an individual. Note that there is no requirement that any individual has the same number of signal and noise observations in the observed counts. There is also no requirement that the posterior predictive analysis for different individuals involve the same numbers of signal and noise observations.

Output inferences

The JASP implementation provides three analyses:

- **discriminability, criterion, and bias marginal posteriors**: the marginal posterior distributions of the discriminability d', bias c, criterion k, and relative bias β for each individual
- **discriminability and bias joint posteriors**: the joint and marginal posterior distribution of the discriminability d' and bias c for each individual
- **hit and false alarm count posterior predictions**: the posterior predictive distribution of the hit count $\hat{k}^{\text{h}}$ and false alarm count $\hat{k}^{\text{f}}$ for each individual from $\hat{n}^{\text{s}}$ signal observations and $\hat{n}^{\text{n}}$ noise observations

The first analysis provides marginal posterior distributions for all four parameters. The second analysis focuses on discriminability and bias,

showing both their marginal and joint posterior distributions. The posterior predictive analysis examines the distribution of predicted hit and false alarm counts for the specified number of signal and noise observations.

2.2 *Wine Tasting*

`wineTasting.jasp`

Abdi (2007) presents an example with a wine taster who is trying to detect the presence of added Gamay in a Pinot Noir wine. According to this source, Gamay is an inferior grape that has dominant flavors including "banana" and "potting soil." The signal detection problem is to identify the possible presence of contaminating Gamay grape as each wine is tasted.

Data

The data Abdi (2007) provides are shown in Table 2.4, and involve 10 contaminated wines and 10 non-contaminated ones.

Table 2.4: Wine tasting data.

	gamay	pure pinot noir
detect gamay	9	2
detect no gamay	1	8

The wine taster detects Gamay for 9 of the 10 contaminated wines, which corresponds to 9 hits and 1 miss. They incorrectly identify Gamay in 2 of the 10 non-contaminated wines, which corresponds to 2 false alarms and 8 correct rejections.

Results

Marginal posterior distributions

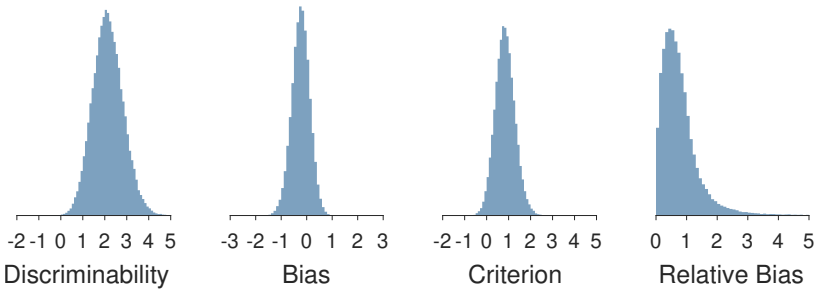


Figure 2.5: The marginal posterior distributions of discriminability d', criterion k, bias c, and relative bias β for the wine tasting data, based on the two-choice signal detection model.

Figure 2.5 shows the marginal posterior distributions for the discriminability d', bias c, criterion k, and relative bias β parameters. The mean posterior discriminability is $d' = 2.13$ and the 95% credible interval is $(0.81, 3.56)$. The mean posterior bias is $c = -0.25$ and the 95% credible interval is $(-0.95, 0.42)$. The mean posterior criterion is $k = 0.81$ and the 95% credible interval is $(-0.03, 1.72)$. The mean posterior relative bias is $\beta = 0.77$ and the 95% credible interval is $(0.06, 2.59)$.

These inferences provide one way to reach conclusions about the Gamay detection abilities of the wine taster. The most important conclusion is that there is large uncertainty about how well the taster can distinguish the presence of Gamay, as reflected in the posterior for discriminability. A $d' = 0.8$ would correspond to quite modest detection ability while a $d' = 3.6$ would correspond to excellent detection ability, and both are possible. Most likely the d' is around 2, which is good detection ability. But, it is impossible to reach a strong conclusion, because of the uncertainty in the posterior. This uncertainty comes from the limited available data: the taster only evaluated 20 wines.

Similarly, it is hard to say anything definitive about the decision-making strategy of the wine taster. This is measured by the bias c, or its re-expression as criterion k or relative bias β. All of these measures also have significant posterior uncertainty. The criterion, for example, could be as low as $k = -0.03$, which below the mean of the noise distribution and would lead to lots of false alarms, or as high as $k = 1.7$, which is well into the right tail of the noise distribution, and would lead to few false alarms.

Joint posterior distribution of discriminability and bias

Figure 2.6 shows the joint and marginal posteriors for discriminability d' and bias c. The marginal distributions at the top and the right are the same as those shown in Figure 2.5. The joint distribution is shown in the main central panel.

Five points a–e in the joint posterior distribution are highlighted. Each of these corresponds to a specific combination of discriminability and bias that is possible, based on the data. To help understand how to interpret joint posterior distributions, each of these points is explicitly represented in terms of the two-choice signal detection model below.

Point a is in the heart of the joint posterior, and is representative of what is the most likely state of affairs. It corresponds to a discriminability of about 2 and a bias just slightly below zero. The interpretation of this combination of parameter values would be that the wine taster has good detection ability, and is slightly more likely to produce false alarms than misses.

The other points b–e are less likely, but remain possible given the limited data. Point b, for example, corresponds to much smaller discriminability and a much greater bias in favor of making false alarms over misses. Point e corresponds to much higher discriminability and a slight bias in favor of misses over false alarms, although with such high discriminability that very few errors are likely. Points b and d complete the extreme possibilities.

A good substantive summary of the wine tasters performance in the task should acknowledge what is likely, but also what is possible given the uncertainty. This summary could be something like:

> "There is some evidence that the wine taster is good at distinguishing Gamay and that they have a bias, when they make errors, towards saying Gamay is present when it is not rather than failing to detect Gamay. The mean posterior

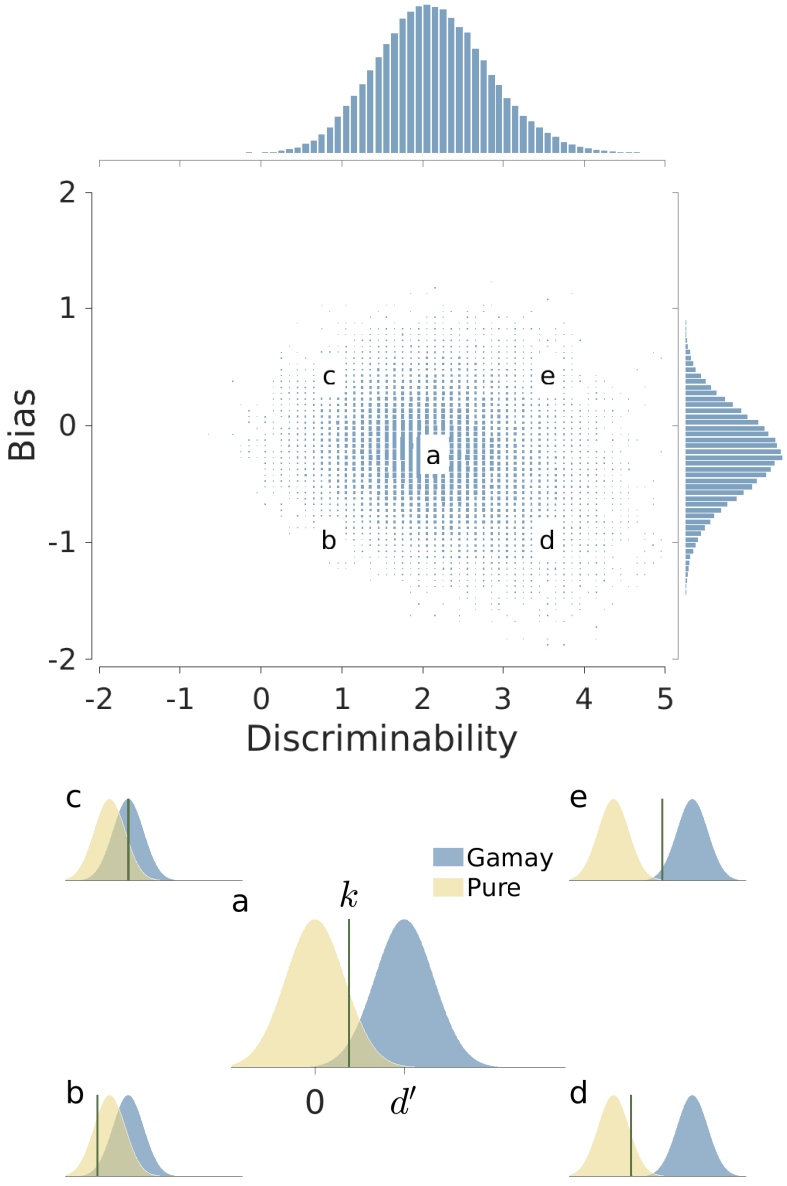


Figure 2.6: The joint and marginal posterior distributions of the discriminability d' and bias c for the wine tasting data, and two-choice signal detection model interpretations of five specific possibilities labeled a–e.

discriminability is $d' = 2.13$ and the mean posterior bias is $c = -0.25$. But the evidence is far from conclusive. The taster could have much better or much worse detection abilities, since the 95% credible interval for discriminability d' is $(0.81, 3.56)$. They could be also using decision-making strategies that are not biased towards false alarms, since the 95% credible interval for bias c is $(-0.95, 0.42)$."

Joint posterior distribution of discriminability and criterion

It is interesting to contrast the results using discriminability and bias in Figure 2.6 with those based on the discriminability and criterion. Figure 2.7 shows this analysis.

One clear difference between the two analyses is that the joint distribution of discriminability and bias is well summarized by the marginal distributions, but the joint distribution between discriminability and

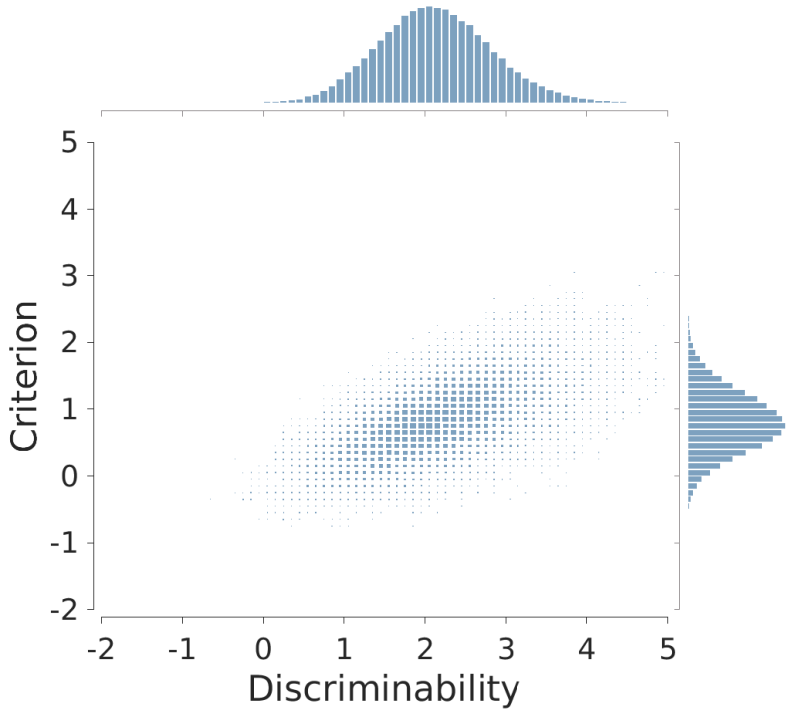


Figure 2.7: The joint posterior distribution between the discriminability d' and criterion k for the wine tasting data, based on the two-choice signal detection model.

criterion is not. For discriminability and bias, if a d' value has significant marginal density and a c value has significant marginal density, then their combination also has significant density in the joint distribution. This is the idea discussed in Chapter 0.2.1 of the joint distribution eing the product of the marginal distributions.

The same is not true for discriminability and criterion. For example, $d' = 3$ and $k = 0$ both have non-negligible posterior densities in the marginal distributions, but their combination has negligible density in the joint distribution. Intuitively, if it turns out that the wine taster is good at distinguishing Gamay, it will turn out they are not using a criterion near zero. The possibility of a such low criterion being used will only be borne out if it also turns out discriminability is low.

The dependency evident in the joint posterior distribution for discriminability and criterion makes the marginals difficult to interpret in isolation. This is why it is often better to use discriminability and bias.

2.3 *The Mentalist*

`mentalist.jasp`
(thanks to Joachim Vandekerckhove for developing this example)

Anderson (2015) considers an example involving a psychic detective (a "mentalist"), who is consulted by the police to help with 50 missing person cases for which bodies have been discovered. The research question is whether the psychic is sensitive to the location of the bodies relative to water.

Data

The mentalist claimed that 31 of the bodies would be discovered close to water. It turned out 28 of the bodies were discovered within 500 meters of a body of water. Of these, 17 were ones predicted by the mentalist, but the remaining 11 were not predicted. That means the mentalist incorrectly predicted 14 bodies would be found near water, and for only 8 bodies did they correctly predicted that the body would not be found near water. Table 2.5 summarizes the data.

	found near water	found not near water
say near water	17	14
say not near water	11	8

Table 2.5: The mentalist data.

In signal detection terms, it is reasonable to treat a body near water as the signal, and bodies not near water as noise. That means, for example, a hit occurs when the mentalist says a body will be found near water, and it turns out that it is. Under this coding, the the mentalist's performance corresponds to 17 hits, 11 misses, 14 false alarms, and 8 correct rejections.

Results

Posterior distributions

Figure 2.8 shows the joint and marginal posterior distributions of discriminability and bias for the mentalist's performance. The joint distribution is well summarized by the two marginal distributions.

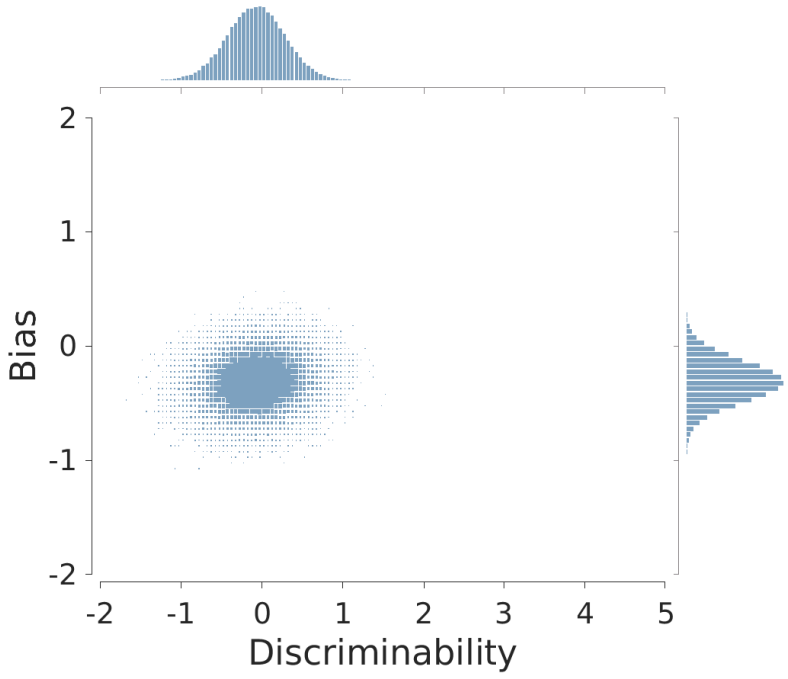


Figure 2.8: The joint and marginal posterior distributions for discriminability d' and bias c for the mentalist data based on the two-choice signal detection model.

One clear result is that the mentalist's ability to detect when bodies will be found near water is poor. The mean posterior discriminability is $d' = -0.07$ with a 95% credible interval $(-0.75, 0.65)$. A second clear result is it is very likely the mentalist has a bias for predicting bodies will be found near water. The mean posterior bias is $c = -0.31$ with a 95% credible interval $(-0.66, 0.03)$.

These two conclusions provide a fairly complete understanding of the mentalist's predictions. They do not seem to know whether bodies will be found near water or not, but they have a bias towards guessing that they will be. This is a good example of how, despite significant remaining uncertainty about the details—for example, it is not clear whether the mentalist has a lot of bias towards predicting water, or just a little bias—the important conclusions can be clear.

Posterior predictive analysis

Figure 2.9 shows the posterior predictive distribution for $\hat{n}^{\mathrm{s}} = 28$ signal and $\hat{n}^{\mathrm{n}} = 22$ noise observations. These counts match the observed data. The area of each square corresponds to the posterior probability for that number of hits and false alarms. The blue square corresponds to the 17 hits and 14 false alarms that were observed.

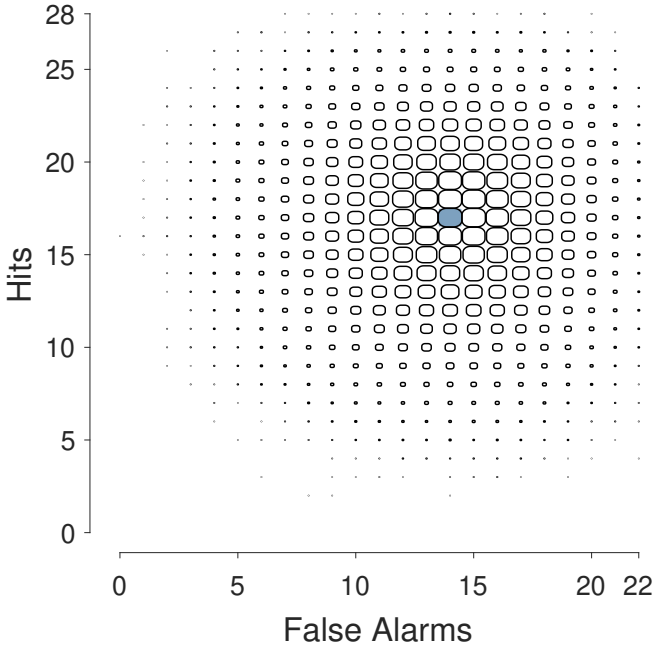


Figure 2.9: Posterior predictive distribution of hits and false alarm counts for the mentalist's performance detecting bodies near water, with the observed data shown in blue.

One use for the posterior predictive analysis is to check the descriptive adequacy of the two-choice signal detection model. The observed data, shown in blue, are given high probability by the posterior predictive distribution, which provides evidence that the model can describe the data. This capability, in turn, gives some basis to trust the inferences about discriminability and bias as being valid.

Another use of the posterior predictive analysis is to think of it in terms of a repeat of the same experiment. Suppose the psychic was asked about another 50 missing person cases, and the truth was again that 28 would be found near water. The posterior predictive distribution

now makes a prediction about the hit and false alarm counts in this replication experiment, based on what we know about them from the completed experiment. From this perspective, the posterior predictive analysis emphasizes the uncertainty coming from the limited data. It is possible, for example, that the outcome of the future experiment could be much better performance, such as 25 hits and 10 false alarms, or much worse performance, such as 10 hits and 20 false alarms.

2.4 *Siblings or Dating*

`siblingsOrDating.jasp`
(thanks to Vinutha Sandadi for developing this example)

"Siblings or dating?" is a popular Instagram account that shows photos of people, and asks viewers to vote as to if the two people are siblings or if are dating. The final vote and the correct answer are eventually provided. One research question that can be examined from the votes is how well the majority vote discriminates between siblings and dating.

Data

We use the majority vote and ground truth for 127 images posted between April 2021 and February 2023. Images that included more than two people, or lead to tied votes, were excluded.

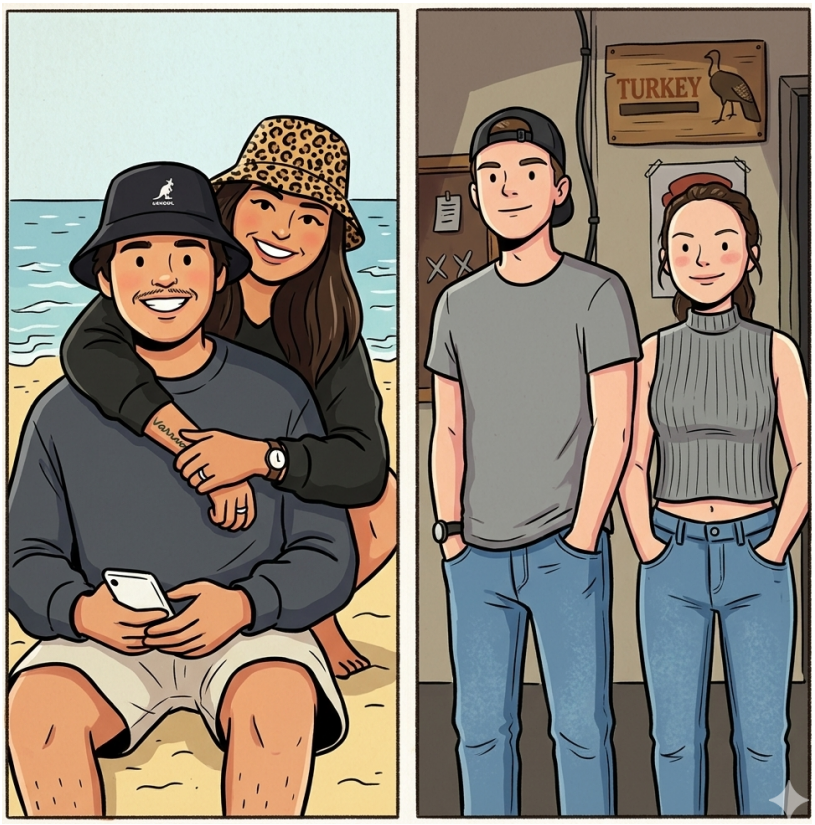


Figure 2.10: Demonstration of close contact (left) and distant posing (right) categories of couples.

The images were also subjectively divided into 50 "close contact" and 77 "distant posing" groups. Figure 2.10 demonstrates the distinction. The left couple is an example of a close images: pairs of people with faces were close together or tilted towards each other, with arms were placed

around each other, with torsos and lower bodies close together, holding hands, and so on. The couple on the right is an example of a distant image: pairs of people with faces spaced apart, with only one (or no) arm around the other's body, with one person in the front and one in the background, selfies, and so on.

Table 2.6: Siblings or dating data.

	truth			
	close contact		distant posing	
decision	dating	siblings	dating	siblings
dating	16	15	16	10
siblings	14	5	36	15

In signal detection theory terms, we treat dating images as the signal and sibling images as noise. This coding leads to the hit, miss, false alarm, and correct rejections for the close contact and distant conditions shown in Table 2.6. It is clear that both conditions include many more errors (misses and false alarms) than correct decisions (hits and correct rejections). In the close condition, the small number of correct rejections is striking, suggesting a bias towards believing that siblings are dating. In the distant condition, the large number of misses is similarly striking, suggesting a bias towards believing dating couples are siblings.

Results

Discriminability and bias

Figure 2.11 shows the joint and marginal posterior distributions of discriminability and bias for the close and distant conditions. It is clear that discriminability is very likely to be negative for both conditions. Conceptually, it means that the center of the signal distribution d' is to the left of the center of the noise distribution. In this situation, no matter where the criterion k is placed, false alarms will always be more likely than hits, and misses will always be more likely than correct rejections. This corresponds to worse-than-chance performance, as observed in the data. One reasonable explanation for this unusual result is that the images being presented are deliberately chosen to be misleading.

The inferences about bias show an interesting difference between the close and distance conditions. The close condition shows a negative bias consistent with favoring a dating response, while the distant condition shows a positive bias, consistent with favoring a siblings response. These biases are consistent with the patterns of errors in the raw data noted in relation to Table 2.6. One interpretation is that the way close contact images are defined is naturally associated with dating, while the nature of the distant posing images is better associated with siblings.

Posterior predictive analysis

Figure 2.9 shows the posterior predictive distribution for the hypothetical balanced experiment with $\hat{n}^{\mathrm{s}} = 40$ signal and $\hat{n}^{\mathrm{n}} = 40$ noise trials for both conditions. The area of each square corresponds to the posterior probability for that number of hits and false alarms in this experiment.

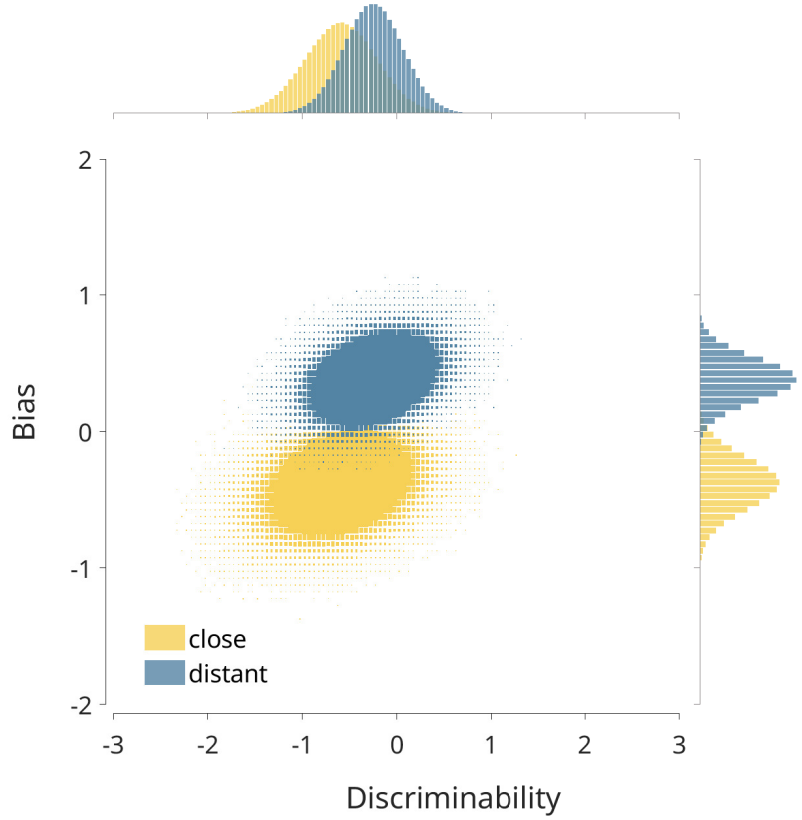


Figure 2.11: The joint posterior distribution between the discriminability d' and bias c for the siblings or dating data.

Besides quantifying the probabilities of outcomes, the posterior predictive distribution visualizes the expected high number of false alarms in the close condition, and the expected low number of hits in the distant condition. These patterns are another way of understanding the changes in bias between the conditions presented in Figure 2.11.

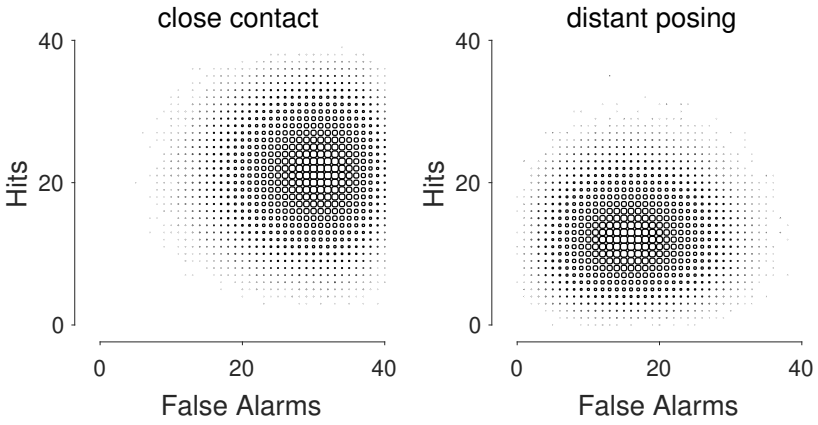


Figure 2.12: Posterior predictive distribution of hits and false alarm counts for a hypothetical balanced siblings or dating experiment.

2.5 *Pilot Landing*

`pilotLanding.jasp`

Salvan et al. (2022) studied the decision making of pilots in a series of landing situations, in which they had to make the decision whether or

not to "go-around" (i.e., abort the landing and continue flying). The two-choice signal detection model provides one way of understanding the pilots' abilities to discriminate safe from dangerous landing conditions, and the sorts of errors that they make.

Data

Salvan et al. (2022) report the decision making of four pilots in a set of about 30 landing situations. On each trial, the pilots were shown a primary flight display interface with 21 features of the situation: headwind, distance to nearest aircraft, pitch, vertical speed, and so on. They had to decide whether they would land the plane in this situation, or execute a "go-around."

A ground truth was established for each landing situation, which allows the results to be presented as a table of counts for each pilot, as shown in Table 2.7. It is worth noting that all of the pilots were mostly given landing situations in which there was no problem, consistent with the low base-rate of go-arounds in the real world.

According to this source the rate of go arounds at San Francisco international airport is about 0.3–0.4%.

Table 2.7: Pilot landing data.

	truth							
	pilot 1		pilot 2		pilot 3		pilot 4	
decision	GA	land	GA	land	GA	land	GA	land
go-around	3	0	2	2	3	3	2	4
land	0	27	1	25	0	24	0	23

In terms of signal detection theory, we treat a go-around situation as a signal and a landing situation as noise. The interpretation is that, when the pilot is coming in to land, there may be some signal or indication that there is a problem. If they detect that problem then they should go-around. In the absence of the problem, they should land the plane as normal. Under this interpretation, a hit is when a pilot executes a go-around when there is a problem, a false alarm is when they execute a go-around when there is no problem, a miss is trying to land the plane when there is a problem, and a correct rejection is landing the plane when there is no problem.

Results

Discriminability and bias

Figure 2.13 shows the joint and marginal posterior distributions of discriminability and bias for all four pilots.

These results suggest some conclusions about how well the different pilots detect problems and how they make go-around decisions.

- Pilot 1 is likely the best able to detect problems, with a mean posterior of $d'_1 = 4.95$ and 95% credible interval $(2.17, 8.86)$. It is very uncertain what decision strategy Pilot 1 uses, with c_1 having a 95% credible interval $(-1.68, 2.09)$. This is because they did not make any errors.

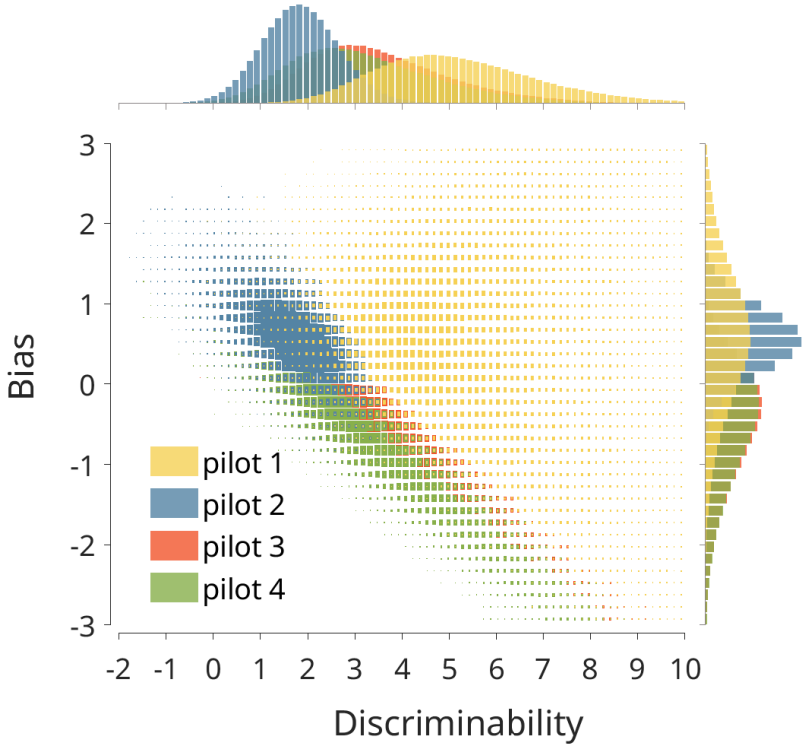


Figure 2.13: The joint posterior distribution between the discriminability d' and bias c for the pilot landing data.

- Pilot 2 is likely the least able to detect problems, with a mean posterior $d'_2 = 1.82$ and 95% credible interval $(0.32, 3.46)$. It is most likely, although not certain, that they have a positive bias, based on their posterior mean of $c_2 = 0.54$ and 95% credible interval $(-0.320, 1.33)$. This corresponds to a propensity for misses rather than false alarms when making errors. Intuitively, even though the pilot had two false alarms and only one miss, there were only three trials in which there was a problem. The fact that they missed one of these three leads to the (highly uncertain) possibility of having too high a threshold for going around when there is a problem.
- Pilots 3 and 4 have very similar data and hence very similar inferences. A small difference is that Pilot 3 received one more signal trial in which there was a problem, but neither pilot missed any of the problems presented to them. The other difference is that Pilot 3 had one fewer false alarm, deciding to go-around 3 times out of 27 noise trials. The false alarms for both pilots lead to a likely inference of a negative bias, with a mean posterior of $c_3 = -0.64$ and 95% credible interval $(-2.61, 0.64)$ for Pilot 3, and mean posterior of $c_4 = -0.61$ and 95% credible interval $(-2.62, 0.74)$ for Pilot 4.

Note that Figure 2.13 violates the general property of the joint distribution of discriminability and bias being well approximated by the product of the marginals. Accordingly, it is important to base results on the joint posterior distribution rather than the marginals. The violation stems from unusual imbalance between signal and noise trials. Making inferences about discriminability depends on having information about decisions on both signal and noise trials. The small number of signal trials in the experiment limits this information.

Relative bias

The behavior of the pilots provides an interesting example of the measure of relative bias provided by β_i. Pilot 2 has a mean posterior of $\beta_2 = 2.76$ and 95% credible interval $(0.47, 8.53)$. So, while it is uncertain, it is likely their relative bias is greater than 1, meaning they set their criterion where noise is more likely than signal. Pilot 4, meanwhile, has a mean posterior of $\beta_4 = 0.57$ and 95% credible interval $(0.00, 2.38)$. So they most likely set their criterion where signal is assumed to be more likely than noise.

There are two ways to interpret these contrasting decisions strategies as measured the relative bias β_i. One interpretation is in terms of the base-rates of signal and noise trials, problems are rare, so $\beta_i < 1$ is well calibrated to the environment. This would argue in favor of Pilot 2's decision strategy. The other interpretation is in terms of the consequences of errors. From this perspective, it seems like a bad idea to land when a go-around is warranted. It seems much less problematic to go-around when it was not strictly necessary. These real-world utilities (costs, payoffs) for miss versus false alarm errors justify Pilot 4's likely use of $\beta_4 > 1$, on the basis of being very sensitive to the possibility of a signal.

Posterior predictive analysis

A final way of understanding the performance of the pilots is provided by posterior predictive analysis. We consider a situation with $\hat{n}^{\mathrm{s}} = 1$ signal and $\hat{n}^{\mathrm{n}} = 99$ noise trials. The motivation is to examine the predicted performance in an expanded set of 100 trials that follow a real-world base rate of 1% signal trials.

Table 3.7 summarizes the mean and 95% credible intervals the hit and false alarm counts for each pilot.

	hits		false alarms	
	mean	95% CI	mean	95% CI
pilot 1	0.90	(0, 1)	0.97	(0, 7)
pilot 2	0.62	(0, 1)	8.26	(0, 22)
pilot 3	0.92	(0, 1)	12.19	(2, 28)
pilot 4	0.89	(0, 1)	15.77	(4, 33)

Table 2.8: Posterior predictive marginal distributions for the hit and false alarm counts with 1 signal and 99 noise trials.

The mean posterior predicted number of hits can be interpreted as the probability the pilot will correctly go-around on the one problem landing. That probability is only 62% for pilot 2, but around 90% for the other three pilots. The mean posterior posterior predicted number of false alarms, and the associated credible intervals, quantify how many times the pilot will mistakenly go-around on the 99 non-problematic landings. Pilot 1 performs the best, almost certainly mistakenly going around fewer than 10 times, with an expectation of 1.44 times. Pilots 3 and 4 fare much worse by this measure, with the possible of going around unnecessary as many as 30 times.

2.6 The Hierarchical Two-Choice Signal Detection Model

2.6.1 Motivation

The basic motivation for extending hierarchical models is explained in detail for the success rate model in Chapter 1. If hierarchical models are unfamiliar, it is worth reading Chapters and 1.6. The hierarchical extension of the two-chocie signal detection model shares the same motivation of providing a group level account of how different individuals are related to each other in their signal detection behavior.

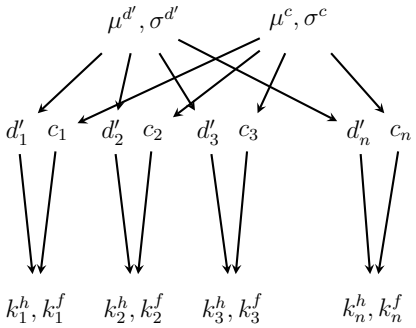


Figure 2.14: Conceptual representation of the hierarchical two-choice signal detection model.

The hierarchical extension is shown schematically in Figure 2.14. Each individual has a discriminability d'_i and bias c_i that generate their observed hit k_i^{h} and false alarm k_i^{f} counts. The topmost level in the hierarchical model ties together the individuals by introducing a group distributions for both discriminability and bias. The group distribution for discriminability is represented by a Gaussian distribution with mean $\mu^{d'}$ and standard deviation $\sigma^{d'}$ and group distribution for bias is represented by a Gaussian distribution with mean μ^c and standard deviation σ^c. The introduction of the group distribution means that the individual discriminabilities and biases are no longer independent of one another, but are connected by their common membership of the group.

The potential advantages of the hierarchical two-choice signal model are the same as those spelled out for the hierarchical success rate model in Chapter 1.6. They are the ability to make inferences about discriminability and bias at both the group and individual level, and the ability to make inferences and predictions related to a representative individual. As for the hierarchical success rate model, different groups are possible, with subsets of the individuals assigned to each group.

2.6.2 Model

Figure 2.15 summarizes the structure of the hierarchical two-choice signal detection model. The multiple individuals are shown with their own discriminabilities d'_i and biases c_i. The group that an individual belongs to is provided by the observed variable g_i. The discriminabilities and biases of individuals in the same group are linked by group distributions with means $\mu^{d'}$ and μ^c, and standard deviations $\sigma^{d'}$ and σ^c.

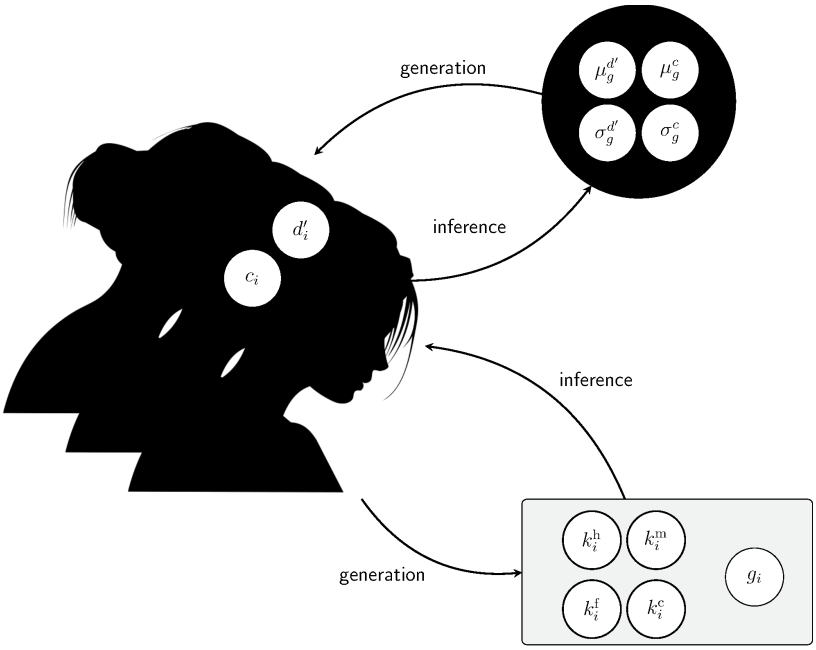


Figure 2.15: The structure of the hierarchical two-choice signal model.

Inferences about the discriminabilities and biases comes from the observed hit k_i^{h}, false alarm k_i^{f}, miss k_i^{m}, and correct rejection k_i^{c} counts. Inferences about the discriminabilities and biases, in turn, allow inferences about the mean and standard deviation of their group distributions.

Inferences about the group distributions allow the discriminability and biases of a representative individual $\tilde{d}'$ and bias $\tilde{c}$ from each group to be generated. This, in turn, allows a predictions to be made about the hit $\hat{k}_i^{\mathrm{h}}$ and false alarm $\hat{k}_i^{\mathrm{f}}$ counts for a representative individual in each group from $\hat{n}^{\mathrm{s}}$ signal and $\hat{n}^{\mathrm{n}}$ noise trials.

2.6.3 Mechanics

Table 2.9 lists the observed variables and latent parameters and variables for the hierarchical two-choice signal detection model, giving both their statistical symbol and their JASP code name. The first set of variables above the dividing line are the core parameters and data included in Figure 2.15. The core parameters are the discriminabilities d_i' and biases c_i and their group means and standard deviations $\mu_g^{d'}$, μ_g^c, $\sigma_g^{d'}$, and σ_g^c . The core observations are the signal detection counts k_i^{h}, k_i^{f}, k_i^{m}, and k_i^{c} for each individual, and the group g_i to which they belong.

Table 2.9 also lists additional parameters and variables. These include the derived criterion k_i and relative bias β_i for each individual. The inclusion of group distributions allow the discriminability $\tilde{d}_g'$ and bias $\tilde{c}_g$ of a representative individual for each group to be inferred. This representative success rate is used, in turn, to make predictions about the number of hits $\hat{k}_g^{\mathrm{h}}$ and false alarms $\hat{k}_g^{\mathrm{f}}$ from $\hat{n}^{\mathrm{s}}$ signal and $\hat{n}^{\mathrm{n}}$ noise trials.

Name	Model	JASP
discriminability	d'_i	`d[i]`
bias	c_i	`c[i]`
group	g_i	`group[i]`
mean discriminability for group	$\mu_g^{d'}$	`muD[g]`
standard deviation for discriminability of group	$\sigma_g^{d'}$	`sigmaD[g]`
mean bias for group	μ_g^{c}	`muC[g]`
standard deviation of bias for group	σ_g^{c}	`sigmaC[g]`
hit count	k_i^{h}	`hit[i]`
false alarm count	k_i^{f}	`fa[i]`
miss count	k_i^{m}	`miss[i]`
correct rejection count	k_i^{c}	`cr[i]`
criterion	k_i	`k[i]`
relative bias	β_i	`beta[i]`
representative discriminability for group	$\tilde{d}'_g$	`dRep[g]`
representative bias for group	$\tilde{c}_g$	`cRep[g]`
posterior predicted hit count for group	$\hat{k}_g^{\text{h}}$	`hitPrd[g]`
posterior predicted false alarm count for group	$\hat{k}_g^{\text{f}}$	`faPrd[g]`
number of signal observations for prediction	$\hat{n}^{\text{s}}$	`nPrdSig`
number of noise observations for prediction	$\hat{n}^{\text{n}}$	`nPrdNoi`

Table 2.9: The meaning of the observed variables and latent parameters in the hierarchical two-choice signal detection model, with their model and JASP names.

Input data

Table 2.3 shows the general JASP data format using the data from the *Amyloid Memory* example.

hit	fa	miss	cr	group	nPrdSig	nPrdNoi
15	0	1	14	1	15	15
13	20	15	1			
10	5	0	15	1		
...	...	...	...	...		
10	5	2	13	2		
10	5	2	13	2		
0	15	1	14	2		
...	...	...	...	...		

Table 2.10: JASP data format for the hierarchical two choice signal detection model.

Table 2.10 shows the JASP data format for the hierarchical success model. The additional columns are `nPrdSig` and `nPrdNoi`, which specify the number of signal and noise trials for the posterior predictive analysis. These are both single numbers that only occupy the first row of their columns.

Output Inferences

The JASP implementation provides four analyses.

- **individual discriminability, criterion, and bias**: the marginal posterior distributions of the discriminability d'_i, bias c_i, criterion k_i, and relative bias β_i for each individual
- **group discriminability and bias**: the joint and marginal posterior distribution of the mean discriminability $\mu_g^{d'}$ and mean bias μ_g^{c} for each group

- **representative individual discriminability and bias**: the joint and marginal posterior distribution of the discriminability $\tilde{d}'_i$ and bias $\tilde{c}_i$ for the representative individual in each group
- **representative posterior predictive analysis**: the marginal posterior predictive distributions of the representative hit count $\hat{k}^{\text{h}}_g$ and false alarm count $\hat{k}^{\text{f}}_g$ from $\hat{n}^{\text{s}}$ signal observations and $\hat{n}^{\text{n}}$ noise trials for the representative individual in each group

2.7 *Grammatical Acceptability*

`grammaticalAcceptability.jasp`

Huang & Ferreira (2020) studied how people make acceptability judgments about the –er nominalization of verbs. For example, adding "er" to the verb "run" gives "runner," which seems grammatically acceptable, but adding "er" to "arrive" gives "arriver" does not. The verb "run" is an example of an *unergative* (agent) verb: other examples are "dance," "hop," and "jog." The verb "arrive" is an example of an *unaccusative* (theme) verb: other examples are "melt," "sink," and "fall." The Unaccusative Hypothesis predicts that only the unergative verbs will lead to acceptable –er words.[†]

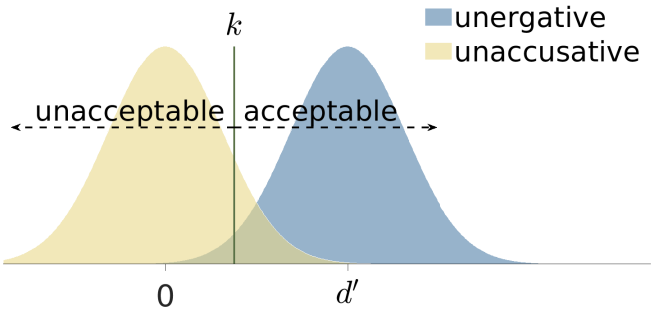


Figure 2.16: The signal detection framework used by Huang & Ferreira (2020) for testing the Unaccusative Hypothesis.

Huang & Ferreira (2020) used signal detection theory as a framework for analyzing their data to test the Unaccusative Hypothesis. As shown in Figure 2.16, they treated unergative verbs as signal stimuli and unaccusative verbs as noise stimuli, with the criterion determining whether a participant judges the –er nominalization to be grammatically acceptable or unacceptable. Using this approach, the discriminability between the unergative and unaccusative distributions measures whether the distinction is useful for explaining people's judgments. A large discriminability is evidence for the Unaccusative Hypothesis. An advantage of using signal detection theory is that it controls for response strategies that people may use, such as a bias towards judging words as grammatically acceptable.

[†] The paper has confusing typographical error at the top of page 6. It says"The subject of the unergative verb (e.g., run) is base-generated as the external argument, whereas the subject of the unergative verb (e.g., arrive) is originally generated as the internal argument." The second "unergative" should be "unaccusative."

Data

The Huang & Ferreira (2020) experiment involved 20 native English speakers judging the acceptability of 60 –er words, made up of 30 unergative and 30 unaccusative verb. Figure 2.17 shows the hit and false alarm counts for the participants. Larger circles indicate that more participants had that combination of hit and false alarm count.

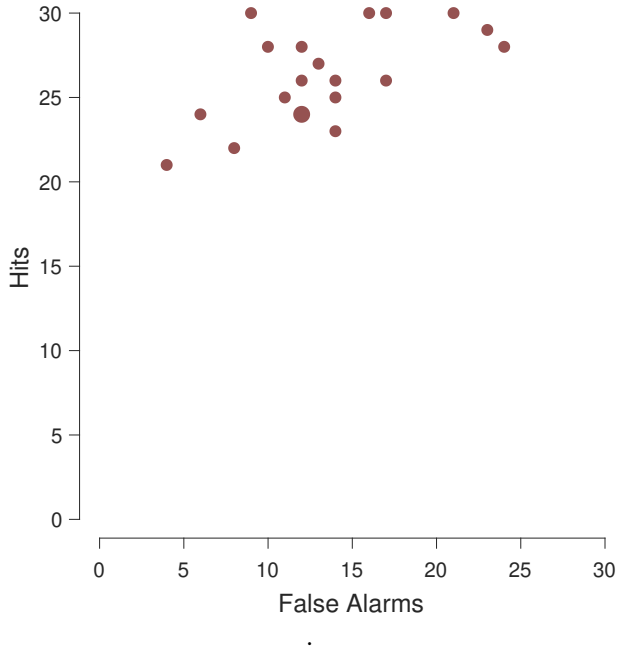


Figure 2.17: Individual hit and false alarm count combinations for the grammatical acceptability data

It is clear from Figure 2.17 that there are individual differences. Some participants made as few as five false alarms, while others made more than 20. The number of hits shows less variation, but still ranges from 20 to 30. Collectively, this suggests that participants tended to judge unergative –er words as grammatically acceptable, but varied in whether they also regarded unaccusative words as acceptable.

Results

Posterior predictive analysis

Figure 2.18 shows the posterior predictive analysis for the hierarchical signal detection theory model. The data are shown by red circles, as in Figure 2.17. The posterior predictive distribution is shown by the gray shading, with darker squares representing hit and false alarm count combinations with greater posterior predictive density. The key result of the analysis is that the posterior predictive distribution gives significant mass to the range over which the data vary. This provides evidence that the hierarchical two-choice signal detection model is descriptively adequate.

Group analysis

Figure 2.19 shows the joint and marginal posterior distribution for the group mean discriminability $\mu^{d'}$ and mean bias μ^{c} in blue, and for the

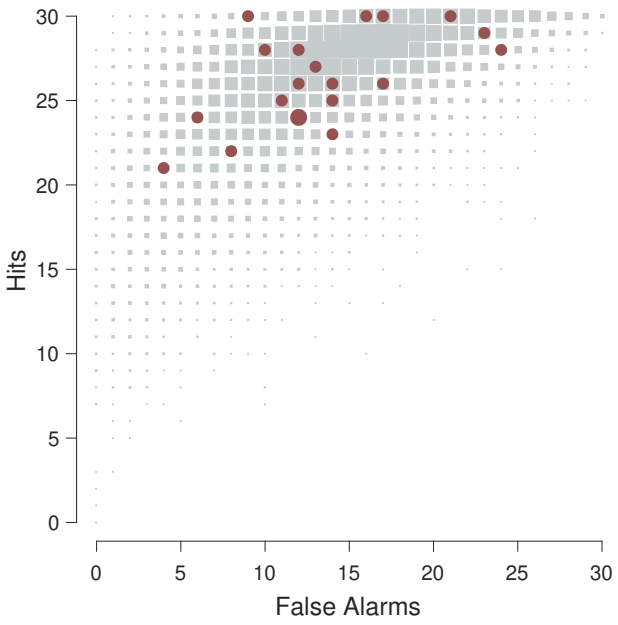


Figure 2.18: Posterior predictive analysis for the hierarchical two-choice signal detection model of the grammatical acceptability data.

representative discriminability $\hat{d}'$ and bias $\hat{c}$ in yellow. The center panel shows the joint posterior distributions, the top panel shows the marginal posterior distributions for the discriminability, and the right panel shows the marginal posterior distributions for the bias.

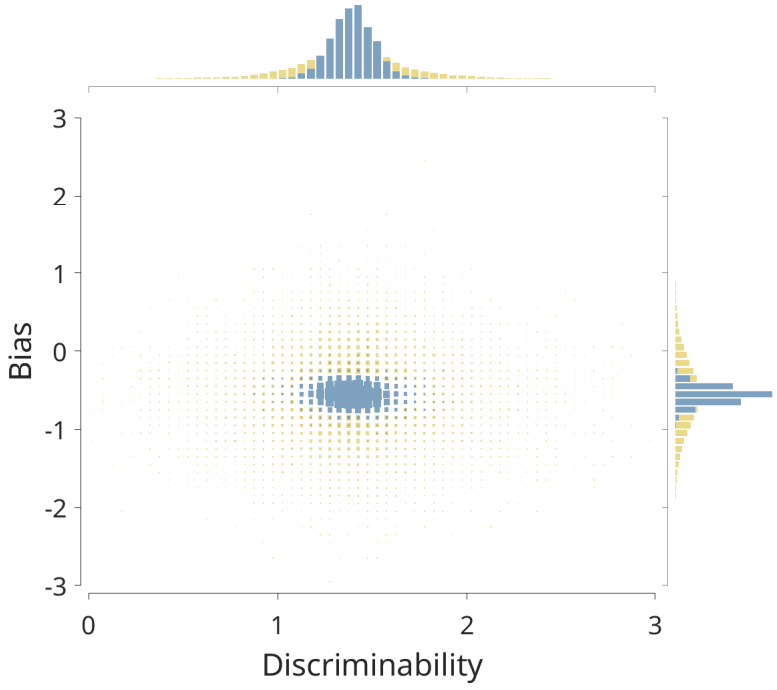


Figure 2.19: The joint and marginal posterior distributions of the mean discriminability $\mu^{d'}$ and mean bias μ^{c} (blue) and the representative individual discriminability $\tilde{d}'$ and representative individual bias $\tilde{c}$ (yellow) for the hierarchical two-choice signal detection model of the grammatical acceptability data.

Figure 2.19 shows that $\mu^{d'} = 1.40$ with a 95% credible interval of $(1.18, 1.62)$. This suggests that the distinction between unergative and unaccusative words does provide some basis for understanding grammaticality judgments, but it does not clearly divide those –er words judged to be acceptable from those that are not. The mean bias is $\mu^{c} = -0.56$ with a 95% credible interval of $(-0.78, -0.35)$. This shows an average

bias towards judging words as grammatically acceptable.

The individual-level representative distribution shows greater uncertainty than the mean, consistent with individual-level variability. The 95% credible interval for $\hat{d}'$ is $(0.73, 2.05)$, and for $\hat{d}'$ is $(-1.44, 0.38)$. This means that it is possible for individuals to judge grammaticality in ways that are weakly consistent or strongly consistent with the Unaccusative Hypothesis, and it is possible for individuals to be strongly biased towards grammaticality or weakly biased against it. In this way, the representative individual analysis provides additional scientific context in using the data to test the hypothesis. It allows more nuanced findings that unpack what it means to claim "people behave according to the predictions of the Unaccusative Hypothesis" by testing that claims across the range of individual differences.

Individual analysis

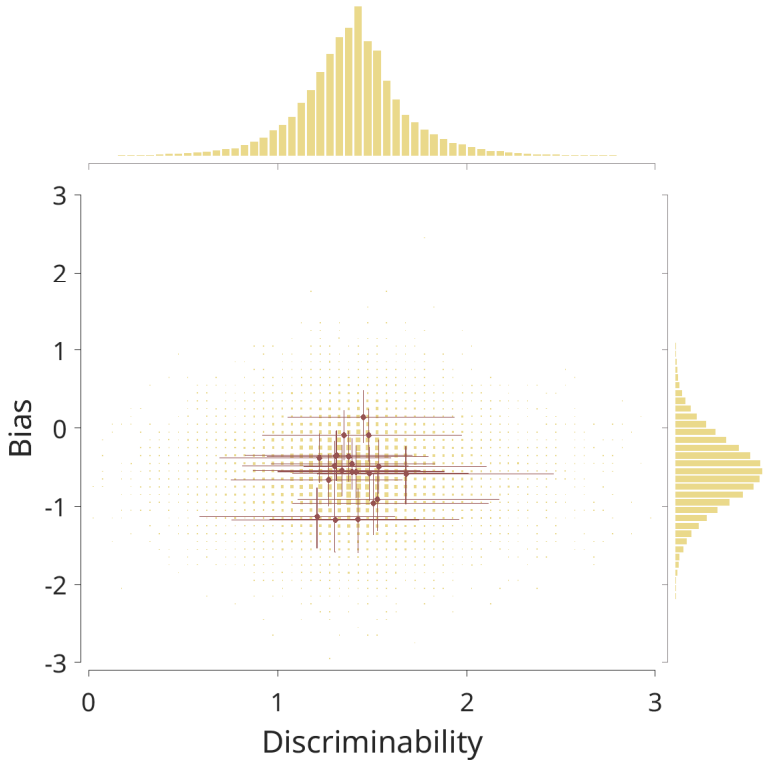


Figure 2.20: The joint and marginal posterior distributions of the mean representative individual discriminability $\tilde{d}'$ and representative individual bias $\tilde{c}$ (yellow), and the individual participant posterior means and 95% credible intervals (blue), for the hierarchical two-choice signal detection model of the grammatical acceptability data.

Figure 2.20 extends the individual analysis. The joint and marginal posterior distributions for the representative individual are again shown in yellow, repeating the information provided in Figure 2.19. Also shown are the posterior mean discriminabilities $d^p rime_i$ and biases c_i and 95% credible intervals for each of the individual participants. Circular blue markers show the posterior means and error bars show the credible intervals.

While Figure 2.19 compares the representative individual to the group means, Figure 2.20 shows how the representative individual relates to the specific individuals for whom data are available. In particular, it is clear that the representative individual spans the range of individual differences observed in the participants, and allows for a slighter wider

range of plausible discriminability and bias. In this sense, the representative individual captures the observed range of individual differences, and allows for the future observation of specific individuals who expand that range.

2.8 *Amyloid Memory*

`amyloidMemorySDT.jasp`

In clinical settings, recognition memory tests are often used to assess people's memory, especially as quick and cheap screening tests for the early detection of Alzheimer's disease and related disorders. This example uses a clinical data from Embic to examine the relationship between an individual's recognition memory performance and their biological level of beta amyloid. Amyloid positivity is thought to be a pre-symptomatic indicator of Alzheimer's disease.

Data

The data involve 200 patients aged over 75 years. Based on a cerebrospinal fluid measurement, 100 of the patients are classified as amyloid positive and the other 100 are amyloid negative.

Every patient completed the Rey auditory verbal learning test (RALVT: Bean, 2011). The RAVLT is a recognition memory task in which a person studies a set of 15 words, before being tested on 30 words, made up of 15 words that were on the study list and 15 words that were not. During the test, the person has to identify whether each presented word is one of the "old" study words or a "new" word.

In signal detection theory terms, it is natural to treat the study list words as signal, and the new words as noise. This means that correctly identifying a word as old is a hit, correctly identifying a word as new is a correct rejection, mistakenly responding a new word was studied is a false alarm, and mistakenly responding that a studied word is new is a miss.

The hit and false alarm counts for the amyloid negative and amyloid positive patients are shown in Figure 2.21. Larger circles indicate that more patients had that combination of hit and false alarm counts. It is clear that there are individual differences within each group, and

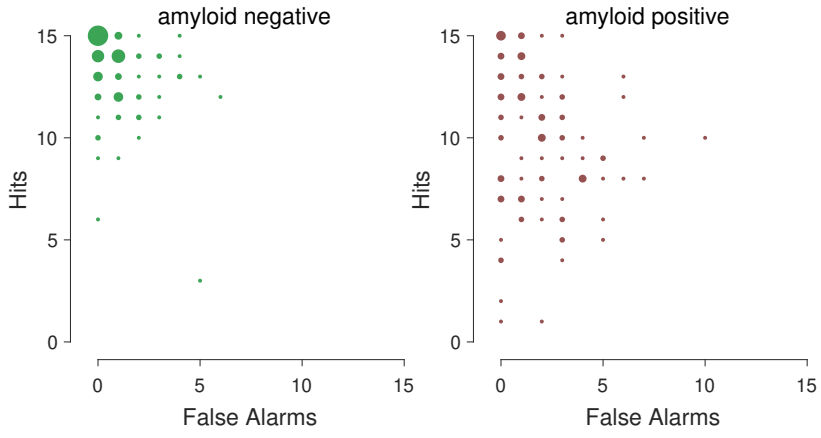


Figure 2.21: Individual hit and false alarm count combinations for the amyloid memory data, for amyloid negative patients in the left panel and amyloid positive patients in the right panel.

differences in the distributions of performance between the two groups. In particular, many amyloid positive patients have fewer hits than are observed in the amyloid negative patients.

Results

Posterior predictive analysis

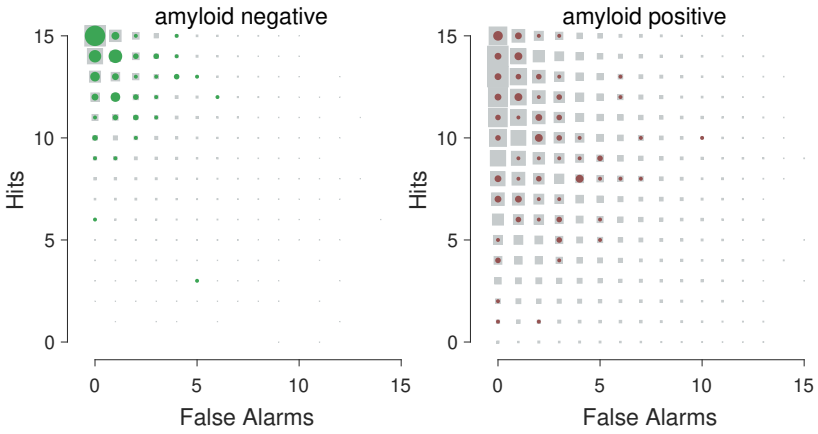


Figure 2.22: Posterior predictive analysis for the hierarchical signal detection theory analysis of the amyloid memory data.

Figure 2.22 shows the posterior predictive analysis. The data are shown by green and red circles for the amyloid negative and amyloid positive groups, as in Figure 2.21. The posterior predictive distributions are shown by the gray shading, with darker squares representing hit and false alarm count combinations with greater posterior predictive density. The key result of the analysis is that the posterior predictive distribution gives significant mass to the range over which the data vary for both the amyloid negative and amyloid positive groups. This provides evidence that the hierarchical model is descriptively adequate.

Group analysis

Figure 2.23 shows the joint and marginal posterior distribution of the group mean discriminabilities and biases, and the posterior representative distribution, for the amyloid negative and amyloid positive groups. The group mean distributions for the amyloid negative group $\mu_1^{d'}$ and μ_1^c are shown in darker green, and for the amyloid negative group $\mu_2^{d'}$ and μ_2^c in darker red. The representative distributions for the amyloid negative group $\tilde{\mu}_1^{d'}$ and $\tilde{\mu}_1^c$ are shown in lighter green, and for the amyloid negative group $\tilde{\mu}_2^{d'}$ and $\tilde{\mu}_2^c$ in lighter red. The joint distributions are shown in the central panel, the marginal distributions for discriminability are shown in the top panel, and the marginal distributions for bias are shown in the right panel.

At the group level, the amyloid positive group clearly has worse mean discriminability than the amyloid negative group. The posterior mean discriminability for the amyloid negative group is $\mu_1^{d'} = 3.19$ with a 95% credible interval $(2.90, 3.50)$, but for the amyloid positive group $\mu_2^{d'} = 1.84$ with a 95% credible interval $(1.61, 2.09)$. The presence of individual differences leads to the 95% credible intervals for the posterior

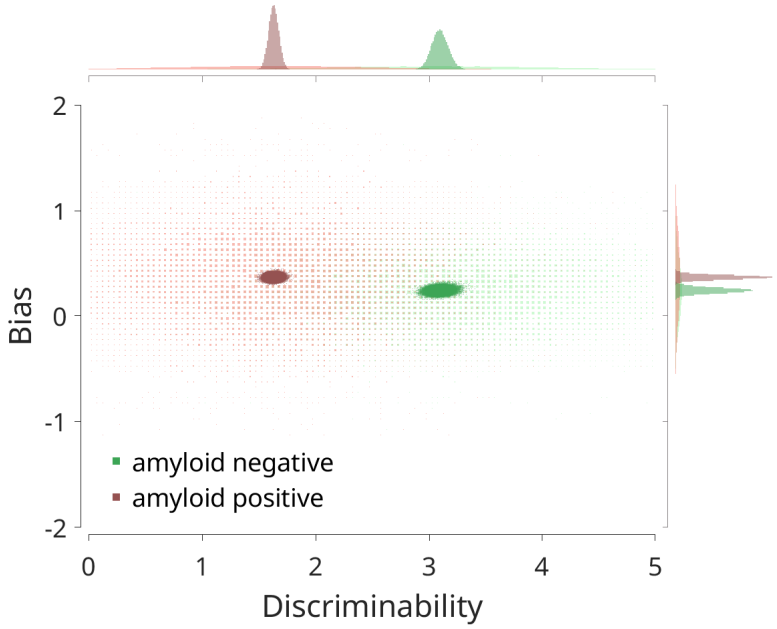


Figure 2.23: Inferences for the discriminability and bias of the amyloid negative and amyloid positive groups, based on the hierarchical two-choice signal detection model.

representative distributions being much wider: $(1.04, 5.32)$ for the amyloid negative group and $(-0.28, 3.90)$ for the amyloid positive group. It seems reasonable to conclude there are meaningfully large difference in discriminability between the two groups as wholes, but those differences do not guarantee that every individual amyloid negative patient has better discriminability than every amyloid positive patient.

The inferences for group mean bias also show evidence of differences between the groups, but the differences are not as large. The posterior mean bias for the amyloid negative group is $\mu_1^c = 0.18$ with a 95% credible interval $(0.09, 0.27)$, but for the amyloid positive group $\mu_2^c = 0.41$ with a 95% credible interval $(0.31, 0.51)$. A reasonable conclusion is that both groups show, on average, a slightly conservative bias to avoid false alarms, and that the amyloid positive group probably has a little more of this bias.

The much more uncertain posterior representative distributions for bias show that this pattern does not reliably apply at the individual level. Their 95% credible intervals are $(-0.31, 0.68)$ for the amyloid negative group and $(-0.33, 1.15)$ for the amyloid positive group.

These results provide good examples of the need to resist the temptation to base conclusions about significant differences on whether credible intervals overlap. The credible intervals for the posterior representative distribution of discriminability, for example, just fail to overlap. This really does not mean anything. Practically, it is likely their 99% credible intervals overlap, and there is nothing special about the 95% bound (or a 99% one). There are also other 95% credible intervals besides those based on the 2.5% and 97.5% percentiles, and some of those will overlap. Most fundamentally, choosing between null and alternative hypotheses using credible intervals is illogical, because the credible intervals already assume the alternative hypothesis is true. This conceptual problem is at the heart of the principle of predictive irrelevance, and Wagenmakers

et al. (2020) show that using credible interval overlap to choose between hypotheses violates this principle.

Individual analysis

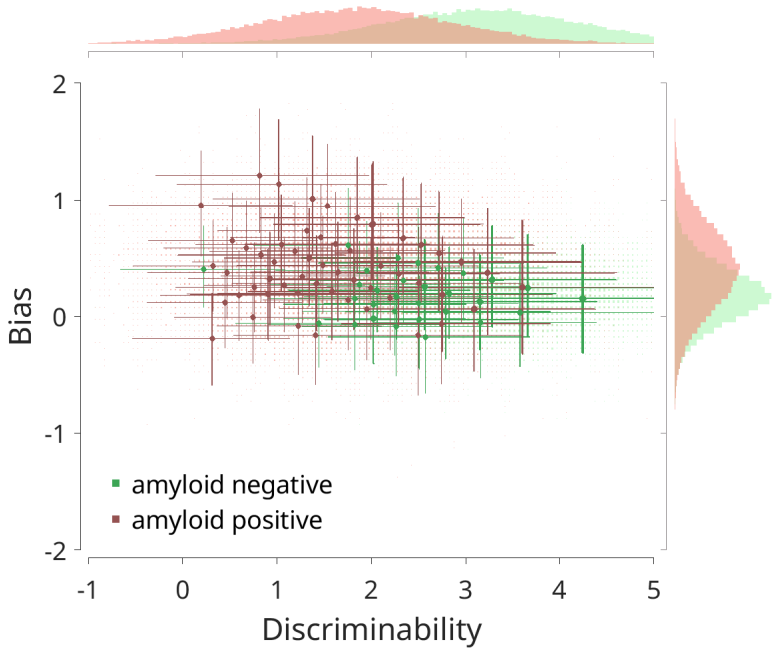


Figure 2.24: The joint and marginal posterior distributions of the mean representative individual discriminability $\tilde{d}'$ and representative individual bias $\tilde{c}$ (yellow), and the individual participant posterior means and 95% credible intervals (blue), for the hierarchical two-choice signal detection model of the amyloid memory data.

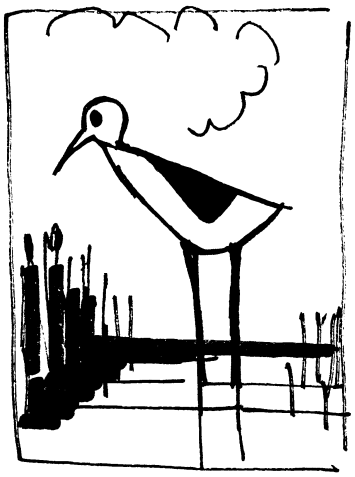
Figure 2.24 extends the individual analysis to show both the representative individual and the specific patients. The joint and marginal posterior distributions for the representative individual are again shown in red and green, respectively, for the amyloid and positive and negative groups. Also shown are the posterior mean discriminabilities $d^p rime_i$ and biases c_i and 95% credible intervals for each of the individual patients, using the same coloring scheme. Circular markers show the posterior means and error bars show the credible intervals.

The marginal posterior distributions for the representative individuals are more legible in Figure 2.24 than Figure 2.23. In particular, it is clear that the distributions for both discriminability and bias have large overlap. Thus, while there are clear differences in mean discriminability at the group level, showing that amyloid positivity reduces memory discriminability, that finding does not necessarily mean that a random amyloid positive patient will have worse memory than a random amyloid negative one. This disconnect is not contradictory in any way. It simply means that the individual differences are large enough that what is true on average is not true in each case. Scientific claims about groups and individuals are answering different questions, and hierarchical models allow both to be addressed simultaneously.

◇

2.9 Formal Details

2.9.1 The Two-Choice Signal Detection Model



Unit normal PDFs and CDFs

To understand how the two-choice signal detection model is formally defined, it is important to understand the probability density function (PDF) and cumulative density functions (CDF) for the "unit Normal" distribution. The unit Normal is the Gaussian distribution with a mean of zero and a standard deviation of one.

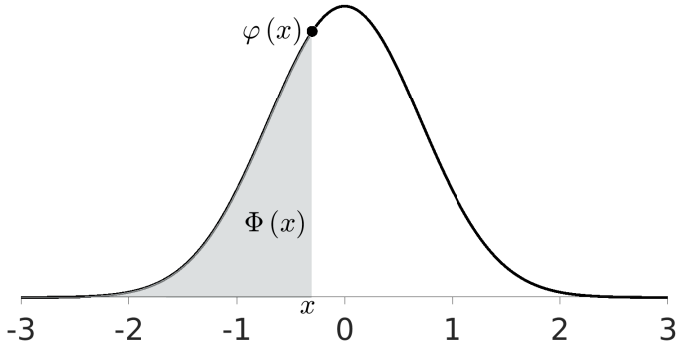


Figure 2.25: Probability density $\varphi(x)$ and cumulative density $\Phi(x)$ functions for the unit Normal distribution evaluated at the point x.

Figure 2.25 shows this Gaussian distribution. The height of the point $\varphi(x)$ is the probability density function at the value x. This is a measure of how relatively likely the value x is within the distribution, so that if $\varphi(x_1)$ is twice as large as $\varphi(x_2)$, x_1 is twice as likely as x_2.

The shaded area $\Phi(x)$ shows the cumulative probability density function for the value x. This is the area under the curve to the left of x. It is the probability of a sampled value from the unit Normal being less than x. The area under the unit Normal is one, so it looks like $\Phi(x) \approx 0.30$, since it involves about 30% of the area.

More formally, the probability density for the unit Normal, which is the Gaussian distribution with mean $\mu = 0$ and standard deviation $\sigma = 1$, at the value x is

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right).$$

The cumulative probability density for the unit Gaussian distribution up to the value x is

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} \exp\left(-\frac{t^2}{2}\right) \mathrm{d}t.$$

Choice probabilities

The CDF for the unit Normal is the key to calculating the probability of hits, misses, correct rejections, and false alarms. The noise distribution is exactly a unit Normal, since it is centered at zeros and has unit variance.

Looking at Figure 2.26, which just presents Figure 2.2 again, the area under the noise distribution to the left of the criterion k is $\Phi(k)$. This corresponds to the probability of a correct rejection. Because the area

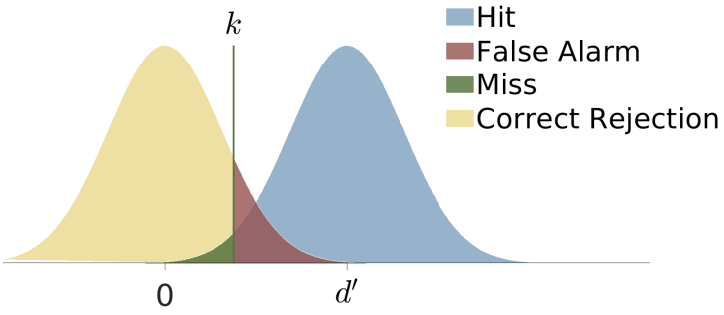


Figure 2.26: Hit, false alarm, miss, and correct rejection decision probabilities according to the two-choice signal detection model.

under any distribution is one, the area to the right of k is $1 - \Phi(k)$. This corresponds to the probability of a false alarm.

The areas for the signal distribution can be calculated in the same way, after accounting for the fact that the distribution is centered at d' rather than zero. The area under the signal distribution to the left of the criterion k is $\Phi(k - d')$. This corresponds to the probability of a miss. The area to the right of k is $1 - \Phi(k - d')$. This corresponds to the probability of a hit. It is a mathematical identity that $1 - \Phi(x) = \Phi(-x)$, so it is also possible to express false alarms as $\Phi(-k)$ and hits as $\Phi(d' - k)$.

Through calculating these areas, the two-choice signal detection model connects the discriminability d' and the criterion k to the hit, miss, false alarm and correct rejection probabilities:

$$\begin{aligned} \theta^{\mathrm{h}} &= 1 - \Phi(k - d') \\ \theta^{\mathrm{f}} &= 1 - \Phi(k) \\ \theta^{\mathrm{m}} &= \Phi(k - d') \\ \theta^{\mathrm{c}} &= \Phi(k). \end{aligned}$$

Priors

There are no widely-agreed priors for the discriminability and criterion parameters. They both lie on the same scale, which is effectively a z-score scale. The noise distribution is a unit Normal, and the signal distribution is expressed in terms of how many unit standard deviations it is centered from the noise distribution. On this scale, a standard rule of thumb in psychology is that three standard deviations is a large difference. On this basis, reasonable priors for both d' and c are Gaussian distributions centered on zero with a standard deviation of 3. JAGS parametrizes the Gaussian distribution in terms of the mean and precision, with precision being simply the reciprocal of the variance, giving priors:

$$\begin{aligned} d' &\sim \text{Gaussian}\left(0, \frac{1}{3^2}\right) \\ c &\sim \text{Gaussian}\left(0, \frac{1}{3^2}\right). \end{aligned}$$

Other priors could also be reasonable. One worth considering is restricting d'_i to be positive, which corresponds to assuming that individuals do not have worse-than-chance ability in the detection task. This is often true, although it is possible to think of exceptions. For example, if the detection task involves trick questions and people's performance is below

chance, discriminability will be negative. More mundanely, if a participant misunderstands the task instructions, and presses the wrong keys to indicate "signal" and "noise," their discriminability will be negative.

Derived parameters

It is straightforward to calculate the criterion k and relative bias β measures of bias from the discriminability d' and bias c. Since the bias c is the difference between the criterion and the unbiased value $\frac{1}{2}d'$, it follows that:

$$k = \frac{1}{2}d' + c.$$

The relative bias β is:

$$\beta_i = \frac{\varphi\left(k_i - d_i'\right)}{\varphi\left(k_i\right)}.$$

It would be possible to place a prior on the criterion k instead of the bias c, and derive c from k instead of the other way around. This is perhaps more natural in terms of the mechanics of signal detection modeling. The reason for the current approach is that the zero-centered prior for the bias c seems better justified. A prior centered on a lack of bias, but allowing some level of bias in either direction, is a reasonable default expectation, and is more easily expressed in terms of the bias c than the criterion k.

Likelihood

The goal of two-choice signal detection model is to describe, explain, and predict data that take the form of k^{h} hits out of $n^{\mathrm{s}} = k^{\mathrm{h}} + k^{\mathrm{m}}$ signal trials and k^{f} false alarms out of $n^{\mathrm{n}} = k^{\mathrm{f}} + k^{\mathrm{c}}$ noise trials. This leads to the likelihood function

$$\begin{aligned} k^{\mathrm{h}} &\sim \text{binomial}\left(\theta^{\mathrm{h}}, n^{\mathrm{s}}\right) \\ k^{\mathrm{f}} &\sim \text{binomial}\left(\theta^{\mathrm{f}}, n^{\mathrm{n}}\right). \end{aligned}$$

2.9.2 JASP Implementation

The JAGS script for the two-choice signal detection model is shown below.

```
# Two-Choice Signal Detection Model
data{
  nIndividuals = length(hit)
  nSignal = hit + miss
  nNoise = fa + cr
}
model{
  for (i in 1:nIndividuals){
    # Likelihood
    hit[i] ~ dbin(thetaHit[i], nSignal[i])
    fa[i] ~ dbin(thetaFA[i], nNoise[i])
    thetaHit[i] = 1 - phi(k[i] - d[i])
    thetaFA[i] = 1 - phi(k[i])
    # Derived parameters
    k[i] = d[i]/2 + c[i]
```

```
    beta[i] = exp(logdensity.norm(k[i], d[i], 1) - logdensity.norm
    (k[i], 0, 1))
    # Priors
    d[i] ~ dnorm(0, 1/3^2)
    c[i] ~ dnorm(0, 1/3^2)
    # Predictive
    hitPrd[i] ~ dbin(thetaHit[i], nPrdSig[i])
    faPrd[i] ~ dbin(thetaFA[i], nPrdNoi[i])
  }
}
```

The JAGS subcode starts with a data block to determine the number of individuals `nIndividuals`, and the number of observed signal `nSignal[i]` and noise `nNoise[i]` observations for each individual.

The main `model` block is organized in terms of the

- **likelihood** that connects the hit `hit[i]` and false alarm `fa[i]` counts to the hit rate `thetaH[i]` and false alarm rate `thetaFA[i]` for the number of signal `nSignal[i]` and noise `nNoise[i]` observations for each individual
- **prior distribution** for the discriminability `d[i]` and bias `c[i]` parameters
- **reparametrization** to calculate the criterion `k[i]` and relative bias `beta[i]`
- **predictive distribution** for generating `hitPrd[i]` hits and `faPrd[i]` false alarms from `nPrdSig[i]` signal and `nPrdNoi[i]` observations

2.9.3 The Hierarchical Two-Choice Signal Detection Model

Priors

The hierarchical two-choice signal detection model has priors for the mean and standard deviation of the discriminability and bias of the group distributions.

$$\begin{aligned} \mu_g^{d'} &\sim \text{Gaussian}\big(0, \frac{1}{3^2}\big) \\ \sigma_g^{d'} &\sim \text{Gaussian}\big(0, \frac{1}{3^2}\big) \\ \mu_g^{c} &\sim \text{Gaussian}\big(0, \frac{1}{3^2}\big) \\ \sigma_g^{c} &\sim \text{Gaussian}\big(0, \frac{1}{3^2}\big). \end{aligned} \tag{2.1}$$

Likelihood

The individual discriminabilities and biases are generated from their group distribution

$$\begin{aligned} d'_i &\sim \text{Gaussian}\big(\mu_{g_i}^{d'}, \frac{1}{\left(\sigma_{g_i}^{d'}\right)^2}\big) \\ c_i &\sim \text{Gaussian}\big(\mu_{g_i}^{c}, \frac{1}{\left(\sigma_{g_i}^{c}\right)^2}\big). \end{aligned} \tag{2.2}$$

The discriminability and bias for the representative individual from each group is generated in the same way:

$$\begin{aligned} \tilde{d}'_g &\sim \text{Gaussian}\left(\mu_g^{d'}, \frac{1}{\left(\sigma_g^{d'}\right)^2}\right) \\ \tilde{c}_g &\sim \text{Gaussian}\left(\mu_g^{c}, \frac{1}{\left(\sigma_g^{c}\right)^2}\right). \end{aligned} \tag{2.3}$$

The process that generates the posterior predicted hit and false alarm counts for each group is based on the discriminability and bias for the representative individual in that group, and is generated with respect to the specified number of signal and noise trials

$$\begin{aligned} \hat{k}_g^{\text{h}} &\sim \text{binomial}\left(\tilde{\theta}^{\text{h}}, \hat{n}^s\right) \\ \hat{k}_g^{\text{f}} &\sim \text{binomial}\left(\tilde{\theta}^{\text{f}}, \hat{n}^n\right). \end{aligned} \tag{2.4}$$

2.9.4 JASP Implementation

The JAGS code for the hierarchical two-choice signal detection model is shown below.

```
# Hierarchical Two-Choice Signal Detection Theory
data{
  nIndividuals = length(hit)
  nSignal = hit + miss
  nNoise = fa + cr
  nGroups = max(group)
}
model{
  for (i in 1:nIndividuals){
    # Likelihood
    hit[i] ~ dbin(thetaHit[i], nSignal[i])
    fa[i] ~ dbin(thetaFA[i], nNoise[i])
    thetaHit[i] = 1 - phi(k[i] - d[i])
    thetaFA[i] = 1 - phi(k[i])
    # Derived parameters
    k[i] = c[i] + d[i]/2
    beta[i] = exp(logdensity.norm(k[i], d[i], 1) - logdensity.norm
    (k[i], 0, 1))
    # Individual priors
    d[i] ~ dnorm(muD[group[i]], 1/sigmaD[group[i]]^2)
    c[i] ~ dnorm(muC[group[i]], 1/sigmaC[group[i]]^2)
  }
  # Representative
  for (g in 1:nGroups){
    dRep[g] ~ dnorm(muD[g], 1/sigmaD[g]^2)
    cRep[g] ~ dnorm(muC[g], 1/sigmaC[g]^2)
    kRep[g] = cRep[g] + dRep[g]/2
    thetaHitRep[g] = 1 - phi(kRep[g] - dRep[g])
    thetaFARep[g] = 1 - phi(kRep[g])
    hitPrd[g] ~ dbin(thetaHitRep[g], nPrdSig[1])
    faPrd[g] ~ dbin(thetaFARep[g], nPrdNoi[1])
  }
  # Group priors
  for (g in 1:nGroups){
```

```
    muD[g] ~ dnorm(0, 1/3^2)
    sigmaD[g] ~ dnorm(0, 1/3^2)T(0,)
    muC[g] ~ dnorm(0, 1/3^2)
    sigmaC[g] ~ dnorm(0, 1/3^2)T(0,)
  }
}
```

High Threshold Models 3

3.1 High Threshold Models

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High threshold models address the same problem as signal detection theory: deciding whether or not there is a "signal" or "target" present in the environment. Chapter 2.1.1 provides motivating examples of signal detection problems. As a reminder, Table 3.1 summarizes the four possible outcomes in a signal detection task. If the decision is that there is a signal when there really is one, the decision is called a *hit*. If the decision is that there is a signal when there really is not, the decision is called a *false alarm*. If the decision is that there is no signal when there really is one, the decision is called a *miss*. If the decision is that there is not a signal and there really is not, the decision is called a *correct rejection*.

Table 3.1: The four outcomes in a signal detection task.

	really signal	really noise
detect signal	hit	false alarm
detect no signal	miss	correct rejection

High threshold models make different cognitive assumptions from signal detection theory. The key difference is that high threshold models do not assume that the representation of stimuli occurs on a uni-dimensional latent "stimulus strength" continuum. Instead, high threshold models characterize the decision making as the culmination of a series of probabilistic cognitive processes. This series takes the form of a decision tree, in which a series of latent cognitive choices eventually lead to an observable behavioral outcome.

The fact that both high threshold models and signal detection theory models provide an account of the same cognitive capability raises the question of the relationship between the two. One perspective is that they can be viewed as competitors. Each makes predictions about the same hit, false alarm, miss, and correct rejection counts, but their predictions are based on different psychological assumptions. It seems reasonable to ask which model makes the better assumptions. This can be assessed by methods for model selection that compare their ability to predict observed data.

Figure 3.1: Different cognitive models as alternative ways of understanding the same reality.

Another perspective is that the models provide different, potentially complementary, and potentially useful ways of analyzing the data. This perspective is depicted in Figure 3.1, through the metaphor of models as

virtual reality glasses that provide a window onto the observed behavior. From this perspective, the challenge is not to identify the correct model. Rather, the challenge is to combine the insights of both models to describe, explain, and predict the psychological phenomena of interest.

Both perspectives are valid and important. But, the models and methods considered in this book are best suited to the second perspective. Methodologically, the focus is on inferring parameters rather than comparing models. The assumption is that the models are already well established theoretically and empirically, and have the potential to be useful. Posterior predictive checking provides a basic measure of reassurance that the models can describe whatever data are being analyzed. If the standard of descriptive adequacy met, the goal is to use the inferences about parameters to understand the observed behavior through the psychological window provided by the model.

3.1.1 Models

High threshold models are a prominent example of a more general class of cognitive models known as multinomial processing trees (Batchelder & Riefer, 1999). Multinomial processing trees are well suited to modeling categorical data, in which observed behavior is distributed across a small set of discrete alternatives. The multinomial processing trees framework has been used to develop models in areas including memory, decision making, and social psychology (Erdfelder et al., 2009).

There are two different high threshold models widely used as cognitive models of signal detection: the one high threshold model and the two high threshold model. Other names are used in the literature: the two high threshold model is also called the "double high threshold" model or the "high low" threshold model. All of these names—indeed, just the use of the word "threshold"—have the unfortunate connotation of referencing a stimulus dimension along which a decision threshold is imposed, in direct conflict with the defining difference with signal detection theory. A better name might be "discrete detection model," but we will use the standard "one high threshold" and "two high threshold" names.

One High Threshold model

As is shown in Figure 3.2, the one high threshold model consists of two decision trees. The left tree corresponds to the situation in which a signal is present. There is a probability α that the decision maker detects the signal. If this happens, they will respond that there is a signal, resulting in an observed hit. If they do not detect a signal, which happens with probability $1-\alpha$, they will produce a response by guessing. Their guesses may be biased, so there is a probability β of a signal response, also leading to a hit, and a probability $1-\beta$ of a no-signal response, leading to a miss.

The right tree corresponds to the situation in which the signal is absent. In this case there is no signal to detect, so responding follows the same guessing process. This means that with probability β a signal response

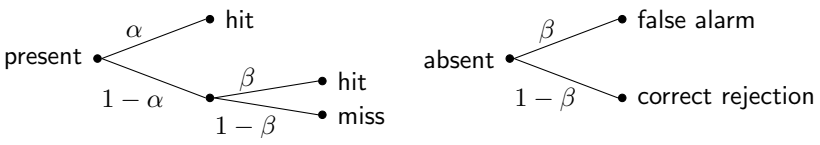


Figure 3.2: Multinomial processing tree for the one high threshold model.

will lead to a false alarm, but with probability $1 - \beta$ a no-signal response will lead to a correct rejection.

The model has two parameters. The α parameter can be interpreted as the probability that a present signal is detected. The β parameter can be interpreted as a guessing bias; it is the probability of guessing that there is a signal if none is detected. Notice that there are two situations in which guessing is needed. One is if a present signal is not detected, and the other is if there is no signal to detect.

The trees in Figure 3.2 define the probability of hits, misses, false alarms, and correct rejections. For example, the probability of a hit is $\alpha + (1 - \alpha)$, which is the probability that either a present signal is detected, or it was not detected but nonetheless guessed to be present.

Two High Threshold model

The two high threshold model makes the additional assumption that there is some probability of detecting that a signal is absent. In some signal detection situations this can be a difficult assumption to justify and interpret. For "needle in a haystack" tasks, like finding treasure on a beach, or deciding whether you have ever seen a person before, it seems hard to be certain of absence without something like an impractical exhaustive search. In other situations, such as knowing a painting is not a forgery, or knowing a word was not on a short study list, detecting absence seems more plausible. There may be simple ways to gather direct evidence for the absence of the signal.

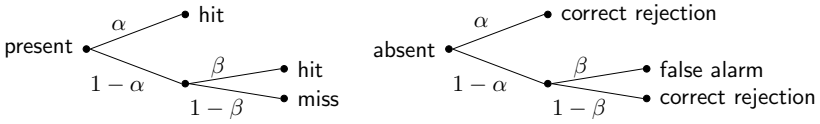


Figure 3.3: Multinomial processing tree for the two high threshold model.

Figure 3.3 shows the decision trees for the two high threshold model. The left tree corresponds to the situation in which a signal is present, and is identical to the corresponding tree for the one high threshold model. The impact of the new assumption occurs in the tree on the right, corresponding to the situation in which signal is absent. There is a probability α this absence will be detected, leading to a correct rejection. Otherwise, with probability $1 - \alpha$, the guessing process occurs, leading to a false alarm or a hit.

Note that the two high threshold model assumes that detecting the presence of a signal and detecting the absence of a signal occur with the same probability α. This assumption seems psychologically implausible in many situations, but is required for model identifiability. If different probabilities were allowed for the two types of detection, counts of hits, misses, false alarms, and correct rejection would generally not provide enough information to make useful inferences.

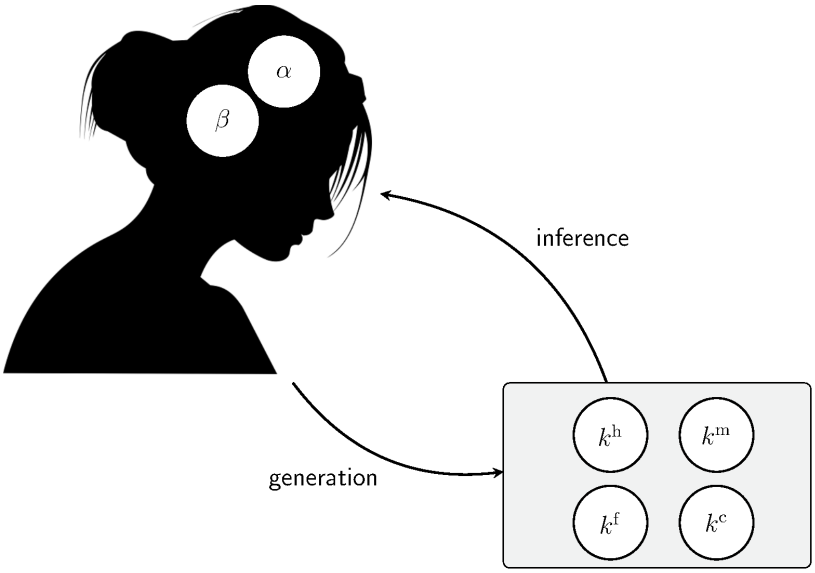


Figure 3.4: The structure of high threshold models.

Figure 3.4 summarizes the structure of both the one and two high threshold models, showing how they relate latent parameters to observed data.

3.1.2 Mechanics

Table 3.2 details the meaning of the parameters and variables in the high threshold models, their name in the model as shown in Figure 3.4, and their name in JASP.

Name	Model	JASP
detection probability	α	`alpha`
guessing bias	β	`beta`
hit count	k^{h}	`hit`
false alarm count	k^{f}	`fa`
miss count	k^{m}	`miss`
correct rejection count	k^{c}	`cr`
predicted hit count	$\hat{k}^{\mathrm{h}}$	`hitPrd`
predicted false alarm count	$\hat{k}^{\mathrm{f}}$	`faPrd`
number of signal observations for prediction	$\hat{n}^{\mathrm{s}}$	`nPrdSig`
number of noise observations for prediction	$\hat{n}^{\mathrm{n}}$	`nPrdNoi`

Table 3.2: The meaning of the observed variables and latent parameters in the high threshold models, with their model and JASP names.

Input data

For each individual, the one and two high threshold models require as input the hit, false alarm, miss, and correct rejection counts, as well as the number of signal and noise trials on which the posterior predictive analysis is based.

In a situation with multiple individuals, these data should be provided on a separate row for each individual. Table 3.3 shows the general data format using the data from the *Head Injury* example, which has four individual experimental conditions.

hit	fa	miss	cr	nPrdSig	nPrdNoi
39	1	6	54	40	60
39	1	6	54	40	60
38	2	14	46	40	60
37	3	14	46	40	60

Table 3.3: Data format for high threshold model analysis.

Output inferences

The JASP implementation provides threee analyses:

- **detection probability and guessing bias posteriors**: marginal posteriors for the detection probabilities α and the guessing bias β for each individual
- **joint posterior of detection and guessing bias**: the joint posteriors for the detection probabilities α and the guessing bias $beta$ for each individual
- **hit and false alarm count posterior predictions**: the posterior predictive distribution of the hit count $\hat{k}^{\text{h}}$ and false alarm count $\hat{k}^{\text{f}}$ for each individual from $\hat{n}^{\text{s}}$ signal observations and $\hat{n}^{\text{n}}$ noise observations

3.2 *Head Injury*

`headInjuryOHT.jasp`
`headInjuryTHT.jasp`

Hannay et al. (1979) studied changes in recognition memory caused by different grades of concussive head injuries. This example applies both of the high threshold models to the recognition memory performance for a healthy group and three injured groups.

Data

In the recognition memory task that Hannay et al. (1979) used, each participant first studied 20 drawings. They then saw five blocks, each with 20 drawings, eight of which were old ones from the study block. The other 12 drawings were new ones. The participant was asked whether each drawing was "old" or "new". Over all of the blocks, the recognition memory experiment involved 100 trials, with 40 old (signal) drawings and 60 new (noise) drawings.

	truth							
	healthy		grade 1		grade 2		grade 3	
decision	old	new	old	new	old	new	old	new
old	39	6	39	6	38	14	37	14
new	1	54	1	54	2	46	3	46

Table 3.4: Head injury data.

One group contains healthy controls. The other groups contain concussive head injury patients, divided into three worsening grades of severity. Table 3.4 shows the results for the median participant in each group Hannay et al. (1979, Table II).

A couple of observations about the data are worthwhile. The performance of the healthy and grade 1 injured partipants is identical. The grade 2 and grade 3 injured participants have worse performance, with more misses and false alarms, but are very similar to each other. The only difference is that the grade 3 median participant has one more miss and one fewer hit than the grade 2 participant.

Results

One high threshold model

Figure 3.5 shows the joint and marginal posteriors for the detection probability α and guessing bias β based on the one high threshold model.

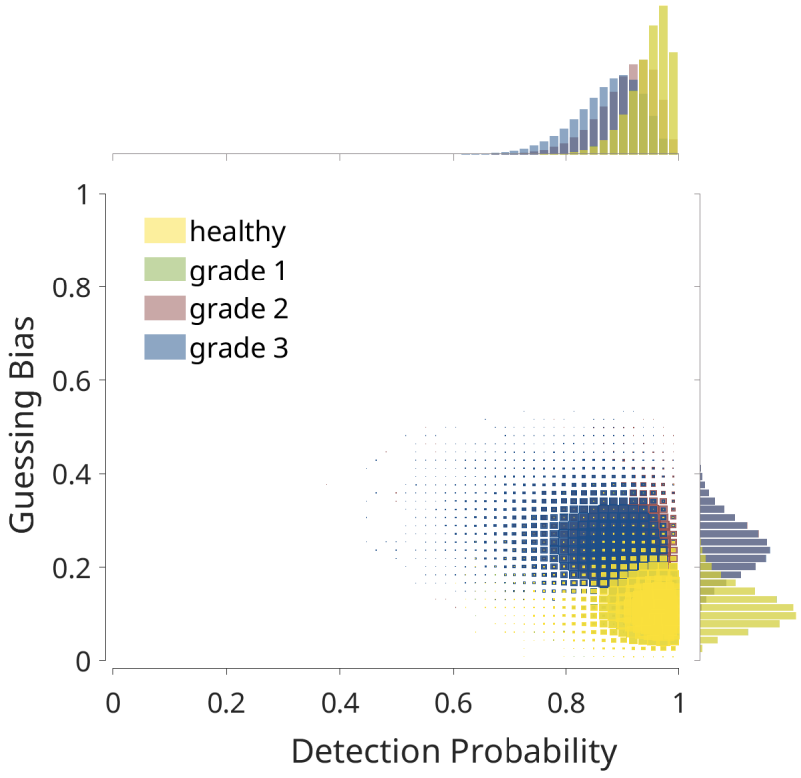


Figure 3.5: The joint and marginal posterior distributions of the detection probabilities α and guessing bias β for the head injury data based on the one high threshold model.

All four groups are inferred to have high detection probabilities, corresponding to high probabilities of remembering studied drawing. This is consistent with the high hit counts in the behavioral data. The probability of guessing that a drawing was studied when it was not remembered is around 10% to 20%. This is consistent with the relatively low false alarm counts relative to correct rejections in the behavioral data.

As also expected from the behavioral data, the groups divide into two clusters: the healthy and grade 1 groups are the same, and the grade 2 and grade 3 groups are very similar. The healthy and grade 1 groups are inferred to be likely to remember drawings better. They are also inferred to be less likely to guess that a drawing that is not remembered was on the study list.

Two high threshold model

Figure 3.6 shows the joint and marginal posteriors for the detection probability α and guessing bias β based on the two high threshold model.

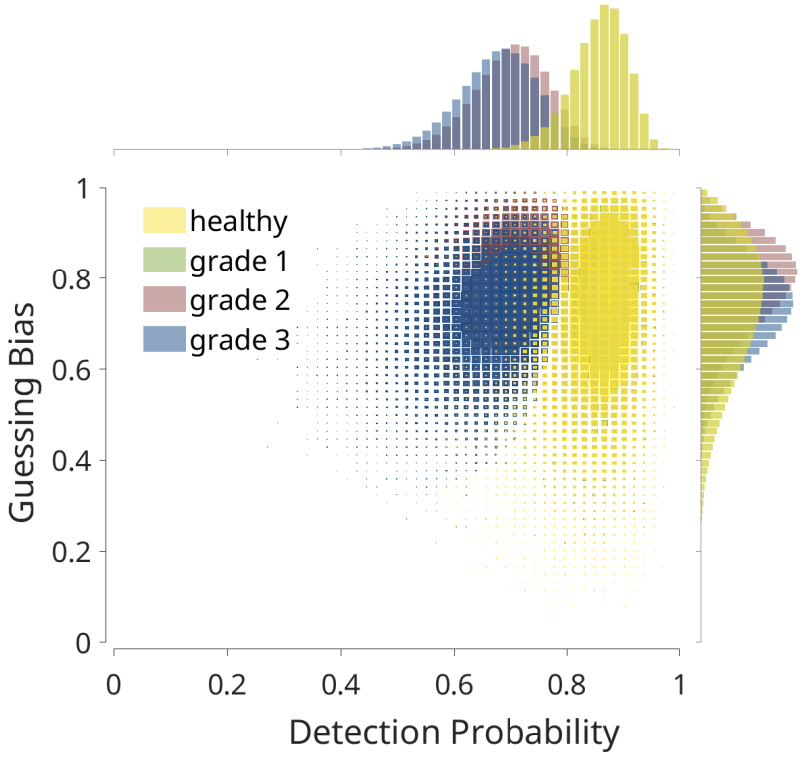


Figure 3.6: The joint and marginal posterior distributions of the detection probabilities α and guessing bias β for the head injury data based on the two high threshold model.

Once again, all four groups are inferred to have high detection probabilities for studied drawings, and the healthy and grade 1 groups are higher than the grade 2 and grade 3 groups. This is qualitatively the same set of results as for the one high threshold model, but there are quantitative differences. The inferred detection probabilities for the two high threshold model are not as high. Intuitively, this is because the detection probabilities are now also responsible for predicting the balance between false alarms and correct rejections when the drawing was not on the study list. The behavioral data shows that new drawings were usually accurately detected, providing additional evidence in favor of a large detection probability.

The inferences about guessing biases show all four groups having a clear bias towards indicating that a drawing was on the study list. There are no clear differences between any of the groups. These results are qualitatively different from the one high threshold model results in Figure 3.5. Intuitively, the two high threshold result comes from the detection probabilities being very high, meaning that guessing is mostly required to account for the errors that are made. The behavioral data show that there were relatively more false alarms than misses, leading to the inference that for the two high threshold model that guessing favors signal responses. The guessing parameter in one high threshold model has to account for all of the behavior on trials for new drawings, not only the errors. Because correct rejections are much more frequent than false

alarms, the guessing bias for the one high threshold model is inferred to favor noise responses.

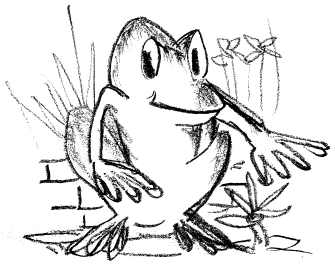
3.3 *Citizen Frogs*

citizenFrogsOHT.jasp
citizenFrogsTHT.jasp

Thorpe et al. (2024) considered data from a "citizen science" project in which non-expert citizens listened to audio recordings of frogs. Using expert judgments to establish the ground truth of whether a frog's call is present in each recording, it is possible to assess the citizens' detection performance.

Performance can be characterized in terms of hit, miss, false alarm, and correct rejection counts, by treating recordings for which experts judge a frog is present as a signal stimulus. Thus, a hit is when a citizen correctly detects a frog, a miss occurs if they do not detect the presence of a frog, a false alarm occurs if they decide there is a frog when none is present, and a correct rejection occurs if they decide there is no frog when none is present.

High threshold models are potentially useful for inferring detection rates and guessing biases. One interesting question is whether one high or two high threshold models are better accounts of the cognitive process. The choice of model depends on whether it is possible, in general, to perceive that a frog is absent from a recording. Without a general answer to this question, a sensible approach is to consider both models.



Data

We use a limited set of data from Thorpe et al. (2024, Study 2), which involved a total of 587 citizens and 1260 audio recordings. Each citizen typically listens to only a small subset of the recordings. We consider the three citizens who listened to most recordings, and one other more representative citizen who listened to a small number of recordings.

Table 3.5: Citizen frogs data.

	expert ground truth							
	citizen A		citizen B		citizen C		citizen D	
decision	present	absent	present	absent	present	absent	present	absent
"there is a frog"	45	1	17	1	28	227	1	9
"there is no frog"	11	488	12	280	22	237	0	1

Table 3.5 summarizes the performance of these four citizens, labeled Citizens A to D. It is worth noting that there is a low base rate, around 10%, of frogs being present on recordings. Citizen A evaluated 545 recordings and made mostly accurate decisions. Their mistakes were more often misses than false alarms. Citizen B evaluated 310 recordings, with a little less accuracy, and produced more misses than false alarms. Citizen C evaluated 514 recordings, with much worse accuracy than Citizens A and B. They produced a very large number of false alarms. Citizen D

evaluated only 11 recordings, only one of which had a frog present. They also produced a relatively high number of false alarms.

Results

One high threshold model

Figure 3.7 shows the joint and marginal posteriors for the four citizen scientists based on the one high threshold model.

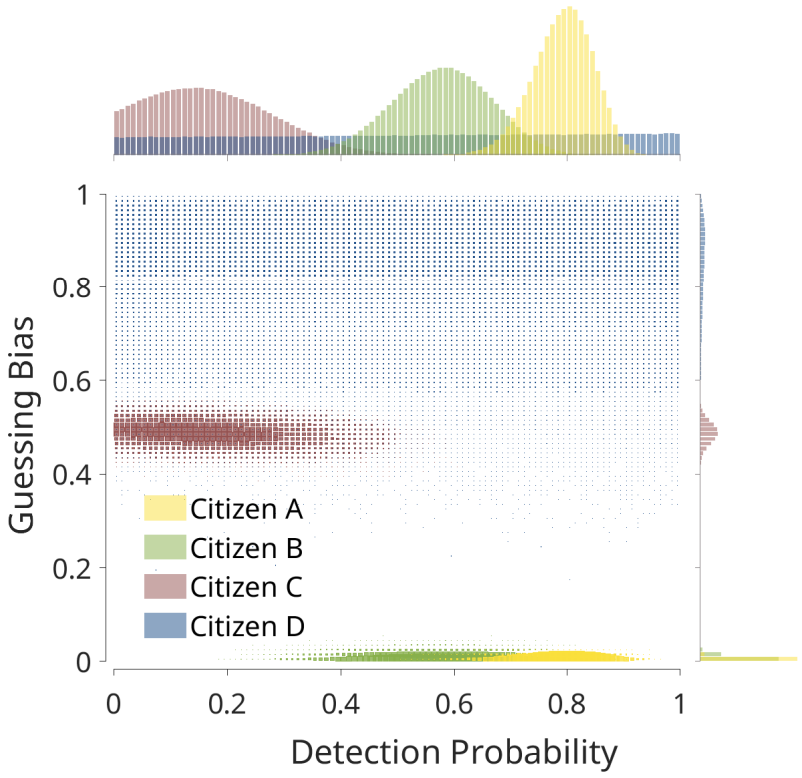


Figure 3.7: The joint and marginal posterior distributions of the detection probabilities α and guessing bias β for the citizen frogs data based on the one high threshold model.

Both Citizens A and B have very low rates of guessing that a frog is present, but Citizen A very likely has a better detection probability. This provides an intuitive account of their good performance and low false alarm rates.

Citizen C, in contrast, guesses a frog is present about half the time when no frog is detected, and their detection probability is poor. This provides an intuitively reasonable explanation of their poor performance and relatively high false alarm rate.

Finally, Citizen D is inferred as being likely to have a high bias towards guessing a frog is present. This explains their high false alarm rate. Almost nothing is learned about the detection probability for Citizen D. The posterior is very similar to the uniform prior. This is because there was only one signal trial, so the behavioral data provide little information about their detection ability.

Two high threshold model

Figure 3.8 shows the joint and marginal posteriors for the four citizen scientists based on the two high threshold model.

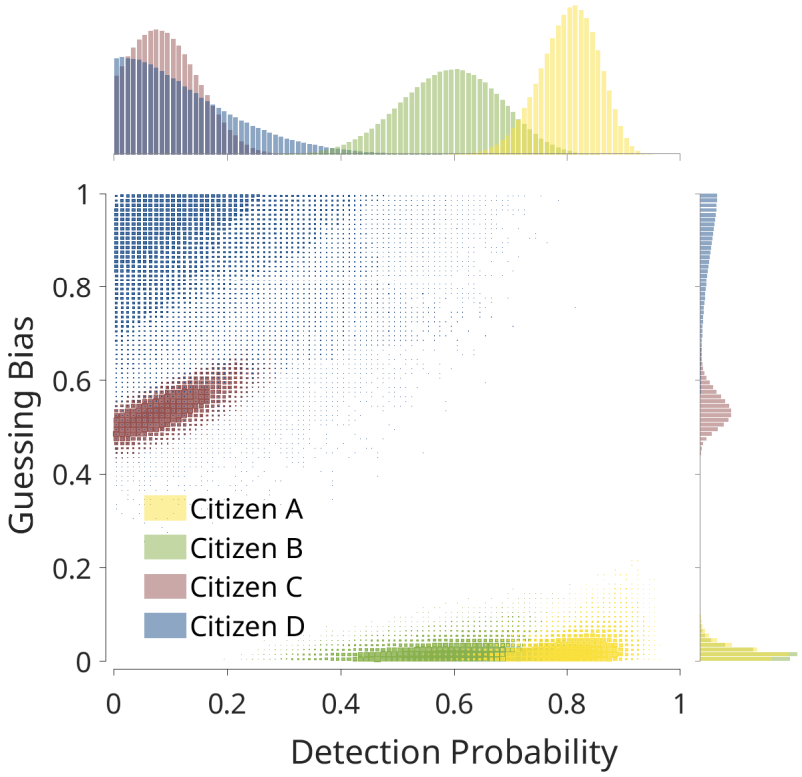


Figure 3.8: The joint and marginal posterior distributions of the detection probabilities α and guessing bias β for the citizen frogs data based on the two high threshold model.

The inferences for Citizens A, B, and C are very similar to those made by the one high threshold model in Figure 3.7.

The inference about the detection probability of Citizen D, however, is very different. The inference is now that they have a low detection probability, with a posterior mean of 0.13 and a 95% credible interval $(0.01, 0.38)$. The behavioral data now significantly update the uniform prior. The reason for this is that the two high threshold model assumes that absences can be detected. This means that both signal and noise trials are relevant to making inferences about detection probability, increasing the relevant number of decisions from 1 to 11.

3.4 *Pilot Landing*

`pilotLandingOHT.jasp`

This example uses the same data as Chapter 2.5. It involves the Salvan et al. (2022) study of the decision making of four pilots in a number of landing situations.

Data

On each trial, the pilots were shown a primary flight display interface with 21 features of the situation, and had to decide whether they would land the plane in this situation, or execute a "go-around". A ground truth

was established for each landing situation, which allows the results to be presented as a table of counts for each pilot, as shown in Table 3.6.

Table 3.6: Pilot landing data.

	truth							
	pilot 1		pilot 2		pilot 3		pilot 4	
decision	GA	land	GA	land	GA	land	GA	land
go-around	3	0	2	2	3	3	2	4
land	0	27	1	25	0	24	0	23

We treat a go-around situation as a signal to be detected. This coding leads to the one high threshold model shown in Figure 3.9. In the left tree, there is objectively no problem with landing, but there is still a guessing bias β that the pilot decides to go-around anyway. In the right tree, there is objectively a problem, which the pilot notices with detection probability α and executes a go-around. If they fail to detect the problem, their behavior is determined by the go-around bias.

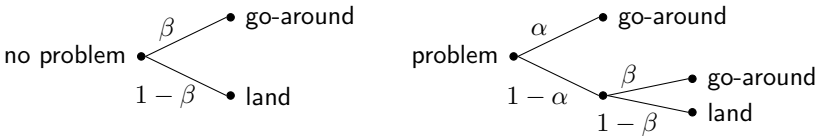


Figure 3.9: One high threshold model applied to the pilot landing data.

Results

Figure 3.7 shows the joint and marginal posteriors for the four pilots based on the one high threshold model. The joint distributions appear to be well summarized by the two marginal distributions, and so we focus the analysis on the marginal distributions for detection probability and guessing bias in turn.

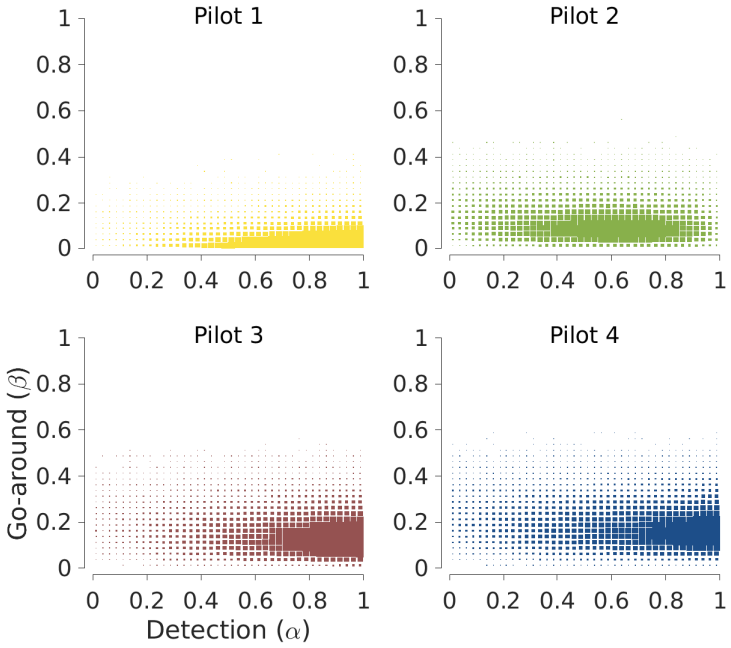


Figure 3.10: The joint and marginal posterior distributions of the detection probabilities α and guessing bias β for the pilot landing data based on the one high threshold model.

Figure 3.11 shows the marginal posteriors for the detection probabilities α based on the one high threshold model.

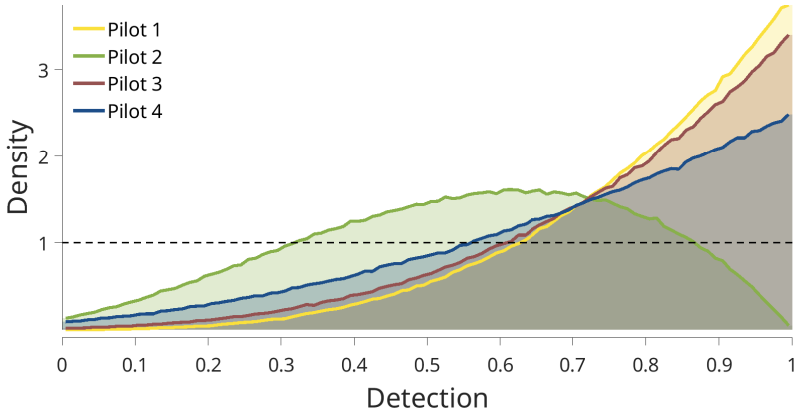


Figure 3.11: The marginal posterior distributions of the detection probabilities α for the pilot landing data based on the two high threshold model.

Pilots 1, 3, and 4 are all likely to have good, and quite possibly excellent, abilities to detect problems. Pilot 2 is more likely to have a lower ability to detect problems. The significant uncertainty for all four pilots is because of the limited number of trials.

Figure 3.12 shows the marginal posteriors for the guessing biases β based on the one high threshold model.

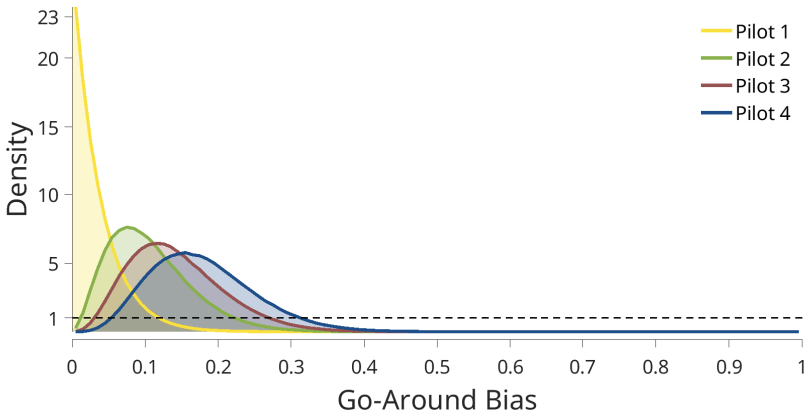


Figure 3.12: The marginal posterior distributions of the guessing biases β for the pilot landing data based on the one high threshold model.

Pilots 2, 3, and 4 have similar probabilities of going around, likely somewhere between 5% and 30%. Pilot 1 is likely to have a lower bias toward going around. This is consistent with the behavioral data. Pilot 1 was the only pilot never to go-around when there was no objective problem.

Posterior predictions

Another way of understanding the performance of the pilots is provided by posterior predictive analysis. We use posterior predictive counts $\hat{n}^{\mathrm{s}} = 1$ for signal trials, and $\hat{n}^{\mathrm{n}} = 99$ for noise trials. The motivation is to examine the predicted performance in an expanded set of 100 trials that still follow a real-world base rate with 1% of landings involving a problem that needs to be detected.

Table 3.7 summarizes the mean and 95% credible intervals the posterior predicted hit and false alarm counts for each pilot. The mean posterior predicted number of hits can be interpreted as the probability the pilot will correctly go-around on the one problem trial. That probability is only 62% for pilot 2, but around 90% for the other three pilots. The mean

	hits		false alarms	
	mean	95% CI	mean	95% CI
pilot 1	0.80	(0, 1)	3.53	(0, 14)
pilot 2	0.60	(0, 1)	10.54	(1, 26)
pilot 3	0.80	(0, 1)	14.14	(3, 31)
pilot 4	0.76	(0, 1)	17.63	(5, 35)

Table 3.7: Posterior predictive marginal distributions for the hit and false alarm counts with 1 signal and 99 noise trials.

posterior posterior predicted number of false alarms, and the associated credible intervals, quantify how many times the pilot will mistakenly go-around on the 99 non-problematic landings. Pilot 1 performs the best, almost certainly mistakenly going around fewer than 10 times, with an expectation of 1.44 times. Pilots 3 and 4 fare much worse by this measure, with the possibility of going around unnecessarily as many as 30 times.

3.5 Hierarchical High Threshold Models

3.5.1 Motivation

The basic motivation for extending hierarchical models is explained in detail for the success rate model in Chapter 1. If hierarchical models are unfamiliar, it is worth reading Sections and 1.6. The hierarchical extension of the high threshold models shares the same motivation of providing a group level account of how different individuals are related to each other in their detection probabilities and guessing biases. Hierarchical high threshold models have potential advantages that are analogous to those spelled out for the hierarchical success rate model. These relate to the ability to make inferences about detection probability and guessing bias at both the group and individual level, and the ability to make inferences and predictions related to a representative individual.

3.5.2 Model

Figure 3.13 summarizes the structure of the hierarchical high threshold models, showing how they relate to latent parameters to observed data. Each individual continues to have their own detection probabilities α_i and guessing biases β_i. For sources in the same group g, these parameters are linked by an overarching group distributions with means μ_g^α and μ_g^β, and standard deviations σ_g^α and σ_g^β. Inference about the detection rates and guessing probabilities come from the hit k_i^{h}, false alarm k_i^{f}, miss k_i^{m}, and correct rejection k_i^{c} counts. Inferences about the individual detection rates and guessing probabilities, in turn, allow inferences about the mean and standard deviation of their group distributions.

The inferences about the group distributions allow the detection rate $\tilde{\alpha}_g$ and guessing probability $\tilde{c}_g$ of a representative individual from each group g to be generated. This, in turn, allows a predictions to be made about the hit $\hat{k}_g^{\text{h}}$ and false alarm $\hat{k}_g^{\text{f}}$ counts for a representative individual in each group.

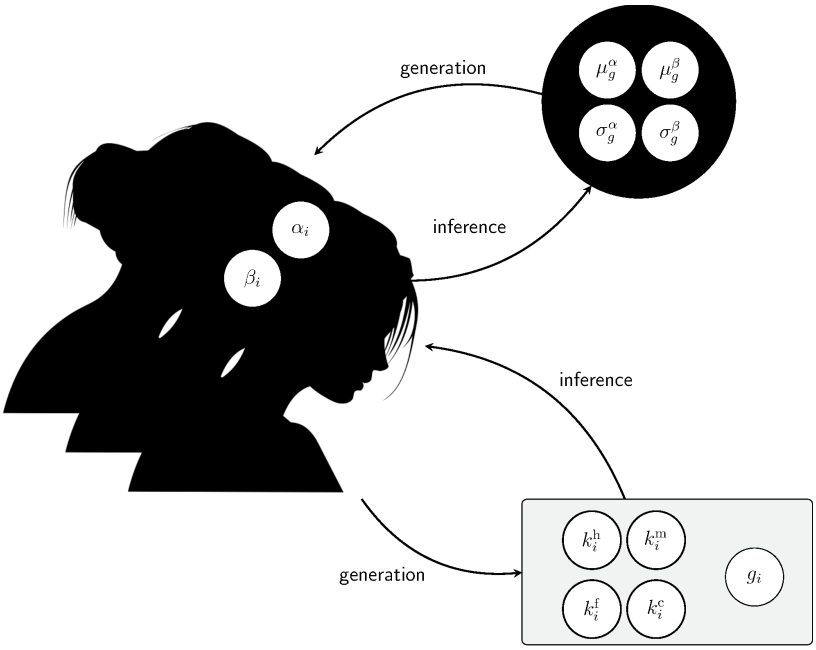


Figure 3.13: The structure of the hierarchical high threshold models.

Table 3.8: The meaning of the observed variables and latent parameters in the hierarchical high threshold models, with their model and JASP names.

Name	Model	JASP
detection probability	α_i	`alpha[i]`
guessing bias	β_i	`beta[i]`
group	g_i	`group[i]`
mean detection for group	μ_g^α	`muAlpha[g]`
standard deviation for discriminability of group	σ_g^α	`sigmaAlpha[g]`
mean bias for group	μ_g^β	`muBeta[g]`
standard deviation of bias for group	σ_g^β	`sigmaBeta[g]`
hit count	k_i^{h}	`hit[i]`
false alarm count	k_i^{f}	`fa[i]`
miss count	k_i^{m}	`miss[i]`
correct rejection count	k_i^{c}	`cr[i]`
representative detection for group	$\tilde{\alpha}'_g$	`alphaRep[g]`
representative bias for group	$\tilde{\beta}_g$	`betaRep[g]`
posterior predicted hit count for group	$\hat{k}_g^{\mathrm{h}}$	`hitPrd[g]`
posterior predicted false alarm count for group	$\hat{k}_g^{\mathrm{f}}$	`faPrd[g]`
number of signal observations for prediction	$\hat{n}^{\mathrm{s}}$	`nPrdSig`
number of noise observations for prediction	$\hat{n}^{\mathrm{n}}$	`nPrdNoi`

3.5.3 Mechanics

Table 3.8 lists the observed variables and latent parameters and variables for the hierarchical high threshold models, giving both their statistical symbol and their JASP code name. The first set of variables above the dividing line are the core parameters and data included in Figure 3.13. The core parameters are the detection probabilities α_i and guessing biases β_i and their group means and standard deviations μ_g^α, μ_g^β, σ_g^α, and σ_g^β. The core observations are the signal detection counts k_i^{h}, k_i^{f}, k_i^{m}, and k_i^{c} for each individual, and the group g_i to which they belong.

The inclusion of group distributions allow the discriminability $\tilde{\alpha}_g$ and bias $\tilde{\beta}_g$ of a representative individual for each group to be inferred. This representative success rate is used, in turn, to make predictions about

the number of hits $\hat{k}_g^{\text{h}}$ and false alarms $\hat{k}_g^{\text{f}}$ from $\hat{n}^{\text{s}}$ signal and $\hat{n}^{\text{n}}$ noise trials.

Input data

Table 3.9 shows the general data format using the data from the *Amyloid Memory* example.

hit	fa	miss	cr	group	nPrdSig	nPrdNoi
13	2	0	15	1	15	15
12	3	1	14	1		
14	1	0	15	2		
...	...	...	...	...		

Table 3.9: JASP data format for the high threshold models.

Output analyses

The JASP implementation provides four analyses:

- **individual detection probability and guessing bias posteriors**: marginal posteriors for the detection probabilities α_i and the guessing bias β_i for each individual
- **group detection probability and guessing bias posteriors**: the joint and marginal posterior distribution of the mean detection probability μ_g^{α} and mean guessing bias μ_g^{β} for each group
- **representative detection probability and guessing bias**: the joint and marginal posterior distribution of the detection probabilities $\tilde{\alpha}_i$ and bias $\tilde{\beta}_i$ for the representative individual in each group
- **representative posterior predictions**: the marginal posterior predictive distributions of the representative hit count $\hat{k}_g^{\text{h}}$ and false alarm count $\hat{k}_g^{\text{f}}$ from $\hat{n}^{\text{s}}$ signal observations and $\hat{n}^{\text{n}}$ noise trials for the representative individual in each group

3.6 *Amyloid Memory*

`amyloidMemoryTHT.jasp`

This example uses the same data as Chapter 2.8. It uses the hierarchical two high threshold model to examine the individual and group differences in recognition memory in groups of people with different levels of beta amyloid.

Data

The data for this application come from a clinical setting, and involve 200 patients aged over 75 years. Based on a cerebrospinal fluid measurement, 100 of the patients are classified as amyloid positive and the other 100 are amyloid negative.

Every patient completed the Rey auditory verbal learning test (RALVT: Bean, 2011). The RAVLT is a recognition memory task in which a person studies a set of 15 words, before being tested on 30 words, made up of 15 words that were on the study list and 15 words that were not. During the

test, the person has to identify whether each presented word is one of the "old" study words or a "new" word.

Results

Posterior predictive analysis

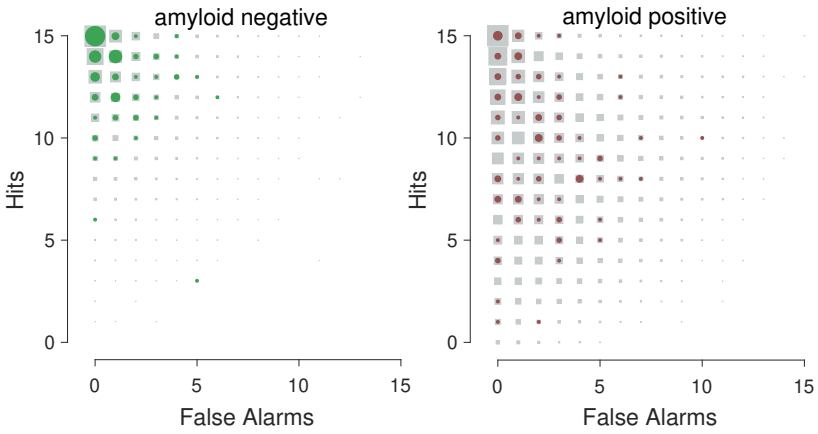


Figure 3.14: Posterior predictive analysis for the hierarchical two high threshold model analysis of the amyloid memory data.

Figure 3.14 shows the posterior predictive analysis for the hierarchical two high threshold model. The data are shown by green and red circles for the amyloid negative and amyloid positive groups, as in Figure 2.21. The posterior predictive distributions are shown by the gray shading, with darker squares representing hit and false alarm count combinations with greater posterior predictive density. The key result of the analysis is that the posterior predictive distribution gives significant mass to the range over which the data vary for both the amyloid negative and amyloid positive groups. This provides evidence that the hierarchical model is descriptively adequate.

Group analysis

Figure 3.15 shows the joint and marginal posterior distribution of the group mean detection probabilities and guessing biases, and the posterior representative distribution, for the amyloid negative and amyloid positive groups. The group mean distributions for the amyloid negative group μ_1^α and μ_1^β are shown in darker green, and for the amyloid negative group μ_2^α and μ_2^β in darker red. The representative distributions for the amyloid negative group $\tilde{\mu}_1^\alpha$ and $\tilde{\mu}_1^\beta$ are shown in lighter green, and for the amyloid negative group $\tilde{\mu}_2^\alpha$ and $\tilde{\mu}_2^\beta$ in lighter red. The joint distributions are shown in the central panel, the marginal distributions for discriminability are shown in the top panel, and the marginal distributions for bias are shown in the right panel.

At the group level, the amyloid negative group probably has better mean detection probability than the amyloid positive group, although there is large uncertainty about both means. The posterior mean detection probability for the amyloid negative group is $\mu_1^\alpha = 0.96$ with a 95% credible interval $(0.88, 1.00)$, while for the amyloid positive group $\mu_2^{d'} = 0.65$ with a 95% credible interval $(0.31, 0.97)$.

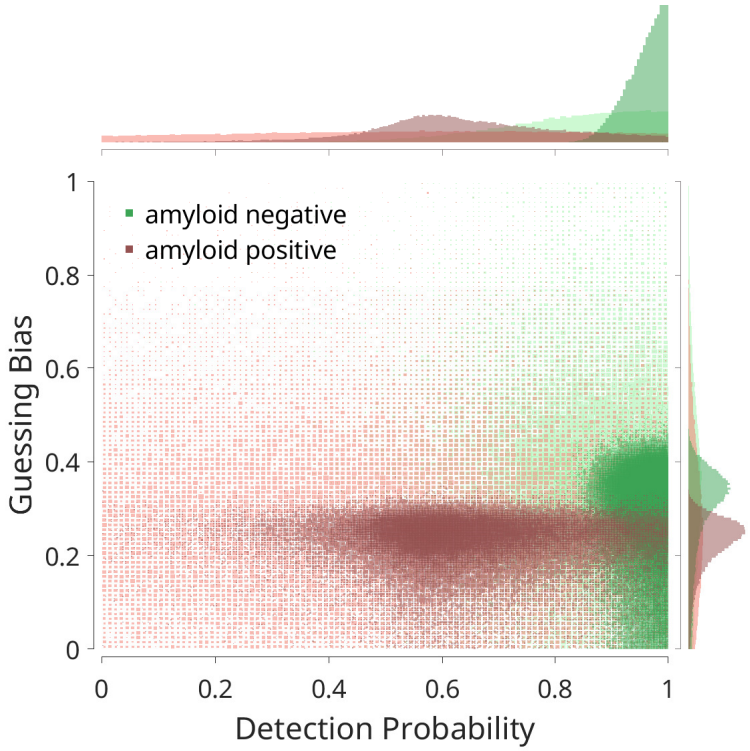


Figure 3.15: Inferences for the detection probability and guessing bias of the amyloid negative and amyloid positive groups, based on the hierarchical two high threshold model.

The inferences for group mean guessing probability do not show clear evidence of differences between the groups. The posterior mean guessing probability for the amyloid negative group is $\mu_1^\beta = 0.31$ with a 95% credible interval $(0.10, 0.42)$, while for the amyloid positive group $\mu_2^\beta = 0.21$ with a 95% credible interval $(0.03, 0.30)$. A reasonable conclusion is that both groups show a tendency to guess new, but it is uncertain how strong those tendencies are.

The inferences for the posterior representative individuals are very wide, and show that conclusions at the level of group means do not apply to individual patients. For example, for detection probabilities, the 95% credible intervals are $(0.51, 0.99)$ for the amyloid negative group and $(0.04, 0.99)$ for the amyloid positive group. This means that amyloid negative individual patients could have detection probabilities anywhere above one-half, and amyloid negative individual patients could have almost any detection probability. This is consistent with the wide range of hit counts seen in the behavioral data, as shown in Figure 3.14.

Individual analysis

Figure 3.16 shows the individual analysis of the patients. Their posterior means and 95% credible intervals are overlayed on the joint posterior distributions for the representative individuals. specific patients. The large individual differences in performance are clear, as is the overlap between the two groups at the individual level.

One reaction to the lack of clear differences between the individual amyloid positive and negative patients is to be disappointed. There is very little that can be concluded with confidence. Another reaction is to

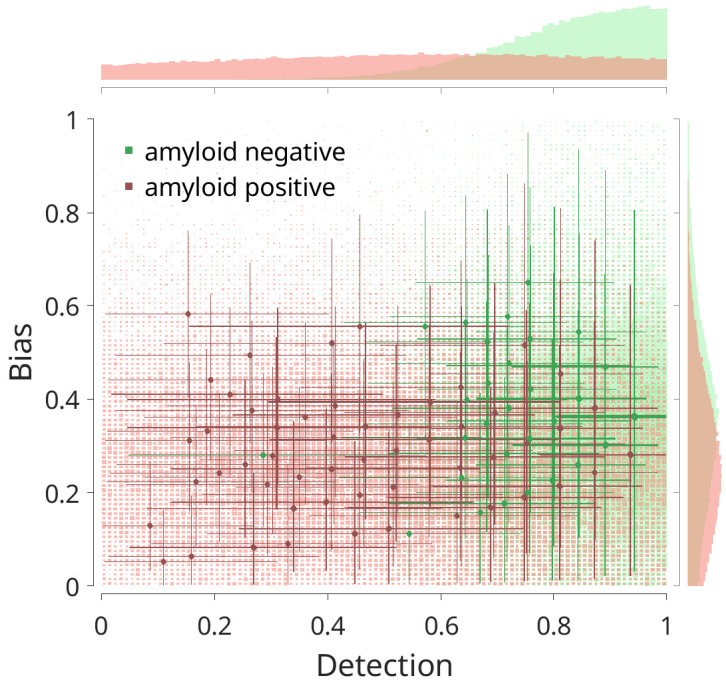
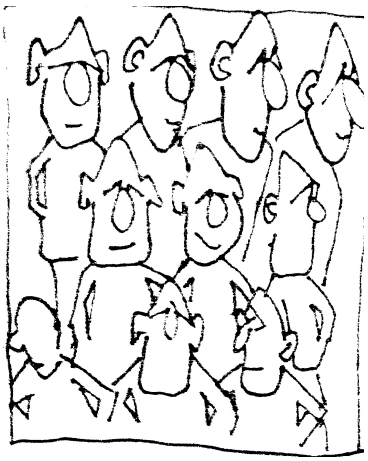


Figure 3.16: Individual analysis of the amyloid memory data using the hierarchical two high threshold model. The joint and marginal posterior distributions of the mean representative individual detection $\tilde{\alpha}$ and representative individual bias $\tilde{\beta}$, for the amyloid positive and negative groups. The circles and error bars show the posterior means and 95% credible intervals for the patients in each group.

appreciate the ability to be able to reach some suggestive conclusions at the group level. In particular, there is enough evidence to pursue the idea that amyloid positivity worsens detection without affecting guessing. The ability to make these inferences at the group level, while respecting and accommodating the large individual differences, is the key strength of the hierarchical model.

◇

3.7 Formal Details



3.7.1 One and two high threshold models

Likelihood

The one high threshold model assume the hit rate θ^{h} and the false alarm rate θ^{f} are

$$\begin{aligned} \theta_i^{\text{h}} &= \alpha_i + (1 - \alpha_i)\,\beta_i \\ \theta_i^{\text{f}} &= \beta_i \end{aligned}$$

The two high threshold model makes the same assumptions about how hits are produced. Either a signal is correctly detected or is guessed to exist. The two high threshold model, however, makes different assumptions about how false alarms are produced. They are made by guessing only

after a failure to detect the signal is absent. This means that the model equations are

$$\begin{aligned} \theta_i^{\text{h}} &= \alpha_i + (1 - \alpha_i)\,\beta_i \\ \theta_i^{\text{f}} &= (1 - \alpha_i)\,\beta_i \end{aligned}$$

For both models, the posterior predictive counts of k_i^{h} hits and k_i^{f} false alarms from $\hat{n}_i^{\text{s}}$ signal observations and $\hat{n}_i^{\text{n}}$ noise observations

$$\begin{aligned} k_i^{\text{h}} &\sim \text{binomial}\big(\theta_i^{\text{h}}, \hat{n}_i^{\text{s}}\big) \\ k_i^{\text{f}} &\sim \text{binomial}\big(\theta_i^{\text{f}}, \hat{n}_i^{\text{n}}\big). \end{aligned}$$

Priors

Both models assume that the detection rate α_i and guessing rate β_i are a priori equally likely to be any possible value, so that

$$\begin{aligned} \alpha_i &\sim \text{uniform}\big(0, 1\big) \\ \beta_i &\sim \text{uniform}\big(0, 1\big). \end{aligned}$$

JASP Implementation

The JAGS script for the one high threshold model is shown below. It starts with a data block that calculates the number of observed signal `nSignal[i]` and noise `nNoise[i]` trials, and calculates the number of individuals `nSources`. The model block defines the likelihood of the hit counts `h[i]` and false alarms `fa[i]` for each individual, and the priors on the detection `alpha[i]` and guessing `beta[i]` parameters. Finally, the posterior predictive hit `hitPrd[i]` and false alarms `faPrd[i]` counts are sampled from the provided number of `nPrdSig[i]` and noise `nPrdNoi[i]` observations.

```
# One High Threshold
data{
  for (i in 1:length(hit)){
    nSignal[i] = hit[i] + miss[i]
    nNoise[i] = fa[i] + cr[i]
  }
  nIndividuals = length(hit)
}
model{
  for (i in 1:nIndividuals){
    hit[i] ~ dbin(thetaHit[i], nSignal[i])
    fa[i] ~ dbin(thetaFA[i], nNoise[i])
    thetaHit[i] = alpha[i] + (1-alpha[i])*beta[i]
    thetaFA[i] = beta[i]
    alpha[i] ~ dunif(0, 1)
    beta[i] ~ dunif(0, 1)
    # Predictive
    hitPrd[i] ~ dbin(thetaHit[i], nPrdSig[i])
    faPrd[i] ~ dbin(thetaFA[i], nPrdNoi[i])
  }
}
```

The JAGS script for the one high threshold model is shown below. The only difference is the likelihood of false alarms `fa[i]` for each individual.

```
# Two High Threshold
data{
  for (i in 1:length(hit)){
    nSignal[i] = hit[i] + miss[i]
    nNoise[i] = fa[i] + cr[i]
  }
  nIndividuals = length(hit)
}
model{
  for (i in 1:nIndividuals){
    hit[i] ~ dbin(thetaHit[i], nSignal[i])
    fa[i] ~ dbin(thetaFA[i], nNoise[i])
    thetaHit[i] = alpha[i] + (1-alpha[i])*beta[i]
    thetaFA[i] = (1-alpha[i])*beta[i]
    alpha[i] ~ dunif(0, 1)
    beta[i] ~ dunif(0, 1)
    # Predictive
    hitPrd[i] ~ dbin(thetaHit[i], nPrdSig[i])
    faPrd[i] ~ dbin(thetaFA[i], nPrdNoi[i])
  }
}
```

3.7.2 Hierarchical one and two high threshold models

Priors

The hierarchical high threshold models have priors for the mean and standard deviation of the detection probability and guessing bias of the group distributions.

$$\begin{aligned} \mu_g^\alpha &\sim \text{uniform}(0,1) \\ \sigma_g^\alpha &\sim \text{uniform}(0,1) \\ \mu_g^\beta &\sim \text{uniform}(0,1) \\ \sigma_g^\beta &\sim \text{uniform}(0,1). \end{aligned} \tag{3.1}$$

Likelihood

The detection probabilities and guessing biases for each individual are generated from truncated Gaussian group distributions

$$\begin{aligned} \alpha_i &\sim \text{Gaussian}_{(0,1)}\left(\mu_{g_i}^\alpha, \frac{1}{\left(\sigma_{g_i}^\alpha\right)^2}\right) \\ \beta_i &\sim \text{Gaussian}_{(0,1)}\left(\mu_{g_i}^\beta, \frac{1}{\left(\sigma_{g_i}^\beta\right)^2}\right). \end{aligned} \tag{3.2}$$

The representative discriminability and bias for each group is generated

in exactly the same way:

$$\begin{aligned} \tilde{\alpha}_g &\sim \text{Gaussian}_{(0,1)}\left(\mu_g^\alpha, \frac{1}{\left(\sigma_g^\alpha\right)^2}\right) \\ \tilde{\beta}_g &\sim \text{Gaussian}_{(0,1)}\left(\mu_g^\beta, \frac{1}{\left(\sigma_g^\beta\right)^2}\right). \end{aligned} \tag{3.3}$$

The process that generates the posterior predicted hit and false alarm counts for each group is based on the representative detection probability and guessing bias for that group, and is generated with respect to the specified number of signal and noise trials

$$\begin{aligned} \hat{k}_g^{\text{h}} &\sim \text{binomial}\left(\tilde{\theta}^{\text{h}}, \hat{n}^s\right) \\ \hat{k}_g^{\text{f}} &\sim \text{binomial}\left(\tilde{\theta}^{\text{f}}, \hat{n}^n\right). \end{aligned} \tag{3.4}$$

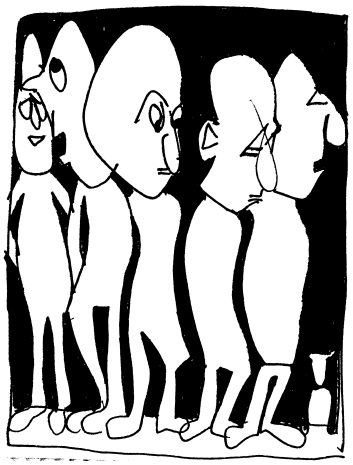
3.8 JASP implementation

The JAGS code for the hierarchical one high threshold model is shown below.

```
# Hierarchical One High Threshold
data{
  nSignal = hit + miss
  nNoise = fa + cr
  nGroups = max(group)
  nIndividuals = length(hit)
}
model{
  for (i in 1:nIndividuals){
    # Observed counts
    hit[i] ~ dbin(thetaHit[i], nSignal[i])
    fa[i] ~ dbin(thetaFA[i], nNoise[i])
    # Two High Threshold Model
    thetaHit[i] = alpha[i] + (1-alpha[i])*beta[i]
    thetaFA[i] = beta[i]
    # Individual priors
    alpha[i] ~ dnorm(muAlpha[group[i]], 1/sigmaAlpha[group[i]]^2)T
    (0, 1)
    beta[i] ~ dnorm(muBeta[group[i]], 1/sigmaBeta[group[i]]^2)T(0,
     1)
  }
  # Representative
  for (g in 1:nGroups){
    alphaRep[g] ~ dnorm(muAlpha[g], 1/sigmaAlpha[g]^2)T(0, 1)
    betaRep[g] ~ dnorm(muBeta[g], 1/sigmaBeta[g]^2)T(0, 1)
    thetaHitRep[g] = alphaRep[g] + (1-alphaRep[g])*betaRep[g]
    thetaFARep[g] = betaRep[g]
    hitPrd[g] ~ dbin(thetaHitRep[g], nPrdSig[1])
    faPrd[g] ~ dbin(thetaFARep[g], nPrdNoi[1])
  }
  # Group priors
  for (g in 1:nGroups){
    muAlpha[g] ~ dunif(0, 1)
    sigmaAlpha[g] ~ dunif(0, 1)
```

```
    muBeta[g] ~ dunif(0, 1)
    sigmaBeta[g] ~ dunif(0, 1)
  }
}
```

The very similar JAGS code for the hierarchical two high threshold model is shown below. The only change is to the definition of the false alarm rate for individuals and the posterior representative individual for each group.



```
# Hierarchical Two High Threshold
data{
  nSignal = hit + miss
  nNoise = fa + cr
  nGroups = max(group)
  nIndividuals = length(hit)
}
model{
  for (i in 1:nIndividuals){
    # Observed counts
    hit[i] ~ dbin(thetaHit[i], nSignal[i])
    fa[i] ~ dbin(thetaFA[i], nNoise[i])
    # Two High Threshold Model
    thetaHit[i] = alpha[i] + (1-alpha[i])*beta[i]
    thetaFA[i] = (1-alpha[i])*beta[i]
    # Individual priors
    alpha[i] ~ dnorm(muAlpha[group[i]], 1/sigmaAlpha[group[i]]^2)T
     (0, 1)
    beta[i] ~ dnorm(muBeta[group[i]], 1/sigmaBeta[group[i]]^2)T(0,
      1)
  }
  # Representative
  for (g in 1:nGroups){
    alphaRep[g] ~ dnorm(muAlpha[g], 1/sigmaAlpha[g]^2)T(0, 1)
    betaRep[g] ~ dnorm(muBeta[g], 1/sigmaBeta[g]^2)T(0, 1)
    thetaHitRep[g] = alphaRep[g] + (1-alphaRep[g])*betaRep[g]
    thetaFARep[g] = (1-alphaRep[g])*betaRep[g]
    hitPrd[g] ~ dbin(thetaHitRep[g], nPrdSig[1])
    faPrd[g] ~ dbin(thetaFARep[g], nPrdNoi[1])
  }
  # Group priors
  for (g in 1:nGroups){
    muAlpha[g] ~ dunif(0, 1)
    sigmaAlpha[g] ~ dunif(0, 1)
    muBeta[g] ~ dunif(0, 1)
    sigmaBeta[g] ~ dunif(0, 1)
  }
}
```

Logistic Choice 4

4.1 The Logistic Choice Model

4.1.1 Motivation

Many psychological tasks ask people to complete a series of trials in which they choose between two alternatives. These are often called two-alternative forced-choice tasks ("2AFC" tasks). As caricatured in Figure 4.1, examples include

- psychophysical tasks, such as "was the line on the left or the right longer?" (e.g., Pelli & Farell, 1995)
- classification tasks, such as "is this face showing the emotion of fear or anger?" (e.g., Wang et al., 2017)
- gambling tasks, such as "would you rather have a 50% chance of winning $10 along with a 50% chance of losing $8, or would you rather not take that bet?" (e.g., Kahneman & Tversky, 1984)

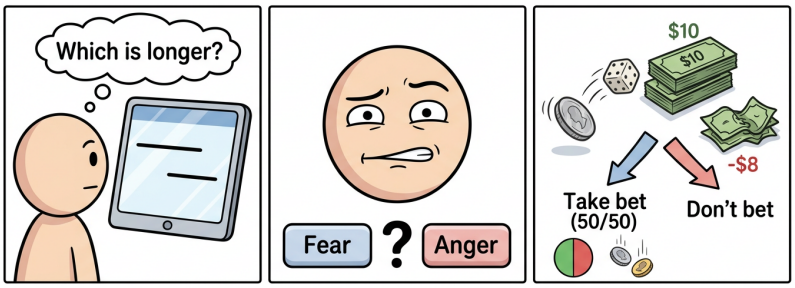


Figure 4.1: Three two-alternative forced-choice tasks.

Many of these tasks use a series of independent trials, so that previous decisions are not relevant to the current decision. Many tasks also do not provide feedback or otherwise involve learning. When these conditions are both met, the logistic choice model provides a simple account of how people make decisions.

The logistic choice model assumes that the relevant information in favor of an alternative can be represented by a single number. If the two alternatives are called "Alternative A" and "Alternative B" then the idea is that, on a specific trial, there are numbers A and B that measure how "good" each alternative is.

For example, in a line length task, A could represent the length of the line on the left and B the length of the line on the right. The longer a line is, the larger the evidence for it. In an emotion classification task, A could represent, in a more abstract way, the number of "fear" features in a face, and B could represent the number of "anger" features. The more that fear features outnumber anger features, the more reason there is to classify the face as fearful. In a gains-and-losses gambling task, A could represent the gains and B the losses associated with the gamble. The more A exceeds B, the more evidence there is to accept the gamble.

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The logistic choice model assumes that two psychological processes affect how the evidence leads to a decision. One process relates to the relative weight a person gives to each source of evidence. For example, a key finding of prospect theory is that people tend to give more weights to losses than gains in risky decision making (Tversky & Kahneman, 1992). This suggests that they give more weight to B than A in the gambling task. In the psychophysical and classification tasks, applying different weights to A and B corresponds to having a bias towards one sort of decision over the other. For example, giving more weight to B would leads to a bias towards saying that the right-hand line is longer, or that the face is angry.

The other psychological process used by the logistic choice model relates to behavioral consistency. If a person is completely consistent, they will choose alternative A every time that A is larger than the (weighted form of) B. This will be true regardless of whether the difference is very small or very large. Often people do not display this strict form of consistency. They may be more likely to choose alternative A, but will not always do so. Often the probability that they choose alternative A depends on the magnitude by which A is larger than the (weighted form of) B.

By incorporating these two processes, the logistic choice model can make inferences about the weights people give to different sources of evidence, and the consistency with which they make decisions.

4.1.2 Model

The logistic choice model gets its name from the logistic function, which is an S-shaped function commonly used in mathematical statistics. A number of logistic functions are shown in Figure 4.2. In general, the logistic function takes as input a real number—a number that can be anywhere from minus infinity to plus infinity—and produces as output a number between 0 and 1. In the logistic choice model, the input is the relative evidence for alternative A and B, and the output is the probability of choosing alternative A.

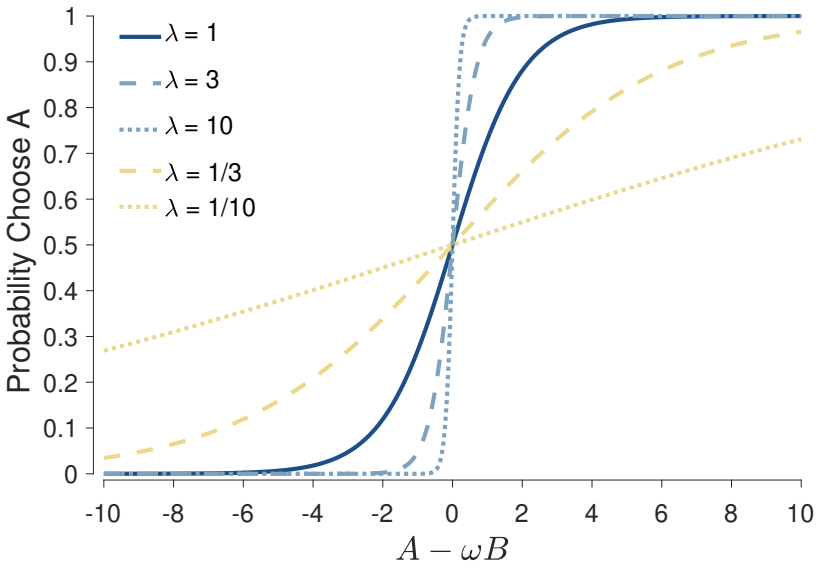

Figure 4.2: Five examples of the logistic choice model using the logistic function to map relative evidence into a choice probability.

The x-axis corresponds to the relative evidence $A - \omega B$, where ω is a parameter that controls how much relative emphasis is given to B. Larger values of ω give more weight to the evidence for alternative B—or, equivalently, less weight to evidence for alternative A. The five different logistic functions shown in Figure 4.2 correspond to different values of a second λ parameter that corresponds to behavioral consistency. As the value of λ increases, choices become more consistent.

If $A - \omega B = 0$ there is equal relative evidence for both alternatives, there is no basis to choose one over the other, and all of the logistic functions produce a choice probability of one half. If the relative evidence is positive, so that there is more evidence for alternative A, the probability of choosing A increases above one half for all five curves. Similarity, if the relative evidence is negative, so that there is more evidence for alternative B, the probability of choosing A decreases below one half. This also makes sense: if there is more evidence for one alternative than the other, it should be more likely to be chosen.

The specific value of the choice probabilities, however, depends on the magnitude of the relative evidence and which logistic curve is used. For example, if the relative evidence $A - \omega B$ is a large positive value such as +8, then the probabilty of choosing alternative A is very close to one for all of the blue logistic functions. These correspond to consistency values of $\lambda = 1,\ 3,$ and 10. For the two yellow logistic functions, with smaller consistency values of $\lambda = \frac{1}{3}$ and $\frac{1}{10}$, the probability of choosing alternative A is between one half and one. This means that if there is strong evidence in favor of alternative A, it will almost always be chosen as long if a person has reasonable consistency. If they are inconsistent, it will still be chosen the majority of the time, but not always.

In this way, the logistic choice model captures the interaction between how much evidence is available and how consistently decisions are made. The shapes of the logistic curves show this interaction. The logistic curve with the least consistency of $\lambda = \frac{1}{10}$ produces a probability of choosing alternative A close to one half regardless of the evidence in the -10 to $+10$ range shown. The logistic curve with the most consistency of $\lambda = 10$ produces a high probability of choosing alternative B when the evidence is negative and favors B, which quickly switches to a high probability of choosing alternative A when the evidence becomes positive and favors A. At intermediate levels of consistency the logistic curves change in an S-shaped way, with more rapid changes for larger values of λ.

Figure 4.3 summarizes the structure of the logistic choice model. An individual has a weight parameter ω and a consistency parameter λ. These parameters control the decisions y_t made on each trial, based on the evidence for alternatives A and B quantified by A_t and B_t.

4.1.3 Mechanics

Table 4.1 lists the variables and parameters used by the logistic model, including their meaning, statistical notation, and JASP code name. The first section corresponds to the observed data and stimulus information, as shown in the observed box in Figure 4.3. The second section corresponds to the latent parameters, as shown in the head in Figure 4.3. The third

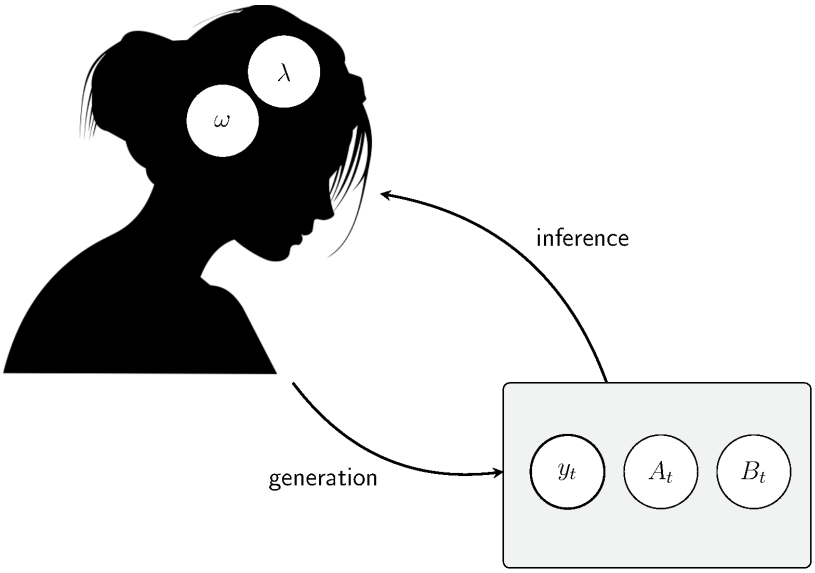

Figure 4.3: The structure of the logistic choice model.

section corresponds to the posterior predictive analysis, which are not shown in Figure 4.3, but mirror the observed information.

Name	Model	JASP
observed decisions	y_t	`decision[t]`
evidence for alternative A	A_t	`A[t]`
evidence for alternative B	B_t	`B[t]`
individual weight	ω_i	`omega[i]`
individual consistency	λ_i	`lambda[i]`
evidence for alternative A on predicted trial	$\hat{A}_t$	`Aprd[t]`
evidence for alternative B on predicted trial	$\hat{B}_t$	`Bprd[t]`
number of predicted observations	$\hat{n}_t$	`nPrd[t]`

Table 4.1: The meaning of the observed variables and latent parameters in the logistic choice model, with their model and JASP names.

Input data

Data are provided to the logistic choice model in JASP using the long format demonstrated in Table 4.2. These data come from the *Mixed Gambles* example.

The first four `individual`, `A`, `B`, and `decision` columns provide information about the trials. They list which individual made each decision, the evidence values A_t and B_t, and the decision $y_t = 1$ if alternative A was chosen and $y_t = 0$ if alternative B was chosen.

The final three columns relate to the posterior predictive analysis. They provide information about the trials for which posterior predictions are inferred for each individual. The `Aprd` and `Bprd` columns provide the evidence values for these trials. The `nPrd` column specifies how often each of the prediction trials is presented to each individual.

Output analyses

The JASP implementation provides three analyses:

individual	A	B	decision	Aprd	Bprd	nPrd
7	10	2.5	1	2	0.5	1
7	6	3.75	1	2	0.75	1
7	16	10	0	2	1	1
...	...	...	...	...	...	...
7	2	1.25	1	16	28	1
7	8	8	0	16	30	1
7	12	13.5	0	16	32	1
...	...	...	...			
3	6	6	0			
3	16	6	1			
3	4	2	1			
...	...	...	...			
11	14	10.5	0			
11	4	1	1			
11	9	6.75	0			
...	...	...	...			

Table 4.2: Data format for the logistic choice model.

- **weight and consistency posteriors**: the marginal posterior distribution of the weight `omega` and consistency `lambda` parameters for each decision source
- **posterior predictions**: the posterior predictive distribution of decisions `decisionPrd` for each individual for the prediction trials specified by `Aprd`, `Bprd`, and `nPrd`

4.2 *Mixed Gambles*

`mixedGambles .jasp`
`mixedGamblesPrediction.jasp`

Stillman (2021) considers a gambling task in which people have to choose between two options. One alternative is to accept a 50-50 gamble between a possible gain and a possible loss, and the other alternative is to reject the gamble. The logistic choice model is useful for understanding how people's risk preferences in weighing gains against losses, and how consistently they apply these risk preferences as they make multiple decisions.

Data

Stillman (2021) provides data for 148 participants who each completed 165 trials. Each trial considered a specific gain G, ranging from \$2 to \$16, and a specific loss L, ranging from \$0.50 to \$32. We consider the behavior of just 12 of the 148 participants, chosen because they represent the range of behavior across all of the participants.

Figure 4.4 provides a visual representation of all the choices made by these participants A–L. Each panel corresponds to a participant. Within each panel, the x-axis corresponds to the loss L and the y-axis corresponds to the gain G. Each colored marker corresponds to a trial involving a gamble with the corresponding gain and loss. Green markers indicate that the participant accepted the gamble and red markers indicate that they rejected the gamble.

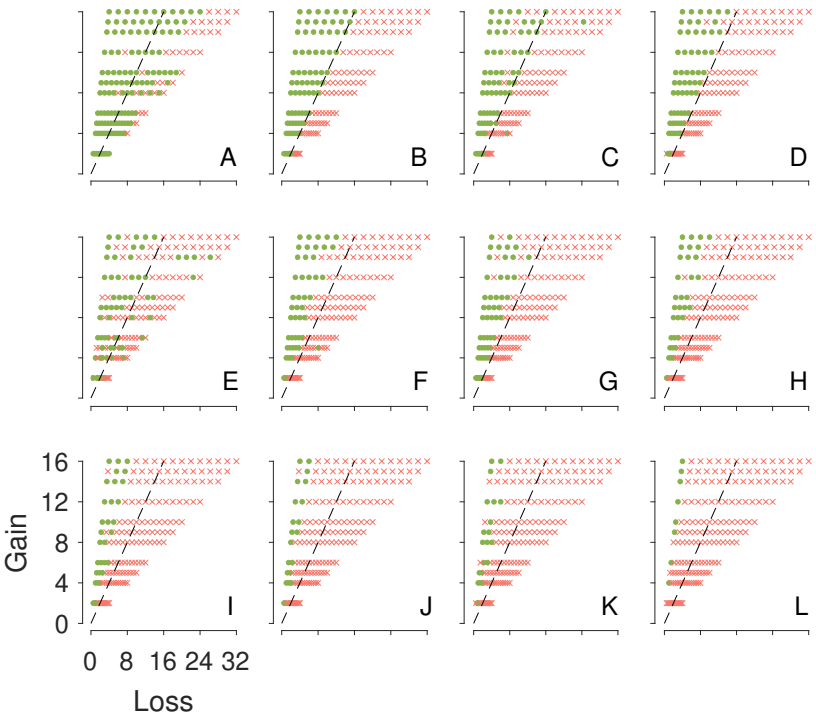

Figure 4.4: Summary of the behavioral data for twelve participants in the mixed gambles task.

The visualization in Figure 4.4 provides some insights into the behavior of the participants. For example, Participant B always accepts gambles for which the potential gain is as large or greater than the potential loss, but rejects all gambles with greater loss. Participant C seems to have similar risk tolerance, but behaves less consistently. They sometimes non-systematically reject gambles with more potential gain than loss, and accept gambles with greater potential loss than gain. By contrast, Participant L is highly and consistently risk averse. They only accept a few gambles for which the potential gain is many times larger than the potential loss. In general, moving from Participant A to Participant L, there is an increase in the prospect theory finding that "losses loom large" (Tversky & Kahneman, 1992).

Results

Descriptive adequacy

Figure 4.5 shows the posterior predictive distribution for the 12 participants, in terms of the 165 trials used in the experiment. At each combination of gain and loss that defines a trial, the color of the square represents the posterior predictive probability of accepting the gamble. The colors range from red to green, representing progressively greater probabilities of accepting. Small black markers are overlaid on the color squares for those gambles that were accepted by the participant in the experiment.

Because the posterior predictive distribution is yoked to the experimental design, the analysis is easily interpreted as testing the descriptive adequacy of the logistic choice model. The model fares well in this regard. For most participants, there is a high posterior predictive probability of accepting the gambles that the participant did accept, and rejecting those

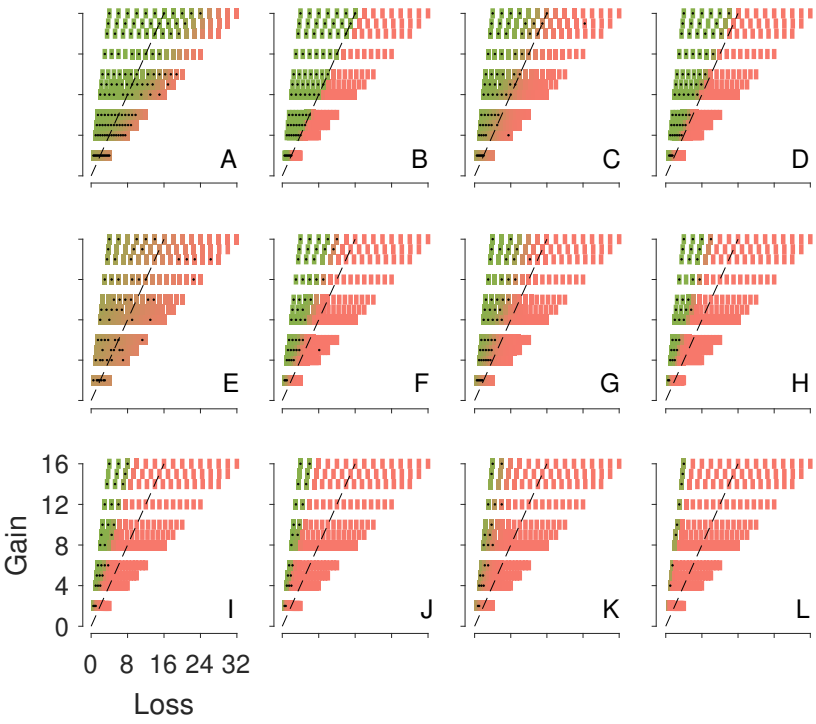

Figure 4.5: Posterior predictive check of descriptive adequacy for twelve participants in the mixed gambles task.

that they did reject. For participant E, who behaved the most inconsistently, the posterior predictive probabilities are often more uncertain. This means that while they do not give high probability to the observed decision, they do not predict the opposite decision.

Loss aversion and consistency

Figure 4.5 shows the joint posterior distribution over the weight ω and consistency λ parameters for all 12 participants. Each panel corresponds to a participant, with their joint posterior distribution highlighted in blue. In each panel the joint posterior distributions for the other participants are show for reference in yellow. The joint posterior distribution for the highlighted participant is also summarized by error bars showing the 95% credible distribution for both parameters.

The inferences for the weight ω are interpretable in terms of the risk sensitivity of the participants. Participants A–D are inferred to give approximately equal weight to gains as losses, with ω values near one. Greater ω values give more weight to losses. For example, the inference for Participant I, with posterior mean $\lambda = 1.83$ and 95% credible interval $(1.69, 1.98)$, means that they weight potential losses about a little under twice as heavily as potential gains.

The inferences for consistency λ are similarly interpretable. Participants such as B and D who are highly consistent are inferred to have large values, while inconsistent participants such as A and E are inferred to have low values. It is worth noting that for these inconsistent participants, the inferences about their weights ω are relatively uncertain. For example, the 95% credible interval for ω for Participant E is $(1.05, 1.93)$. This makes conceptual sense: if a person is not behaving consistently, it is difficult to be certain what their underlying attitude is to risk.

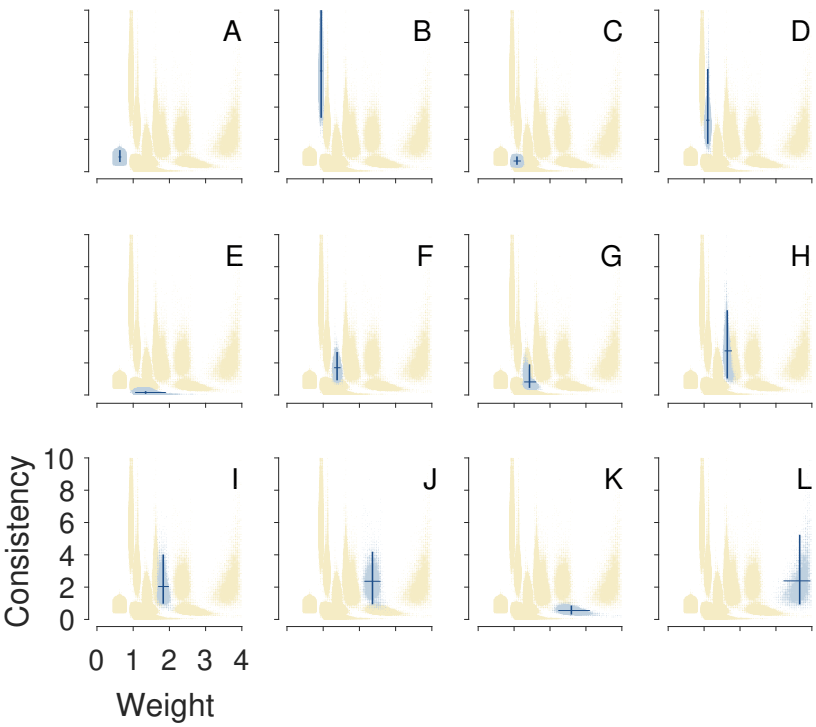

Figure 4.6: Summary of the behavioral data for twelve participants in the mixed gambles task.

Generalization

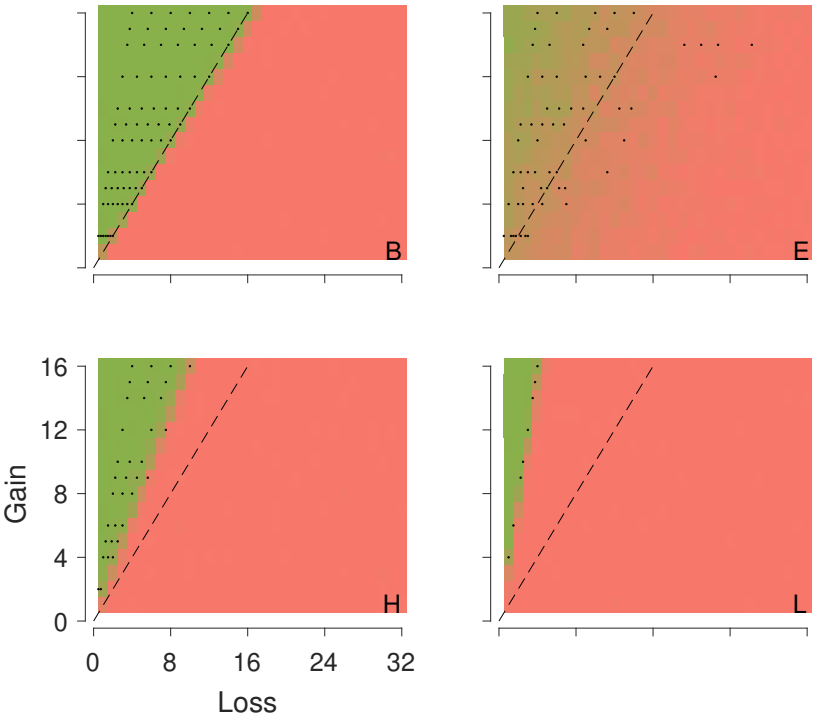

Figure 4.7: Posterior predictive distribution over the stimulus space for four participants in the mixed gambles task.

Figure 4.7 focuses on participants B, E, H, and L to demonstrate a different posterior predictive analysis. The posterior predictive distribution is now inferred for all gains from \$1 to \$16 combined with all losses from \$1 to \$32. Once again, the colors represent the probability of accepting the gamble, and black markers indicate the gambles that were accepted by the participant in the experiment.

This posterior predictive analysis is not restricted to the 165 gambles that

were used in the experiment. It considers all of the possible gambles that can be constructed from the range of possible gains and losses. In this sense, the posterior predictive analysis goes beyond described what was observed, and makes predictions about other possible future experiments. This is a form of *generalization* test (Busemeyer & Wang, 2000), made possible by the use of the logistic choice model. For each participant, the inferences made about their bias ω and consistency λ from their observed experimental behavior allow predictions to be made about whether they would accept gambles that were not part of the experiment.

The generalizations shown in Figure 4.7 seem intuitively reasonable. For participants B, H, and L, there is a clear dividing line between predicted acceptance and rejection of gambles, showing progressively greater risk aversion across the three participants. For participant E, there is much more uncertainty for a large set of gambles about whether they will be accepted or rejected. This uncertainty is caused by the inferred lack of behavioral consistency for participant F.

4.3 *Duration Perception*

`durationPerception.jasp`
`durationPerceptionPrediction.jasp`

van Driel et al. (2014) studied how the brain integrates temporal information across modalities using a psychophysical duration perception task. On each trial, participants were presented with a briefly illuminated light or briefly sounded tone. They had to decide if the stimulus duration was longer or shorter than a 500ms standard. The logistic choice model provides a psychophysical function (Kuss et al., 2005) for understanding how precisely participants can make this decision, and what biases they may have in perceiving durations as being longer or shorter than they really are.

Data

The van Driel et al. (2014) experiment involved 19 participants each completing 240 trials in both the visual and auditory modalities. There were 20 stimuli, each presented 12 times that varied in duration from 335ms to 830ms and were always compared to the 500ms standard.

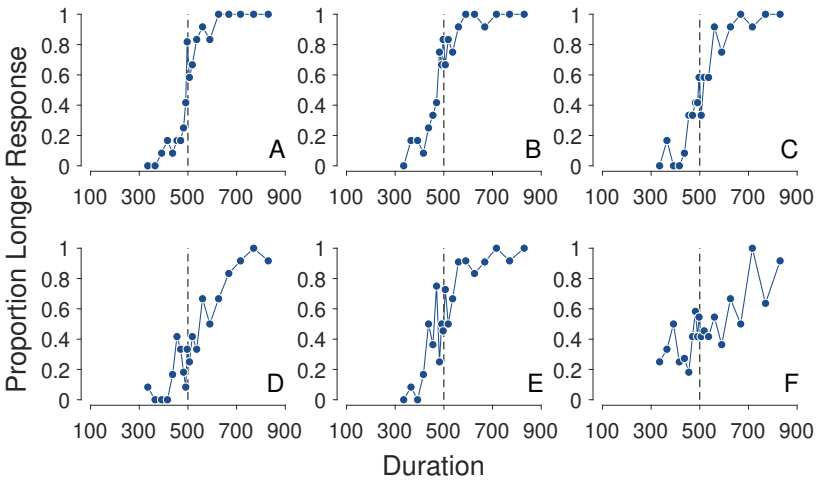

Figure 4.8: Summary of the behavioral data for six participants in the duration perception task.

We focus on the same six participants considered in the analysis of these data presented by Lee (2018). The behavior of these participants A–F on the visual stimuli is summarized in Figure 4.8. Circular markers correspond to the proportion of times each stimulus duration was perceived as longer than 500ms.

It is clear there are individual differences in the participants' abilities to make accurate duration discriminations. For example, Participant A answered 84% of the trials correctly, while Participant F answered only 59% correctly. Figure 4.8 provides the insight that there are two, not mutually exclusive, ways in which participants can be inaccurate.

One way to make errors is by being inconsistent in their responses. Any proportion of responses that is not 0 or 1 means that the participant responded differently to the same stimulus on different trials. For example, Participant F has many response proportions between 0.2 and 0.8.

The other way to make errors is to be biased towards one response over the other. For example, Participant D does not consistently have a probability above 50% of responding long until stimuli have much greater duration than 500 ms. This means they are mistakenly responding "shorter" to many stimuli that are longer than 500 ms.

Results

In the *Mixed Gambles* example, the weight parameter ω was naturally interpreted as risk sensitivity, corresponding to the emphasis given to losses relative to gains. One way to understand the situation is that the gains provide evidence for accepting the gamble, but this evidence is lessened by losses. The parameter ω quantifies how much losses matter in combining those two competing sources of evidence.

The current *Duration Perception* example requires a different interpretation of the ω parameter. Now, the evidence for a "longer" response comes from the duration of the presented stimulus. This means that the stimulus durations on each trial are the A values. The countering evidence in favor of a shorter response is the 500ms standard that the stimulus duration must exceed. This means that the B value is now 500 on every trial. If $\omega = 1$ the standard is given the objectively correct emphasis relative to the stimulus duration, and the resulting decisions are not biased. If $\omega > 1$, the standard is given too much emphasis, and the overall evidence $A - \omega B$ will be smaller than it should be. This will lead to a bias towards "shorter" responses. Conversely if $\omega < 1$ there will be a bias towards "longer" responses. Following this logic, the ω parameter is naturally interpreted as a measure of response bias.

Descriptive adequacy

Figure 4.9 shows a posterior predictive check of descriptive adequacy for participant B. The posterior predictive distribution matches the experimental design, with 12 presentations of each of the 20 stimulus duration. The area of the blue squares represents the probability that each stimulus duration is predicted to lead to a long response on $0, 1, \ldots, 12$ of the presentations. The observed number of times participant B did

respond longer in the experiment is highlighted with a dark border, and connected by lines across the stimulus durations.

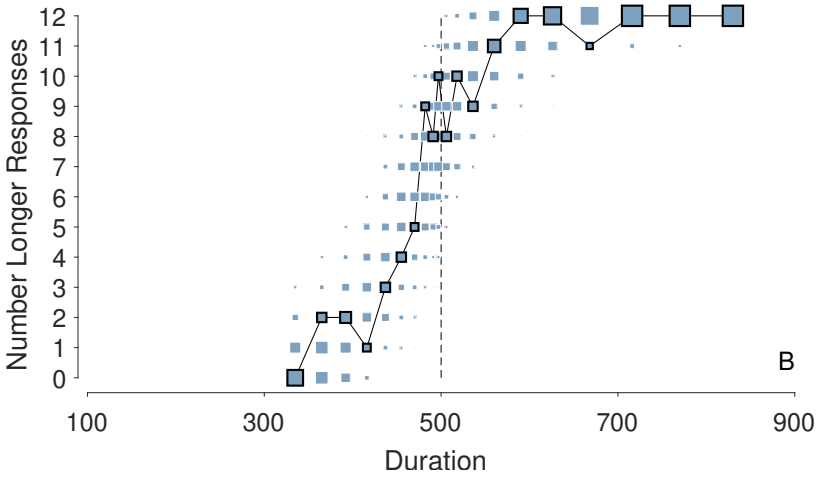

Figure 4.9: Posterior predictive check of descriptive adequacy for participant B in the duration perception task.

This analysis suggests the model describes participant B's behavior well. The observed number of times they made long responses for each stimulus is given relatively high probability. The distribution over possible sets of observations also seems consistent with the specific observed pattern. For example, participant B responded that the 506ms stimulus was longer than the 500ms standard 8 out of the 12 times it was presented in the experiment. The posterior predictive distribution gives approximately equal probability to 8, 9, or 10 long responses, and smaller but non-negligible probability to 6, 7, 11, or 12 long responses. These seem like reasonable expectations, in the sense that if the participant did the experiment again, these are plausible outcomes given their slight bias towards long responses and their relatively good consistency.

Similar analyses of descriptive adequacy could be undertaken for the other five participants.

Response bias and consistency

Figure 4.10 shows the joint posterior distribution over the bias ω and consistency λ parameters for all 12 participants. Each panel corresponds to a participant, with their joint posterior distribution highlighted in blue. In each panel the joint posterior distributions for the other participants are show for reference in yellow. The joint posterior distribution for the highlighted participant is also summarized by error bars showing the 95% credible distribution for both parameters.

The inferences about bias show that the participants do not have large response biases, with ω values near one. The three most likely exceptions are participants D, F, and B. Participant D has posterior mean $\omega = 1.13$ and 95% credible interval $(0.99, 1.05)$. Participant F has posterior mean $\omega = 1.09$ and 95% credible interval $(0.99, 1.21)$. These inferences suggest that both participants have some bias towards short responses, consistent with their behavioral summaries shown in Figure 4.8. Participant B, in contrast, has posterior mean $\omega = 0.94$ and 95% credible interval $(0.90, 0.97)$. This suggests they have a bias towards long responses.

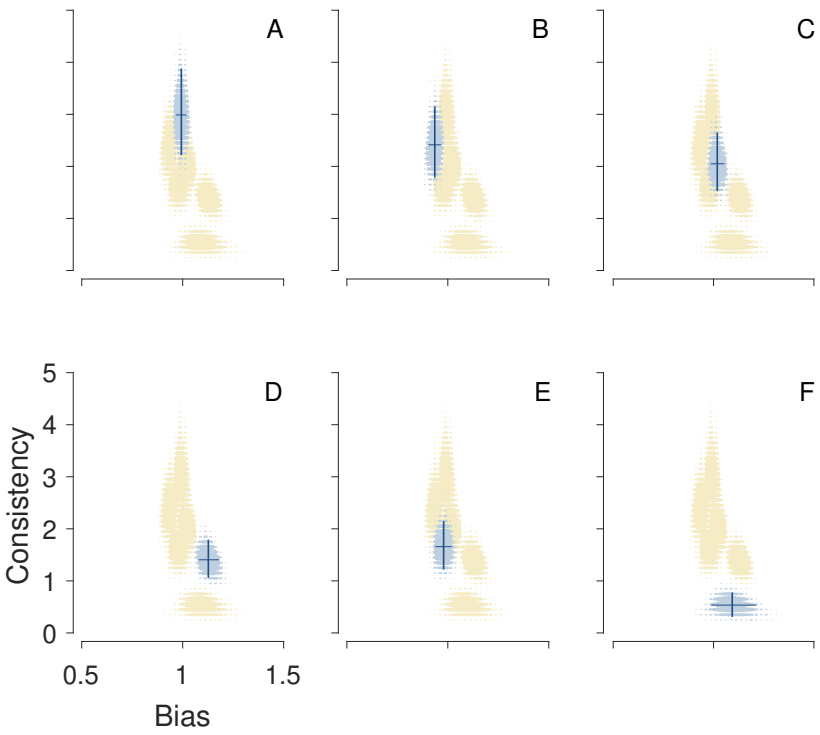

Figure 4.10: Joint posterior distributions of the response bias and determinism parameters for six participants in the duration perception data..

The inferences about consistency show that, for example, participants A, B, C are likely more consistent than participants D and E who, in turn, are likely more consistent than participant F. These inferences also agree with the behavioral summaries in Figure 4.8.

Generalization

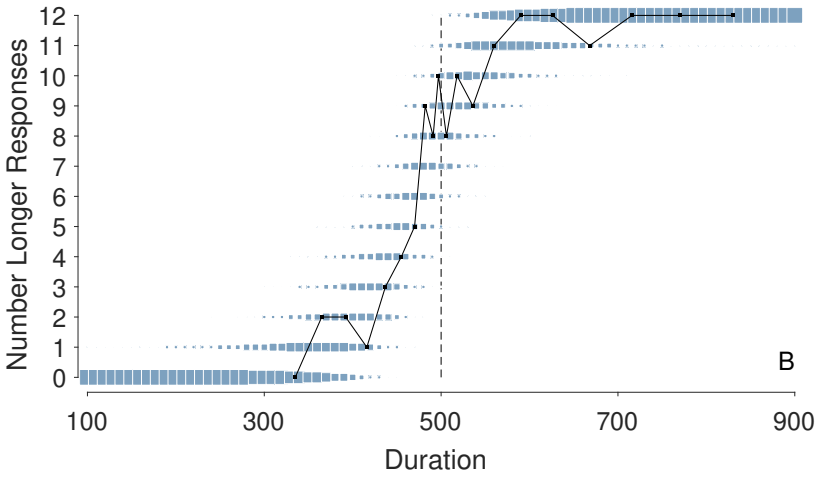

Figure 4.11: Posterior predictive distribution over the stimulus space for participant B in the duration perception task.

Figure 4.11 shows a more general posterior predictive analysis for participant B that considers stimulus durations not presented in the experiment. All durations from 100ms to 900ms in increments of 10ms are considered. It is still assumed that each stimulus is presented 12 times. The observed behavior, relating to the 20 stimuli actually presented, is superimposed. This analysis provides a more general characterization of the expectations for participant B's ability to discriminate visual durations accurately. It uses what is known about their bias and consistency to make predictions for a wide range of possible stimulus durations.

4.4 The Hierarchical Logistic Choice Model

4.4.1 Motivation

The hierarchical extension of the logistic choice model assumes that each individual belongs to a specified group. There can be one group or multiple groups across the individuals. This allows inferences to be made about both individual weights and consistencies and group weights and consistencies.

4.4.2 Model

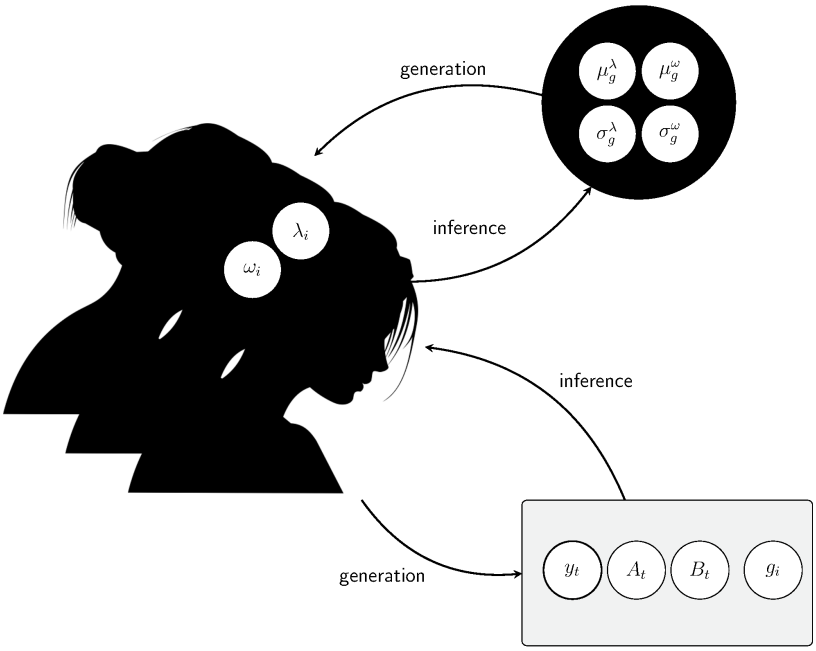

Figure 4.12: The structure of the hierarchical logistic choice model.

Figure 4.12 summarizes the structure of the hierarchical logistic choice model. The multiple individuals are shown as multiple people, with individual weight ω_i and consistency λ_i parameters generated from the group distributions. The group membership of the decision source g_i is provided as observed data.

The individual weight ω_i and consistency λ_i for individual i is assumed to comes from a specified group g_i. The group distributions are assumed to be Gaussian, with means μ_g^ω and μ_g^λ and standard deviations σ_g^ω and σ_g^λ for group g.

From the group distribution, it is also possible to sample the representative weight $\tilde{\omega}_g$ and consistency $\tilde{\lambda}_g$. These inferences are at the individual level, but not tied to a specific individual from that group. Instead, they correspond to the properties of a representative (or random) individual from the group.

4.4.3 Mechanics

Table 4.3 lists the variables and parameters used by the hierarchical logistic model, including their meaning, statistical notation, and JASP

code name. The first section corresponds to the observed data and stimulus information, as shown in the observed box in Figure 4.12. The second section corresponds to the latent parameters, as shown in the head in Figure 4.12. The third section corresponds to the posterior predictive analysis, which are not shown in Figure 4.12, but mirror the observed information.

Table 4.3: The meaning of the observed variables and latent parameters in the hierarchical logistic choice model, with their model and JASP names.

Name	Model	JASP
observed decisions	y_t	`decision[t]`
evidence for alternative A	A_t	`A[t]`
evidence for alternative B	B_t	`B[t]`
group membership	g_i	`group[i]`
individual weight	ω_i	`omega[i]`
individual consistency	λ_i	`lambda[i]`
mean weight for group	μ_g^ω	`muOmega[g]`
mean consistency for group	μ_g^λ	`muLambda[g]`
standard deviation of weight for group	σ_g^ω	`sigmaOmega[g]`
standard deviation of consistencyt for group	σ_g^λ	`sigmaLambda[g]`
representative weight for group	$\tilde{\omega}_g$	`omegaRep[g]`
representative consistency for group	$\tilde{\lambda}_g$	`lambdaRep[g]`
evidence for alternative A on predicted trial	$\hat{A}_t$	`Aprd[t]`
evidence for alternative B on predicted trial	$\hat{B}_t$	`Bprd[t]`
number of predicted observations	$\hat{n}_t$	`nPrd[t]`
predicted decision for group representative	$\hat{y}_{gt}$	`decisionPrd[g,t]`

Table 4.3 lists the observed variables and latent parameters and variables for the hierarchical logistic choice model, giving both their statistical symbol and their JASP code name.

Input data

Data are provided to the hierarchical logistic choice model in JASP using the long format demonstrated in Table 4.4. These data come from the *Modality Duration Perception* example. All but one of the columns are the same as for the non-hierarchical logistic choice model, as demonstrated in Table 4.4. The one additional `group` column has as many rows as there are individual, and specifies to which group each individual belongs.

Output analyses

The JASP implementation provides four analyses:

- **individual weight and consistency posterior**: the marginal posterior distribution of the weight `omega` and consistency `lambda` parameters for each decision source
- **group weight and consistency posterior**: the marginal and joint posterior distribution of the mean and standard deviation of the weight `omega` and consistency `lambda` parameters for each group

individual	A	B	decision	group	Aprd	Bprd	nPrd
1	4.16	5.00	0	1	1.00	5.00	12
1	6.26	5.00	1	1	1.10	5.00	12
1	7.16	5.00	1	1	1.20	5.00	12
...	...	...	...	...	...	...	...
1	4.37	5.00	0	2	2.90	5.00	12
1	3.65	5.00	0	2	3.00	5.00	12
1	3.35	5.00	0	2	3.10	5.00	12
...	...	...	...		...	...	...
1	4.82	5.00	1		4.80	5.00	12
1	6.26	5.00	1		4.90	5.00	12
1	5.60	5.00	1		5.00	5.00	12
...	...	...	...				
20	5.18	5.00	1				
20	8.30	5.00	1				
20	3.92	5.00	0				
...	...	...	...				

Table 4.4: Data format for the hierarchical logistic choice model.

- **representative weight and consistency posterior**: the marginal and joint posterior distribution of the weight `omega` and consistency `lambda` parameters for the representative individual from each group
- **posterior predictions**: the posterior predictive distribution of decisions `decisionPrd` for the representative individual from each group for the prediction trials specified by `Aprd`, `Bprd`, and `nPrd`

4.5 *Modality Duration Discrimination*

`modalityDurationPerception.jasp`

This example uses data from the same van Driel et al. (2014) experiment as the *Duration Perception* example in Section 4.3. The experiment studied how the brain integrates temporal information across modalities using a psychophysical duration perception task. The hierarchical logistic choice model allows both individual and overall behavior between the two modality conditions to be analyzed.

Data

Figure 4.13 summarizes the data for all 19 participants in both the visual and auditory modality conditions. Each panel corresponds to a participant, with participants A–F being the same as for the *Duration Perception* example. Circular markers correspond to the proportion of times each stimulus duration was perceived as longer than 500ms, with blue markers corresponding to the visual condition and red markers corresponding to the auditory condition.

There is clearly variability in consistency and response bias, across participants for the same modality, and within participants across modalities. In terms of the overall comparison of the modalities, many of the participants, such as participants H and O, seem to show greater consistency for the auditory condition. Few, if any, participants, show behavioral patterns with clearly greater consistency for the visual condition.

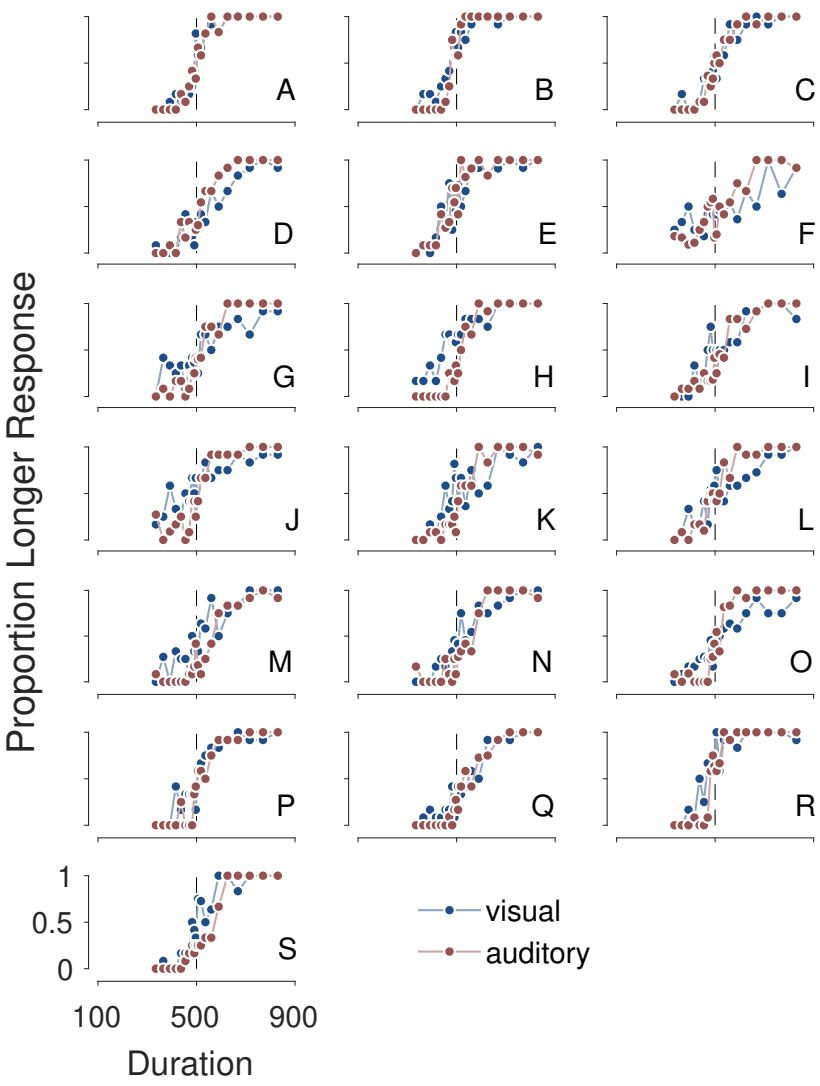

Figure 4.13: Summary of the behavioral data for the participants in the modality duration perception experiment.

Results

Individual consistency and response bias

Figure 4.14 shows the joint posterior distribution for response bias and consistency for all of the participants. Each participant corresponds to a panel, and the colored distributions show the joint posterior distributions without marginal distributions.

These results make clearer the insight that the participants are often, but not always, more consistent in making auditory duration discriminations. Evidence for this is provided by auditory distributions that are above visual distributions. As anticipated, participants H and O show this pattern. It also seems possible there is a slightly greater bias towards "shorter" responses in the auditory condition, Evidence for this is provided by auditory distributions that are to the right of visual distributions. Good examples are participant H and S, but there are clear exceptions like participant D.

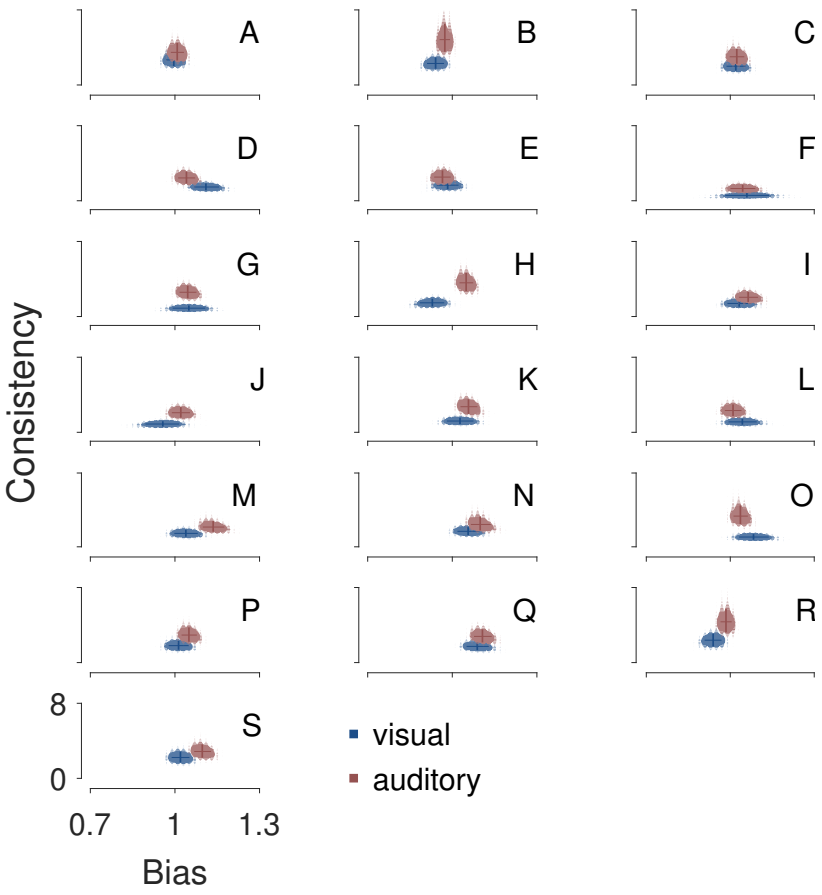

Figure 4.14: Joint posterior distributions of the response bias and consistency parameters for individual participants in the modality duration perception data.

Group consistency and response bias

Figure 4.15 shows the joint posterior distribution for the group means for response bias and consistency for each modality condition, and the joint posterior distribution for the representative individual corresponding to each modality. The group distributions are shown in darker red and blue, while the representative distributions are shown in lighter red and blue.

The group mean posterior distributions show that auditory discrimination is very likely more consistent, on average, than visual discrimination. The visual group mean has expectation $\mu^{\lambda}_{\text{vis}} = 1.50$ with 95% credible interval $(0.91, 1.87)$, while the auditory group mean has expectation $\mu^{\lambda}_{\text{aud}} = 2.75$ with 95% credible interval $(2.10, 3.34)$. Both modalities show close to unbiased responding, with only a suggestion that auditory discrimination has a little more bias in favor of long responses.

Representative consistency and response bias

The representative posterior distributions shown in Figure 4.15 quantify how consistency and response bias differ between the modalities at the level of individual participants, including the impact of individual differences. The representative visual participant's consistency has expectation $\tilde{\lambda}_{\text{vis}} = 1.57$ with 95% credible interval $(0.26, 3.09)$, while the auditory participant's consistency has expectation $\mu^{\lambda}_{\text{aud}} = 2.85$ with 95% credible interval $(0.69, 5.08)$. A reasonable conclusion is that it is still likely

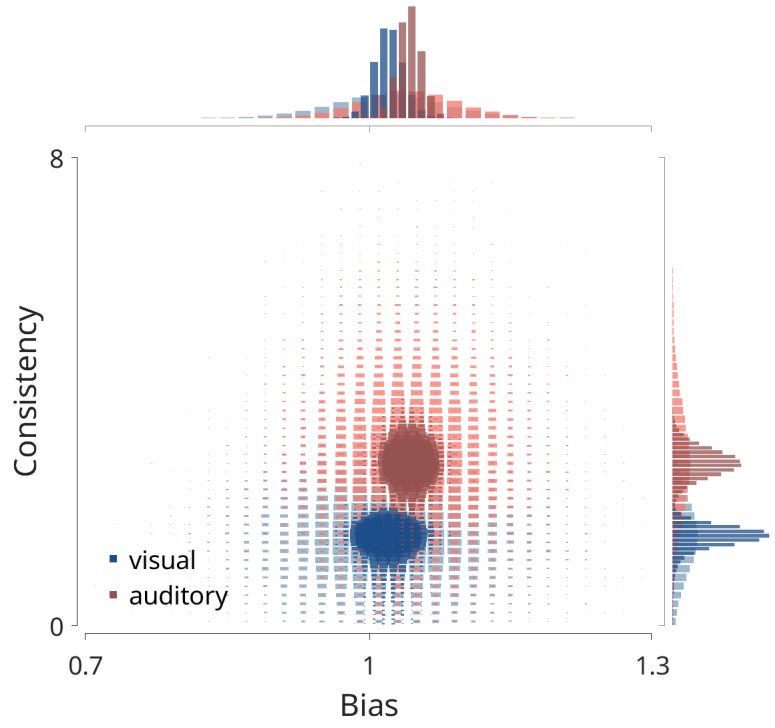

Figure 4.15: Joint posterior distributions of the response bias and consistency parameters for representative individuals in the visual and auditory modalities.

that individual participants are more consistent in auditory duration discrimination, but it is far from certain.

The representative posterior distributions for response bias are very similar. Both expectations are about 1, with 95% credible intervals about $(0.90, 1.15)$. A reasonable conclusion is that there is no evidence of systematic bias for responses in either modality, nor any evidence of differences in response bias between the modalities.

Representative individual posterior predictions

Figure 4.16 shows the posterior predictions for all stimulus durations from 100ms to 900ms in increments of 10ms, in situation in which each stimulus is presented 12 times.

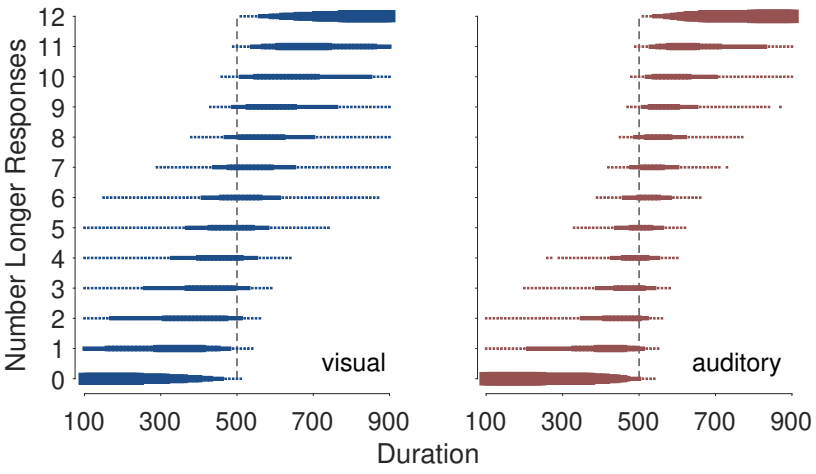

Figure 4.16: Posterior predictive distributions for representative visual and auditory modality participants.

The lower consistency of the representative individual in the visual modality are evident in the greater uncertainty about response outcomes. For example, consider the posterior predictive probabilities for six long responses. Predicting six long responses means also predicting six short responses, and corresponds to maximally inconsistent behavior. The visual posterior predictive for six long responses has non-negligible probability for stimulus durations as small as about 150ms and as large as almost 900ms. In the auditory modality, in contrast, the posterior predictive distribution has non-negligible probability only for stimuli much closer to the 500ms: from about 400ms to 650ms. This difference applies throughout the posterior predictive distribution, for other possible counts of long responses.

◇

4.6 Formal Details

4.6.1 Logistic choice model

Likelihood

The logistic choice function that maps the evidence $A - \omega B$ to the probability θ of choosing alternative A is given by

$$\theta = \frac{1}{1 + \exp\{-\lambda\,(A - \omega B)\}}, \tag{4.1}$$

where ω is the weighting parameter that gives the relative emphasis to evidence for the two alternatives, and λ controls the behavioral consistency of decision making.

The observed behavior is $y = 1$ if alternative A is chosen and $y = 0$ if alternative B is chosen. This is modeled directly in terms of the choice probability, so that

$$y \sim \text{binomial}\big(\theta\big). \tag{4.2}$$

Priors

We use a prior for the relative weights of

$$\omega \sim \text{Gaussian}\big(0, \frac{1}{10^2}\big). \tag{4.3}$$

This standard deviation of this Gaussian prior commits to a scale for the evidence, and so has implications for the scale on which A and B are measured when applying the model. Roughly, the scale is consistent with the x-axis shown in Figure 4.2. Evidence values within an order of magnitude of the -10 to $+10$ shown are consistent with the prior assumptions, and should produce the expected behavior of the model.

We use the prior for behavior consistency shown in Figure 4.17, which takes the form

$$\lambda \sim \text{gamma}\big(2, 1\big). \tag{4.4}$$

This prior gives support to positive values of λ, consistent with increasing evidence corresponding to increasing choice probabilities. It has a mode

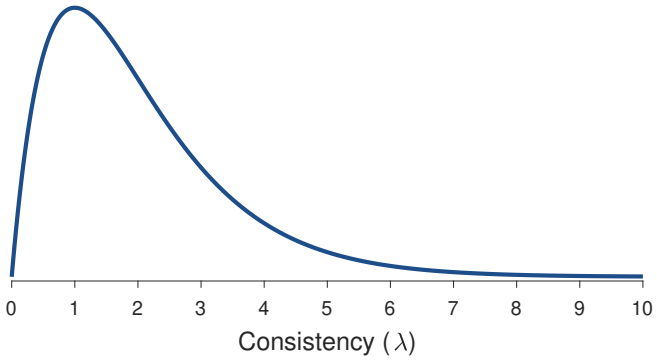

Figure 4.17: The prior probability for consistency λ assumed by the logistic choice model.

at $\lambda = 1$, which is theoretically sensible, because there is a natural symmetry between reciprocal values $\lambda = a$ and $\lambda = \frac{1}{a}$.

Most importantly, the gamma$(2, 1)$ prior is quantitatively consistent with the scale set by the weight prior. The near step-increase logistic function when $\lambda = 10$ shown in Figure 4.2, which corresponds to close-to-deterministic choice behavior, seems about as extreme as is needed to describe people's decision making. This is the λ value at which the prior starts giving near zero prior probability. Note that there is relatively little difference in the behavior of the logistic choice model as consistency increases from $\lambda = 3$ to $\lambda = 10$. Thus, it makes sense that only a fraction of the prior probability for λ lies in this range. Most of the prior probability is in a range from (roughly) $\lambda = \frac{1}{3}$ to $\lambda = 3$, which corresponds to the large changes in what the logistic choice model predicts. A less carefully designed prior, such as $\lambda \sim$ uniform$(0, 10)$, would give most of the prior probability to λ values that corresponded to step functions. This would amount to the prior assumption that decision making was very likely deterministic, which is theoretically implausible, and would the practical effect of making the model difficult to apply in useful ways.

4.6.2 JASP implementation

The JASP implementation of the logistic choice model is shown below. The constant `epsilon` in line 6 ensures that the response probabilities `theta` and `thetaPrd` are not so close to 0 and 1 as to prevent numerical problems. The definition of `nTotalPrd` in line 2 is designed to avoid counting blank entries in the `nPrd` column resulting from the long format of the data.

```
1  # Logistic choice
2  data{
3    nTotal = length(individual)
4    nTotalPrd = sum(nPrd <= max(nPrd))
5    nIndividuals = max(individual)
6    epsilon = 0.001
7  }
8  model{
9    # priors
10   for (i in 1:nIndividuals){
11     omega[i] ~ dnorm(0, 1/10^2)
12     lambda[i] ~ dgamma(2, 1)
```

```
  }
  # observed decisions
  for (t in 1:nTotal){
    utility[t] = A[t] - omega[individual[t]]*B[t]
    theta[t] = min(1-epsilon, max(epsilon, 1/(1 + exp(-lambda[
    individual[t]]*utility[t]))))
    decision[t] ~ dbern(theta[t])
  }
  # posterior predictive
  for (i in 1:nIndividuals){
    for (t in 1:nTotalPrd){
      utilityPrd[i, t] = Aprd[t] - omega[i]*Bprd[t]
      thetaPrd[i, t] = min(1-epsilon, max(epsilon, 1/(1 + exp(-
    lambda[i]*utilityPrd[i, t]))))
      decisionPrd[i, t] ~ dbin(thetaPrd[i, t], nPrd[t])
    }
  }
}
```

4.6.3 Hierarchical logistic choice model

Priors

The hierarchical logistic choice model has priors for the mean and standard deviation of the weight and consistent of the group distributions, set as follows:

$$\begin{aligned} \mu_g^{\omega} &\sim \text{uniform}\big(0, \frac{1}{3^2}\big) \\ \sigma_g^{\omega} &\sim \text{uniform}\big(0, \frac{1}{3^2}\big) \\ \mu_g^{\lambda} &\sim \text{uniform}\big(0, \frac{1}{3^2}\big) \\ \sigma_g^{\lambda} &\sim \text{uniform}\big(0, \frac{1}{3^2}\big). \end{aligned} \tag{4.5}$$

These priors are consistent with those placed directly on the ω and λ parameters in the non-hierarchical model. Those priors, in turn, were set based on the evidence scale assumed by the model, as discussed in Section 4.6.1.

Likelihood

The weights and consistencies for each individual decision source are generated from Gaussian group distributions, which are truncated to be positive for the consistency parameters:

$$\begin{aligned} \omega_i &\sim \text{Gaussian}\big(\mu_{g_i}^{\omega}, \frac{1}{\left(\sigma_{g_i}^{\omega}\right)^2}\big) \\ \lambda_i &\sim \text{Gaussian}_{(0,\infty)}\big(\mu_{g_i}^{\lambda}, \frac{1}{\left(\sigma_{g_i}^{\lambda}\right)^2}\big). \end{aligned} \tag{4.6}$$

The representative weight and consistency for each group is generated

in exactly the same way:

$$\begin{aligned} \tilde{\omega}_g &\sim \text{Gaussian}\left(\mu_g^{\omega}, \frac{1}{\left(\sigma_g^{\alpha}\right)^2}\right) \\ \tilde{\lambda}_g &\sim \text{Gaussian}_{(0,\infty)}\left(\mu_g^{\lambda}, \frac{1}{\left(\sigma_g^{\lambda}\right)^2}\right). \end{aligned} \tag{4.7}$$

The probability of choosing alternative A on predictive trial t with evidence A_t and B_t for and against alternative A, for the representative individual in group g is

$$\tilde{\theta}_g = \frac{1}{1 + \exp\left\{-\tilde{\lambda}_g\left(\hat{A}_t - \tilde{\omega}_g \hat{B}_t\right)\right\}}. \tag{4.8}$$

The predicted count of the number of times alternative A is chosen from $\hat{n}_t$ trials with these evidence values is

$$\hat{y}_g \sim \text{binomial}\left(\tilde{\theta}_g, \hat{n}_t\right). \tag{4.9}$$

4.6.4 JASP implementation

```
# Hierarchical logistic choice
data{
  nTotal = length(person)
  nPeople = max(person)
  nGroups = max(group)
  epsilon = 0.001
}
model{
  for (g in 1:nGroups){
    muOmega[g] ~ dnorm(0, 1/3^2)
    sigmaOmega[g] ~ dnorm(0, 1/3^2)T(0, )
    muLambda[g] ~ dnorm(0, 1/3^2)T(0, )
    sigmaLambda[g] ~ dnorm(0, 1/3^2)T(0, )
  }
  for (i in 1:nPeople){
    omega[i] ~ dnorm(muOmega[group[i]], 1/sigmaOmega[group[i]]^2)
    lambda[i] ~ dnorm(muLambda[group[i]], 1/sigmaLambda[group[i
    ]]^2)T(0, )
  }
  for (t in 1:nTotal){
    decision[t] ~ dbern(theta[t])
    theta[t] = min(1-epsilon, max(epsilon, 1/(1 + exp(-lambda[
    person[t]]*utility[t]))))
    utility[t] = A[t] - omega[person[t]]*B[t]
  }
  for (g in 1:nGroups){
    omegaRep[g] ~ dnorm(muOmega[g], 1/sigmaOmega[g]^2)
    lambdaRep[g] ~ dnorm(muLambda[g], 1/sigmaLambda[g]^2)T(0, )
  }
}
```

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5.1 Multiple-Choice Signal Detection Model

5.1.1 Motivation

The basic two choice signal detection model is covered in Chapter 2. If you are not familiar with signal detection models, it is recommended that you read that chapter first.

The signal detection model in Chapter 2 allows for only two choice alternatives. In one way, this is reasonable, since it designed for situations in which there really are two possible states of the world: "signal" or "noise." In another way, this is a limiting assumption, because it is often useful to allow decision makers to express their level of confidence in their decision.

One common approach is to use ratings scales with verbal expressions of confidence, such as "definitely noise," "probably noise," "unsure", "probably signal," or "definitely signal." Another common approach is to use a numerical rating scale ranging from, for example, 1 to 7, with 7 indicating the highest confidence in there being a signal and indicating the highest confidence in there not being a signal.

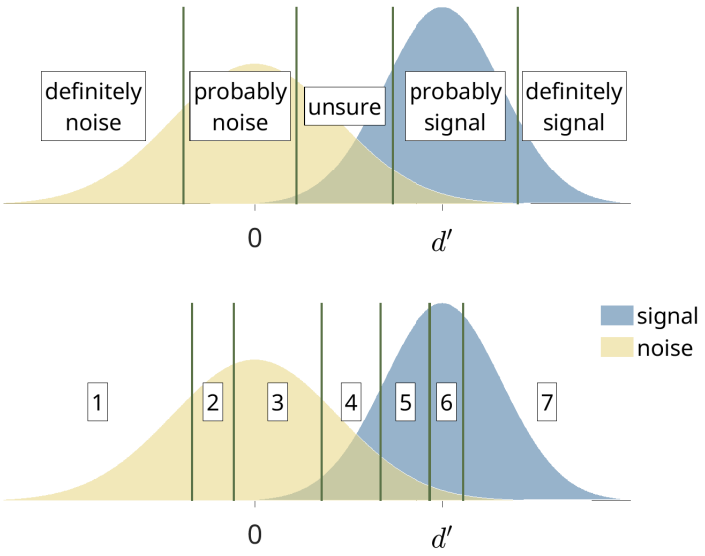

Figure 5.1: The multiple-choice SDT modeling framework.

Figure 5.1 shows a signal detection model that allow for multiple responses. There is still one signal and one noise distribution, but there are now multiple decision criteria. On each trial, a strength value is still sampled from either the signal or noise distribution, depending on whether there really is a signal. Where that sampled value occurs in relation to the decision criteria determines the response.

5.1.2 Model

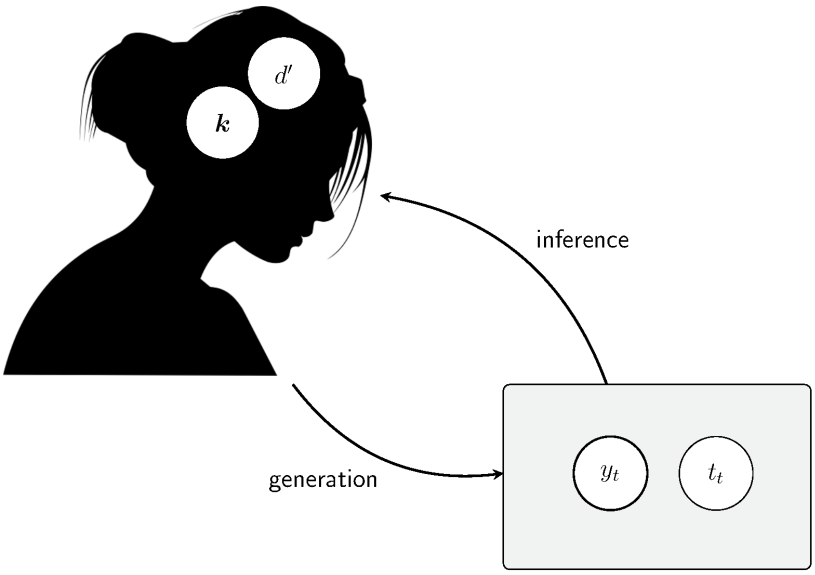

Figure 5.2: The structure of the multiple-choice signal detection model.

Figure 5.2 summarizes the structure of the multiple-choice signal detection model. An individual has a discriminability d' and set of criteria $\boldsymbol{k}$. The bold $\boldsymbol{k}$ indicates that there is a vector of criteria. This means that $\boldsymbol{k} = (k_1, k_2, \ldots)$ with as many criteria as are needed for the number of response options. The two-choice signal detection model has a single criterion for an individual, since there are only two possible responses.

The data for the multiple-choice signal detection model no longer are expressed in terms of hit, miss, false alarm, and correct rejection counts: since there are no longer single "signal" and "noise" response options, those counts are not well defined. Instead, the data take the form of the responses y_t on each trial, and the true state t_t for each trials, indicating whether it was signal or noise.

5.1.3 Mechanics

Table 5.1 details the meaning of the parameters and variables in the multiple-choice signal detection model, their name in the model as shown in Figure 5.2, and their name in JASP.

Name	Model	JASP
response	y	`response`
signal or noise truth	t	`truth`
number of signal observations for prediction	$\hat{n}^{\mathrm{s}}$	`nPrdSig`
number of noise observations for prediction	$\hat{n}^{\mathrm{n}}$	`nPrdNoi`
discriminability	d'	`d`
criteria	$\boldsymbol{k}$	`k[1], k[2], ...`
predicted signal response count	$\hat{\boldsymbol{k}}^{\mathrm{h}}$	`respSigPrd`
predicted noise response count	$\hat{\boldsymbol{k}}^{\mathrm{f}}$	`respNoiPred`

Table 5.1: The meaning of the observed variables and latent parameters in the multiple-choice signal detection model, with their statistical and JASP names.

Input data

Table 5.5 shows the general data format using some of the data from the *Visual Metacognition* example. The data are in a long format, with each row representing a trial.

person	truth	response	nPrdSign	nPrdNoi
1	2	4	120	120
1	1	7	120	120
1	2	1	120	120
1	1	2		
1	2	3		
...	...	...		
2	1	7		
2	2	2		
2	2	5		
...	...	...		

Table 5.2: JASP data format for the multiple-choice signal detection model.

The `person` column indicates which decision source made the response on the trial. The `truth` column indicates whether the stimulus for the trial is signal (`truth=1`) or noise (`truth=2`). The `response` column provides the response that the decision source made for the stimulus. The `response` data must be coded from 1 to the total number of possible choices.

The `nPrdSig` and `nPrdNoi` columns fill only as many rows as there are decision sources. They provide the number of signal and noise trials to be used to define the posterior predictive distribution for each decision source.

Output analyses

The JASP implementation provides three analyses.

- **discriminability and criteria marginal posteriors**: the marginal posterior distributions of the discriminability d' and the criteria $\boldsymbol{k}$ for each individual
- **posterior predictions**: the marginal posterior predictive distribution for each possible response for signal $\hat{y}_j^{\text{s}}$ and noise $\hat{y}_j^{\text{n}}$ stimuli, based on $\hat{n}^{\text{s}}$ signal observations and $\hat{n}^{\text{n}}$ noise observations

5.2 *Visual Metacognition*

`visualMetacognition.jasp`

Shekhar & Rahnev (2018) studied the role of different brain regions in visual metacognition. They were particularly interested in the dorsolateral prefrontal cortex and the anterior prefrontal cortex. In their experiment, people completed a simple task choosing beween two possible orientations of Gabor stimuli, and were subjected to transcranial magnetic stimulation to interfere with the generation of decisions and confidence ratings.

Data

We use the data from just three participants, chosen because they show differences in their ability to detect orientations and their use of the confidence scale. We only consider the control condition, in which the transcranial magnetic stimulation was applied to the somatosensory cortex.

For each participant, there were 120 trials in which a right-titled Gabor was presented, and 120 trials in which a left-titled Gabor was presented. Decisions and confidence responses were collected simultaneously with a single keypress. The four left-hand fingers were mapped to four confidence responses for left-oriented decisions, and the four right-hand fingers were mapped to four confidence responses for right-oriented decisions. The finger mappings were designed so that, for each hand, the index finger indicated the lowest confidence of 1 and little finger indicated the highest confidence of 4.

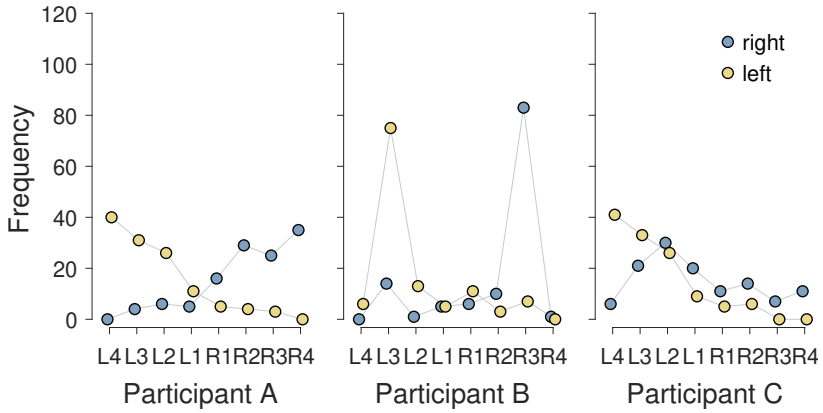

Figure 5.3: Summary of the behavioral data for three participants in the visual metacognition task.

The performance of the three participants is summarized in Figure 5.3. The circular markers show how often each of the eight possible responses is made, with yellow markers corresponding to left-oriented stimuli and blue markers corresponding to right-oriented stimuli.

Participant A seems to be able to distinguish between the two types of stimuli. They generally give "right" responses to right-oriented stimuli and "left" responses to left-oriented stimuli. The most common confidence rating is 4 for both decisions, although the other levels of confidence are also frequently used. Participant B also generally makes correct decisions, but shows an interesting pattern of often judging their confidence to be 3. Participant C makes relatively inaccurate decisions, especially in failing to detect right-oriented stimuli.

Results

Posterior predictive analysis

There is no danger of the two-choice signal detection model failing to describe the data, as long as there is a discriminability and bias (or criterion) parameter for every set of hit, miss, false alarm, and correct rejection counts being modeled. This is because the model essentially reparameterizes the hit and false alarm counts in terms of the discriminability

and bias parameters. The purpose of the model is not to compress the data into a simpler underlying parameterization, but to express observed counts in terms of psychologically meaningful concepts.

Sometimes this sort of model is called a *saturated model*. Because saturated models can always describe data, the value of posterior predictive analysis lies not in checking descriptive adequacy, but in making predictions about behavior in different circumstances. The examples in Chapter **??** applied posterior predictive analysis to examining the variability in replications of an experiment, or to situations with different numbers of signal and noise trials.

The multiple-choice signal detection model, however, is not saturated. It is possible for it to fail to describe observed data, because the same set of criteria have to match the response distribution for both signal and noise stimuli. This means that posterior predictive analysis is useful for checking descriptive adequacy. Accordingly, it makes sense to prioritize the posterior predictive analysis. If the multiple-choice signal detection model fails to be descriptively adequate, it is difficult to know what value lies in the parameter posterior distributions.

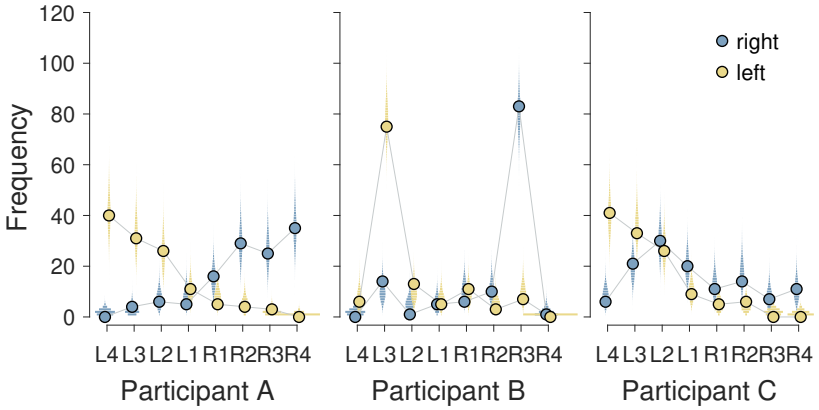

Figure 5.4: Posterior predictive analysis of the visual metacognition data using the multiple-choice signal detection model.

Figure 5.4 shows a posterior predictive check of descriptive adequacy, based on the summary of the behavioral data presented in Figure 5.3. The vertical histograms—sometimes called violin plots or ballerina plots—indicate the posterior predictive mass for each possible number of counts at every level of confidence for both right-oriented and left-oriented stimuli. Because the number of right-oriented and left-oriented stimuli chosen for posterior predictive analysis match the experimental design, the posterior predictive distributions provide a test of the model's descriptive adequacy for the observed data.

There is clear agreement between the posterior predictive distribution and the data. All of the data counts are given non-negligible posterior predictive mass, and often are close to the counts with the greatest mass.

Posterior distributions

Figure 5.5 shows the joint and marginal posterior distributions of discriminability and the seven criteria for the three participants. One clear result is that participant A, with a posterior mean discriminability $d'_{\mathrm{A}} = 2.35$ and 95% credible interval (1.91, 2.85), and participant B, with a posterior

mean discriminability $d'_B = 1.88$ and 95% credible interval $(1.54, 2.20)$, are very likely to have better discriminability than participant C, with a posterior mean discriminability $d'_C = 1.09$ with a 95% credible interval $(0.81, 1.35)$.

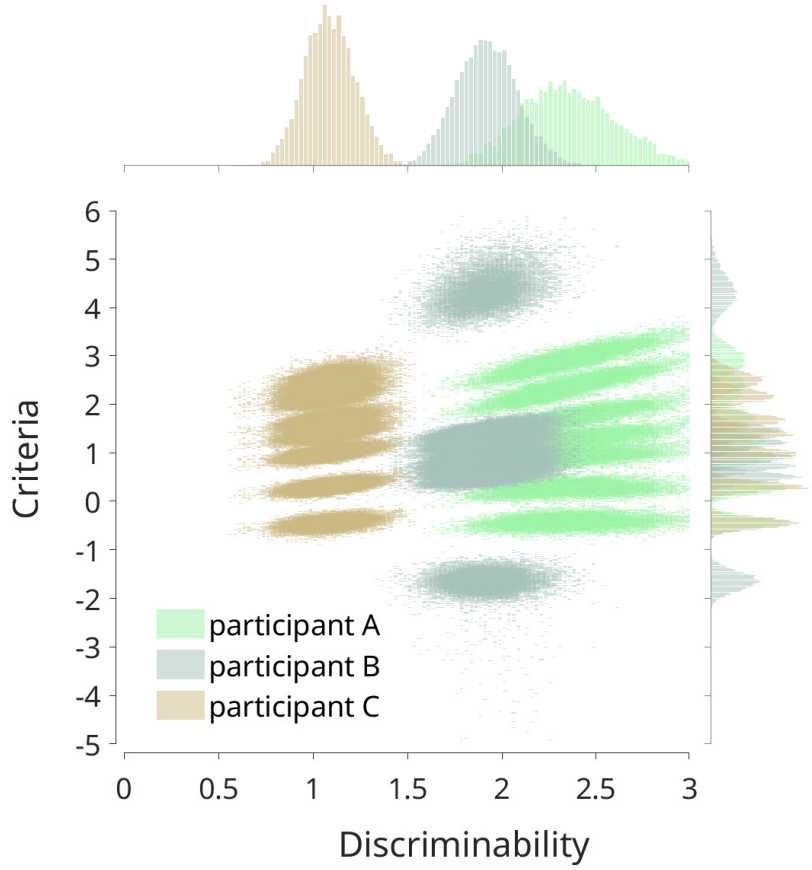

Figure 5.5: The joint and marginal posterior distributions for discriminability d' and criteria $\boldsymbol{k}$ for the visual metacognition data based on the multiple-choice signal detection model.

The criteria are a little more difficult to interpret. For example, many of the criteria for Participant B have very similar values near 1. This actually explains participant B's frequent use of the confidence rating 3, but that may not be intuitively clear in Figure 5.5.

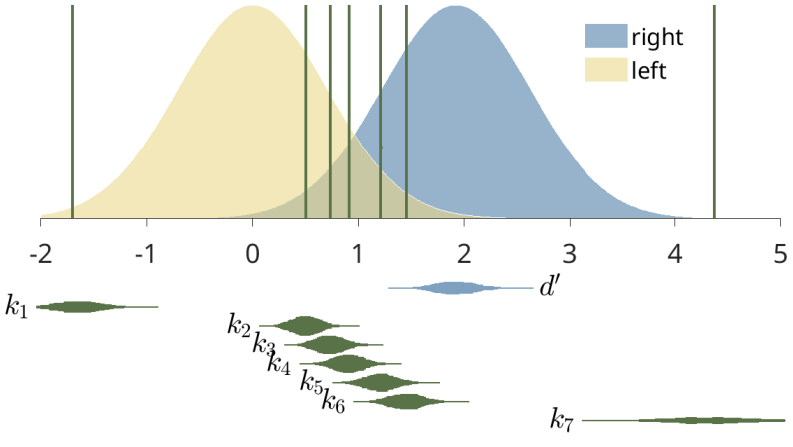

Figure 5.6: Multiple-choice signal detection model presentation of the discriminability d' and criteria $\boldsymbol{k}$ for participant B.

An alternative visualization of the inferences for participant B is provided

in Figure 5.6. The top part of the figure shows a visualization based on the posterior means for the stimulus discriminability and response criteria.The left-oriented stimulus distribution is centered at zero and the right-oriented stimulus distribution is centered at $d'_{\mathrm{B}} = 1.88$. The criteria are placed at the posterior means for $k_1, \ldots, k_7$.

This visualization makes clear how the inferred criteria lead to the frequent use of confidence ratings of 3. The leftmost and rightmost criteria are set at very stringent levels, making it unlikely that stimuli will be sampled from the stimulus distributions that fall in the region corresponding to confidence ratings of 4. The other criteria are placed very near each other, making the gaps between them small, and limiting the possibility of confidence ratings of 1 or 2. The net result is that there a high probability of a sample from the right distribution leading to a "right" decision with confidence 3, and a similarly high probability of a sample from the left distribution leading to a "left" decision with confidence 3.

Figure 5.6 also shows the marginal posterior distributions for d' and $k_1, \ldots, k_7$. These distributions provide information about uncertainty. For example, although the right-oriented stimulus distribution is set at the posterior mean of 1.88, it could be that the true discriminability is as small as about 1.5 or as large as about 2.5.

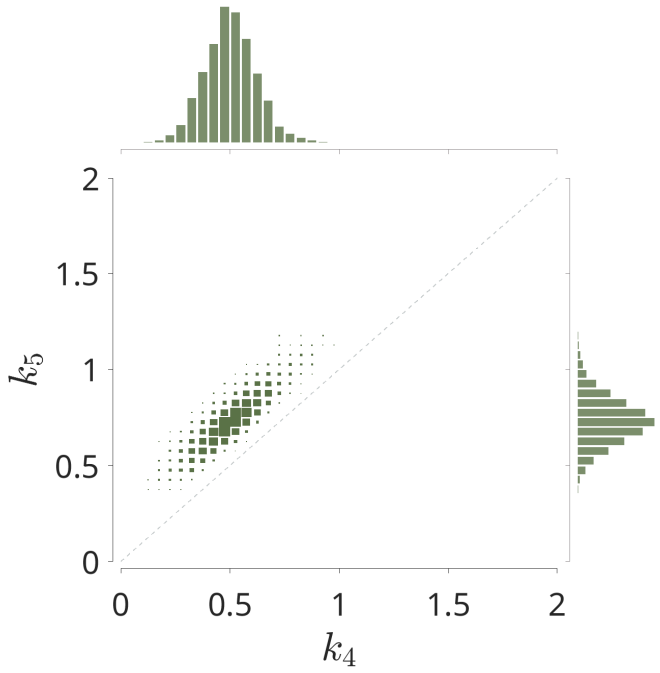

Figure 5.7: The joint and marginal posterior distributions for criteria k_2 and k_3 for participant C.

The marginal distributions for the criteria provide also useful information about uncertainty. It is clear, for example, that k_1 is much smaller than k_2 and k_7 is much larger than k_6. In general, however, it is important to appreciate the limitations of using the marginal distributions of the criteria to communicate uncertainty. For example, the multiple-choice signal detection model constrains the criteria so that $k_1 \leq k_2 \leq \ldots k_7$. This seems contradicted by the overlap in the marginal posteriors for $k_2, \ldots, k_6$ in Figure 5.6. The problem is that the underlying joint posterior distribution for the criteria is not well approximated by the product of

its marginal distributions. Figure 5.7 helps demonstrate the issue, by showing the joint and marginal distribution for just for criteria k_2 and k_3. It is clear in the joint distribution that the constraint $k_2 \leq k_3$ is satisfied, since all of the posterior density is above the diagonal line. But it is also true that the marginal distributions overlap.*

With this caveat in place, the visualization in Figure 5.7 is useful for interpreting the model inferences for participant B. It shows they have reasonably good discriminability, but an unusual strategy for assigning confidence to their decisions that can be explained in terms of criteria placement by the multiple-choice signal detection model.

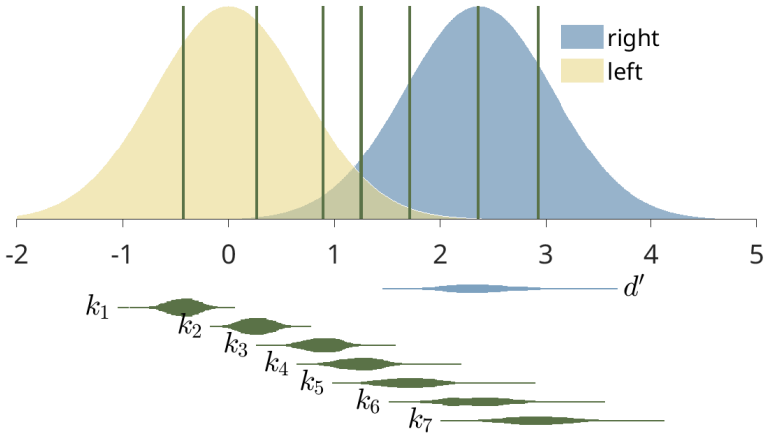

Figure 5.8: Multiple-choice signal detection model presentation of the discriminability d' and criteria $\boldsymbol{k}$ for participant A.

Figure 5.8 presents the same visualization for participant A. It shows that they are able to discriminate the two types of stimuli very well. It also shows that they place the criteria defining the confidence rating regions relatively evenly along the stimulus representation dimension. These findings are consistent with the behavioral data for Participant A, who generally made accurate decisions but expressed confidence over the full range of possible responses.

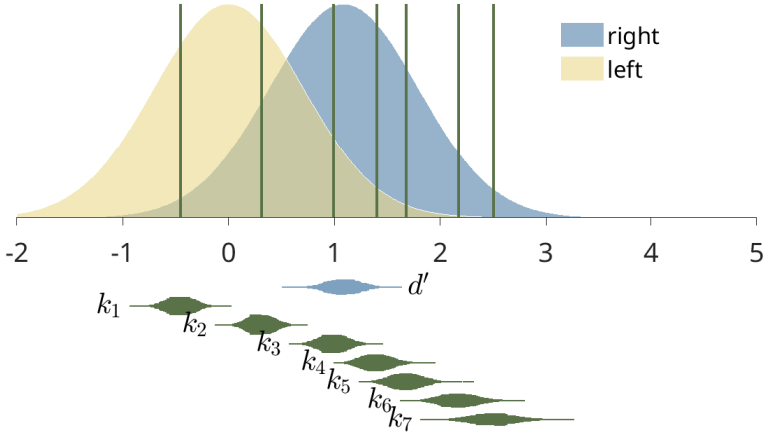

Figure 5.9: Multiple-choice signal detection model presentation of the discriminability d' and criteria $\boldsymbol{k}$ for participant C.

* The issue of joint distributions potentially have more information than marginal distributions was originally discussed in Chapter 0.2.1. If the result in Figure 5.6 is not clear, it may be helpful to review that discussion at this point.

In contrast, the visualization for participant C in Figure 5.9 shows that they have relatively poor discriminability. There is significant overlap between the distributions for the right and left stimuli, which provides an explanation for their inaccurate decisions. Their criteria are relatively evenly spaced relative to one another, but are shifted to the right with respect to the stimulus distributions. In particular, the criterion k_4 that separates "left" from "right" responses is very likely to be greater than the point near 0.5 at which the two stimulus distributions have equal likelihood. This explains the bias towards responding "left" evident in participant C's behavioral data.

5.3 *Conditioned Recognition*

`conditionedRecognition.jasp`
`conditionedRecognitionTwoChoice.jasp`

Vijayakumar (2024) studied the interaction between of recognition memory and conditioning social and monetary rewards. In the recognition memory test, a two-stage response procedure was used. Participants first deciced whether the presented item was old or new, and then rated their confidence in their decision.

This response procedure is not exactly aligned with the single-response assumption of the multiple-choice signal detection model. Accordingly, this example considers how to make a two-stage response procedure compatible with the model. It also compares the inferences based on the multiple-choice signal detection model with those based on the two-choice signal detection model using just the initial binary decision.

Data

Vijayakumar (2024) studied the recognition memory of 44 participants. The study task involved three phases, called the preconditioning, conditioning, and post-conditioning phases. In the preconditioning phase, participants were shown images and had to categorize them as animals or tools. In the conditioning phase participants were presented with 48 forced-choice recognition trials, each including one image from preconditioning and a new image. In a between-participants manipulation, monetary and social rewards were provided for correct answers involving either animal or tool images. In the post-conditioning phase, as for preconditioning, participants viewed 48 new images and had to classify them as animals or tools.

The recognition memory task occurred 24 hours later. Each of 240 trials presented a single image, coming from the 144 old images from the preconditioning, conditioning, and post-conditioning phases, and 96 new images. On each trial, participants provided a binary response as to whether the image was old or new. They then rated their confidence in their decision on all 11-point scale numbered from 0 ("'not at all confident") to 10 ("extremely confident").

The obvious way to combine these two responses to produce one response is to consider an "old" decision with confidence 10 as the response +10, an "old" decision with confidence 9 as the response +9, and so on. Similarly, we consider a "new" decision with confidence 10 as the response -10, a "new" decision with confidence 9 as the response -9,

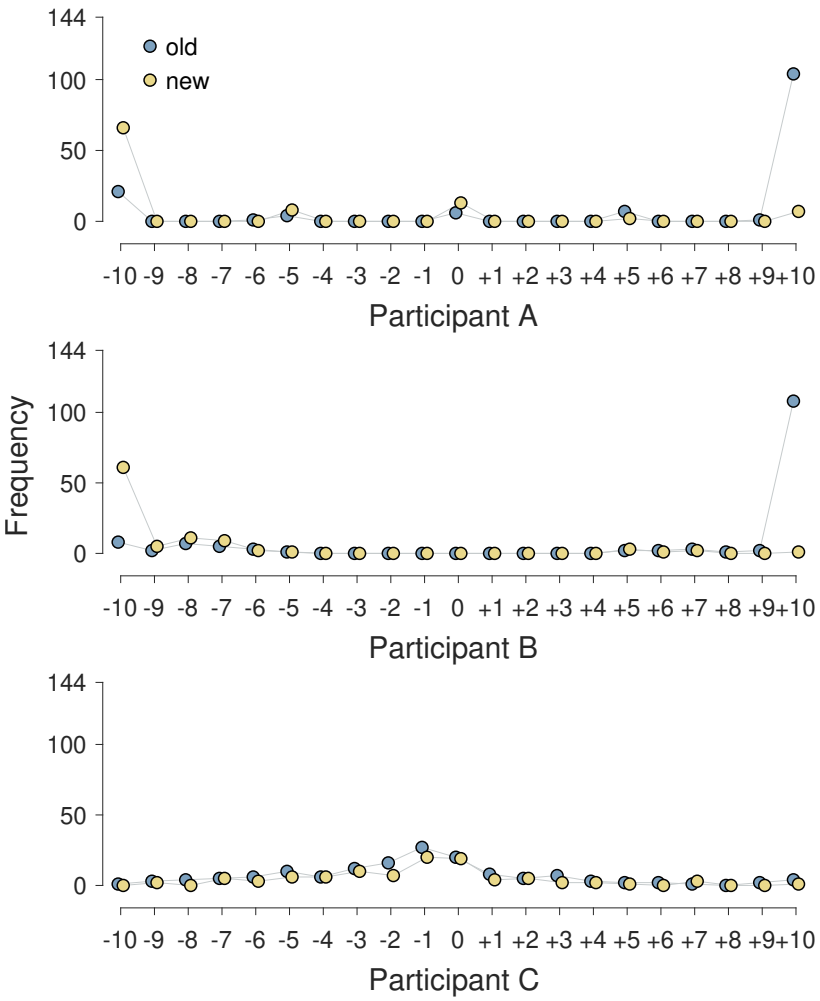

Figure 5.10: Summary of the behavioral data for three participants in the conditioned recognition task.

and so on. If the confidence is 0, it does not matter whether the initial decision was "old" or "new." In this way, each trial has a single response $-10, -9, \ldots -1, 0, +1, \ldots, +9, +10$.

We consider only three participants, whose behavior is summarized in Figure 5.10. The circular markers show how often each of the 21 possible responses is made, with blue markers corresponding to old images and yellow markers corresponding to new images.

Participant A mostly responds with a confidence of 10, for both "old" and "new" decisions. Occasionally they respond with a confidence of 0. Even less often, they respond with a confidence of 5. These three confidence ratings cover 238 of their 240 responses. The only exceptions—which are only just perceptible in the visualization—are a confidence of 6 for a "new" decision and a confidence of 9 for an "old" decision. Their high-confidence decisions are often correct, but do contain errors. The +10 response values are mostly for old images, but not always, indicating a number of false alarms. Their -10 decisions are more often for new images, but there are also some misses.

Participant B responds with maximum confidence even more often than participant A. About three-quarters of their confidence ratings are 10, and the lowest level of confidence they provide is 5. Their decisions are also more accurate. There are very few false alarms and misses.

Participant C, in contrast, uses all of the confidence ratings for at least a few trials, but mostly provides low levels of confidence. About two-thirds of their responses have confidence 3 or below. Their decisions are also much less accurate. Most of the 21 responses are provided about equally often for old as for new images, which is consistent with near-chance performance.

Results

Posterior predictive analysis

Figure 5.11 shows a posterior predictive check of the descriptive adequacy of the multiple-choice signal detection model. The vertical histograms show the posterior predictive mass for each possible number of counts for every response for both old and new images. The number of old and new words chosen for posterior predictive analysis match the experimental design.

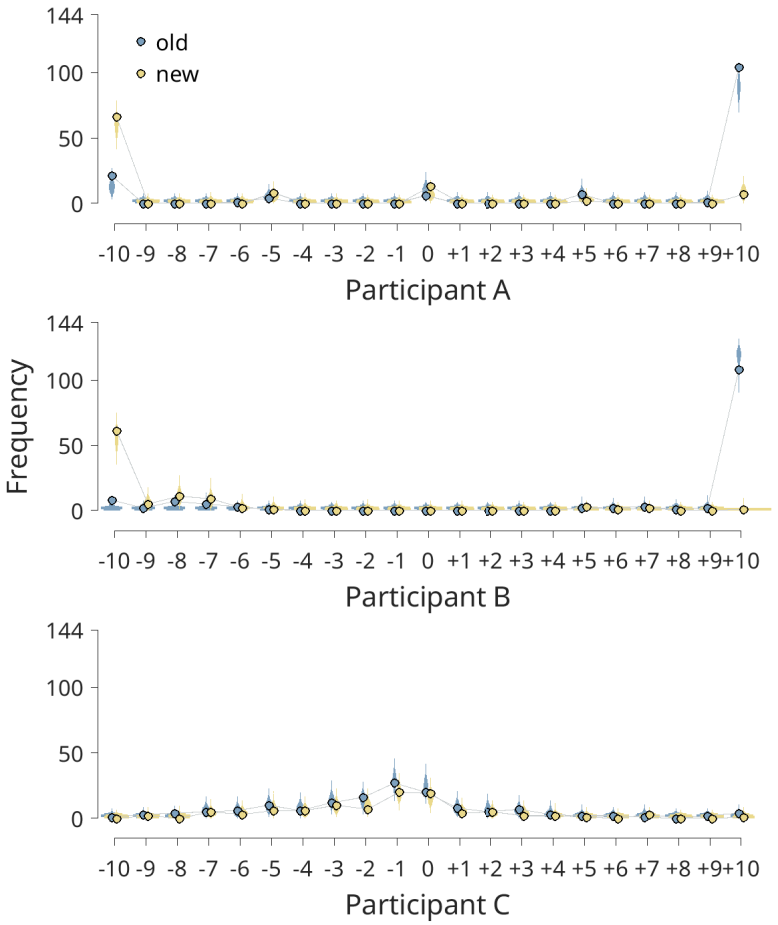

Figure 5.11: Posterior predictive analysis of the conditioned recognition data using the multiple-choice signal detection model.

The agreement between the predictive distributions and the observed counts is not perfect. For example, the predictive distribution for +10 responses to old images for participant A gives most of its mass to counts smaller than the 104 observed. Overall, however, the predictive

distributions are close to the observed counts, and follow the important patterns of change. The predictive distributions capture the reliance on confidence ratings of 0, 5, and 10 for participant A, the near-exclusive use of the confidence rating of 10 for participant B, and the use of a range of low confidence ratings by participant C.

Posterior distributions

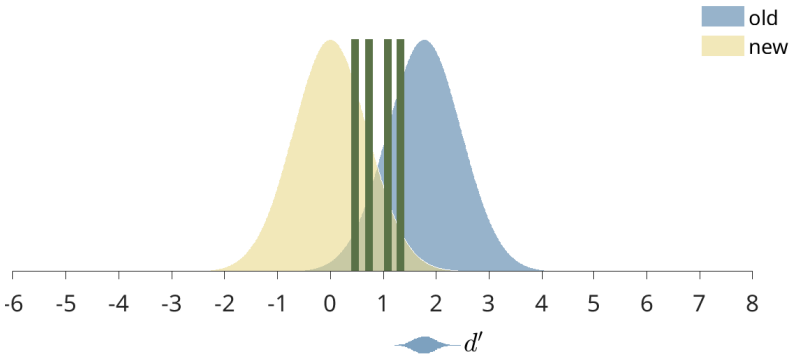

Figure 5.12: Multiple-choice signal detection model presentation of the discriminability d' and criteria $\boldsymbol{k}$ for participant A.

Figure 5.12 presents the multiple-choice signal detection model visualization for participant A. The top panel shows the signal and noise distributions corresponding to old and new images based on the posterior mean for discriminability. It also shows the 20 criteria, based on their posterior means. The lower panel shows the marginal posterior distribution for discriminability. The marginal distributions for the criteria are not shown, because of the limitations discussed in the *Visual Metacognition* example in Chapter 5.2.

Figure 5.12 shows that participant A is inferred to have modest discriminability with a posterior mean $d' = 1.78$ and 95% credible interval $(1.45, 2.10)$.

Figure 5.12 also shows how the multiple-choice signal detection model explains participant A's use of confidence ratings of 0, 5, and 10. Most of the noise distribution is smaller than the smallest criterion, which makes "new" decisions with confidence 10, corresponding to -10 responses, very likely when new images are presented. Similarly, +10 responses are very likely for old images.

Most of the criteria are clustered tightly together. The only visible gaps correspond to responses of -5, 0, and +5. Note that confidence ratings of 0, corresponding to the central region between the criteria, have roughly equal density under the signal and noise distributions. This is consistent with the observed behavior of providing confidence ratings of 0 about equally often for old as for new images.

Figure 5.13 shows the signal detection visualization for participant B. Although the visualization shows extreme separation of the signal and noise distributions, based on the posterior mean of $d' = 5.60$, there is large uncertainty in the separation, with a 95% credible interval of $(3.28, 8.75)$. Participant B's frequent confidence of 10 for old images is captured by the largest criterion being at the left edge of the signal distribution, but it is important to interpret this in the context of the

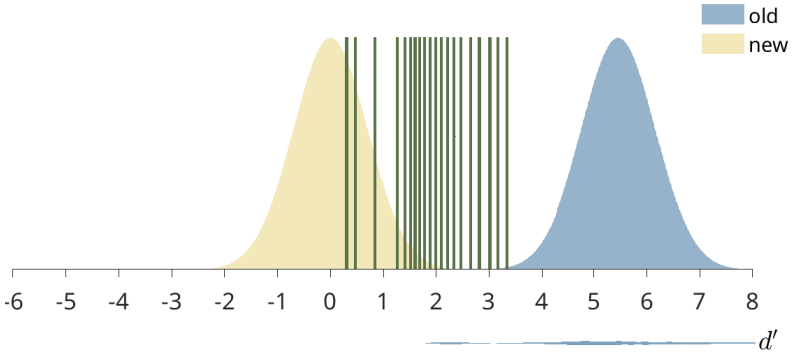

Figure 5.13: Multiple-choice signal detection model presentation of the discriminability d' and criteria $\boldsymbol{k}$ for participant B.

uncertainty for d'. The discriminability could be a smaller value such as 3 or 4, which would introduce overlap between the largest criteria and the signal distribution.

A reasonable inference is that participant B discriminates very well between old and new images, and sets their criteria such that high confidence responses for old images are likely. In the same way, it is clear that their confidence for new images is sometimes not quite as high, since there are more high density areas of the noise distribution between the leftmost criteria.

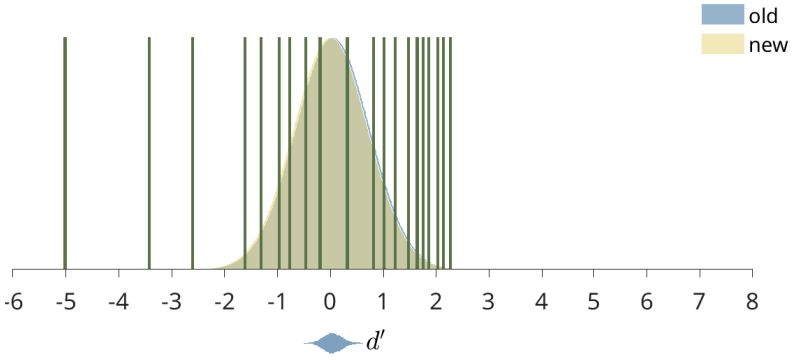

Figure 5.14: Multiple-choice signal detection model presentation of the discriminability d' and criteria $\boldsymbol{k}$ for participant C.

Finally, Figure 5.14 shows the signal detection visualization for participant C. The obvious result is that discriminability is near zero with a posterior mean $d' = 0.05$ and 95% credible interval $(-0.27, 0.36)$. This corresponds to the signal and noise distributions not being separated. The criteria are located so that low confidence responses are likely for both old and new images, with many of the largest criteria to the right of the distributions and many of the smallest criteria to the left.

Comparison to the two-choice signal detection model

Because the experimental design involved responses being collected in two stages, it is possible to apply the two-choice signal detection model to the initial binary decisions, ignoring the confidence ratings. It is interesting to compare the inferences about discriminability for participants A and B made this way with those made by the multiple-choice signal detection model incorporating the confidence ratings.

Table 5.3 shows the signal detection counts for participants A and B based on the initial binary decisions. The patterns of counts are quite similar for the two participants. Both show good performance, with high hit and correct rejection counts, although participant B performs a little better. Both participants' errors tend to be misses rather than false alarms.

	truth			
	participant A		participant B	
decision	old	new	old	new
old	113	11	118	7
new	31	85	26	89

Table 5.3: Signal detection counts of the initial binary decisions for Participants A and B in the conditioned recognition task.

Based on these counts, the two-choice signal detection model infers participant A's discriminability to have a posterior mean of $d' = 2.00$ with a 95% credible interval $(1.57, 2.39)$. The inference for participant B is $d' = 2.38$ with a 95% credible interval $(1.93, 2.84)$. Thus, both participants are inferred to have similar discriminability, although participant B's is likely a little better, consistent with the signal detection counts.

It is striking that the conclusions reached for the two participants using their intitial binary decisions are not mirrored by what was found by including confidence ratings and using the multiple-choice signal detection model. As discussed above, the multiple-choice model found participant A to have a posterior mean $d' = 1.78$, consistent with the two-choice inferences, but found participant B to have a much higher posterior mean of $d' = 5.60$ with a 95% credible interval of $(3.28, 8.75)$.

Thus, it seems clear that including confidence and using the multiple-choice signal detection model changes the inference about the discriminality of participant B, but not participant A. Intuitively, this is because the confidence ratings provide additional information about how well the participants could discriminate old from new words. The confidence ratings provided by participant A include many 0 and 5 responses, which are consistent with difficulties in discrimination. They also express a confidence of 10 in the majority of their misses, as shown by the blue marker for the response -10 in Figure 5.10. This is also strong evidence for failure in discrimination. The confidence ratings provided by participant B, in contrast, are all high, and are mostly the maximum of 10. Very few of their errors, however, are made with full confidence. In this way, the confidence ratings provide additional evidence of good discrimination.

One way to understand the diverging inferences about discriminability for the two participants under the two models is that participant A made many initial binary decisions that were correct, but had low confidence (often zero confidence) in them. Participant B, in contrast, was most confident in their correct decisions. The multiple-choice signal detection model is able to use this additional evidence to make more informed inferences about discriminability.

5.4 Hierarchical Multiple-Choice Signal Detection Model

5.4.1 Motivation

In the same way as the two-choice signal detection model, the hierarchical extension of the multiple-choice signal detection models has the motivation of providing a group level account of how individuals are related to each other in their discriminability and criteria. For the multiple-choice signal detection model the focus is often more on discriminability than criteria, especially in making comparisons across different groups. Experimental manipulations, or other ways of forming groups, often systematically affect discriminability rather than criteria. The hierarchical model, however, places group structure on both discriminability and every criterion, so inferences about both are available.

5.4.2 Model

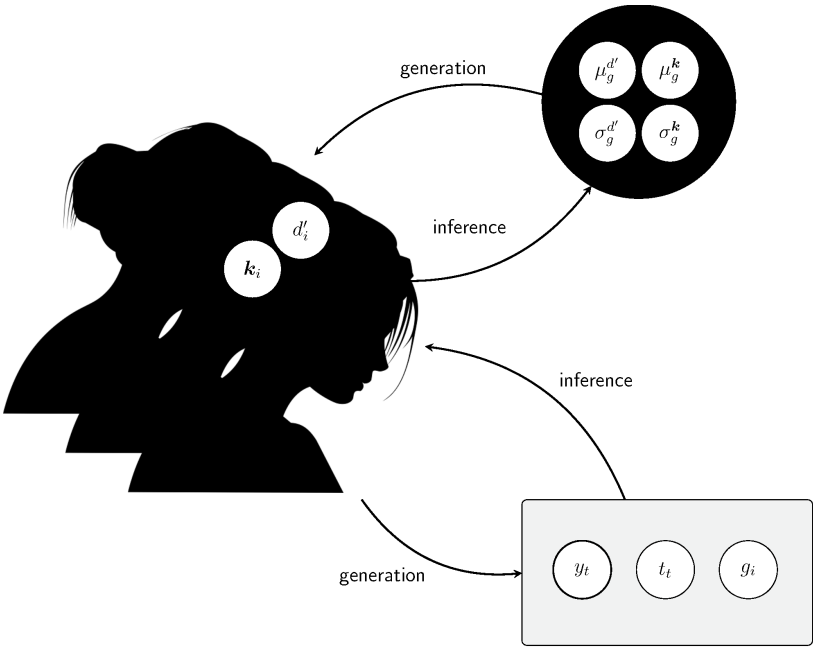

Figure 5.15: The structure of the hierarchical multiple-choice signal detection model.

Figure 5.15 summarizes the structure of the multiple-choice signal detection model. The multiple individuals are shown with their own discriminabilities d_i' and criteria $\boldsymbol{k}_i = (k_{i1}, k_{i2}, \ldots)$, with as many criteria as are needed for the number of response options. For individuals in the same group, these parameters are linked by an overarching group distributions with means $\mu^{d'}$ and $\mu^{\boldsymbol{k}}$, and standard deviations $\sigma^{d'}$ and $\sigma^{\boldsymbol{k}}$.

The data take the form of the responses y_{it} for each individual on each trial, and the true state t_{it} for each trial, indicating whether it was signal or noise. The additional observed information is the group membership g_i of each individual.

The inferences about the group distributions allow the discriminability and biases of a representative individual $\tilde{d}_g'$ and criteria $\tilde{\boldsymbol{k}}_g$ from each

group to be generated. This, in turn, allows a predictions to be made about the responses $\hat{y}_t$ for a representative individual in each group.

5.4.3 Mechanics

Table 5.4 details the meaning of the parameters and variables in the multiple-choice signal detection model, their name in the model as shown in Figure 5.15, and their name in JASP.

Name	Model	JASP
response	y_t	`response[t]`
signal or noise truth	t_t	`truth[t]`
group membership	g_i	`group[i]`
number of signal observations for prediction	$\hat{n}^{\text{s}}$	`nPrdSig[i]`
number of noise observations for prediction	$\hat{n}^{\text{n}}$	`nPrdNoi[i]`
individual discriminability	d'_i	`d[i]`
individual criteria	$\boldsymbol{k}_i$	`k[i,]`
individual bias	c_i	`c[i]`
group discriminability mean	$\mu_g^{d'}$	`muD[g]`
group discriminability standard deviation	$\sigma_g^{d'}$	`sigmaD[g]`
group mean criteria	μ_g^{k}	`muK[g,]`
group criteria standard deviation	σ_g^{k}	`sigmaK[g,]`
group representative discriminability	$\tilde{d}'$	`dRep`
group representative criteria	$\tilde{\boldsymbol{k}}_g$	`kRep[g,]`
group predicted signal responses	$\hat{y}_g^{\text{s}}$	`respSigPrd[g,]`
group predicted noise responses	$\hat{y}_g^{\text{n}}$	`respNoiPrd[g,]`

Table 5.4: The meaning of the observed variables and latent parameters in the multiple-choice signal detection model, with their statistical and JASP names.

Input data

Table 5.5 shows the general data format for some of the data from the *Baggage Screeners* example. The first three column are in a long format, with each row representing a trial. The `person` column indicates the individual making the decision, the `truth` column indicates whether a signal or noise stimulus is presented, and the `response` column indicates whether a signal or noise response was made.

The fourth `group` column corresponds to individuals. It has as many entries as there are individuals, and indicates the group to which each individual belongs. The final two `nPrdSig` and `nPrdNoi` columns correspond to groups, and indicate the number of predicted signal and noise trials for the posterior predictive analysis.

Output analyses

The JASP implementation provides four analyses.

- **individual discriminability, criterion, and bias**: the marginal posterior distributions of the discriminability d'_i, criteria $\boldsymbol{k}_i$, and biases $\boldsymbol{c}_i$ for each individual

person	truth	response	group	nPrdSig	nPrdNoi
1	1	6	1	100	150
1	2	2	1	100	150
1	1	6	1		
1	4	2	1		
...	...	...	...		
2	2	3	2		
1	2	2	2		
...	...	...			
20	2	3			
20	1	4			

Table 5.5: JASP data format for the multiple-choice signal detection model.

- **group discriminability and criteria**: the marginal posterior distribution of the mean discriminability $\mu_g^{d'}$ and mean criteria μ_g^{k} for each group
- **representative individual discriminability and bias**: the marginal posterior distribution of the representative discriminability $\tilde{d}'_g$ and criteria $\tilde{k}_g$ for each group
- **representative posterior predictions**: the marginal posterior predictive distributions of the representative trial responses $\hat{y}_{gt}$ from $\hat{n}^{\text{s}}$ signal observations and $\hat{n}^{\text{n}}$ noise trials for each group

5.5 *Baggage Screening*

`baggageScreening.jasp`

Enright & McCarley (2019) studied people's decision making in a mock baggage screening task, examining how well individual people and pairs of people working together could detect a knife in an x-ray image of baggage.

Data

The experiment conducted by Enright & McCarley (2019) baggage screening involved sixteen pairs of people. Each person completed a screening task individually as well as in pair. The task involved 250 trials, with 100 of the images containing a knife and the other 150 not containing a knife. The image was presented throughout the trial and there was no time limit for making decisions.

On each trial, the person or people responded on a 6-point scale with labels "definitely no," "probably no," "guess no," "guess yes," "probably yes," and "definitely yes." Because of this labeling, the first three responses can be treated as "no" responses and the last three treated as "yes" responses. This two-choice summary was used to provide feedback on every trial.

Figure 5.16 summarizes the behavioral data. The left panel relates to individual decisions and the right panel relates to the decisions made by pairs of people. The circular markers show how often each of the six possible responses is made, with lines connecting the response frequencies for the same individual or pair. Lines with blue markers correspond to stimuli in which a knife is present and yellow markers corresponding to stimuli in which no knife is present.

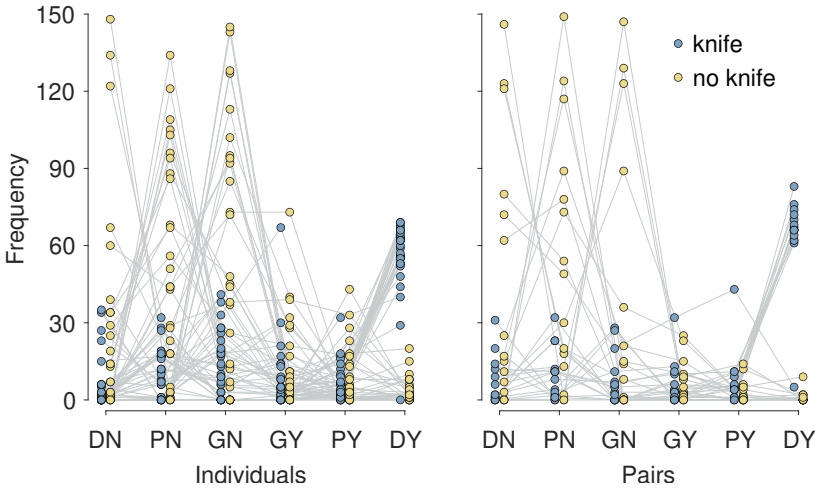

Figure 5.16: Summary of the baggage screening behavioral data.

This visual summary of the results makes clear that there is variability across both different individuals and different pairs. For example, some individuals almost always respond "guess no" when no knife is present. This is a correct decision, but expressed with low confidence. Some other individuals almost always respond "probably no," and still others almost always respond "definitely no." Thus, there appear to be individual differences in the confidence people are willing to express when they believe there is no knife. The same variability is evident for pairs of people.

In contrast, both individuals and pairs often respond "definitely yes" for images that do have a knife, and "probably yes" and "maybe yes" are must less often used. This suggests there is an asymmetry in how confidently people and pairs of people can detect the presence versus absence of a knife.

Figure 5.16 does not show obvious large differences between the decisions and confidence of individual people and pairs of people. One hint of a difference is that individuals seem to respond with "no" decisions for images with knives relatively more often. This suggests that pairs of people may be better at detecting knives when they are present.

Results

Posterior predictive analysis

Figure 5.17 shows a posterior predictive check of descriptive adequacy. It overlays the posterior representative distribution on the summary of the behavioral data presented in Figure 5.16. The posterior representative distribution is appropriate, because it is at the level of a single individual or single pair, but applies to all of the observed individuals and pairs. An alternative approach to checking descriptive adequacy would be to compare the posterior predictive distribution for each individual and each pair to their specific behavioral data. This would have the advantage of assessing descriptive adequacy in a more detailed way, but would be impossible to present in a consolidated visualization.

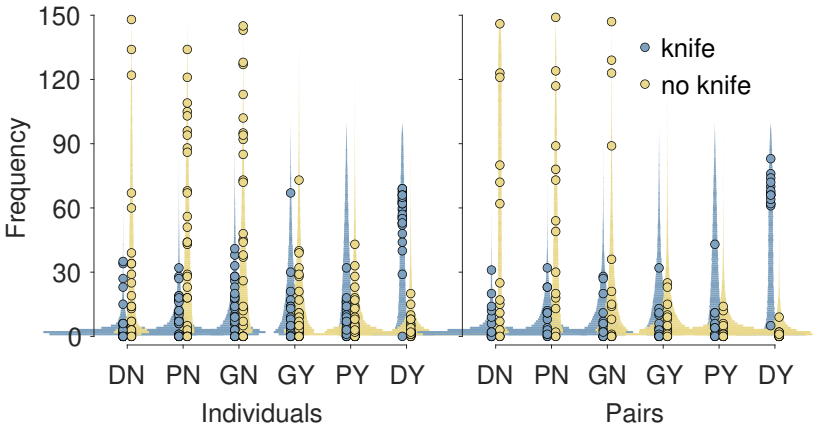

Figure 5.17: Posterior predictive analysis of the baggage screening data using the multiple-choice signal detection model.

The posterior predictive check in Figure 5.17 serves the purpose of providing confirmation of the descriptive adequacy of the multiple-choice signal detection model. It shows that the expected behavior of a representative individual or pairs spans the range of behavior observed across all individuals and pairs. Most importantly, the posterior representative distributions capture the important regularities of the behavioral data. For example, they expect "no" responses to images with knives to be rare, and they allow for a wide range of frequencies for "no" responses with different levels of confidence when images do not have knives.

Group results

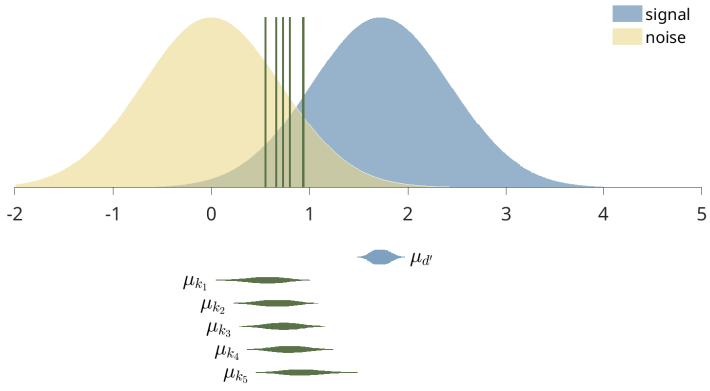

Figure 5.18: Multiple-choice signal detection model presentation of the group discriminability d' and criteria $\boldsymbol{k}$ for people completing the baggage screening task as individuals.

Figures 5.18 and 5.19 the inferences about group discriminability and criteria for the individual and pair task conditions, respectively. The clear result is that the pairs of people have greater mean discrimability. Their mean discriminability has posterior mean $\mu_{d'} = 2.31$ with 95% credible interval (2.12, 2.52), while the mean discriminability for the individuals is $\mu_{d'} = 1.73$ with 95% credible interval (1.59, 1.86).

The results are much less clear for the criteria. One interesting possibility is that pairs may set more conservative criteria than individuals, as shown by criteria being further to the right, requiring more evidence before being confident there is a signal. But, this is not a clear result. The clear result is that the criteria are so uncertain as to make a group-level characterization difficult. There are several possible reasons for this,

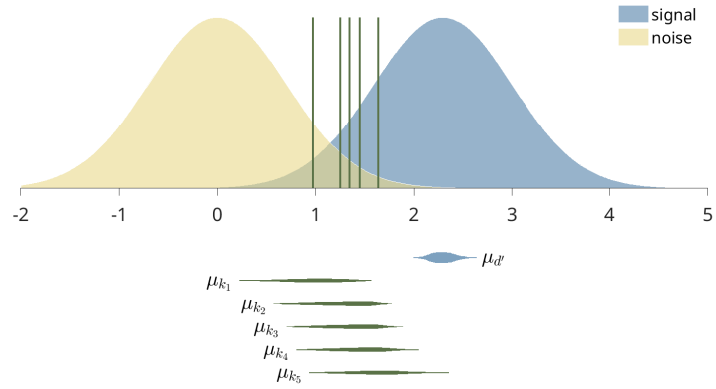

Figure 5.19: Multiple-choice signal detection model presentation of the group discriminability d' and criteria $\boldsymbol{k}$ for people completing the baggage screening task in pairs.

including large homogeneity in the criteria used by different individuals and pairs.

Representative results

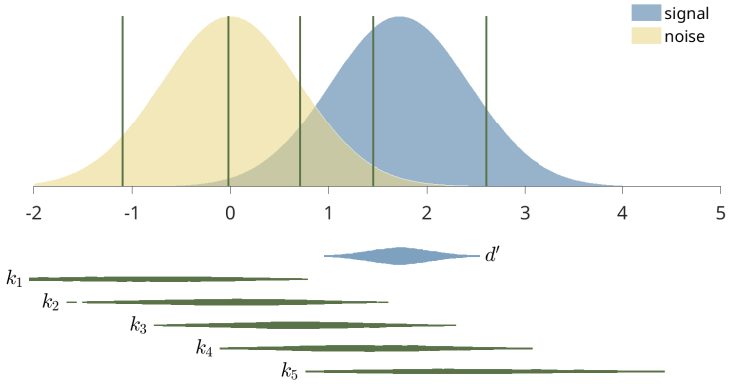

Figure 5.20: Multiple-choice signal detection model presentation of the discriminability d' and criteria $\boldsymbol{k}$ for the representative individual person.

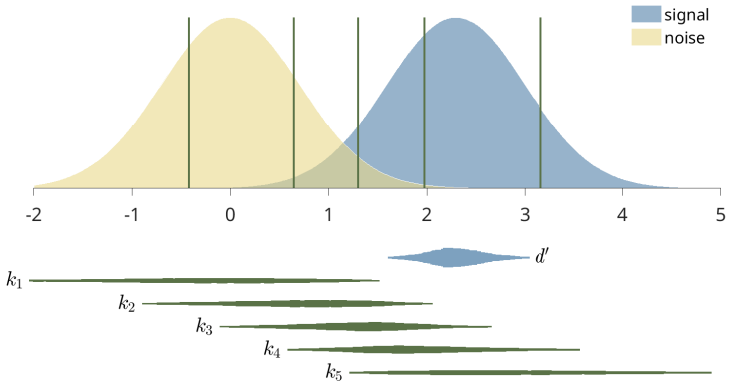

Figure 5.21: Multiple-choice signal detection model presentation of the discriminability d' and criteria $\boldsymbol{k}$ for the representative pair of people.

Figures 5.20 and 5.21 show the inferences about individual discriminability and criteria for the representative individual and the representative pair of people, respectively. At the representative individual level, with its sensitivity to individual differences, there is so much uncertainty about the criteria that there is little that can be concluded. The interesting results relate to discriminability. The group mean discriminabilities shown in

Figures 5.18 and 5.19 are well separated. At the level an individual person or pair of people, however, there is more uncertainty and more overlap in the posterior distributions. The representative individual discriminability has posterior mean $\mu_{d'} = 1.72$ with 95% credible interval $(1.08, 2.34)$, while the representative pair discriminability has has posterior mean $\mu_{d'} = 2.31$ with 95% credible interval $(1.60, 3.04)$. This is because of the additional uncertainty again introduced by individual differences in the representative analysis.

5.6 *Fake News*

`fakeNews.jasp`

Luo et al. (2022, Experiment 1) studied people's ability to distinguish between real and fake news headlines, sourced from social media. People's judgments were collected on a 7-point scale.

5.6.1 Data

The 337 participants were randomly assigned to different conditions corresponding to different news content areas: health, politics, or science. Each participant was shown ten headlines, five of which were real and five or which were fake. For each headline, they provided a response on a seven-point scaling ranging from 1 = "definitely fake" to 7 = "definitely real."

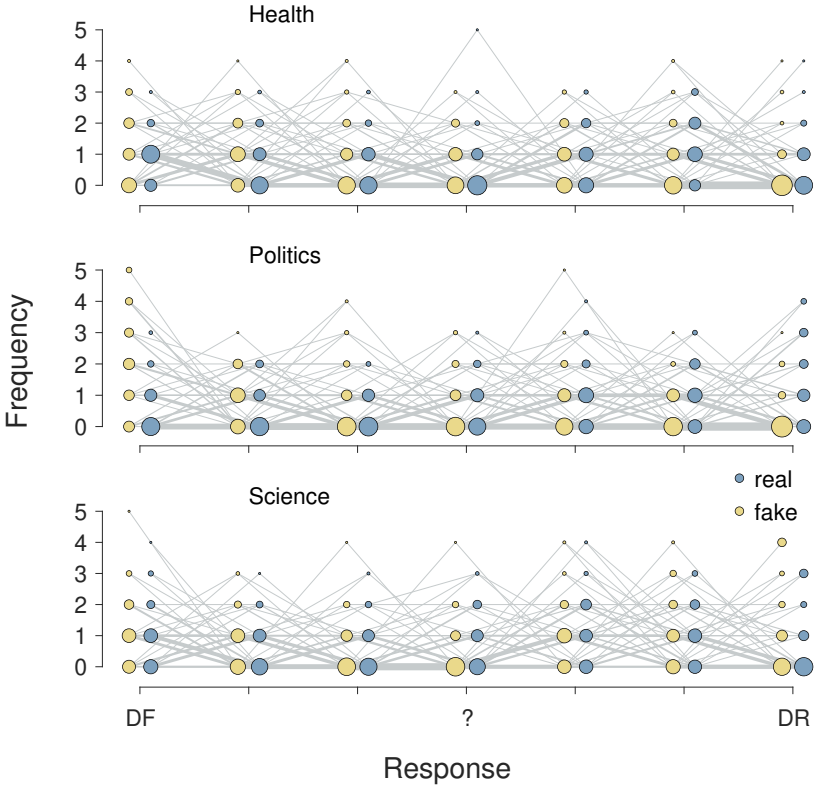

Figure 5.22: Summary of the fake news behavioral data.

Figure 5.22 summarizes the behavioral data. The three panels relate the three content topics. The area of the circular markers show how often an individual participant made each of the seven possible responses, for both real and fake headlines. The responses range from definitely

fake "DF" to definitely real "DR." The middle response is shown as "?" since it is not clear whether it indicated the decision of a real or fake headline. Luo et al. (2022) give these ranking an accuracy score of 0.5 in their analyses. Blue markers correspond to real headlines and yellow markers corresponding to fake headlines. The connecting lines show the frequencies connecting individual participants across the response frequencies.

This visual summary is not easy to interpret, but does show that participants did not discriminate very well between the real and fake headlines. Most response options have similar distributions over the possible frequencies for both real and fake headlines. It is also clear that few individuals were consistent in their behavior. It is rare that a participant used the same response for either real or fake headlines all five times they were presented.

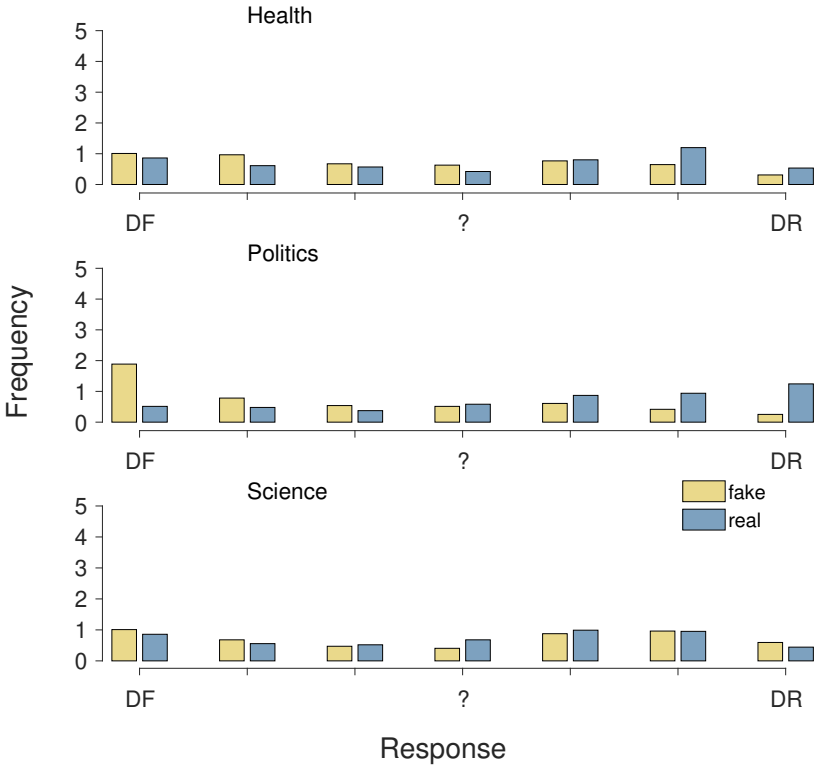

Figure 5.23: A simpler summary of the fake news behavioral data.

Figure 5.23 provides a simpler summary of the data that does not consider individual differences. It shows the frequency of all responses between the bounds of 0 and 5, averaged over all participants in each condition, for both real and fake headlines. This visualization continues the show the all of the responses used for both real and fake headlines, but provides the insight that headlines are better distinguished in the politics condition.

5.6.2 Results

Posterior predictive analysis

An advantage of the simpler data summary in Figure 5.23 is that it provides a way to visualize the posterior predictive distribution. Figure 5.24 overlays the means and 95% credible intervals of the posterior predictive

distribution for the representative individual in each condition. It is clear the posterior predictive distributions are uncertain, but their means track the changes in the distributions of observed responses. This provides some evidence that the multiple-choice signal detection model achieves a basic level of descriptive adequacy.

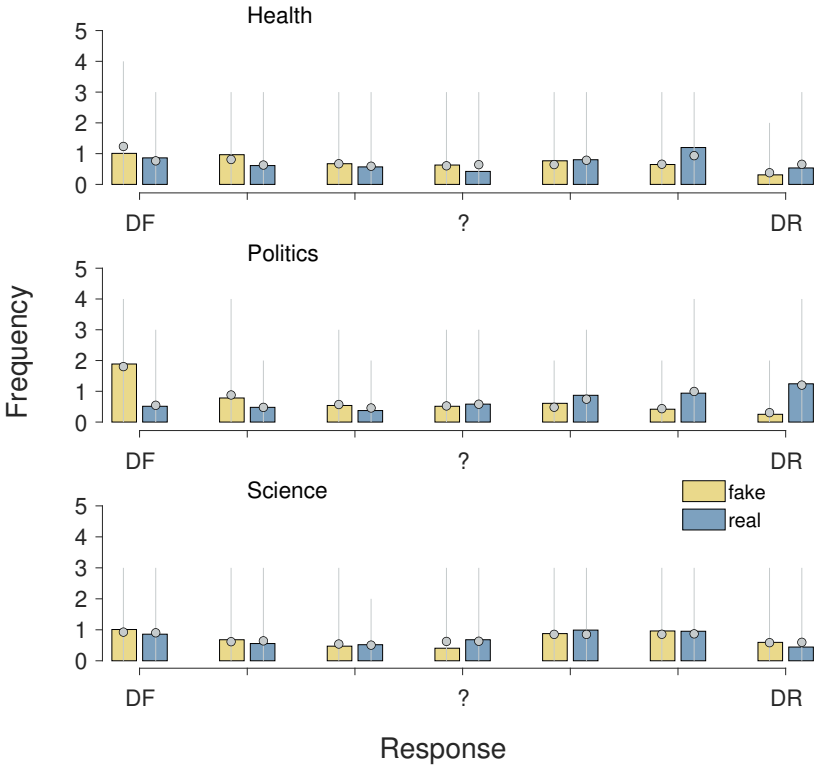

Figure 5.24: Posterior predictive analysis of the representative individual against average individual behavior for the fake news data.

Group means

Figure 5.25 shows the joint posterior distribution of the group mean discriminability and group mean criteria for both three conditions. Participants were best able to detect fake politics headlines, with $\mu_{d'_2} = 0.85$ and 95% credible interval $(0.85, 1.05)$. They also showed some ability to detect fake health headlines, with $\mu_{d'_1} = 0.36$ and 95% credible interval $(0.20, 0.47)$. There is no evidence they were able to detect fake science headlines with $\mu_{d'_3} = 0.02$ and 95% credible interval $(-0.10, 0.15)$.

The multiple-choice signal detection model provides all sort of other inferences that could be used in a more extensive analysis. It is possible to look at the individual discriminabilities and criteria of participants or representative participants from each of the three groups. Sometimes, however, endless possibilities can lead to a loss of focus on the results that answer the motivating research question. The primary goal of Luo et al. (2022, Experiment 1) is to learn about people's general ability to detect fake news for the three content areas. This is answered directly by the inferences about the group-level mean discriminability parameters.

Presenting an abbreviated set of results does not mean, however, that conceptually simpler model than the multiple-choice signal detection model could or should have been used. The criteria parameters are needed to make inferences about discriminability. Without them, the

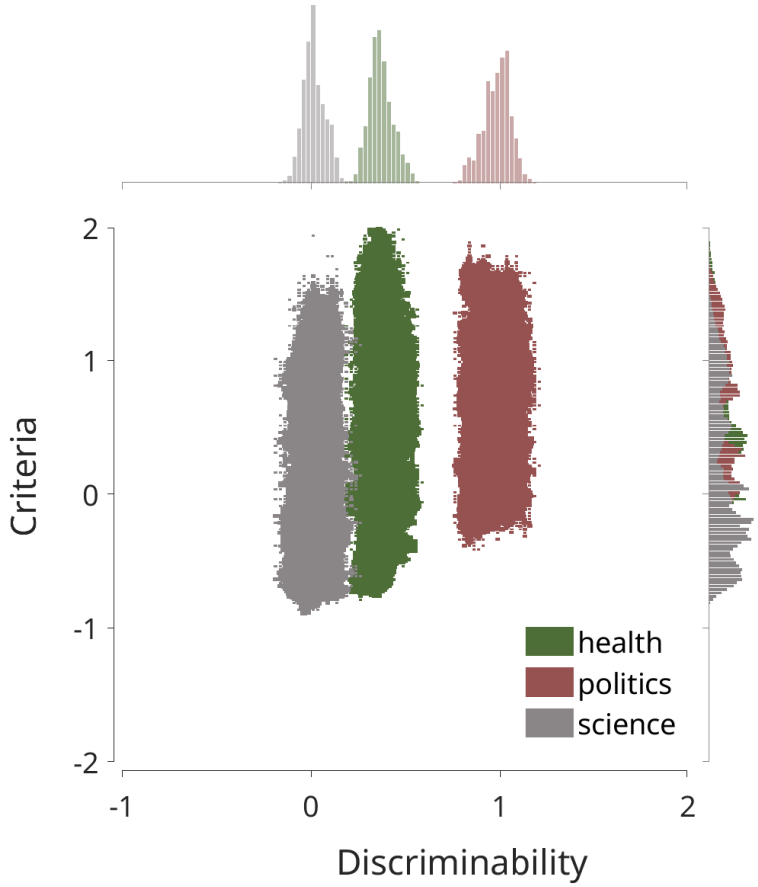

Figure 5.25: Marginal posteriors for the mean discriminability of the three content areas for the fake news data.

signal detection model is incomplete and does not make predictions about the experimental data. In this context, the criteria can be regarded as what are sometimes called *nuisance* parameters (Hotelling, 1940): parameters that are necessary to define the model, but not of inherent interest in the current scientific context. In the same way, the use of a hierarchical model is necessary to make realistic assumptions about individual differences for the fake news data. But inferences about the individual-level parameters themselves are not of interest, especially because there are so few trials per participant. Their role is to make sure that inferences about the group discriminability parameters are founded on good assumptions.

◇

5.7 Formal Details

5.7.1 Multiple-choice signal detection model

Priors

For an m-choice task, there are $m-1$ criteria, and these must satisfy the order constraint that $k_1 \leq k_2 \leq \ldots \leq k_{(m-1)}$ to make the model identifiable. The prior for the discriminability d' is the Gaussians centered on zero with a standard deviation of 3, consistent with the priors used for the

standard two-choice signal detection model. The marginal distributions of the criteria are given the same distribution, so that

$$\begin{aligned} d' &\sim \text{Gaussian}\big(0, \frac{1}{3^2}\big) \\ k_1, k_2, \ldots, k_{(m-1)} &\sim \text{Gaussian}\big(0, \frac{1}{3^2}\big) : k_1 \leq k_2 \leq \ldots \leq k_{(m-1)}. \end{aligned}$$

Likelihood

The probability of each response is given by the geometry shown in Figure 5.1 as the area under either the signal or noise distribution between the relevant adjacent criteria. For the leftmost response, this means that the response probabilities θ^s_{i1} and θ^n_{i1} for signal and noise trials, respectively, are

$$\begin{aligned} \theta^s_1 &= \Phi\big(k_1 - d'\big) \\ \theta^n_1 &= \Phi\big(k_1\big). \end{aligned}$$

Analogously, the rightmost response has probabilities

$$\begin{aligned} \theta^s_m &= 1 - \Phi\big(k_{(m-1)} - d'\big) \\ \theta^n_m &= 1 - \Phi\big(k_{(m-1)}\big). \end{aligned}$$

The probabilities for the intermediate responses are

$$\begin{aligned} \theta^s_j &= \Phi\big(k_j - d'\big) - \Phi\big(k_{(j-1)} - d'\big) \\ \theta^n_j &= \Phi\big(k_j\big) - \Phi\big(k_{(j-1)}\big). \end{aligned}$$

The observed response y_t on the trial t follows these response probabilities, depending on whether the stimulus is signal or noise, so that

$$y_t \sim \begin{cases} \text{categorical}\big(\theta^s_1, \theta^s_2, \ldots \theta^s_m\big) & \text{if } t_t = 1 \\ \text{categorical}\big(\theta^n_1, \theta^n_2, \ldots \theta^n_m\big) & \text{if } t_t = 2. \end{cases}$$

5.7.2 JASP implementation

The JAGS code is shown below.

```
# multiple choice SDT
data{
  nTrials = length(individual)
  nIndividuals = max(individual)
  nResponses = max(response)
  nCriteria = max(response)-1
  lo = 0.01
}
model{
  for (t in 1:nTrials){
    response[t] ~ dcat(theta[t, 1:nResponses, truth[t]])
    # signal
    theta[t, 1, 1] = max(lo, phi(k[individual[t], 1] - dPrime[
     individual[t]]))
    for (j in 2:(nResponses-1)){
```

```
      theta[t, j, 1] = max(lo,phi(k[individual[t], j] - dPrime[
    individual[t]]) - phi(k[individual[t], j-1] - dPrime[
    individual[t]]))
    }
    theta[t, nResponses, 1] = 1 - max(lo, phi(k[individual[t],
    nCriteria] - dPrime[individual[t]]))
    # noise
    theta[t, 1, 2] = max(lo, phi(k[individual[t], 1]))
    for (j in 2:(nResponses-1)){
      theta[t, j, 2] = max(lo, phi(k[individual[t], j]) - phi(k[
    individual[t], j-1]))
    }
    theta[t, nResponses, 2] = 1 - max(lo, phi(k[individual[t],
    nCriteria]))
  }
  for (i in 1:nIndividuals){
    dPrime[i] ~ dnorm(0, 1/3^2)
    for (j in 1:nCriteria){
      kTmp[i, j] ~ dnorm(0, 1/3^2)
    }
    k[i, 1:nCriteria] = sort(kTmp[i, 1:nCriteria])
    c[i, 1:nCriteria] = k[i, 1:nCriteria] - dPrime[i]/2
    # posterior predictive
    thetaPred[i, 1, 1] = max(lo, phi(k[i, 1] - dPrime[i]))
    for (j in 2:(nResponses-1)){
      thetaPred[i, j, 1] = max(lo, phi(k[i, j] - dPrime[i]) - phi(
    k[i, j-1] - dPrime[i]))
    }
    thetaPred[i, nResponses, 1] = 1 - max(lo,phi(k[i, nCriteria] -
     dPrime[i]))
    thetaPred[i, 1, 2] = max(lo,phi(k[i, 1]))
    for (j in 2:(nResponses-1)){
      thetaPred[i, j, 2] = max(lo, phi(k[i, j]) - phi(k[i, j-1]))
    }
    thetaPred[i, nResponses, 2] = 1 - max(lo, phi(k[i, nCriteria])
    )
    respSigPrd[i, 1:nResponses] ~ dmulti(thetaPred[i, 1:nResponses
    , 1], nPrdSig[1])
    respNoiPrd[i, 1:nResponses] ~ dmulti(thetaPred[i, 1:nResponses
    , 2], nPrdNoi[1])
  }
}
```

5.7.3 Hierarchical multiple-choice signal detection model

Priors

The hierarchical multiple-choice signal detection model has priors for the group means for the discriminability and criteria that follow the non-hierarchical model, including the order constraints on the criteria:

$$\begin{aligned} \mu_g^{d'} &\sim \text{Gaussian}\left(0, \frac{1}{3^2}\right) \\ \mu_{g1}^k, \ldots, \mu_{g(m-1)}^k &\sim \text{Gaussian}\left(0, \frac{1}{3^2}\right) : \mu_{g1}^k \leq \ldots \leq \mu_{g(m-1)}^k. \end{aligned}$$

The group standard deviations are

$$\begin{aligned} \sigma_g^{d'} &\sim \text{Gaussian}\big(0, \frac{1}{3^2}\big) \\ \sigma_{g1}^{k} \dots \sigma_{g(m-1)}^{k} &\sim \text{Gaussian}\big(0, \frac{1}{3^2}\big). \end{aligned} \tag{5.1}$$

Likelihood

The discriminabilities and criteria for each individual are generated from their group distribution

$$\begin{aligned} d'_i &\sim \text{Gaussian}\big(\mu_{g_i}^{d'}, \frac{1}{\left(\sigma_{g_i}^{d'}\right)^2}\big) \\ k_{ix} &\sim \text{Gaussian}\big(\mu_{g_i x}^{k}, \frac{1}{\left(\sigma_{g_i}^{k} x\right)^2}\big). \end{aligned} \tag{5.2}$$

The representative discriminability and bias for each group is generated in exactly the same way:

$$\begin{aligned} \tilde{d}'_g &\sim \text{Gaussian}\big(\mu_g^{d'}, \frac{1}{\left(\sigma_g^{d'}\right)^2}\big) \\ \tilde{k}_{gx} &\sim \text{Gaussian}\big(\mu_{gx}^{k}, \frac{1}{\left(\sigma_{gx}^{k}\right)^2}\big). \end{aligned} \tag{5.3}$$

The process that generates the posterior predicted hit and false alarm counts for each group is based on the representative discriminability and bias for that group, and is generated with respect to the specified number of signal and noise trials

$$\begin{aligned} \hat{k}_g^{\text{h}} &\sim \text{binomial}\big(\tilde{\theta}^{\text{h}}, \hat{n}^s\big) \\ \hat{k}_g^{\text{f}} &\sim \text{binomial}\big(\tilde{\theta}^{\text{f}}, \hat{n}^n\big). \end{aligned} \tag{5.4}$$

The probability of each response is the same as for the non-hierarchical model, based on areas under the signal and noise distributions between adjacent criteria. Equations 5.1 to 5.1 provide the probabilities for each individual on each trial. The likelihood observed responses y_t also are the same as for the non-hierarchical model, and are given by Equation 5.1.

The predicted signal and noise responses for the representative individual from group g are generated the same way

$$\begin{aligned} \hat{\boldsymbol{y}}_g^{\text{s}} &\sim \text{categorical}\big(\theta_{g1}^{s}, \theta_{g2}^{s}, \dots \theta_{gm}^{s}\big) \\ \hat{\boldsymbol{y}}_g^{\text{n}} &\sim \text{categorical}\big(\theta_{g1}^{n}, \theta_{g2}^{n}, \dots \theta_{gm}^{n}\big). \end{aligned} \tag{5.5}$$

5.7.4 JASP implementation

The JAGS code is shown below.

```
# hierarchical multiple choice SDT
data{
  nTrials = length(individual)
```

```
  nIndividuals = max(individual)
  nResponses = max(response)
  nCriteria = max(response)-1
  nGroups = max(group)
  lo = 0.01
}
model{
  for (t in 1:nTrials){
    response[t] ~ dcat(theta[t, 1:nResponses, truth[t]])
    # signal
    theta[t, 1, 1] = max(lo, phi(k[individual[t], 1] - dPrime[
    individual[t]]))
    for (j in 2:(nResponses-1)){
      theta[t, j, 1] = max(lo, phi(k[individual[t], j] - dPrime[
    individual[t]]) - phi(k[individual[t], j-1] - dPrime[
    individual[t]]))
    }
    theta[t, nResponses, 1] = 1 - max(lo, phi(k[individual[t],
    nCriteria] - dPrime[individual[t]]))
    # noise
    theta[t, 1, 2] = max(lo, phi(k[individual[t], 1]))
    for (j in 2:(nResponses-1)){
      theta[t, j, 2] = max(lo, phi(k[individual[t], j]) - phi(k[
    individual[t], j-1]))
    }
    theta[t, nResponses, 2] = 1 - max(lo, phi(k[individual[t],
    nCriteria]))
  }
  # Individual parameters
  for (i in 1:nIndividuals){
    dPrime[i] ~ dnorm(muD[group[i]], 1/sigmaD[group[i]]^2)
    for (j in 1:nCriteria){
      kTmp[i, j] ~ dnorm(muK[group[i], j], 1/sigmaK[group[i], j
    ]^2)
    }
    k[i, 1:nCriteria] = sort(kTmp[i, 1:nCriteria])
    c[i, 1:nCriteria] = k[i, 1:nCriteria] - dPrime[i]/2
  }
  # Group priors
  for (g in 1:nGroups){
    muD[g] ~ dnorm(0, 1/3^2)
    sigmaD[g] ~ dnorm(0, 1)T(0, )
    for (j in 1:nCriteria){
      muKtmp[g, j] ~ dnorm(0, 1/3^2)
      sigmaK[g, j] ~ dnorm(0, 1)T(0, )
    }
    muK[g, 1:nCriteria] = sort(muKtmp[g, 1:nCriteria])
    for ( j in 1:nCriteria){
      muC[g, j] = muK[g, j] - muD[g]/2
    }
  }
  # Representative
  for (g in 1:nGroups){
    dPrimeRep[g] ~ dnorm(muD[g], 1/sigmaD[g]^2)
    for (j in 1:nCriteria){
      kRepTmp[g, j] ~ dnorm(muK[g, j], 1/sigmaK[g, j]^2)
    }
    kRep[g, 1:nCriteria] = sort(kRepTmp[g, 1:nCriteria])
```

```
    cRep[g, 1:nCriteria] = kRep[g, 1:nCriteria] - dPrimeRep[g]/2
    thetaRep[g, 1, 1] = max(lo, phi(kRep[g, 1] - dPrimeRep[g]))
    for (j in 2:(nResponses-1)){
      thetaRep[g, j, 1] = max(lo, phi(kRep[g, j] - dPrimeRep[g]) -
     phi(kRep[g, j-1] - dPrimeRep[g]))
    }
    thetaRep[g, nResponses, 1] = 1 - max(lo, phi(kRep[g, nCriteria
    ] - dPrimeRep[g]))
    thetaRep[g, 1, 2] = max(lo, phi(kRep[g, 1]))
    for (j in 2:(nResponses-1)){
      thetaRep[g, j, 2] = max(lo, phi(kRep[g, j]) - phi(kRep[g, j
    -1]))
    }
    thetaRep[g, nResponses, 2] = 1 - max(lo, phi(kRep[g, nCriteria
    ]))
    respSigRep[g, 1:nResponses] ~ dmulti(thetaRep[g, 1:nResponses,
     1], nPrdSig[1])
    respNoiRep[g, 1:nResponses] ~ dmulti(thetaRep[g, 1:nResponses,
     2], nPrdNoi[1])
  }
}
```

Rasch Model 6

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6.1 The Rasch Model

6.1.1 Motivation

The Rasch model is the simplest form of a more general class of psychometric models known as item response theory models (Wilson et al., 2008). These models are motivated by the problem of measuring people's ability on the basis of their performance in evaluative tests.

As a a concrete example, Figure 6.1 shows the grading of a multiple choice exam. A total of 101 students completed the exam, which consisted of 40 compulsory four-alternative questions. Each column represents a student, each column represents a question, and a blue square means the person answered the question correctly.

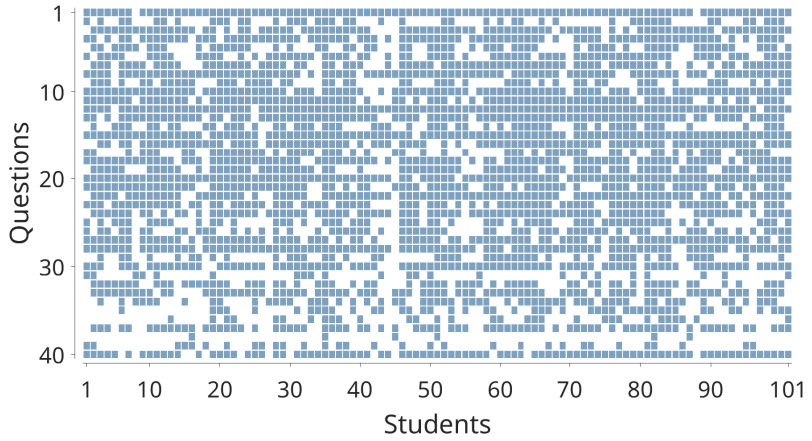

Figure 6.1: The performance of 101 students on an exam with 40 questions.

It is clear from Figure 6.1 that some students performed better on the exam than others. For example, student 81 answered almost all of the questions correctly, but student 1 in the first column made many more incorrect answers. It is also clear that the students found some questions more difficult than others. For example, the first question in the top row was answered correctly by most of the students, but question 38 near the bottom row was rarely answered correctly.

The intuition underlying the Rasch model is that the difference in question difficulties has implications for assessing student performance. If two students both have 37 correct answers, but one of them made mistakes for three easy questions, while the other made mistakes for three hard questions, it may be that their abilities are not the same. Perhaps the first student just made trivial errors on the easy questions, and understands the material well, but the second student has failed to understand some of the more difficult material being assessed. Other interpretations may be reasonable too: the point is that the total correct is not necessarily the complete story, and question difficulty matters.

Considerations of question difficulty become even more important if different students complete different sets of questions. For example, if one version of an exam contains easier questions, then a score of 37 does not indicate as much ability as the same score on a more difficult version of the exam. The Rasch model addresses this issue by making simultaneous inferences about how both a person's ability and the difficulty of the question interact to determine the likelihood of a correct answer.

6.1.2 Model

The Rasch model assumes that each person has an latent *ability* and each question has a latent *difficulty*. The ability of the person p is α_p and the difficulty of the question q is β_q. When a person answers a question, the ability of the person and the difficulty of the question combine to determine the probability they will answer the question correctly. The observed data take the form $y_{pq} = 1$ is the person p answered the question q correctly, and $y_{pq} = 0$ if they did not.

In principle the "person" could be some sort of individual decision source that is not a person: an artificially intelligent agent, a group of people, or something else. For most of the other models in this book, we have used the term "individual" to capture the generality of potential decision sources. But terms like "person ability" are so deeply embedded in the psychometric literature that we refer to the individuals as "people" for the Rasch model.

Dominance and response probability

The Rasch model assumes that the probability person p answers question q correctly depends on the difference between the person's ability and the question's difficulty. This difference $\alpha_p - \beta_q$ is called the *dominance*
.

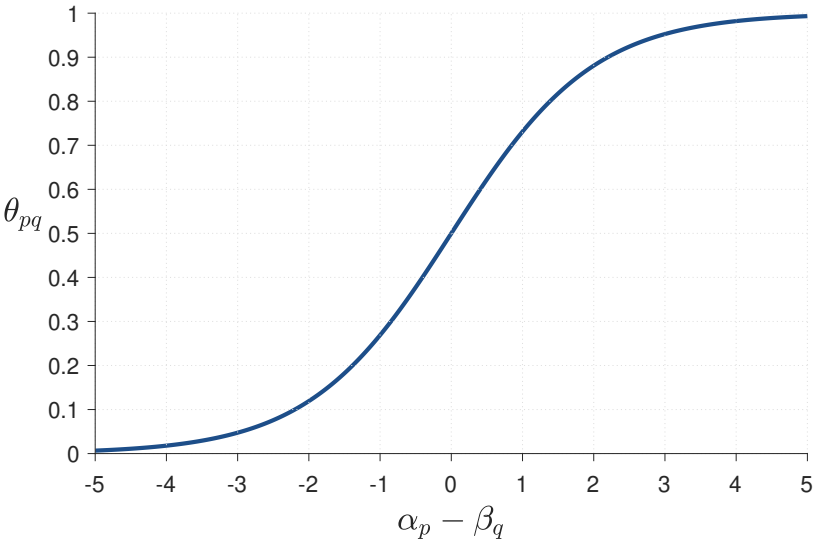

Figure 6.2: The logistic link function relating the dominance to the probability of a correct response in the Rasch model.

A logistic link function is used to translate the dominance to a probability θ_{pq} that the person answers the question correctly. This link function is the curve shown in Figure 6.2. If the difference is zero—that is, if the

person's ability is the same as the question's difficulty—the probability of a correct answer is 0.5. The more a person's ability exceeds a question's difficulty, making the dominance a larger positive number, the greater the probability of a correct answer. For example, if the dominance is +3 the probability of a correct answer is about 95%. In the other direction, the more a question's difficulty exceeds a person's ability, making the dominance a more negative number, the smaller the probability of a correct answer. For example, is the dominance is −1, the probability of a correct answer is about 30%.

Guttman ideal assumptions

The Rasch model assumes that person abilities and question difficulties are fixed numbers that lie on the same scale, and are directly comparable. This means that the ability of a person is the same for all questions, and the difficulty of a question is the same for all people. This situation is shown in Figure 6.3. The abilities of four people $\alpha_1, \ldots, \alpha_4$ and the difficulties of three questions $\beta_1, \ldots, \beta_3$ are shown. The probability a person answers a question correctly depends only on the difference between the corresponding α and β_q values.

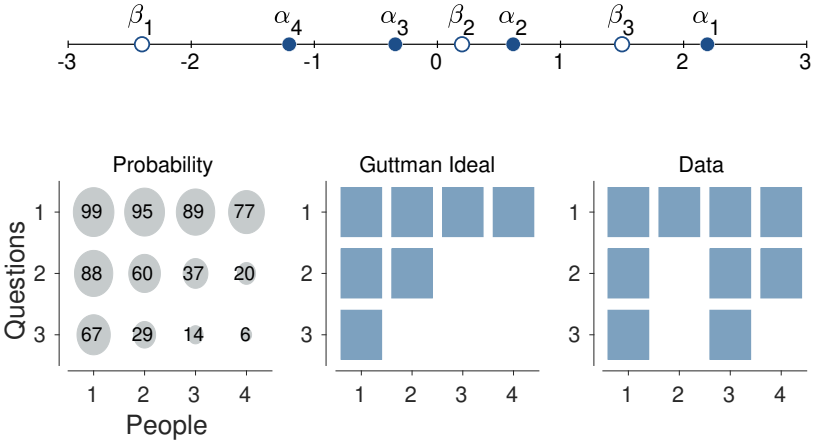

Figure 6.3: The consequences of the Guttman ideal assumptions in the Rasch model.

One consequence of these assumptions is that, because person 1 has more ability than person 2—α_1 is greater than α_2–they have a greater probability of answering each of the questions correctly. Applying this consequence to all of the people and questions leads to a constraint on the probabilities of correct answers known as the *Guttman ideal*. It is demonstrated in the bottom-left panel of Figure 6.3, in which the people are ordered from highest to lowest ability from left to right, and the questions are ordered from easiest to hardest from top to bottom. The probability of a correct response has to be maximal at the top left, where the person with the highest ability (person 1) answers the easiest question (question 1). The probabilities of correct responses are then constrained to decrease moving to the right or downwards, as people with lower ability answer questions having greater difficulty. In terms of observed responses, the Guttman ideal is shown in the bottom-center panel. There is a boundary that separates a solid block of correct responses from the incorrect responses. The constraint, however, applies to the underlying probabilities, not the observed responses. This means that response

patterns like those shown in the bottom-right panel are possible, even when the ideal is met.

The reason the Guttman ideal matters is that it impacts what sort of data are appropriately analyzed by the Rasch model. The assumptions of fixed and commensurable abilities and difficulties means that the model is not naturally applied to situations in which questions span different topics of expertise, or people's abilities vary across the topics covered by questions. If one person is good at history questions and another is good at geography questions, that could lead to a situation in which person 1 is better at question 1 but person 2 is better at question 2, violating the Guttman ideal. Fundamentally, because the Rasch model uses a single representational dimension to measure ability and difficulty, it assumes that there is a single type of knowledge or other cognitive capability being assessed.

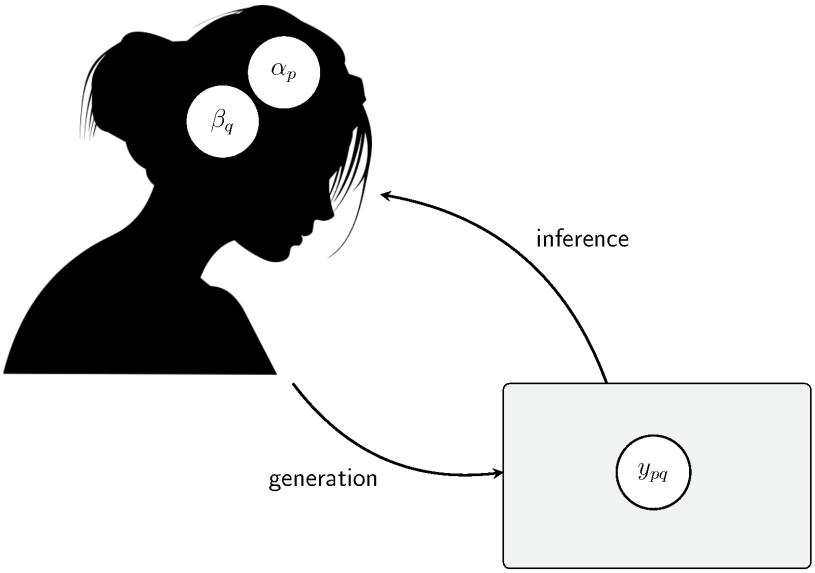

Figure 6.4: The structure of the Rasch model.

Figure 6.4 summarizes the latent parameters and observed data for the Rasch model. The latent parameters are the person abilities α_p for the person p and the question difficulties β_q for the question q. The observed data are the correct and incorrect responses y_{pq}.

6.1.3 Mechanics

Table 6.1 details the meaning of the parameters and variables in the Rasch model, their name in the model as shown in Figure 6.4, and their name in JASP.

Name	Model	JASP
person ability	α	`alpha[p]`
question difficulty	β_q	`beta[q]`
accuracy of observed answer	y_{pq}	`y[p, q]`
probability of a correct answer	θ	`theta[p, q]`

Table 6.1: The meaning of the parameters and variables in the Rasch model, with their model and JASP names.

Input data

Table 6.2 shows the general data format using some of the data from the *Class Exam* example.

person	question	correct
1	1	1
1	2	1
1	3	1
1	4	1
...	...	...
2	1	1
2	2	1
2	3	1
2	4	1
...	...	...

Table 6.2: JASP data format for Rasch model.

Note that a *long* rather than *wide* data format is used. This is because the JASP data structure needs to be defined with a fixed number of columns. A wide format would have different numbers of columns depending on the number of questions. A long data format always uses the same three columns, with each row corresponding to the observed accuracy of a person answering a question.

The three columns are

- `person`: the person who answered the question
- `question`: the question that was answered
- `correct`: the value 1 if the questions was answered correctly and 0 if it was answered incorrectly

The identifying numbers for people and questions are required to start at 1 and be numbered consecutively. There is no requirement that the rows be entered in any specific order. There is also no requirement that every person is observed to answer every question. If there are missing data, that combination of person and question should not be included as a row in the data.

Output analyses

The JASP implementation provides three analyses.

- **person ability and question difficulty posteriors**: the posterior distributions of the person abilities α_p and question difficulties β_q
- **posterior predictions**: the posterior means of probabilities θ_{pq} for every person answering every question correctly

6.2 *Class Exam*

`classExam.jasp`

A first application of the Rasch model considers the class exam data presented in Figure 6.1, in which 101 students answered 40 questions.

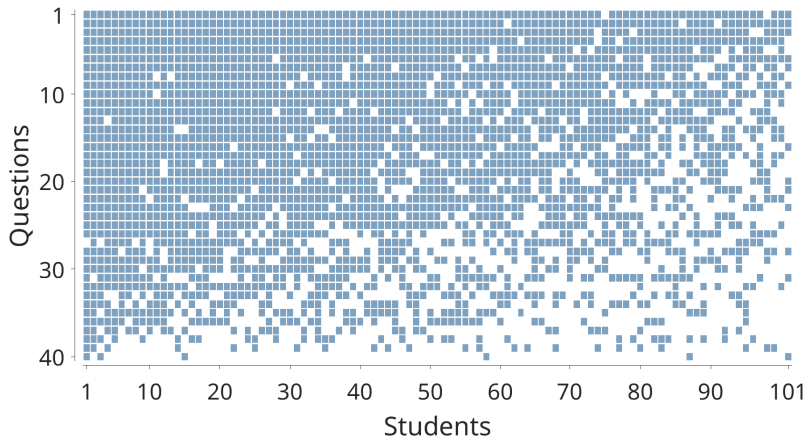

Figure 6.5: The performance of 101 students on an exam with 40 questions.

Data

Figure 6.5 shows the same data as Figure 6.1, but with a different organization. In Figure 6.5, students are ordered in terms of decreasing numbers of correct answers from left to right, and questions are ordered in terms of increasing numbers of correct answers from top to bottom.

Structured this way, it is possible to assess visually whether the data are roughly consistent with the Guttman ideal. The presence of predominantly correct answers in the top left and predominantly incorrect answers in the bottom right provide some evidence of consistency.

Results

Question difficulties

Figure 6.6 summarizes the marginal posterior distributions of the question difficulties abilities. Because the questions have been ordered in terms of the number of students who answered them correctly, it is not surprising that their inferred difficulties increase. One notable feature of the results is that question 40 is very likely to be something of an outlier, in the sense that it is probably much more difficult than all of the other questions.

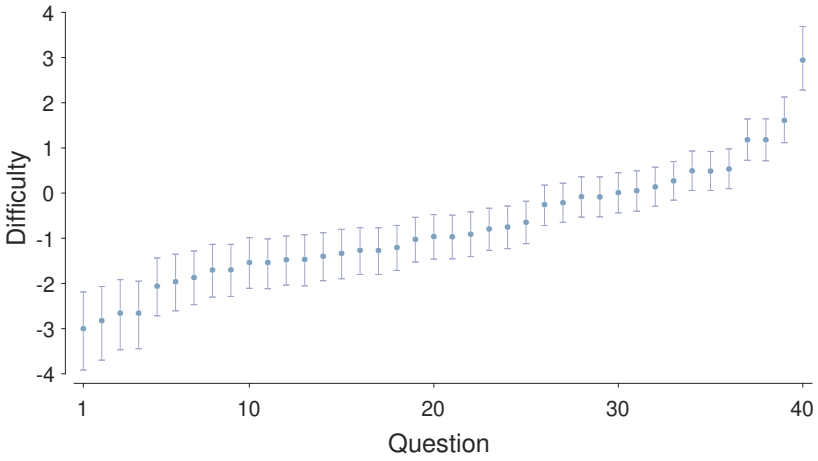

Figure 6.6: The posterior means and 95% credible intervals of the posterior distributions of the question difficulties β_q for the class exam data.

Person abilities

Figure 6.7 summarizes the marginal posterior distributions of the student abilities. The circular markers show the posterior mean and the error bars show the 95% credible interval for each student. Because the students have been ordered in terms of the number of questions they answered correctly, it is not surprising that their inferred abilities decrease. In fact, the inferred abilities track the total number of correct answers for each student. For example, student 1 got 39 correct answers, student 2 got 28 correct answers, students 3, 4, and 5 got 36 correct answers and so on. Even though students 3, 4, and 5 got the same total by getting slightly different sets of questions correct and incorrect, Figure 6.7 shows that their abilities are very similar.

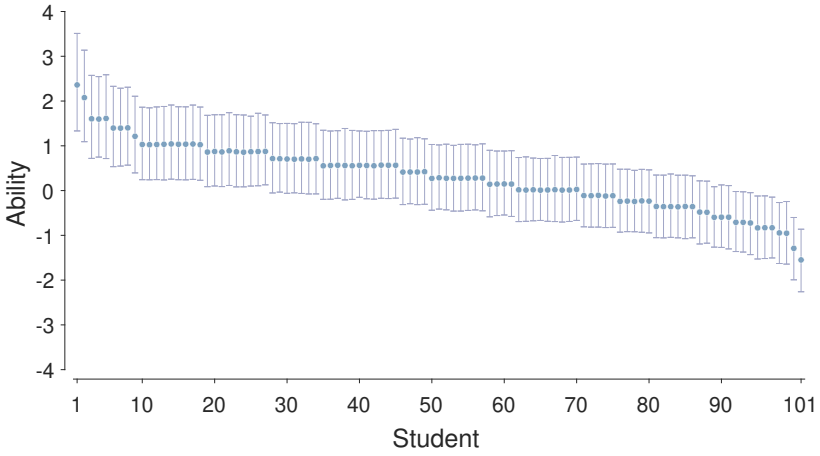

Figure 6.7: The posterior means and 95% credible intervals of the posterior distributions of the person abilities α for the class exam data.

6.3 *Risk Perception*

`riskPerception.jasp`

This example involves people's subjective assessment of risk, using the Domain-Specific Risk-Taking (Adult) scale (DOSPERT: Blais & Weber, 2006). The DOSPERT is a psychometric test for measuring people's risk taking propensities and their perception of risk. It involves 30 statements that describe different situations, such as "going camping in the wilderness" and "piloting a small plane," that potentially involve risk, and asks people to make personal judgments about the level of risk for each. Because there are no correct answers to these sorts of judgments, this example is different from the class exam example, and shows how opinions or preferences can be inferred by the Rasch model.

Data

The data involve the the responses of 207 people to the risk perception part of the DOSPERT, in which people are asked "For each of the following statements, please indicate how risky you perceive each situation," and respond on a seven-point Likert scale that ranges from "not at all risky" to "extremely risky."

Figure 6.8 shows their thresholded responses, with blue lines representing responses of 5, 6, or 7 ("risky", "very risky", and "extremely risky") that

correspond to a belief there is risk involved. The people are ordered from left to right in terms of an increasing number of risky responses. The situations are ordered from top to bottom in terms of increasing numbers of people perceiving risk.

Figure 6.8: The responses of 207 people for the DOSPERT risk perception questions. Blue lines indicate that a person perceived the activity to involve risk.

The pattern of responses in Figure 6.8 provide some evidence of consistency with the Guttman ideal, which means that the Rasch model can reasonably be applied. In this context, what is usually called the ability of each person is better conceived as their risk propensity, and the difficulty of each situation is better conceived as its inherent appearance of riskiness.

Results

Situation risk appearances

Figure 6.9 summarizes the marginal posterior distributions of the appearance of risk β_q for the situations. It is clear that the inferred appearance of risk increases from top to bottom as the situation is perceived by more people to involve risk. It may seems counter-intuitive that the β_q values decrease as the appearance of risk increases. One way to think about it is that lower values of β_q correspond to easier questions that are more

likely to be answered correctly which, in this context, means situations for which it is easier for people to respond that they perceive risk.

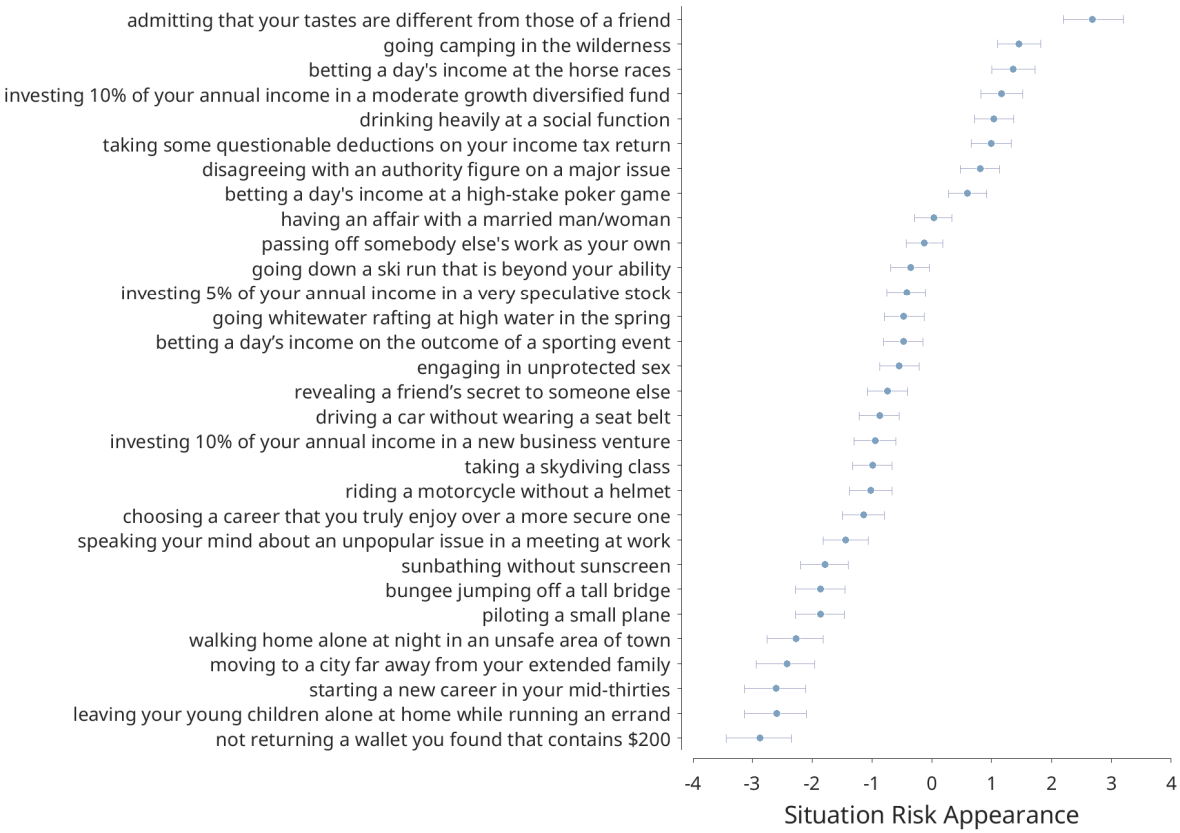

Figure 6.9: The posterior means and 95% credible intervals of the posterior distributions of the situation risk appearances β_q for the risk perception data.

Person risk propensities

Figure 6.10 summarizes the marginal posterior distributions of people's risk propensities. The circular markers show the posterior mean for α_p and the error bars show the 95% credible interval. As expected, the risk propensities increase from left to right, as the number of situations people perceived as risky increases.

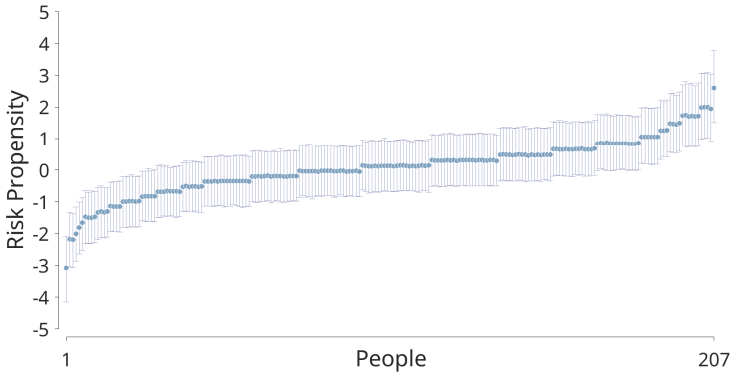

Figure 6.10: The posterior means and 95% credible intervals of the posterior distributions of the risk propensities α_p for the risk perception data.

6.4 *Cricket Santa*

`cricketSanta.jasp`

This example involves naturally occurring data from a social media quiz. In the quiz, people to identify twelve cricket bowlers, based on line drawings of their bowling actions. Figure 6.11 shows the line drawings. Why the bowlers are dressed as Santa Claus is unclear.

Figure 6.11: Drawings of twelve bowlers presented in the online "cricket Santa" quiz.

Data

A number of people answered the quiz in the comments section of the social media post, writing a name for one or more of the 12 cricketers. Figure 6.12 shows the performance of 127 people collated by hand from the social media site. A green rectangle indicates the person provided the correct answer, a red rectangle indicates the person provided an incorrect answer, and a blank space indicates the person did not attempt an answer.

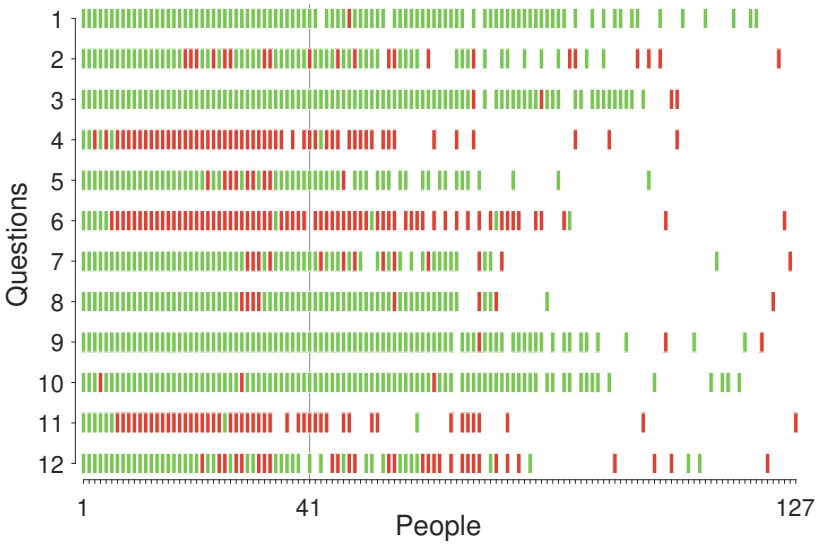

Figure 6.12: The responses of 127 people to the cricket Santa quiz.

The questions are ordered from top to bottom in the same order as the quiz image shown in Figure 6.11. The people are ordered from left to right first in terms of the number of answers they provided and secondly in terms of their number of correct answers. This means that all people who answered all 12 questions are shown first, all people who answered

11 of the 12 questions are shown second, and so on, and that, for example, among the people who answered all 12 questions, those who answered the most correctly are shown first.

The behavioral data in Figure 6.12 suggests that the questions have different levels of difficulty. Cricket fans likely appreciate this from looking at Figure 6.11. For example, player 1 is pretty clearly Jasprit Bumrah, but player 6 is less obvious (Waqar Younis, Jeff Thomson, . . . ?). In terms of the behavioral data in Figure 6.12, questions 4, 6, and 11 seem to be difficult, with many incorrect answers. Questions 1, 3, 9, and 10, on the other hand, appear to be relatively easy.

Since different people answered different questions, it seems intuitively important that the difficulty of the questions they answered affects any assessment of the people's abilities to identify the cricketers. This is the sort of challenge the Rasch model is especially well suited for.

Question difficulties

Figure 6.13 shows the marginal posterior distributions for the 12 question as vertical violin plots. The width of the violin corresponds to the marginal density of β_q at each difficulty value on the latent scale. The inferences match intutions from Figure 6.12, with questions 4, 6, and 11 having greater difficulty than questions 1, 3, 9, and 10.

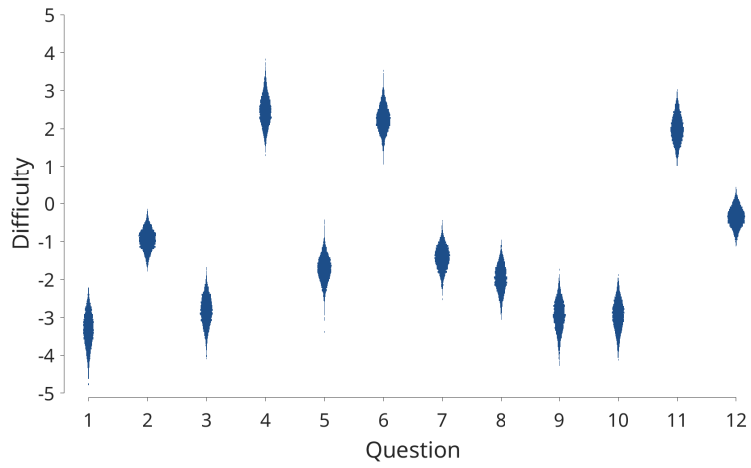

Figure 6.13: The posterior means and 95% credible intervals of the posterior distributions of the question difficulties β_q for the cricket Santa data.

Person abilities

Figure 6.14 summarizes the marginal posterior distributions of the person abilities. The circular markers show the posterior mean and the error bars show the 95% credible intervals for the α ability parameters.

The ability inferences in Figure 6.14are sensitive to—in fact, are inferred jointly with—the question difficulties shown in Figure 6.13. One concrete example of this co-dependence is provided by the last six people, from person 122 to person 127. As the behavioral data in Figure 6.12, each of these people attempted exactly one question, but answered it incorrectly. Because they only answered one question, the inferences about their abilities has large uncertainty. But, there are small variations in the posterior means that reflect the difficulty of the single question they

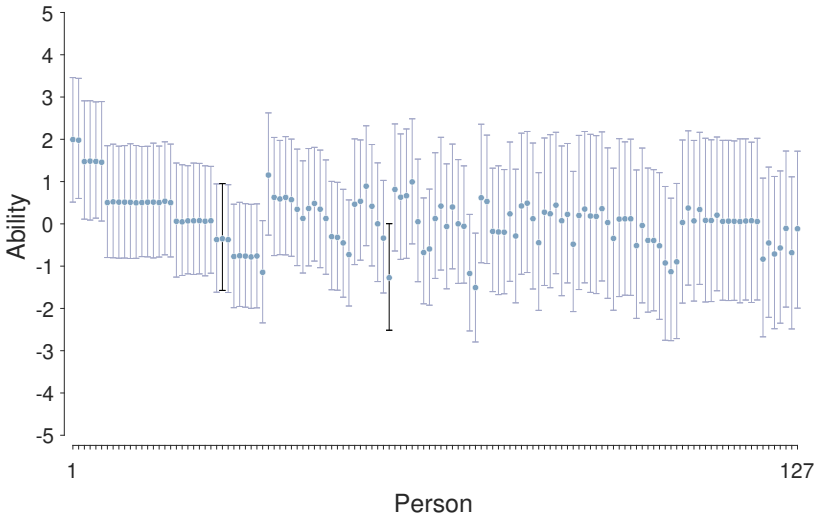

Figure 6.14: The posterior means and 95% credible intervals of the posterior distributions of the person abilities α for the cricket Santa data.

got wrong. For example, person 127 got question 11 wrong, which is a difficult question, while person 122 got the easier question 9 wrong. Accordingly, the posterior mean ability for person 127 is larger than that for person 122.

Another concrete example relates to the highlighted person 41 and person 42 immediately following. Figure 6.12 shows that both people answered 11 questions with 3 incorrect responses. Person 42 made mistakes on questions 4, 6, and 11, which are the most difficult questions, and did not answer question 12. Person 41 also made mistakes on questions 4 and 11, but did not attempt to answer the difficult question 6, and made a mistake with the easy question 2. These patterns suggest that person 42 has greater ability, despite their identical correct and incorrect counts. The posterior distributions represent this possible difference.

Posterior predictions

The Rasch model makes posterior predictions for the probability that every person answers every question correctly. For the 925 answers actually provided, these posterior predictions provide a way to assess the descriptive adequacy of the model. For the remaining 599 person-question combinations that were not answered, the posterior predictions provide an expectation about whether they would have been answered accurately.

Figure 6.15 provides a visualization of this posterior predictive analysis. The background colors represent the posterior predictive mean of the probability each person answers each question correctly. Green colors represent high probabilities of correct decisions and red colors represent high probabilities of incorrect decisions. The overlaid rectangles in Figure 6.15 show the observed data, in the same way as Figure 6.12. There are observed data that the model does not describe well. For example, the fourth person answered question 10 incorrectly, but the posterior predictive expectation is a high probability of a correct answer. For the most part, however, the model describes the data well.

The posterior predictive expectations use the Rasch model to generalize beyond the observed data. There are some green and red horizontal

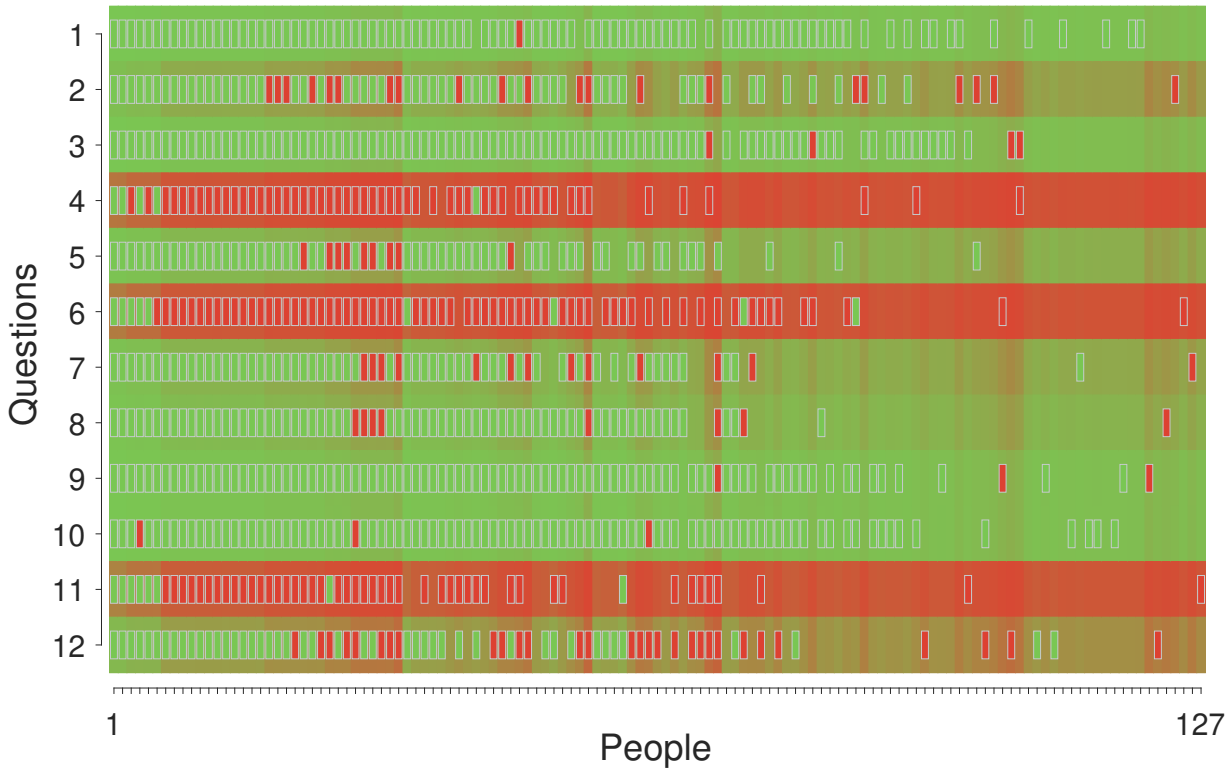

Figure 6.15: The posterior predictive expectations of accuracy for every person-question combination, with the observed accuracy data overlaid.

bands for questions that are especially easy or difficult, corresponding to the expectation that most people will answer these questions correctly or incorrectly. For questions with intermediate difficulty, the bands change from green towards red moving from left to right, corresponding to the expectation that people with higher ability will answer these questions correctly, but people with lower ability will not. For example, although person 127 did not answer question 4, the expectation is that they would have answered it incorrectly.

6.5 Hierarchical Rasch Model

Motivation

The non-hierarchical Rasch model measures person abilities and question difficulties. A hierarchical extension allows for group structure among people or among questions (or both). In the context of the *Class Exam* example in which students completing questions, the students could be grouped in terms of age ranges, and the questions could be grouped in terms of the different content areas they address. The assumption is that each group has its own mean ability or difficulty, allowing the groups to be compared to one another.

The hierarchical Rasch model is related to what are known as *explanatory* item response theory models (Wilson et al., 2008, Table 4). In terms of the nomenclature in this literature, the basic Rasch model is "doubly descriptive" because the parameters simply measure person ability and

question difficulty. They do not attempt to account for why people have different abilities, or why questions have different difficulties. Models that are "person explanatory" incorporate observed information about people—such as their age group—and model the relationship between this covariate information and their ability. Model that are "item explanatory" adopt the same approach in relation to questions. Models are "doubly explanatory" do both, as well as modeling the interaction. The hierarchical Rasch model we use has a different statistical definition, but the same conceptual goals, as the doubly-descriptive Rasch models. One key difference is that the hierarchical model does not attempt to model interactions.

6.5.1 Model

Figure 6.16 summarizes the structure of the hierarchical Rasch model. The observed variables include the responses y_{pq} and the group membership of each person g_p^α and each question g_q^β. The hierarchical parameters μ_g^α and μ_g^β correspond to the mean ability and mean difficulty of the groups of people and questions. Note that there are no group standard deviations. Although not shown in Figure 6.16, the group distributions are also used to sample representative group abilities $\tilde{\alpha}_g$ and group difficulties $\tilde{\beta}_g$.

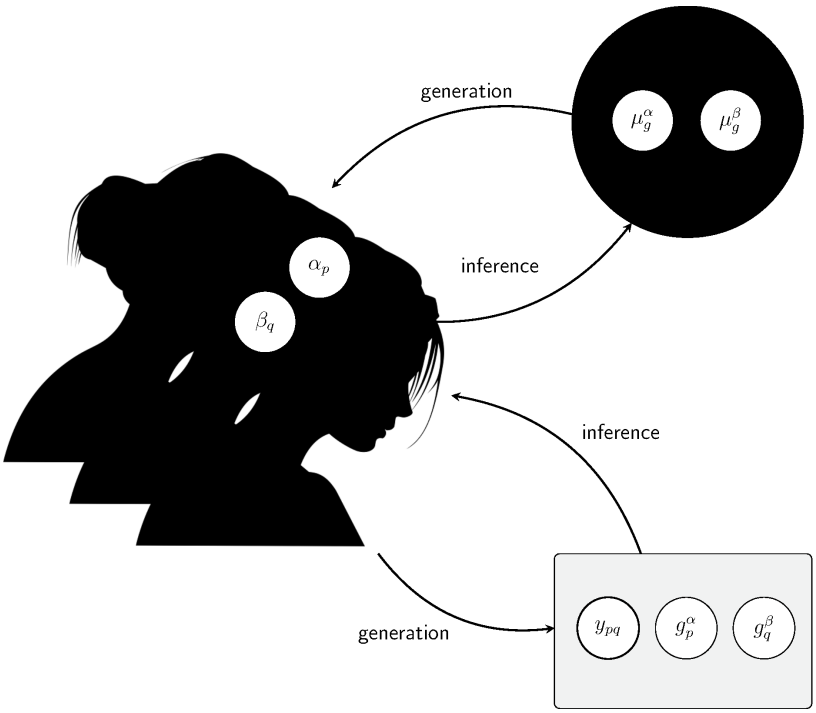

Figure 6.16: The structure of the hierarchical Rasch model.

6.5.2 Mechanics

Table 6.3 details the meaning of the parameters and variables in the explanatory Rasch model, their name in the model as shown in Figure 6.4, and their name in JASP. The new variables are the assignments of people g_p^α and questions g_q^β to groups, the mean person ability μ_g^α and question

difficulty μ_g^β for each group, and the representative person ability $\tilde{\alpha}_g$ and question difficulty $\tilde{\beta}_g$ for each group.

Name	Model	JASP
person ability	α_p	`alpha[p]`
question difficulty	β_q	`beta[q]`
probability of a correct answer	θ_{pq}	`theta[p,q]`
person group	g_p^α	`personGroup[p]`
question group	g_q^β	`questionGroup[q]`
mean group ability	μ_g^α	`muAlpha[g]`
mean question difficulty	μ_g^β	`muBeta[g]`
representative group ability	$\tilde{\alpha}_g$	`alphaRep[g]`
representative question difficulty	$\tilde{\beta}_g$	`betaRep[g]`

Table 6.3: The meaning of the parameters and variables in the explanatory Rasch model, with their model and JASP names.

Input data

Table Table 6.2 shows the general data format using some of the data from the *Critic Content* example.

person	question	correct	personGroup	questionGroup
1	1	0	1	1
1	2	0	1	1
1	5	0	2	1
1	6	0	1	1
...	...	...	...	...
2	7	0	1	3
2	14	0	1	3
2	19	1	2	3
2	24	0	1	3
...	...	...	...	...

Table 6.4: JASP data format for the explanatory Rasch model.

The `person`, `question`, and `correct` data continue to be provided in a long format. The new `personGroup` and `questionGroup` data are provided once for each person and question. Table shows part of an example with six people and four questions. This means that the `personGroup` column has six entries and the `questionGroup` column has four entries. If there is only one person or question group, then all of the people or questions should be assigned to group `1`.

Output analyses

The JASP implementation provides four analyses.

- **individual ability and difficulty posteriors**: the posterior distributions of the person abilities α_p and question difficulties β_q
- **group ability and difficulty**: the marginal and joint posterior distributions for the group means of person ability μ_g^α and question difficulty μ_g^β for each group
- **representative ability and difficulty**: the marginal and joint posterior distributions for the representative person ability $\tilde{\alpha}_g$ and question difficulty $\tilde{\beta}_g$ for each group

- **posterior predictions**: the posterior means of probabilities θ_{pq} for every person answering every question correctly

6.6 *Critic Content*

criticContent.jasp

Rotten tomatoes is a review aggregation website for movies. Both general users and expert movie critics can rate movies as "fresh" or "rotten." This application involves a data set collected from the rotten tomatoes site. It is the same data set as was used for the *Movie Critics* application in Chapter 1.8.

The Rasch model is well suited to understanding the latent tendencies of critics and the latent qualities of movies. Person ability can be interpreted as something like a critic's standards for giving a fresh rating. Question difficulty can be interpreted as something like the inherent quality ("freshness") of the movie.

The hierarchical Rasch model is additionally useful in measuring the ability and difficulty of different groups of critics and movies. This application distinguishes between two groups of critics and three groups of movies. The critics are divided into regular critics and those identified by the Rotten tomatoes site as "top critics." The top critics are chosen by a selection panel to be those who are well established, influential and prolific. The movies are divided by their content classifications that provide guidance on the appropriate reviewing audience. The common "general audience" (G), "parental guidance recommended" (PG), and "restricted" (R) are used to create groups.

6.6.1 Data

Figure 6.17 shows the pattern of fresh and rotten ratings for 200 critics and 60 movies. A blue rectangle indicates the critic gave the movie a fresh rating, a red rectangle indicates a rotten rating, and blank space indicates that the critic did not rate the movie. It is clear that the ratings of movies are relatively sparse: most critics do not rate most movies. This makes the ability of the Rasch model especially useful, because it is able to make inferences about critic standards and movie quality with data that have complicated pattern of missingness.

The groups of critic and movies are shown by dividing dashed black lines in Figure 6.17. The first 137 critics are regular critics and the final 63 are top critics. The critics were chosen to be the 200 with the most ratings in the entire data set, although not necessarily those who rated the 60 movies most often. The division between regular and top critics roughly matches the proportion in the entire data sets.

The first 10 movies are rated G, the next 30 are rated PG, and the final 20 are rated R. These proportions also approximately follow the content ratings of the entire data set. The movies were chosen in a deliberately in terms of the proportion of fresh ratings in each group, to make the application illustrative. In particular, as is clear from the pattern of fresh and rotten ratings in Figure 6.17, the G-rated movies were chosen to be

Figure 6.17: Fresh ratings for 137 regular critics (left) and 63 top critics (right) for a set of G rated (top), PG rated (middle) and R rated (bottom) movies. Blue rectangles indicate a fresh rating and red rectangles indicate rotten rating.

disproportionately rotten, and the R-rated movies were chosen to be disproportionately fresh.

6.6.2 Results

Movie qualities

Figure 6.18 shows the marginal posterior distributions for the quality of the 60 movies, presented in the same order as Figure 6.17. The circles correspond to the posterior mean of β_q and the error bars show 95% credible intervals. The circles are colored according to the G, PG, or R content classification of the movie. It is worth emphasizing that larger values of movie quality make it more difficult for a critic to evaluate the movie as fresh, so thinking of the x-axis as "lack of movie quality" may be helpful for interpretation.

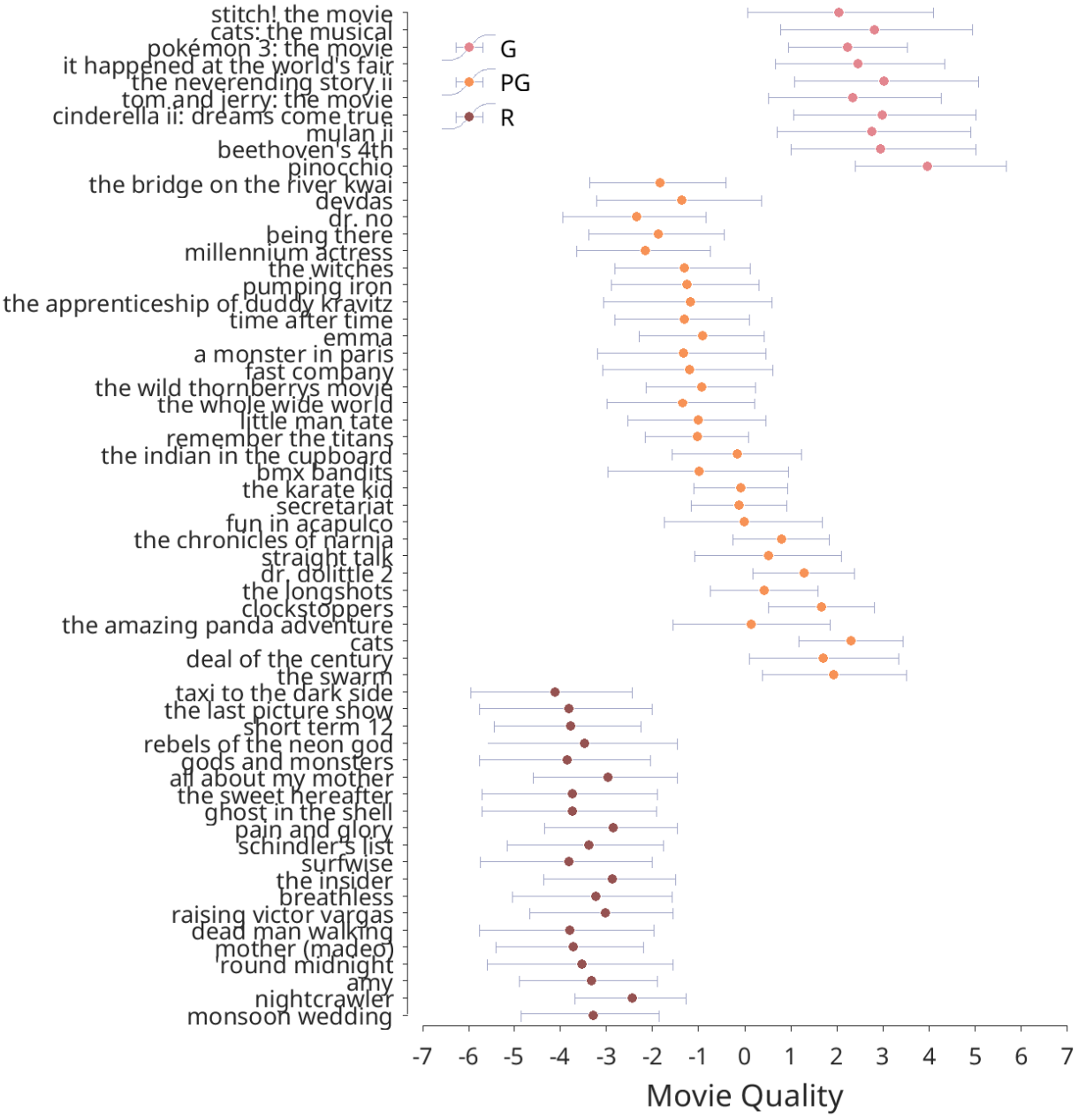

Figure 6.18: The posterior means and 95% credible intervals of the movie qualities β_q for the critic Content data, colored according to content rating.

It is clear that the movie qualities generally cluster into the three content groups. Some of the worst rated PG movies, such as "cats," "deal of the century," and "swarm" have similar quality as the G movies, but the clustering is clear. This is consistent with the way the movies were selected, and the pattern of fresh and rotten ratings shown in Figure 6.17.

The hierarchical structure of the model can be used to make direct inferences about the quality of the G, PG, and R movie groups. Figure 6.19 shows the marginal posterior means for both the group mean qualities μ_g^β and the representative quality $\tilde{\beta}_g$ of an individual movie from each group.

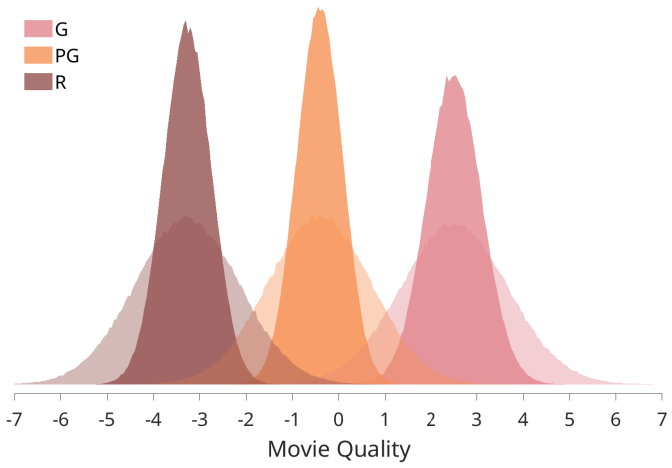

Figure 6.19: The posterior distributions of the movie quality group means (darker shading) and representative individual movie from each group (lighter shading).

The posteriors for the group means show the ordering of average quality increasing from G to PG to R movies. The representative distributions have more uncertainty, consistent with their goal of inferring the distribution of possibility quality for any movie belonging to each of the groups. Taken together, these data analyzed by the hierarchical Rasch model provide evidence for an average group ordering from G to PG to R, but show that this does not guarantee individual movies in these groups will follow that order. The representative distributions show that there is a non-neglible probability that a PG movie could have higher quality than an R one, and that a G movie could have higher quality than a PG one.

Critic standards

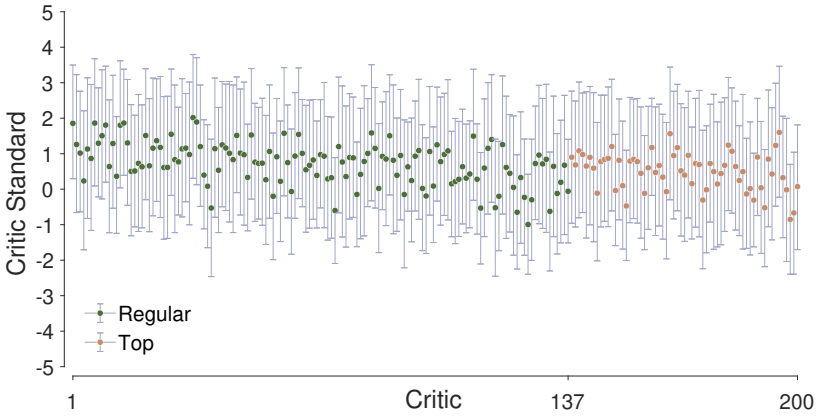

Figure 6.20: The posterior means and 95% credible intervals of the critic standards α_q for the critic Content data, colored according to regular or top critic status.

Figure 6.20 summarizes the marginal posterior distributions for the standards of the 200 critics. The circular markers show the posterior mean of α_p and the error bars show the 95% credible intervals. The circles are colored according to whether the critic is a regular or top critic.

There is significant uncertainty about the standards of all of the critics. This is because of the relatively small number of movies each rated. There is also no obvious difference between the regular and top critics at the individual level.

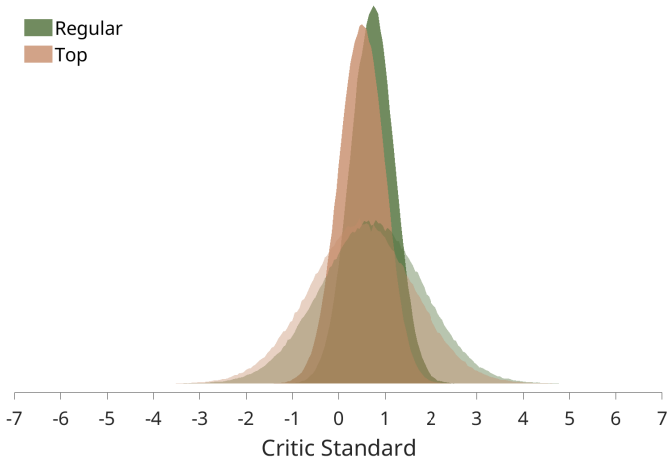

Figure 6.21: The posterior means and 95% credible intervals of the posterior distributions of the movie qualities β_q for the critic Content data. Each movie is colored according to its content rating.

The hierarchical structure of the model can again be used to make direct inferences about the standards of the regular and top critic groups. Figure 6.21 shows the marginal posterior means for both the group mean standards μ_g^α and the representative standard $\tilde{\alpha}_g$ of an individual critic from each group. The mean and representative distributions are both extremely similar for both critic groups, providing evidence that critic status does not have an influence on the standards they require to rate the movies as fresh.

◇

6.7 Formal Details

6.7.1 Rasch model

Priors

The prior for the ability α_p is of person p is a unit Gaussian

$$\alpha_p \sim \text{Gaussian}(0, 1), \tag{6.1}$$

as is the prior for the difficulty β_q of question q

$$\beta_q \sim \text{Gaussian}(0, 1). \tag{6.2}$$

Likelihood

The probability the person p answers question q correctly is modeled by a logistic link function that maps the dominance $\alpha_p - \beta_q$ to the probability θ_{pq}:

$$\theta_{pq} = \frac{1}{1 + \exp\left\{-\left(\alpha_p - \beta_q\right)\right\}}. \tag{6.3}$$

JASP implementation

The JAGS code is shown below. Note that is structured to accommodate the long data format. The constant `nTotal` is the total number of observed responses in the data set. The constants `nPeople` and `nQuestions` are the total number of people and questions.

```
# Rasch model
data{
  nTotal = length(correct)
  nPeople = max(person)
  nQuestions = max(question)
}
model{
  # Likelihood in long format
  for (t in 1:nTotal){
    correct[t] ~ dbern(1/(1 + exp(-(alpha[person[t]] - beta[
    question[t]]))))
  }
  # Priors
  for (p in 1:nPeople){
    alpha[p] ~ dnorm(0, 1)
  }
  for (q in 1:nQuestions){
    beta[q] ~ dnorm(0, 1)
  }
  # Posterior predictions
  for (p in 1:nPeople){
    for (q in 1:nQuestions){
      thetaPrd[p, q] = 1/(1 + exp(-(alpha[p] - beta[q])))
    }
  }
}
```

6.7.2 Hierarchical Rasch model

In the hierarchical model, person ability and question difficulty are sampled from Gaussian distributions with the mean that corresponds to the person's or question's group, as follows:

$$\alpha_p \sim \text{Gaussian}\left(\mu^{\alpha}_{g^{\alpha}_p}, 1\right) \tag{6.4}$$

$$\beta_q \sim \text{Gaussian}\left(\mu^{\beta}_{g^{\beta}_q}, 1\right). \tag{6.5}$$

Representative abilities and difficulties for each group are defined in the same way

$$\tilde{\alpha}_g \sim \text{Gaussian}\big(\mu^{\alpha}_{g^{\alpha}_p}, 1\big) \tag{6.6}$$

$$\tilde{\beta}_g \sim \text{Gaussian}\big(\mu^{\beta}_{g^{\beta}_q}, 1\big). \tag{6.7}$$

The group means for both ability and difficulty have unit Gaussian priors

$$\mu^{\alpha}_g \sim \text{Gaussian}\big(0, 1\big) \tag{6.8}$$

$$\mu^{\beta}_g \sim \text{Gaussian}\big(0, 1\big). \tag{6.9}$$

Note that, unlike many of the hierarchical extensions of other models in this book, the group distributions for person ability and question difficulty are only characterized by their means. Fixing the standard deviation to 1 is done to make the model well identified for computational Bayesian inference. In non-Bayesian inference, the dependence of the probability θ_{pq} on the dominance $\alpha_p - \beta_q$ leads to a lack of identifiability without additional constraints. For example, since adding the same constant to both α_p and β_q does not change the dominance, constraints such as a zero mean across all abilities need to be imposed. The non-hierarchical Rasch model does not need these constraints because the unit-Gaussian priors on ability and difficulty are informative enough to support stable computational inference. That property is preserved in a hierarchical model that allows different mean abilities and difficulties, but is weakened significantly by also allowing different group standard deviations.

6.7.3 JASP Implementation

The JAGS code is shown below

```
# Hierarchical Rasch model
data{
  nPeople = max(person)
  nQuestions = max(question)
  nTotal = length(person)
  nPersonGroups = max(personGroup)
  nQuestionGroups = max(questionGroup)
}
model{
  # Likelihood in long format
  for (t in 1:nTotal){
    correct[t] ~ dbern(1/(1 + exp(-(alpha[person[t]] - beta[
    question[t]]))))
  }
  # Priors
  for (p in 1:nPeople){
    alpha[p] ~ dnorm(muAlpha[personGroup[p]], 1)
  }
  for (g in 1:nPersonGroups){
    muAlpha[g] ~ dnorm(0, 1)
  }
  for (q in 1:nQuestions){
    beta[q] ~ dnorm(muBeta[questionGroup[q]], 1)
```

```
  }
  for (g in 1:nQuestionGroups){
    muBeta[g] ~ dnorm(0, 1)
  }
  # Posterior predictions
  for (p in 1:nPeople){
    for (q in 1:nQuestions){
      thetaPrd[p, q] = 1/(1 + exp(-(alpha[p] - beta[q])))
    }
  }
  # Posterior Representative
  for (g in 1:nPersonGroups){
    alphaRep[g] ~ dnorm(muAlpha[g], 1)
  }
  for (g in 1:nQuestionGroups){
    betaRep[g] ~ dnorm(muBeta[g], 1)
  }
}
```

Thurstone Scaling 7

7.1 Motivation

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The Thurstone scaling model has its origins in the foundational psychophysical model known as the "law" of comparative judgment Thurstone (1927a). This law can be understood as a model of how people make simple perceptual decisions about physical stimuli. Examples include judging which of two lights is brighter, which of two lights is flashed on longer, which of two sounds is louder, or which of two weights is heavier.

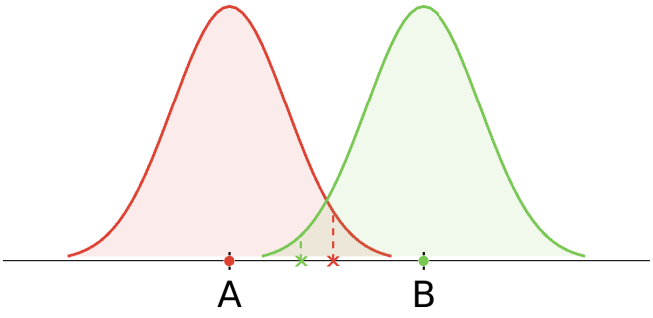

Figure 7.1: The framework for the law of comparative judgment.

The basic assumptions of the law of comparative choice are shown in Figure 7.1, which considers the situation of deciding which of two stimuli is "greater." Both stimuli are represented by values on a psychological scale that represents the brightness, duration, loudness, heaviness, or whatever the perceptual property is. The true value of the red stimulus on this scale is A while the true value of the green stimulus is B, with $B > A$.

According to the law, however, both stimuli have an inherent "discriminal" variability that applies moment to moment. This means that they are not always perceived perfectly by the human senses, nor always in exactly the same way. At each instant in time, different perceptions of their brightness, duration, loudness, heaviness, or other physical properties are possible. These different perceptions fall within the green and red distributions. Examples of specific momentary perceptions are shown by cross markers.

Given these representational assumptions, the law of comparative judgment adds the decision-making assumption that the the probability a person chooses the green stimulus is the probability that its momentary perception is greater than the momentary perception of the red stimulus. In Figure 7.1 it is very likely that the green stimulus will correctly be judged to be greater, but it is not certain. The two specific momentary perceptions shown in Figure 7.1 correspond to a case in which the red stimulus would be judged to be greater.

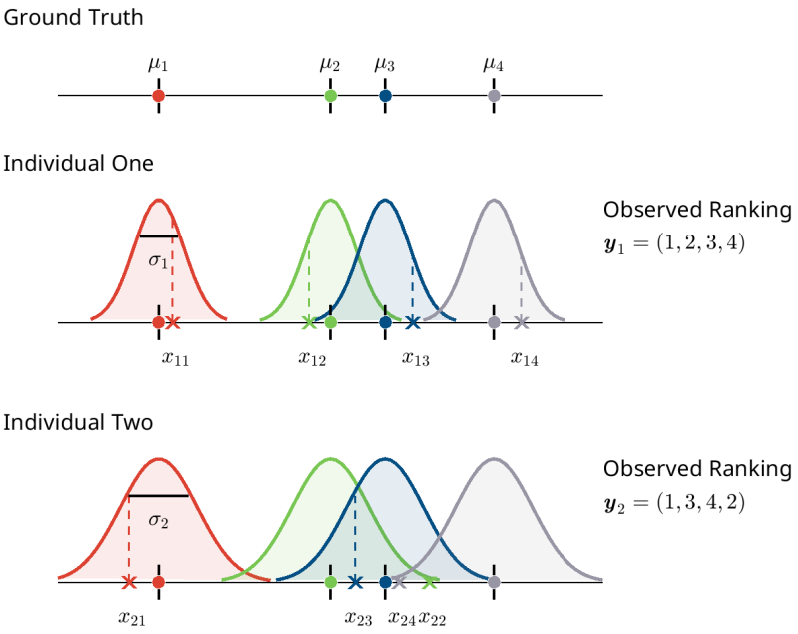

Figure 7.2: Thurstone scaling model for inferring ground truths and expertise from order data.

The Thurstone scaling model extends these representational and decision-making assumptions to multiple stimuli. Instead of choosing between two stimuli, behavioral judgments are made in the form of orders of many stimuli. Accommodating order data was an explicit motivation for the original development of the law of comparative judgment Thurstone (1927a,b). The early reliance on using only choices between pairs of stimuli, which is the most limited form of order data, was because of computational considerations rather than inherent limitations of the model.

7.1.1 Model

The Thurstone scaling model is shown in Figure 7.2. The example shows a situation in which two individuals are order four stimuli. The ground truth locations of the stimuli along the single dimension are represented by a parameter μ_j. These stimulus locations are assumed to be the same for all people, consistent with the idea that there is a ground truth.

Following the law of comparative judgment, when an individual orders stimuli they are assumed to experience mental samples that vary around the true value from moment to moment. The samples come from Gaussian distributions that are centered at the true values, but with a level of variability that is determined by a standard deviation σ_i for the individual i.

The model assumes that the ordering of the mental samples determines the decisions made by each individual. In Figure 7.2 individual 1 has mental samples for red, green, blue, and gray from left to right, and so they provide the order $\boldsymbol{y}_1 = (1, 2, 3, 4)$. Individual 2, in contrast, has mental samples for red, blue, gray, and green, and so they provide the order $\boldsymbol{y}_2 = (1, 3, 4, 2)$.

Individual 1 in Figure 7.2 samples with less variability than individual 2. Their smaller standard deviation means they are more likely to produce

the correct order, because their mental samples will deviate less from the ground truth. For this reason, it is natural to interpret the standard deviation parameter σ_i as a measure of expertise, with smaller values reflecting greater expertise.

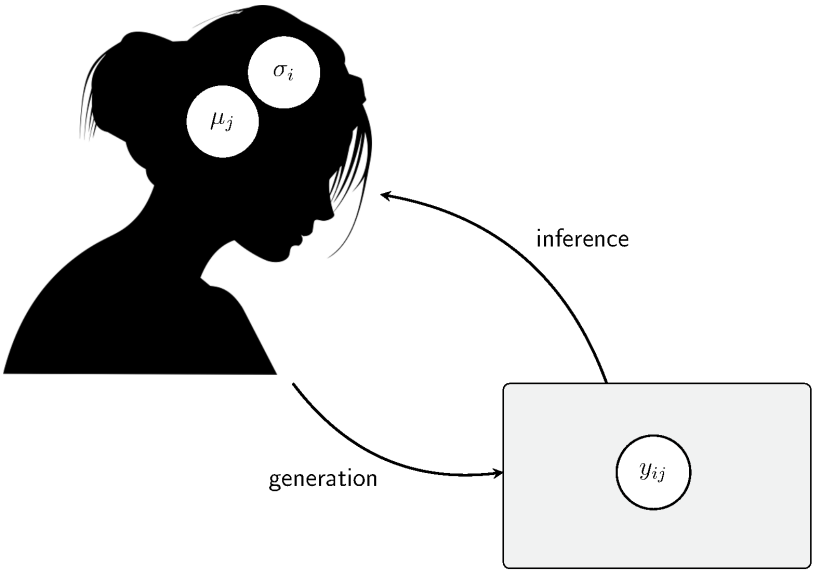

Figure 7.3: The structure of the Thurstone scaling model.

Figure 7.3 summarizes the latent parameters and observed data for the Thurstone scaling model. The latent parameters are the stimulus locations μ_j and individual expertises σ_i. The observed data are the orders $\boldsymbol{y}_i$ provided by the individuals.

It is important to understand that the Thurstone scaling model does not have access to the true values. The only information with which it is provided are the orders provided by individuals. This makes its ability to infer expertise potentially useful. The inference is not based on comparing individuals' order to true orders. Instead, the inference comes from the pattern of agreement among individuals' orders. Intuitively, if a subset of individuals all provide similar orders, that is very unlikely to be by chance, because there are so many possible orders. The model implicitly uses agreement between individuals as a sign of their expertise, and gives the orders of more expert individuals more weight in inferring the aggregate group order.

The lack of requiring a true values also means that the Thurstone scaling model can operate in practical situations in which the ground truth is yet not known, such as when people are asked to make predictions, or for situations in which there is no objective ground truth, such as the expression of socially or culturally determined preferences.

7.1.2 Mechanics

Table 7.1 details the meaning of the parameters and variables in the Thurstone scaling model, their name in the model as shown in Figure 7.3, and their name in JASP.

Name	Model	JASP
item location	μ_j	`mu[j]`
individual expertise	σ_i	`sigma[i]`
ranked position of item	y_{pq}	`y[p, q]`
item location comparison	π_{jk}	`pi[j, k]`

Table 7.1: The meaning of the parameters and variables in the Thurstone scaling model, with their model and JASP names.

Input data

The data are represented in JASP in a long format, with each row specifying the position that a person placed an item. As a concrete example, consider the orders of the two individuals in Figure 7.2. Individual 1 provides the order red, green, blue, gray. Individual 2 provides the order red, blue, gray, green.

Assuming the colors are coded according to the true values, so that `1=red`, `2=green`, `3=blue`, and `4=gray`, this leads to the observed orders $\boldsymbol{y}_1 = (1, 2, 3, 4)$ and $\boldsymbol{y}_2 = (1, 3, 4, 2)$ shown in Figure 7.2. The JASP data format, however, does not use these orders directly. Instead, it uses the *rank* form of the same information.* The rank form asks what position each item was placed in, not the order of the items. Formally, the ith element of a ranking gives the position of item i in the order. It does *not* give the number of the item that occurs in the ith position.

Concretely, for individual 1, the first item (red) was placed in the first position, the second item (green) was placed in the second position, and so on. So, in this case, the rank representation is $\boldsymbol{p}_1 = (1, 2, 3, 4)$, which happens to match the order representation. For individual 2, the first item (red) was placed in the first position, the second item (green) was placed in the fourth position, the third item (blue) was placed in the second position, and the fourth item (gray) was placed in the third position. So the rank representation is $\boldsymbol{p}_2 = (1, 4, 2, 3)$. This is different from the order representation $\boldsymbol{y}_2 = (1, 3, 4, 2)$ shown in Figure 7.2.

The long format for the rank representations is shown in Table **??**. The first row says that individual 1 ranked the first item in the first position. The last row says that individual 2 ranked the fourth item in the second position.

person	item	ranked
1	1	1
1	2	2
1	3	3
1	4	4
2	1	1
2	2	3
2	3	4
2	4	2

Table 7.2: Data format for the Thurstone scaling model.

It is important to provide the rows in the sequence shown in Table 7.2, with all of the items for the individual 1 listed first, then all the items for the individual 2, and so on. It is also necessary to arrange the items

* The difference between the *order* and *rank* representation is conceptually simple, but notoriously difficult to keep straight. It is well explained here.

consecutively within each individual, starting from the first item and finishing with the last one.

Output analyses

The JASP implementation provides three analyses:

- **item locations**: the marginal posterior distributions of the item locations μ_j
- **person expertise**: the marginal posterior distributions of the person expertises σ_i
- **item location comparisons**: the comparison probabilities π^{β}_{jk} that the location of item j is to the left of (corresponds to a higher rank than item k

7.2 *US States*

`usStates.jasp`

Lee et al. (2014) studied the wisdom of the crowd (Surowiecki, 2004) in various general knowledge and prediction using the Thurstone scaling model.

Data

The data reported by Lee et al. (2014) involve 78 participants ordering 10 US states geographically from the west to the east. The states were chosen so that there was no ambiguity arising from overlaps between the states. The states in their correct order are Oregon, Utah, Nebraska, Iowa, Alabama, Ohio, Virginia, Delaware, Connecticut, and Maine.

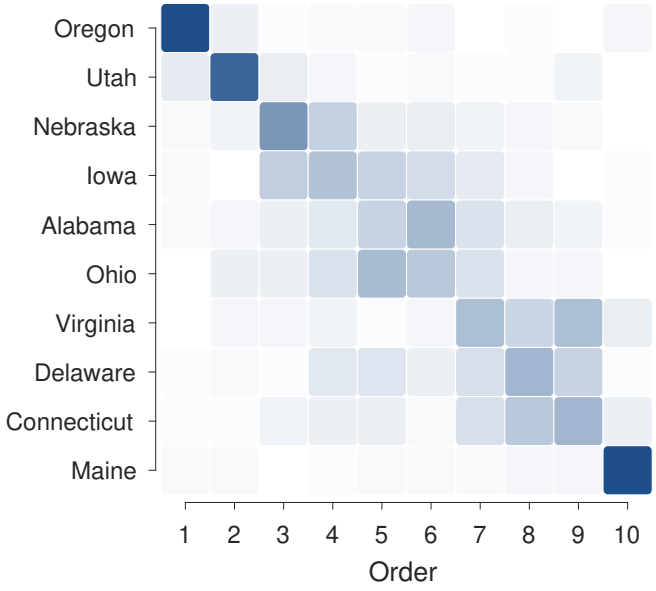

Figure 7.4: Summary of US states data. Darker shading corresponds to more participants placing a state in a position in their order.

Figure 7.4 summarizes the orders provided by all of the participants. States are shown in rows and positions in the orders are shown in columns. Darker shading represents more participants placed a state in a position within their order. The states are arranged from top to bottom in terms of

their true geographic locations from west to east. It is clear, for example, that Oregon was usually placed in position 1, Utah in position 2, and Maine in position 10. For the other states, there is a broader distribution over the possible positions within the orders provided.

Results

Inferred representation

Figure 7.5 shows the marginal posterior distributions for the μ_j location parameters based on the Thurstone scaling model. The states are again arranged from top to bottom in terms of the ground truth. One way to summarize the inferences is to create a group order based on the posterior means of the μ_j locations. This aggregate order is correct to the extent the posterior means increase to the right moving down the states, which is almost the case. The only error is that Ohio is inferred to be to the west of Alabama, but it is actually to the east.

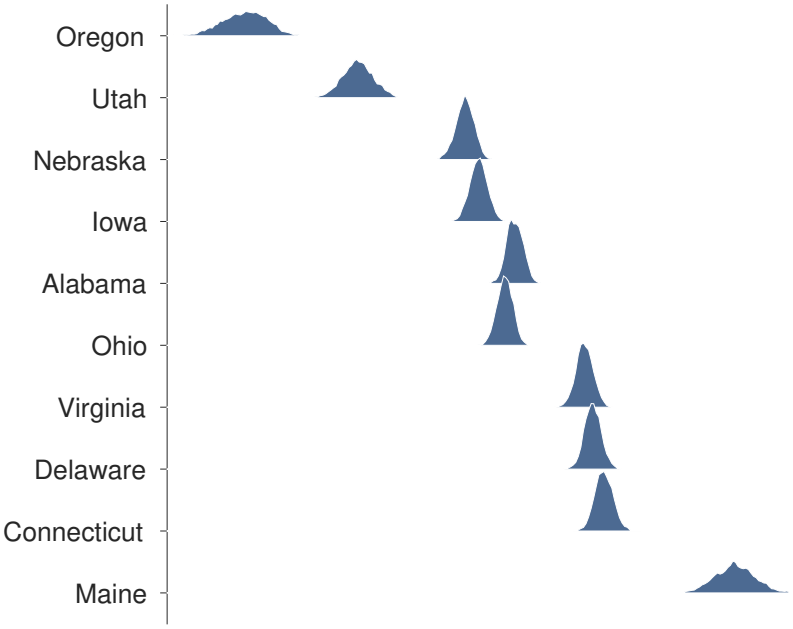

Figure 7.5: Inferences about the location of the US states in the aggregate ranking.

One way to measure the accuracy of an order is using a measure known as the tau distance (Kendall, 1938, see van Doorn et al. 2021 for an overview aimed at social science researchers). The tau distance counts the number of pairwise swaps between two orders, and can be interpreted as an error measure. Because only the adjacent Ohio and Alabama in the aggregate order need to be swapped to produce the true order, the tau distance of the aggregate order is one. Less accurate orders of the states would require more swaps of adjacent pairs and increase the tau disance measure.

It is tempting to use the marginal posterior distributions of μ_j to reach conclusions about the uncertainty of the group order. The visual overlap seen in Figure 7.5 can, however, be misleading. For example, the marginal posteriors for Ohio and Alabama have significant overlap, suggesting that the group order is uncertain about which is further west. This is

not, in fact, the case, because the joint posterior distribution has more information than the marginal distributions.

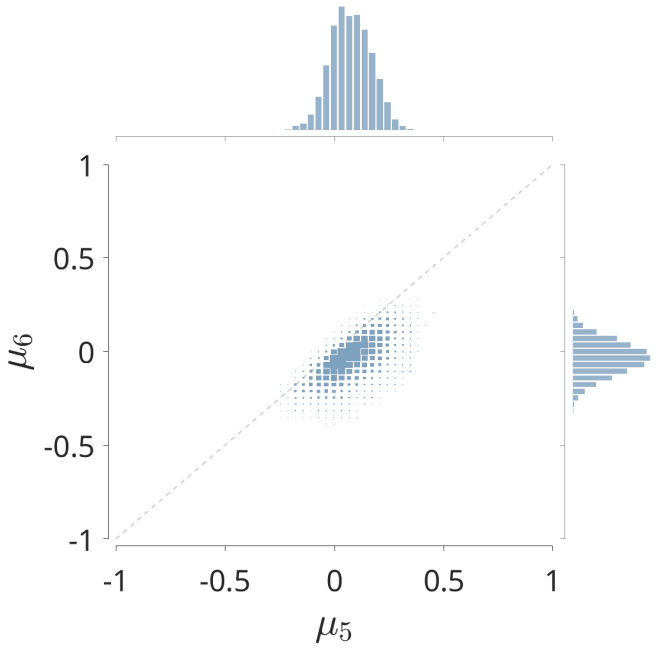

Figure 7.6: The joint and marginal posterior distributions for μ_5 and μ_6.

Figure 7.6 demonstrates the issue for the relationship between Alabama and Ohio represented by the μ_5 and μ_6 parameters. The central panel shows the joint posterior distribution and the top and right panels show the marginal distributions. The dotted diagonal line in the central panel shows where $\mu_5 = \mu_6$. There is a clear correlation in the joint posterior distribution that leads to μ_5 almost always being greater than μ_6, despite the marginals having much greater overlap. The issue of joint distributions potentially have more information than marginal distributions was originally discussed in Chapter 0.2.1, and may be helpful if the idea conveyed by Figure 7.6 is unclear.

The uncertainty in the group ranking is accurately quantified by the π_{jk} probabilities shown in Table 7.3, which are calculated with respect to the joint posterior distribution. The states are now arranged from top to bottom (and left to right) in terms of the posterior means of the μ_j parameters. This means that they follow the aggregate order rather than the ground truth. The entries below the diagonal show the probability that the state in the row is earlier in the aggregate order (i.e., judged to be to the west of) the state in the column. Most of these probabilities are 1.00, indicating strong certainty in the group order. The largest exception is in the comparison of Alabama and Ohio, which shows a probability of 0.97 that Ohio is further west. This corresponds to the joint distribution in Figure 7.6, which shows that only 3% of the joint posterior samples lie above the diagonal line.

Individual expertise

The inferences about the expertise of each individual are provided by the posterior distributions for σ_i. As an example of these inferences, consider the first two participants, who provided the following rankings:

	Ore	Uta	Neb	Iow	Ohi	Ala	Vir	Del	Con	Mai
Oregon	–	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Utah	1.00	–	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Nebraska	1.00	1.00	–	0.02	0.00	0.00	0.00	0.00	0.00	0.00
Iowa	1.00	1.00	0.98	–	0.00	0.00	0.00	0.00	0.00	0.00
Ohio	1.00	1.00	1.00	1.00	–	0.03	0.00	0.00	0.00	0.00
Alabama	1.00	1.00	1.00	1.00	0.97	–	0.00	0.00	0.00	0.00
Virginia	1.00	1.00	1.00	1.00	1.00	1.00	–	0.02	0.00	0.00
Delaware	1.00	1.00	1.00	1.00	1.00	1.00	0.98	–	0.01	0.00
Connecticut	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99	–	0.00
Maine	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	–

Table 7.3: Probabilities π_{ij} that each US state is to the west of each other state.

- *participant 1*: Oregon, Utah, Nebraska, Iowa, Delaware, Connecticut, Ohio, Alabama, Virginia, Maine
- *participant 2*: Connecticut, Alabama, Oregon, Delaware, Virginia, Iowa, Ohio, Nebraska, Utah, Maine

The posterior mean for participant 1 is $\sigma_1 = 0.98$ with a 95% credible interval $(0.41, 2.00)$, and for participant 2 the posterior mean is $\sigma_2 = 2.14$ with a 95% credible interval $(1.49, 2.63)$. Thus, participant 1 was inferred to be much more expert than participant 2, which is consistent with the much more accurate order they gave. It is worth emphasizing again that the Thurstone scaling model makes these predictions about individual expertise *without* access to the ground truth.

7.3 *Piano Advertisements*

`pianoAdvertisements.jasp`

Knight (1921) studied individual and group judgments in a 1921 thesis, with tasks ranging from orderings the lengths of lines to ordering the intelligence of people from photographs. This example focuses on judgments of the effectiveness of different advertisements.

Data

Knight (1921, Table 3)[†] reports the orders of 40 people who were shown 15 advertisements of the "Virtuolo Player piano." The people were asked to

> "Arrange the advertisements in the order of the degree in which they would influence you to buy the product advertised. Put the one that would have the strongest influence in that direction on top, the next strongest second, etc."

An objective measure of the advertisements was determined based on their commercial returns. It seems unlikely the people making the judgments were aware of these returns, so their task can be viewed as a prediction of a ground truth.

[†] A scanned version of the table is here. As it turns out, the rankings provided in the table for two people are problematic. Person 16 has two entries for "4" and none for "12", while person 32 has two entries for "14" and none for "15". This leaves 38 people with valid available rankings.

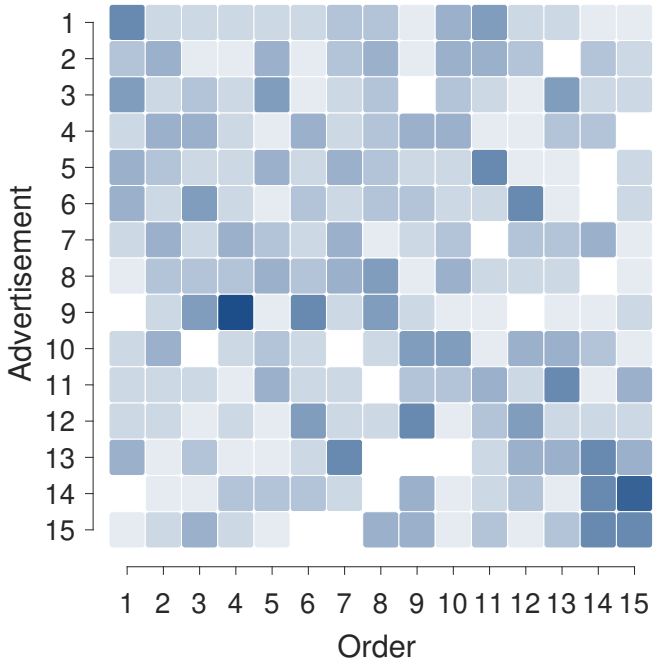

Figure 7.7: Summary of piano advertisements data. Darker shading corresponds to more participants ranking a state in a position.

Figure 7.7 summarizes the predictions people made. Advertisements are shown in rows and predicted rankings are shown in columns. Darker shading represents more people predicting an advertisement in a rank position. The advertisements are arranged from top to bottom according to the true commercial returns. The data show significant variability in the judgments of people. There is no clear agreement for most of the advertisements. Perhaps one eception is that the lowest-return advertisements 13, 14, and 15 are often placed near the end of people's orders.

Results

Figure 7.8 shows the marginal posterior distributions for the μ_j location parameters based on the Thurstone scaling model. The advertisements are arranged in terms of the true commercial returns from top to bottom. These inferences show an interesting pattern. Advertisements 1 to 9, which have the highest commercial returns, are positioned first in the aggregate order. Within this subset, however, the aggregate order does not relate systematically to the true returns. Advertisements 10 to 14, which have the lowest commercial returns, are positioned last in the aggregate order. There is meaningful ordering within this subset of advertisements, largely following the true returns.

The Kendall tau distance again provides a way to quantify the accuracy of the aggregate prediction. The order provided by the posterior means of the μ_j parameters requires 16 pairwise swaps to match the true ranking in terms of commercial returns. One person was equally as accurate as the aggregate order, but the other 37 made less accurate judgments. The mean tau distance across all people is about 45.

Table 7.4 shows the probability each advertisement is positioned ahead of every other advertisement in the aggregate order. The advertisements are now arranged from top to bottom in terms of the posterior means

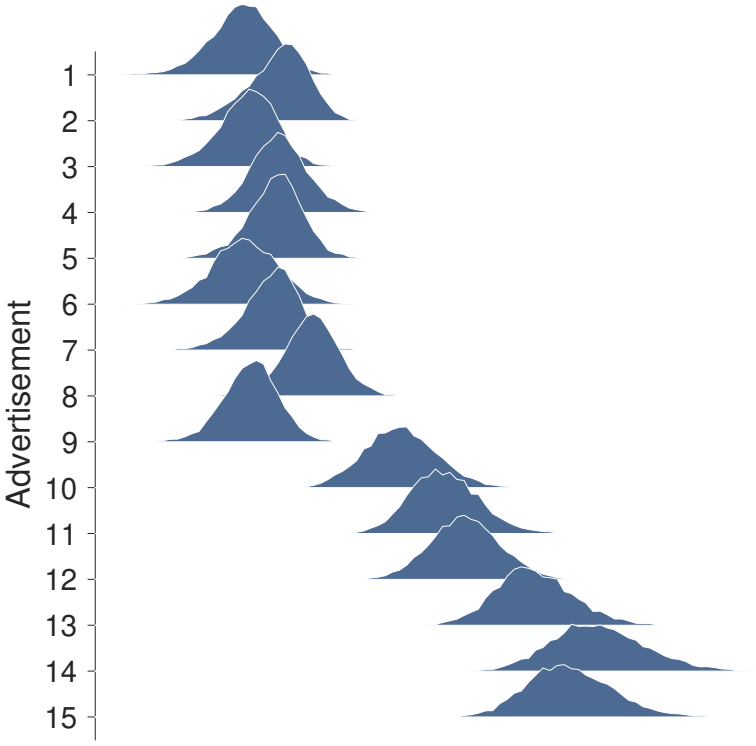

Figure 7.8: Inferences about the position of the 15 piano advertisements in the aggregate order.

Table 7.4: Probability that each advertisement is positioned ahead of each other advertisement in the aggregate order.

	1	6	3	9	7	5	2	4	8	10	11	12	13	15	14
1	–	0.52	0.41	0.38	0.22	0.17	0.19	0.13	0.05	0.00	0.00	0.00	0.00	0.00	0.00
6	0.48	–	0.45	0.39	0.26	0.21	0.24	0.17	0.07	0.00	0.00	0.00	0.00	0.00	0.00
3	0.59	0.55	–	0.50	0.22	0.19	0.17	0.17	0.05	0.00	0.00	0.00	0.00	0.00	0.00
9	0.62	0.61	0.50	–	0.25	0.21	0.23	0.19	0.06	0.00	0.00	0.00	0.00	0.00	0.00
7	0.78	0.74	0.78	0.75	–	0.50	0.39	0.47	0.11	0.00	0.00	0.00	0.00	0.00	0.00
5	0.83	0.79	0.81	0.79	0.50	–	0.42	0.49	0.14	0.00	0.00	0.00	0.00	0.00	0.00
2	0.81	0.76	0.83	0.77	0.61	0.58	–	0.55	0.17	0.00	0.00	0.00	0.00	0.00	0.00
4	0.87	0.83	0.83	0.81	0.53	0.51	0.45	–	0.20	0.00	0.00	0.00	0.00	0.00	0.00
8	0.95	0.93	0.95	0.94	0.89	0.86	0.83	0.80	–	0.02	0.00	0.00	0.00	0.00	0.00
10	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.98	–	0.15	0.09	0.00	0.00	0.00
11	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.85	–	0.27	0.03	0.01	0.00
12	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.91	0.73	–	0.04	0.01	0.00
13	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.97	0.96	–	0.19	0.08
15	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.99	0.81	–	0.22
14	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.92	0.78	–

of the μ_j parameters, following the aggregate order rather than the true commercial returns. These probabilities quantify the cluster structure observed in Figure 7.8. There is uncertainty about the order of each pair of advertisements in the subset of advertisements 1 to 9. There is much more certainty for the ordering of advertisements 10 to 14.

7.4 *Basketball Predictions*

`basketballPredictions.jasp`

This example also uses data from the Lee et al. (2014) wisdom of the crowd study, using data from a prediction task involving teams in the US National Basketball Association's Eastern Conference

Data

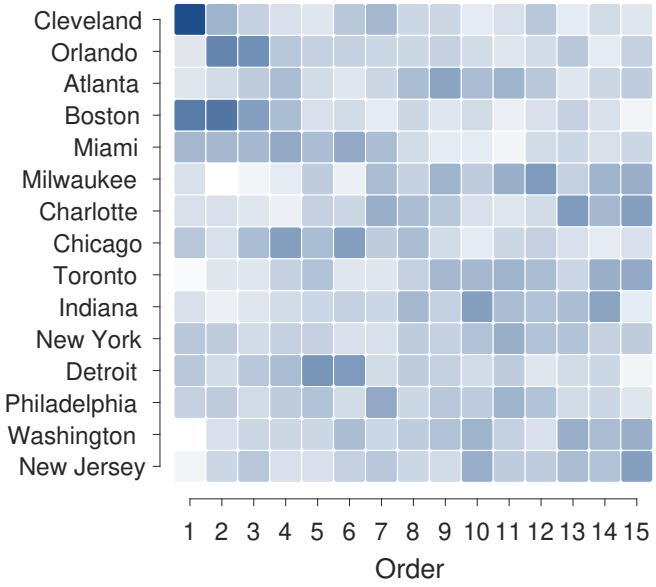

Figure 7.9: Summary of basketball prediction data. Darker shading corresponds to more participants ranking a state in a position.

The data were collected from 148 participants early in the 2010 season. Each participants was asked to predict the final standings of the 15 teams at the end of the regular season, before the playoffs. Figure 7.9 summarizes the predictions people made. Teams are shown in rows and predicted final standings are shown in columns. Darker shading represents more participants predicted a team will finish in a position. The teams are ordered from top to bottom according to their eventual final standings. This is a ground truth that was not known at the time the predictions were made, but later become available once the regular season was completed.

Figure 7.9 shows that Cleveland was often predicted to finish top of the standings, and Boston and Orlando were often predicted to finish in the top three. It also shows teams like New Jersey, Washington, Toronto, Charlotte and Milwaukee were predicted to finish bottom. In general, though, it seems clear there is no strong consensus across the predictions. Most teams are predicted to finish in most final positions by a non-negligible number of participants.

Results

Figure 7.10 shows the marginal posterior distributions for the μ_j location parameters based on the Thurstone scaling model. The teams are again arranged in terms of the true final standings from top to bottom. This means that the aggregate predicted order based on the posterior means is correct to the extent they increase to the right moving down the final standings. Visually, Figure 7.10 shows that this generally happens, but there are many misorderings. For example, the aggregate prediction for the top four is Cleveland, Boston, Orlando, and Miami. These teams finished first, fourth, second, and fifth, respectively, which identifies many of the top teams, but does not perfectly predict their final standings.

The Kendall tau distance again provides a way to quantify the accuracy of the aggregate prediction. The order provided by the posterior means

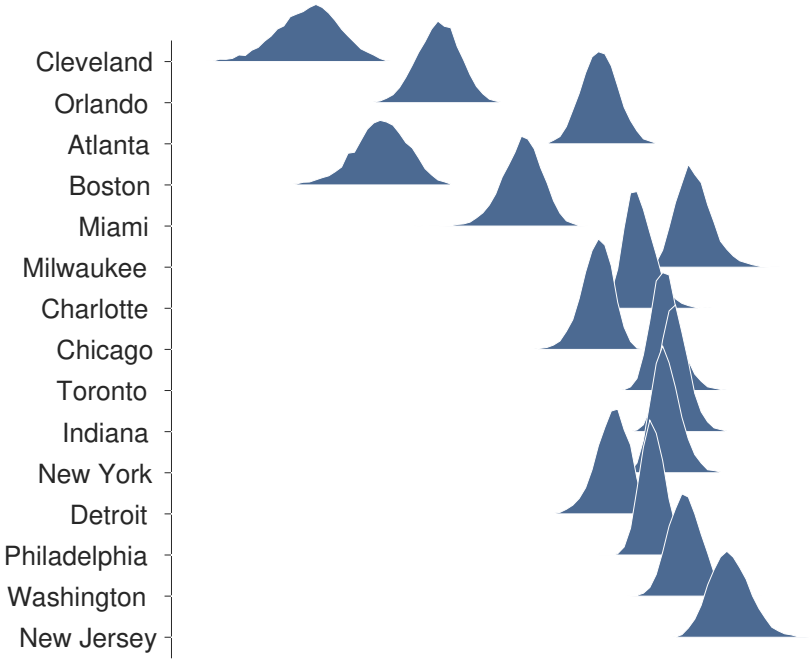

Figure 7.10: Inferences about the location of the NBA teams in the aggregate prediction of final standings.

of the μ_j parameters requires 21 pairwise swaps to match the true final standings. The predictions of 10 participants were closer to the truth, but the other 138 made less accurate predictions. The mean tau distance across all people is about 46.

Table 7.5: Probability that each NBA team is predicted to finish above each other team by the aggregate order.

	Cle	Bos	Orl	Mia	Chi	Atl	Det	Cha	Phi	Tor	New	Ind	Was	Mil	New
Cleveland	–	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Boston	1.00	–	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Orlando	1.00	1.00	–	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Miami	1.00	1.00	1.00	–	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Chicago	1.00	1.00	1.00	1.00	–	0.49	0.21	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Atlanta	1.00	1.00	1.00	1.00	0.51	–	0.33	0.01	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Detroit	1.00	1.00	1.00	1.00	0.79	0.67	–	0.14	0.01	0.00	0.00	0.00	0.00	0.00	0.00
Charlotte	1.00	1.00	1.00	1.00	0.99	0.99	0.86	–	0.21	0.07	0.07	0.04	0.02	0.00	0.00
Philadelphia	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.79	–	0.22	0.19	0.10	0.06	0.01	0.00
Toronto	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.93	0.78	–	0.51	0.27	0.15	0.03	0.00
New York	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.93	0.81	0.49	–	0.29	0.18	0.05	0.00
Indiana	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.96	0.90	0.73	0.71	–	0.27	0.12	0.00
Washington	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.98	0.94	0.85	0.82	0.73	–	0.37	0.01
Milwaukee	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.97	0.95	0.88	0.63	–	0.05
New Jersey	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.95	–

Table 7.5 shows the probability each team is predicted to finish ahead of every other team by the inferred group ordering. The teams are now arranged from top to bottom in terms of the posterior means of the μ_j parameters, following the aggregate prediction rather than the final standings.

Once again, for the reasons explained in Chapter 7.2, the probabilities in Table 7.5 show that there is much more certainty than the overlap of the marginal posterior distributions of the μ_j parameters in Figure 7.10 suggests. There are, however, parts of the aggregate prediction that

are uncertain. For example, there is only a 51% probability Atlanta is predicted to finish ahead of Chicago.

7.5 *Star Wars*

starWars.jasp

This Kaggle data repository involves a survey of fans of the Star Wars movie franchise. One of the questions asks people to provide a preference order for the first six movies in the franchise. This example is one in which there is no ground truth, since it involves subjective movie preferences. The aggregate order can be thought of as a fan consensus opinion.

Data

The available data involve the order provided by 834 fans for the six movies: "A New Hope", "The Empire Strikes Back," "The Return of the Jedi," "The Phantom Menace," "The Attack of the Clones," and "The Revenge of the Sith." We use just the subset of the preferences provided by the first 50 fans in the overall repository.

Figure 7.11 summarizes the orders people provided. Movies are shown in rows and preference rankings are shown in columns. Darker shading represents more people placing a movie in a position within their order. The movies are arranged from top to bottom in the same way as the original data set, which follows the chronology of the Star Wars universe. The data show significant variability in people's preferences. There is no clear agreement for most of the movies, although "The Phantom Menace," is often placed last and "The Empire Strikes Back" is often placed first. There is some suggestion of a two-cluster structure, with the first three movies generally being placed lower than the last three movies.

Results

Figure 7.12 shows the marginal posterior distributions for the μ_j location parameters based on the Thurstone scaling model. The movies continue to be arranged in terms of the Star Wars chronology. These results show the two-cluster structure suggested by Figure 7.11. It also seems clear that "The Empire Strikes Back" is the consensus favorite movie, followed by "A New Hope" and "The Return of the Jedi." There is no way of assessing the accuracy of this aggregate order, since it constitutes a consensus fan preference and does not have an objective ground truth.

Table 7.6 shows the probability that each movie is placed ahead of every other movie in the aggregate order. The movies are now arranged from top to bottom in terms of the posterior means of the μ_j parameters, following the consensus order inferred by the Thurstone scaling model. The probabilities in Table 7.6 make clear that there is considerable certainty in the consensus order. The least certain components of the aggregate order are 97% certain. These small uncertainties could involve swapping "A New Hope" with "The Return of the Jedi" or "The Attack of the Clones" with '"The Phantom Menace."

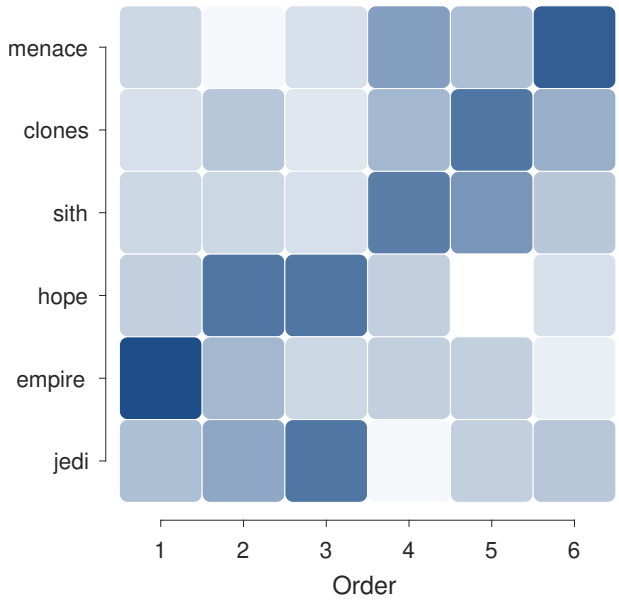

Figure 7.11: Summary of Star Wars data. Darker shading corresponds to more participants ranking a movie in a position.

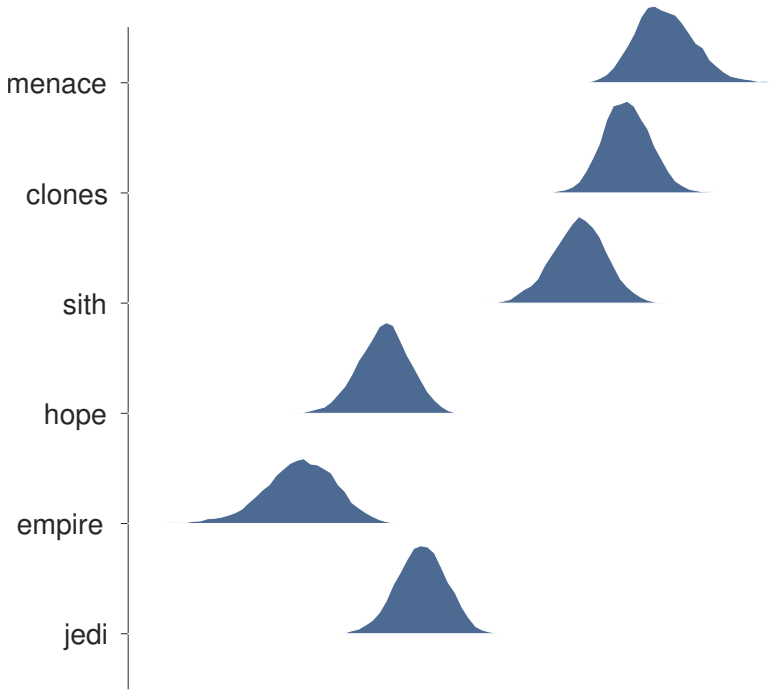

Figure 7.12: Inferences about the location of the six Star Wars movies in the aggregate ranking.

◇

	empire	hope	jedi	sith	clones	menace
empire strikes back	–	0.00	0.00	0.00	0.00	0.00
new hope	1.00	–	0.03	0.00	0.00	0.00
return of jedi	1.00	0.97	–	0.00	0.00	0.00
revenge of sith	1.00	1.00	1.00	–	0.01	0.00
attack of the clones	1.00	1.00	1.00	0.99	–	0.03
phantome menace	1.00	1.00	1.00	1.00	0.97	–

Table 7.6: Probability that each Star Wars movie is placed ahead of each other movie in the aggregate order.

7.6 Formal Details

7.6.1 Thurstone scaling model

Priors

The first set of parameters of the Thurstone scaling model are the item location parameters μ_j for the jth item, indicating their true position on an assumed underlying single dimension. These are given a uniform prior distribution over a wide range

$$\mu_j \sim \text{uniform}\big(-50, 50\big). \tag{7.1}$$

A constraint is also imposed on the μ_j parameters, to control for location invariance. The issue is that if all of the μ_j parameters are shifted by some additive constant, the behavior of the model is identical. Imagine the distributions in Figure 7.2 all moving to the right. This clearly does not change the probabilities of mental samples being greater or lesser than each other, and so the probabilities of the different possible orderings are unchanged. When models systematically make identical predictions for different parameter values in this sort of way, the model is called non-identifiable. It prevents meaningful inference for the μ_j parameters, because there are infinitely many possibilities equally consistent with the observed data.

To overcome this identifiability problem, the μ_j parameters are constrained to have an arithmetic mean of zero, so that

$$\sum_j \mu_j = 0. \tag{7.2}$$

This constraint is sometimes called "zero centering." It prevents the μ_j parameters from being able to move to the left or right, and makes the model identifiable with respect to this translation.

The second set of parameters in the Thurstone scaling model is the expertise parameters σ_i for the ith person, indicating how accurately they represent information about each stimulus. These are given a uniform prior distribution over the unit interval

$$\sigma_i \sim \text{uniform}\big(0, 1\big). \tag{7.3}$$

A constraint is also imposed on the σ_i parameters, to control for scale invariance. The issue is that if all the σ_i parameters were multiplied by the same constant, the behavior of the model could be made identical with a compensating adjustment in the μ_j location parameters. Imagine all of the standard deviations being doubled, so that the distributions

overlapped more, but then all of their means being stretched left and right, so that the original pattern of overlap is recovered. This could be done even with the zero centering constrain on the μ_j. This lack of scale invariance also makes the model non-identifiable.

To overcome this, the σ_i parameters are constrained to have an arithmetic mean of one, so that

$$\frac{1}{m}\sum_i \sigma_i = 1, \tag{7.4}$$

where m is the number of people. This constraint is sometimes called "unit scaling."

Likelihood

The decision process assumes that person i takes a mental sample x_{ij} for each of the $j = 1, 2, \ldots, n$ items. The mental sample is drawn from a Gaussian distribution centered on the true location of the item, with a standard deviation determined by their expertise, so that

$$x_{ij} \sim \text{Gaussian}\big(\mu_j, \frac{1}{\sigma_i^2}\big). \tag{7.5}$$

The order of the mental samples then determines the order $\boldsymbol{y}_i$ that the person produces

$$\boldsymbol{y}_i \sim \text{rank}\left(\boldsymbol{x}_i\right). \tag{7.6}$$

7.7 JASP implementation

The JAGS code for the Thurstone scaling model is shown below. The `data` block organizes the ranking data into a `person` by `item` matrix `y`. The zero centering and unit scaling of the `mu` and `sigma` parameters are achieved using temporary variables. The mental samples in `x` are samples from the appropriate Gaussian distribution.

The most interesting feature is the use of the `dinterval` distribution to impose the censored ordering implied by the ranking data on line 23. The statement `y[i, j]~dinterval(x[i, j], sort(x[i, ]))` can be interpreted as "the mental sample person i has for item j must fall in the position among their ordered mental samples specified by where they ranked it." For example, with reference to Figure 7.2, it requires that the mental sample individual 2 has for the gray item `x[2,4]` is in position `y[2,4]=3` among the sorted ascending of all of their mental samples `x[2,1]`, `x[2,2]`, `x[2,3]`, and `x[2,4]`.

The final part of the code calculates the probabilities $\pi_{jk} = p\left(\mu_j < \mu_k\right)$ in the matrix `pi`. Because smaller values on the dimension correspond to items at the beginning of the ordered list, this is a measure of the probability that item j is placed before the item k. These probabilities are useful for understanding how confident the inferences are about the ordering of the items.

```
1  # Thurstone scaling
2  data{
3    nIndividuals = max(individual)
4    nItems = max(item)
5    for (idx in 1:length(individual)){
```

```
    y[individual[idx], item[idx]] = rnkd[idx]-1
  }
}
model{
  # Latent truth
  for (j in 1:nItems){
    muTmp[j] ~ dunif(-50, 50)
    mu[j] = muTmp[j] - mean(muTmp)
  }
  # Expertise
  for (i in 1:nIndividuals){
    sigmaTmp[i] ~ dunif(0, 1)
    sigma[i] = sigmaTmp[i]/mean(sigmaTmp)
  }
  # Data
  for (i in 1:nIndividuals){
    for (j in 1:nItems){
      x[i, j] ~ dnorm(mu[j], 1/sigma[i]^2)
      y[i, j] ~ dinterval(x[i, j], sort(x[i, ]))
    }
  }
  # Probability comparisons
  for (j in 1:nItems){
    for (k in 1:nItems){
      pi[j, k] = step(mu[k] - mu[j])
    }
  }
}
```

8 EZ Diffusion

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8.1 The EZ Diffusion Model

Response times (also called response latencies) are arguably the most ubiquitous behavioral measure in psychology. As Luce (1986) points out, every behavior and every sequence of behaviors that people produce has a time course. The time it takes to produce behavior can provide a lot of information about underlying psychological processes. This potential makes psychological models that predict not just behaviors, but also the response times associated with them, very useful.

Different sorts of tasks are used in the study of response times. One important distinction is between *simple* and *choice* response time tasks. Simple response time tasks only have one possible response. A good example is a vigilance task in which the goal is to press a button every time a tone sounds. Choice response time tasks involve an associated decision between two or more discrete alternatives, or choosing along some continuum of possible responses. An example involving discrete choice is deciding whether a person is a friend or foe. An example involving continuous choice is pointing in the direction that a sound is coming from.

Another important distinction in studying response times is between *speeded* and *non-speeded* tasks. Speeded tasks require a response within a second or two. An example of a speeded simple response time task is ducking when an object is thrown at you. An example of a speeded choice response time task is deciding whether to duck or catch an object thrown at you. Non-speeded tasks involve decisions that happen over longer time frames, and usually have a much heavier reliance on memory, reasoning, or other cognitive capabilities beyond the immediate decision-making process.

8.1.1 Model

A general framework for modeling response time is provided by sequential sampling models. The idea is that a person collects information from a stimulus over time, and accumulates that information as evidence for making a response. Once some critical level of evidence has been accumulated, the person makes a response. It is helpful to think of this visually in terms of a running tally reaching a boundary that represents the critical threshold. Different specific sequential sampling models make different assumptions about how evidence is sampled (Busemeyer & Townsend, 1993), how evidence is accumulated (Brown & Heathcote, 2008; Vickers, 1979), how evidence is maintained (Usher & McClelland, 2001), and whether boundaries change over time (Ratcliff et al., 2016).

In discrete choice tasks there are different boundaries for each of the possible decisions. For example, there may be an upper boundary corresponding to the "friend" response and a lower boundary corresponding

to the "foe" response. The decision that is made depends on which boundary the tally crosses first. The response time depends on how many evidence samples were required for the crossing to happen. In continuous tasks, a continuous boundary is needed. One type of sequential sampling model for continuous choice tasks uses a circular boundary surrounding the evidence accumulation tally (Smith, 2016). The point at which the tally crosses the circle defines the continuous response, as in the example of pointing in the direction that a sound is coming from. Once again, the response time depends on how many evidence samples were required for the crossing to happen.

The Wiener diffusion model

The most popular and widely-used sequential sampling model of response times is the *drift diffusion model* (Ratcliff, 1978; Ratcliff & McKoon, 2008). It is usually applied to speeded tasks with discrete choices between two possible decisions. The Wiener diffusion process that underlies the drift diffusion model is shown in Figure 8.1. In Wiener diffusion, momentary samples of evidence are repeatedly drawn from a Gaussian distribution. These evidence samples are combined in a single tally until it reaches one of two boundaries.

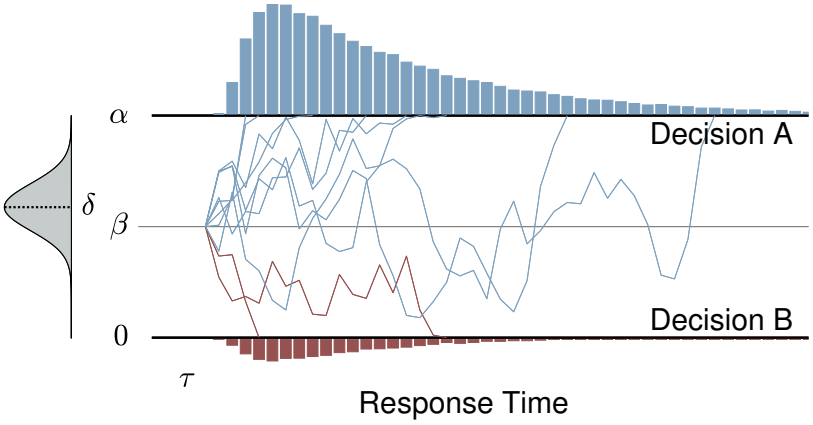

Figure 8.1: The Wiener diffusion model.

Figure 8.1 shows the Normal distribution at the left, shaded in gray. It has a mean of δ. This is known as the *drift rate*. The two boundaries are shown as associated with decision A and decision B. They are separated by a distance α. This is known as the *boundary separation*. A number of possible evidence tallies are shown by the blue and red lines. These show the cumulative values that result from summing independent samples from the Normal distribution. The tallies all start at β. This is known as the *bias*. Time progresses from left to right as the evidence values are samples. The tallies do not start until a time τ has already elapsed. This is known as the *non-decision time*.

The evidence tallies colored blue all eventually cross the decision A boundary, while the red tallies cross the decision B boundary. The blue and red histograms at the boundaries show the distribution of how long it takes for the tallies to reach the boundaries. Both show a positive skew that is typical of empirical response time distributions. Most tallies reach the boundaries relatively quickly, but a few take much longer. Figure 8.1 shows two blue evidence tallies that take much longer to reach the decision A boundary than the others.

Technically, neither the blue nor red histograms are distributions, because they do not represent all possibilities. The total area of the blue histogram is not one, but is a measure of the probability of a decision A response. Similarly, the total area of the red histogram measures the probability of a decision B response. This means that, together, the red and blue histograms form a bi-variate distribution that represents both the probability of each decision, and the response time taken to make each sort of decision.

The four key parameters in the Wiener diffusion model have natural interpretations in terms of predictions about decision probabilities and response times.

- The drift rate δ is the mean evidence for a decision provided by the evidence samples from the stimulus. Often, drift rates are best conceived as properties of the stimuli. For example, stimuli that are easy to make decisions about have large drift rates in favor of the decision boundary. This will mean that evidence tallies will generally quickly cross the boundary, corresponding to high probabilities of accurate responses and fast response times. More difficult stimuli, in contrast, have drift rates closer to zero. This will mean that the evidence tallies they generate are more likely to cross either boundary, and will generally take longer to do so. This corresponds to response probabilities closer to one-half and longer response times.
- The boundary separation α is the difference in evidence between the boundaries for the decisions. Often boundary separations are best conceived as something the decision maker can control. For example, if a person is being cautious in their decision making, and emphasizing accuracy over speed, it makes sense to increase the boundary separation so that more evidence is required to make a decision. The probability of an unlucky set of evidence samples leading to a tally that crosses the wrong boundary decreases, but it will generally take longer to make decisions. In contrast, a small boundary separation will lead to faster decisions, but with choice probability closer to one-half.
- The non-decision time τ is the time taken to respond that does not involve the evidence accumulation process. It is sometimes called the time for encoding and responding. The idea is that it corresponds to the time it takes people to perceive the sensory information provided by the stimulus to initiate a decision-making process, and to produce the motor movements needed to respond. This means, for example, it is reasonable to expect non-decision time to be greater in older people with slower perceptual and motor systems.
- The bias β is the starting point for the evidence tally relative to the boundaries. There are two ways to interpret bias. One interpretation is as prior information given by the base-rate of the two types of stimuli. If, for example, the last 100 trials have involved 99 stimuli in favor of decision A and only one in favor of decision B, it is reasonable to have a starting bias favoring decision A. The other interpretation of the starting point is in terms of the utilities for the decisions. If there is a high payoff for correctly identifying decision A stimuli, but a much smaller payoff for the correct

identification of decision B stimuli, then it is again reasonable to have a bias towards decision A.

The EZ diffusion model

The EZ diffusion model is an approximation to the drift diffusion model that is easy to apply computationally (Wagenmakers et al., 2007). The approximation does not include the bias parameter. As shown in Figure 8.3, the EZ diffusion model makes the simplifying assumption that the starting point is unbiased with $\beta = \alpha/2$.

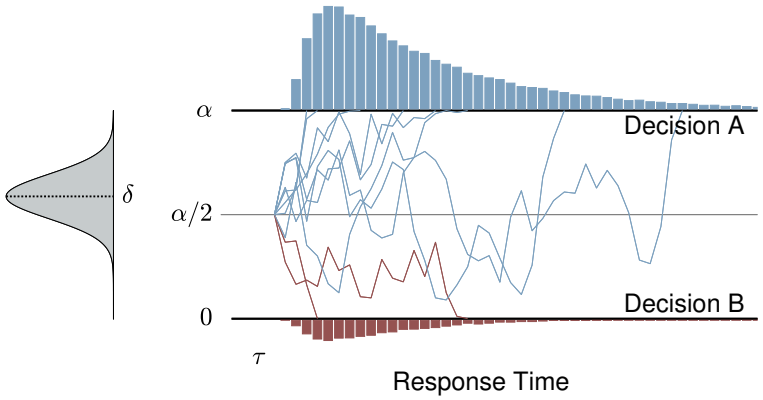

Figure 8.2: The EZ diffusion model.

The original development of the EZ diffusion model by Wagenmakers et al. (2007) provided a method for obtaining point estimates of the drift rate, boundary separation, and non-decision time parameters. It was not inherently probabilistic, and so not immediately amenable to Bayesian inference. Chávez De la Peña & Vandekerckhove (2025) developed a probabilistic implementation of the EZ diffusion model that does allow for Bayesian inference.

The omission of the bias parameter limits the usefulness of the EZ diffusion model for some applications. We use the model because it has practical advantages. One is that it is easier to implement and use in JAGS and JASP than the Wiener diffusion model. The other advantage is that the EZ diffusion model does not require trial-level information about decisions and response times. The data it requires are summary measures of the decision and response times.

Figure 8.3 summarize the structure of the EZ diffusion model, showing how it relates latent parameters to observed data. The parameters are the drift rate δ, boundary separation α, and the non-decision time τ. The observed data are the number of correct decisions k, the number of decisions, the mean of the response times for correct decisions m, and the variance of the response times for correct decisions v.

8.1.2 Mechanics

Table 8.1 details the meaning of the parameters and variables in the EZ diffusion model, their name in the model as shown in Figure 8.3, and their name in JASP.

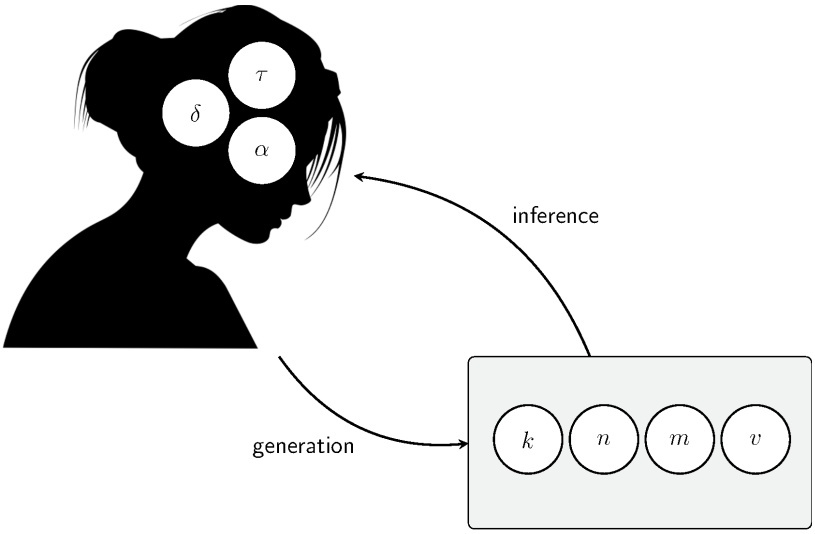

Figure 8.3: The structure of the EZ diffusion model.

Table 8.1: The meaning of the parameters and variables in the EZ diffusion model, with their model and JASP names.

Name	Model	JASP
drift rate	δ	`delta`
boundary separation	α	`alpha`
non-decision time	τ	`tau`
number of correct decisions	k	`k`
number of decisions	n	`n`
mean of response times for correct decisions	m	`m`
variance of response times for correct decisions	v	`v`

Input data

In some tasks or other data generating settings, there are correct answers. In other cases, however, there are two alternative choices, but neither is objectively correct. For example, if the task is to decide whether a person is a friend or a foe, there may be no ground truth. In these situations, one of the alternatives (say "friend") should nominally be treated as "correct." This would mean that the number of correct responses is calculated as the number of "friend" responses, and the mean and variance of the correct response times are similarly determined from the "friend" responses.

In a situation with multiple individual decision sources, the four measures should be provided on a separate row for each individual. An example is shown in Table 8.2, which shows the data for the *Random Dot Motion* application Chapter 8.2. There are five individual data sources, corresponding to five experimental conditions. Each has 384 trials but different numbers of correct responses and different means and variances for the response times of the correct responses.

Table 8.2: Data format for EZ diffusion analysis.

trial	correct	meanRT	varRT
384	250	2.99	1.44
384	291	2.83	1.08
384	342	2.51	0.57
384	384	2.19	0.19
384	384	2.07	0.11

Output analyses

The JASP implementation provides a single analysis:

- **drift, boundary, and non-decision posteriors**: the marginal posterior distributions of the drift rate δ, boundary separation α, and non-decision time τ parameters for each individual

8.2 *Random Dot Motion*

`randomDotMotion.jasp`

A widely used task for study speeded choice response time involves detecting random dot motion. Figure 8.4 shows the type of stimuli used in the task. Each stimulus contains a set of moving points, with a subset that are moving in the same direction while the others move in their own random directions. The goal of the task is to identify the direction of movement. Different stimuli have different proportions of dots moving in the same direction. This proportion is called the coherence of the stimuli. Greater coherence generally makes detecting the direction of motion easier.

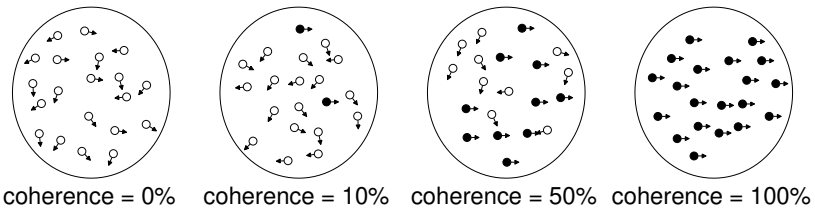

Figure 8.4: Stimuli with different levels of coherence in a random dot motion task. This figure is based on online documentation for the Empirica system for designing human experiments (Almaatouq et al., 2021).

Data

Hellmann et al. (2023, Experiment 2) used a random dot motion task in which 12 participants had to decide whether the direction of motion was up or down. The task involved five within-participant levels of coherence, with 1.6%, 3.2%, 6.4%, 12.8%, or 25.6% of dots moving coherently.

Figure 8.5 shows the correct and error response time distributions for one participant across the five coherence conditions. The participant makes many more error decisions for the conditions with lower coherence. Their decisions were always correct in the two conditions with highest coherence. The mean response time become faster as coherence increases, decreasing from 2.99 to 2.07 seconds over the five conditions. Part of this overall increase is speed is due the reduction in long response times in the tails of the distributions. For example, response times over 5 seconds occur only in the three conditions with lowest coherence.

Results

Figure 8.6 shows the marginal posterior distributions for the drift rate, boundary separation, and non-decision time parameters for the five coherence conditions.

The drift rate likely increases as the random dot motion stimuli become more coherent. The three conditions with lowest coherence clearly have

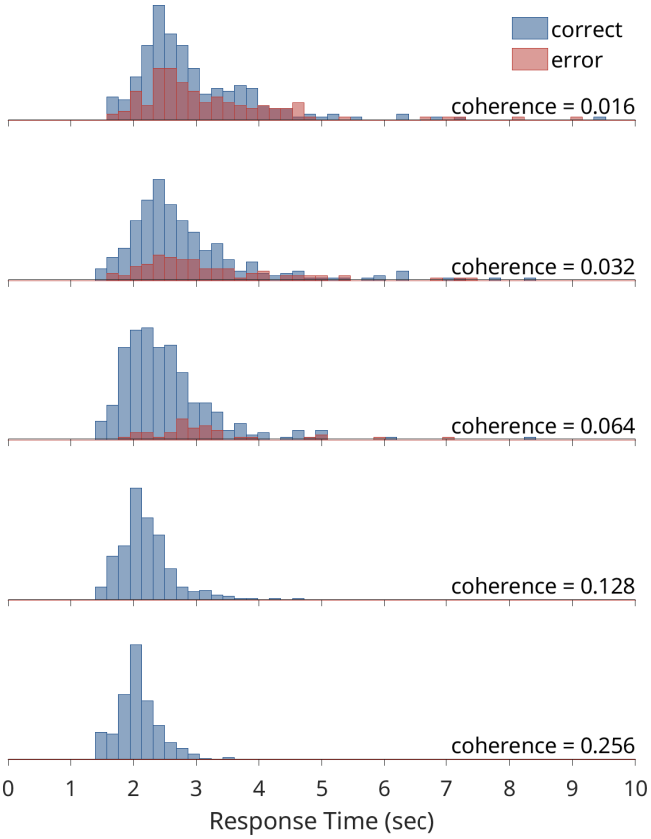

Figure 8.5: The correct and error response time distributions for a single participant in the random dot motion task.

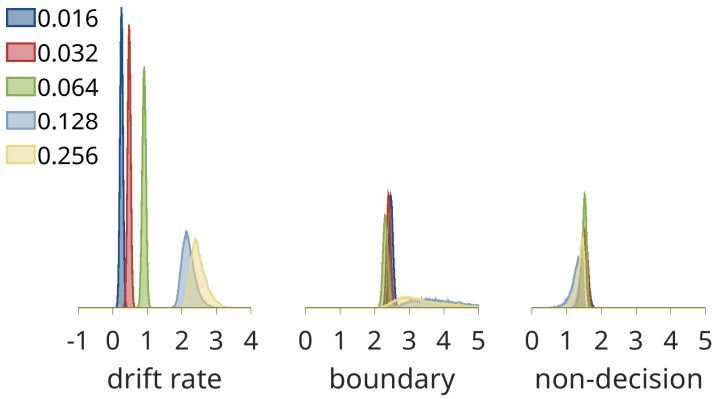

Figure 8.6: Inferences for the EZ diffusion model parameters for the random dot motion data.

lower drift rates than the two highest-coherence conditions. This is consistent with the interpretation of drift rate as measuring the information provided by stimuli. Higher coherence provides more evidence in terms of the up-or-down decision the participant was required to make. The uncertainty about the relationship between the 0.128 and 0.256 coherence conditions is likely a ceiling effect: the stimuli were easy enough that the participant made no errors and responded as quickly as possible in both conditions.

The boundary separation does not clearly change over the five conditions. For the three conditions with lowest coherence the marginal posterior

distributions are very similar. There is more uncertainty for the two conditions with highest coherence, but they allow for values around 2.5 consistent with the other conditions. A lack of change in boundary separation is consistent with its interpretation as a participant-specific level of caution. The uncertainty in the inference for the 0.128 and 0.256 coherence conditions is again likely related to the ease of these conditions. The lack of error responses means that the behavioral data provide little information about the separation of the upper and lower boundaries for correct and error decisions.

Finally, the non-decision time also shows no clear evidence of change. This is consistent with its interpretation as related to participant-specific perceptual and motor speed.

8.3 *Speed-Accuracy Tradeoff*

`speedAccuracyTradeoffA.jasp`
`speedAccuracyTradeoffB.jasp`
`speedAccuracyTradeoffC.jasp`

Grange (2022) studied differences in decision making between depressed and non-depressed participants in a perceptual identification task. The task involved deciding whether a simple cartoon drawing of a face had a long mouth, extending to edge of the eyes, or a short mouth, contained entirely within boundary of the eyes. A within-participant experimental manipulation alternated instructions emphasizing either making fast decisions or accurate decisions. Accordingly, this application considers the *speed-accuracy tradeoff*, which is a central idea in understanding decision making, and a common application of diffusion and other sequential sampling models.

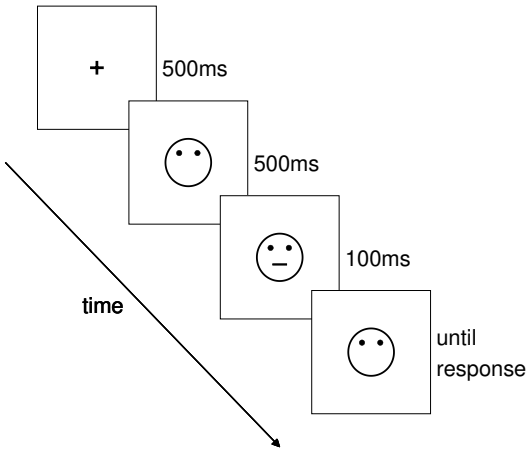

Figure 8.7: Stimuli and trial structure used by Grange (2022).

Data

We consider the pilot data from 50 prolific participants Grange (2022) used as a manipulation check. In the accuracy emphasis condition participants were instructed to respond "as accurately as possible without taking too long to respond." If participants made an error they saw the message "error!" presented centrally in red font for 1500ms.

In the speed emphasis condition participants were instructed to respond "as quickly as possible without making too many mistakes. DO NOT GUESS." If participants responded more slowly than 500ms they saw the message "TOO SLOW!" for 1500ms.

Each participant completed 8 blocks of 48 trials, alternating between speed emphasis and accuracy emphasis blocks. Reminders of the speed or accuracy instructions were presented before each block. For all participants and all trials, 99% of the response times are between 0.1 seconds and 1 second. As a simple approach to decontamination, we consider that majority of the trials.

Individual participant analyses

We analyze the behavior of three of the participants using the EZ diffusion model. The first of these participants shows a classic speed-accuracy tradeoff, and the inferences of the EZ diffusion model explain the trade-off in terms of increased caution leading to greater boundary separation. The other two participants are presented as examples of different sorts of empirical regularities and patterns of EZ diffusion inferences found among the participants. These variations emphasize that neat theoretical accounts—such as the explanation of the speed-accuracy trade-off in terms of deliberate changes to caution—are not always cleanly reflected in model-based empirical science. Mike Tyson was on to something when he said "Everyone has a plan until they get punched in the mouth."

Participant A

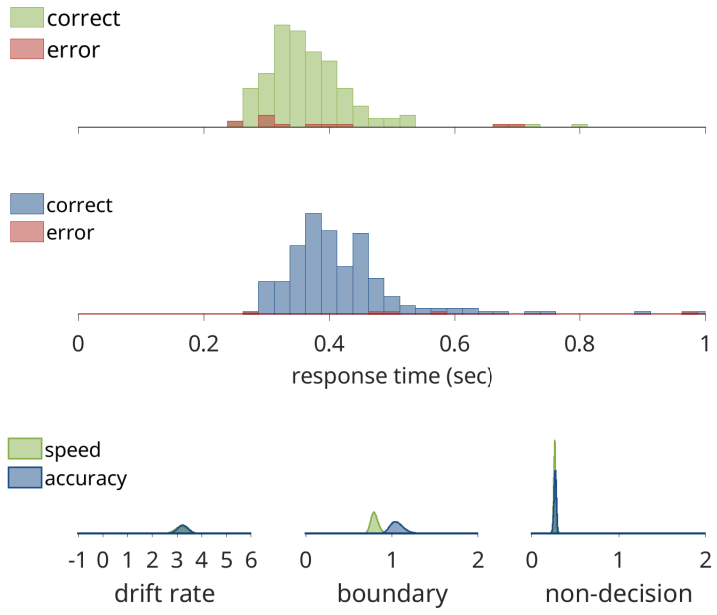

Figure 8.8: The correct and error response time distributions for participant A in the speed (top panel) and accuracy (bottom panel) of the mouth length task, and the (bottom panel) EZ diffusion model inferences.

The top two panels of Figure 8.8 show the correct and error response time distributions for participant A in the speed and accuracy conditions. The participant did not make many error decisions, but there are a few more in the speed condition. There is also an increase in the response time distribution for the accuracy condition.

The bottom panel of Figure 8.8 shows the inferences of the EZ diffusion model. The posterior distributions for the drift rate and non-decision time parameters are extremely similar across the speed and accuracy conditions. The posterior distributions for the boundary separation, however, are different, with a larger boundary likely in the accuracy condition. This pattern of results suggests the change from speed to accuracy instructions does not affect the ability of the participant to gather evidence from the stimulus, nor their perceptual or motor speed. Instead, the impact of the accuracy instructions is to cause them to increase their caution, requiring more evidence before making a "long" or "short" response. This is the standard sequential sampling and diffusion model account of the speed-accuracy trade-off.

Participant B

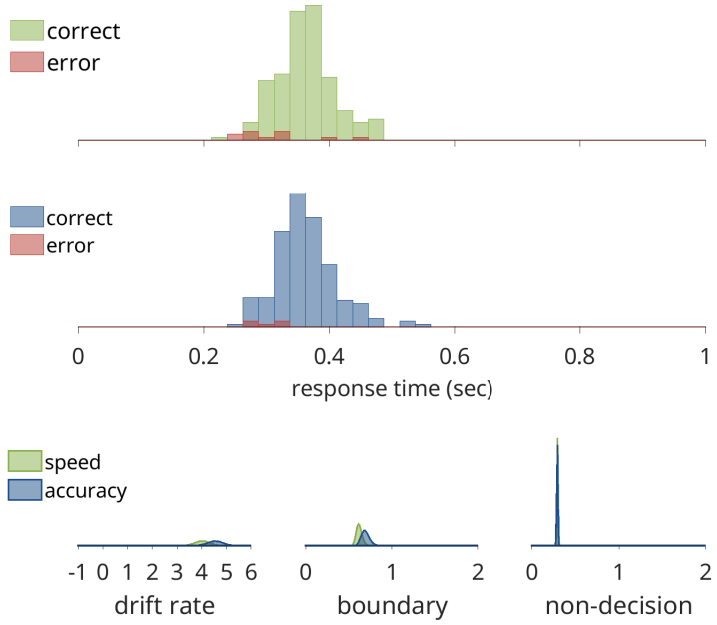

Figure 8.9: The correct and error response time distributions for participant B in the speed (top panel) and accuracy (bottom panel) of the mouth length task and the (bottom panel) EZ diffusion model inferences.

The top two panels of Figure 8.9 show the correct and error response time distributions for participant B in the speed and accuracy conditions. Their accuracy is similar to participant A, with a few more error decision in the speed condition, but not many overall. There is little evidence, however, of a change in response time between the conditions. In other words, there is no clear evidence of a speed-accuracy trade-off.

The bottom panel of Figure 8.9 shows the inferences of the EZ diffusion model. There is some evidence of an increase in both the drift rate and boundary separation for the accuracy condition. One interpretation of these results is that the participant was able to extract more information from the stimulus in the accuracy condition, but also was more cautious in reaching a decision. This is consistent with making fewer errors in the accuracy conditions without shortening the time to make those decisions.

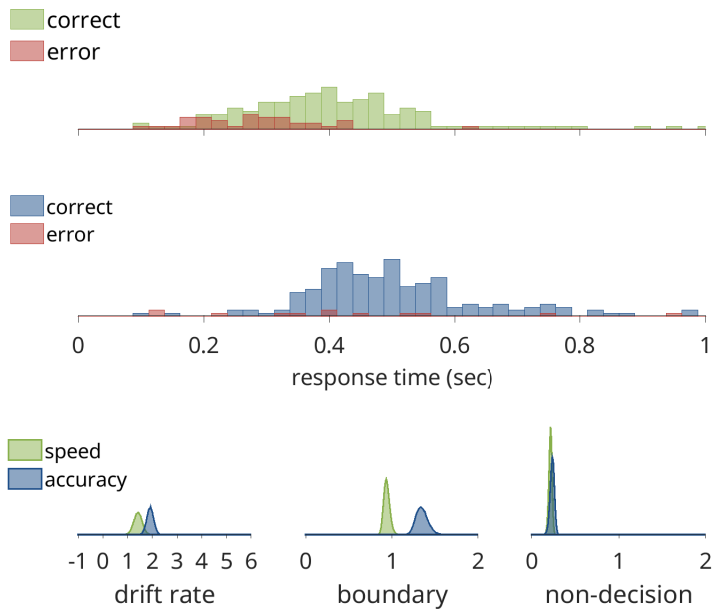

Figure 8.10: The correct and error response time distributions for participant C in the speed (top panel) and accuracy (bottom panel) of the mouth length task and the (bottom panel) EZ diffusion model inferences.

Participant C

The top two panels of Figure 8.10 show the correct and error response time distributions for participant C in the speed and accuracy conditions. Participant C makes many more error decisions than participants A or B, especially in the speed condition, and shows a clear increase in response times in the accuracy condition.

The bottom panel of Figure 8.10 shows the inferences of the EZ diffusion model. Despite the speed-accuracy trade-off, the inferences do not suggest only the boundary separation changed between conditions. In the same way as for participant B, both the drift rate and boundary separation increase in the accuracy condition. One way to interpret these results is that the larger number of errors in the speed condition is due to the participant being both less cautious and getting less information from the stimuli.

It is worth contrasting the absolute values of the inferred drift rates and boundary separations across the three participants, and relating to those inferences to the proportions of errors and the absolute values of response times. Participant C makes made more errors and generally takes longer to respond. This is reflected in their smaller drift rate and larger boundary separation values.

8.4 *Book Pages*

`bookPages.jasp`

The website Story Graph collects people's book reading activity, and their evaluations of what they have read. This example considers using fiction versus non-fiction books as two alternative "decisions," and the number of pages in the books as "response times," so that the EZ diffusion model can be applied. Nothing sensible or useful follows, which is the point.

Data

The data come from the first 50 books listed in the community notes of Story Graph on January 14, 2025. The top panel of Figure 8.11 shows the distribution of the number of pages in the 45 fiction and 5 non-fiction books sampled. These distributions look enough like response time distributions that the EZ diffusion model can be applied.

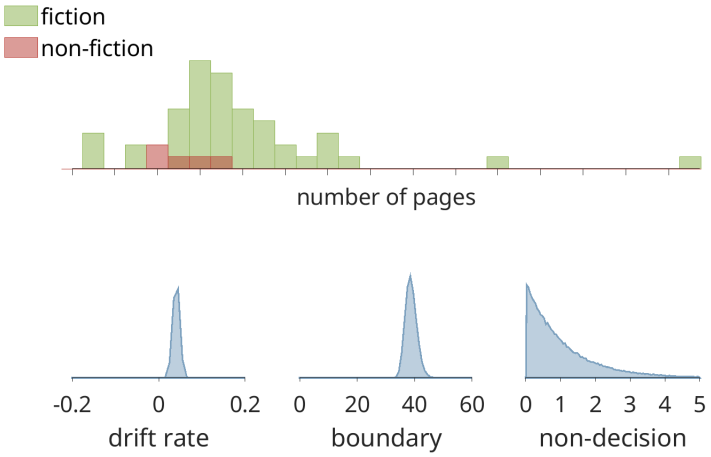

Figure 8.11: The number of pages in a sample of fiction and non-fiction books (top panel) and the EZ diffusion model inferences (bottom panel) .

Results

The bottom panel of Figure 8.11 shows the marginal posterior distributions for the three EZ diffusion model parameters. The drift rate is positive, he boundary separation is around 40, and the non-decision parameter is somewhere below 5.

Interpreting the parameters is a challenge. The decision is to choose whether a book is fiction or not, so the momentary samples are measures of evidence for "fictionness," whatever that is. The boundary separation measures how much fictionness is needed to make a decision. The number of evidence samples needed to reach the threshold measures the number of pages in a book. The non-decision parameter is something like the number of pages books have regardless of going through the process of distinguishing fiction from fact.

If none of this makes sense to you, you are not alone. The EZ diffusion model is designed to provide a drift rate that measures the evidence stimuli provide for decision making, a boundary separation that measures the caution of the decision maker, and a non-decision time that measures the time it takes to perceive the stimulus and act. The book pages data come from a context that does not map to these interpretations, making the application of the model problematic. It is difficult to say what is learned from the parameter inferences, other than perhaps the positive drift rate is consistent with 45 of the 50 books being fiction books, and this could have been modeled much more simply with a success rate model.

The moral of this example is that just because a cognitive model can statistically be applied to data does not mean that it is useful to do so. One

of the key advantages of using cognitive models to analyze behavioral data is that the inferred parameters have psychological meaning. While regression analysis returns inferences about regression coefficients and other parameters that primarily have statistical interpretations, cognitive models return inferences about psychological properties. If the psychological interpretations are lost, much of the rationale for using cognitive models is also lost. This moral may seem obvious, but it is not hard to find questionable applications of cognitive models in journals and at conferences.*

8.5 The Hierarchical EZ Diffusion Model

8.5.1 Motivation

A hierarchical extension of the EZ diffusion models assumes that each parameter for an individual comes from a group distribution, and allows for different groups. The hierarchical model can make inferences about drift rates, boundary separations, and non-decision times at both the individual and group level, including inferences about a representative individual for each group.

8.5.2 Model

The hierarchical EZ diffusion model assumes that the δ_i, α_i, τ_i parameters for the individual i come from a Gaussian group distribution with a mean and standard deviation. There are separate group distributions for each of the three parameters. This means that there are group-level means μ_g^δ, μ_g^α, and μ_g^τ and group-level standard deviations σ_g^δ, σ_g^α, and σ_g^τ. The group membership of the individual i is assumed to be known and is specified by g_i. The parameters for the representative individual of group g are $\tilde{\delta}_g$, $\tilde{\alpha}_g$, $\tilde{\tau}_g$. They are modeled as samples from the Gaussian group distributions.

Figure 8.12 summarizes the structure of the hierarchical EZ diffusion model, showing how it relates latent parameters to observed data. The group level distributions generate the individual drift rate, boundary separation, and non-decision time parameters. These individual parameters, in turn, generate the observed behavior that is summarized by counts of correct responses and the mean and variance of their response times. The data also specify the group membership for each individual.

8.5.3 Mechanics

Table 8.3 details the meaning of the parameters and variables in the hierarchical EZ diffusion model, their name in the model as shown in Figure 8.12, and their name in JASP.

* For some reason diffusion models are among the most over-applied. My best guess is the desire to use the diffusion model in all circumstances comes from a kind of intellectually prepubescent physics envy, but I don't really know. Observing the "diffusion models, is there anything they cannot do?" attitude reached a nadir for me recently when a friend, given the opportunity of an extended conversation with Charlize Theron, spent most of it talking about diffusion models.

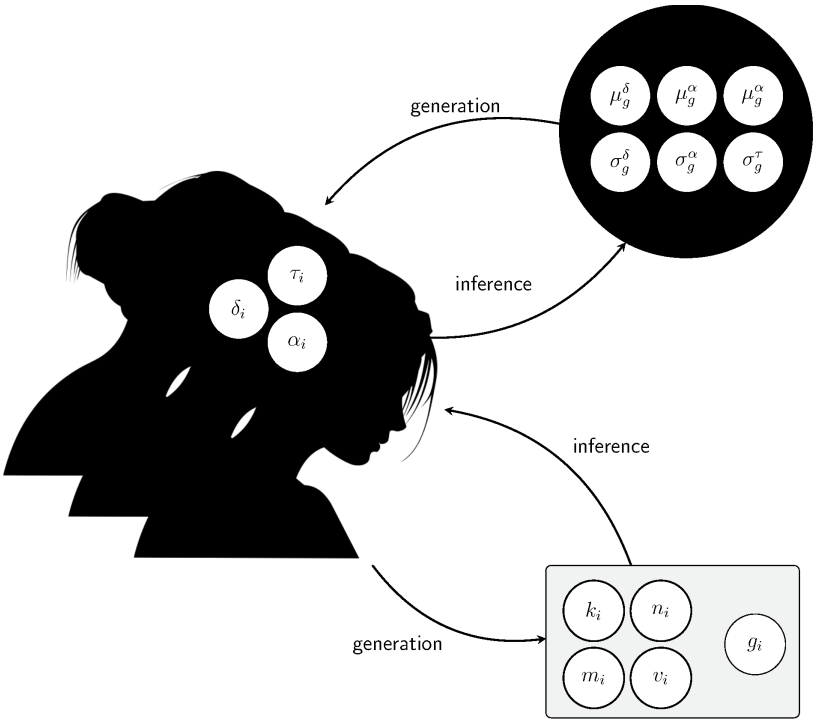

Figure 8.12: The structure of the hierarchical EZ diffusion model.

Table 8.3: The meaning of the parameters and variables in the explanatory Rasch model, with their model and JASP names.

Name	Model	JASP
group membership	g_i	`group[i]`
number of correct decisions	k_i	`k[i]`
number of decisions	n_i	`n[i]`
mean of response times for correct decisions	m_i	`m[i]`
variance of response times for correct decisions	v_i	`v[i]`
individual drift rate	δ_i	`delta[i]`
individual boundary separation	α_i	`alpha[i]`
individual non-decision time	τ_i	`tau[i]`
group drift rate mean	μ_g^δ	`muDelta[g]`
group drift rate standard deviation	σ_g^δ	`sigmaDelta[g]`
group boundary mean	μ_g^α	`muAlpha[g]`
group boundary standard deviation	σ_g^α	`sigmaAlpha[g]`
group non-decision time mean	μ_g^τ	`muTau[g]`
group non-decision time standard deviation	σ_g^τ	`sigmaTau[g]`
group representative drift rate	$\tilde{\delta}_g$	`deltaRep[g]`
group representative boundary	$\tilde{\alpha}_g$	`alphaRep[g]`
group representative non-decision time	$\tilde{\tau}_g$	`tauRep[g]`

Input data

An example is shown in Table 8.4 using part of the data from the *Numerosity* example.

Output analyses

The JASP implementation provides three analyses:

- **individual** : the marginal posterior distributions of the drift rate δ, boundary separation α, and non-decision time τ parameters for

trial	correct	meanRT	varRT	group
3344	3167	0.386	0.005	1
3344	2865	0.413	0.009	1
3344	3081	0.378	0.008	1
...	...	...	...	...
3344	3128	0.492	0.079	2
3344	3328	0.493	0.011	2
3344	3320	0.427	0.009	2
...	...	...	...	...
3040	2966	0.489	0.008	3
3344	3291	0.509	1.127	3
3344	3090	0.544	0.055	3
...	...	...	...	...
3040	3022	0.622	0.063	4
3344	3302	0.505	0.006	4
3344	3269	0.714	0.592	4
...	...	...	...	...

Table 8.4: Data format for hierarchical EZ diffusion analysis.

each individual

- **group drift, boundary, and non-decision posteriors**: the marginal posterior distributions of the mean drift rates μ_g^δ, mean boundary separations μ_g^α, and mean non-decision times μ_g^τ for each group
- **representative drift, boundary, and non-decision posteriors**: the marginal posterior distributions of the mean drift rates $\tilde{\delta}_g$, mean boundary separations $\tilde{\alpha}_g$, and mean non-decision times $\tilde{\tau}_g$ for the representative individual from each group

8.6 *Numerosity*

`numerosity.jasp`

Childers (2023) provides data from a numerosity experiment. On each trial, a participant was presented with visual array containing asterisk symbols at random location is presented, and had to decide whether more or fewer than 50 asterisks were present. Childers (2023) used both younger and older participant groups, and each participant completed blocks of trials under both speed and accuracy instructions.

In this example, we use the hierarchical EZ diffusion model to make individual and group inferences for the four groups defined by the experimental design: younger participants with speed instructions, younger participants with accuracy instructions, older participants with speed instructions, and older participants with accuracy instructions. The hierarchical model is well suited to this analysis, because it makes inferences at both the group and individual levels. Comparisons of the group-level parameters provide information about differences due to age and task instructions. Inferences at the individual level provide information about what group-level differences mean for individual performance.

Data

Childers (2023) used 19 younger and 19 older participants. Most participants completed 7040 trials in both speed and accuracy conditions. We

use a subset of the data that correspond to "easy" trials, in which the true numerosity is less than 40 or greater than 60. About half the trials in the experiment meet these criteria.

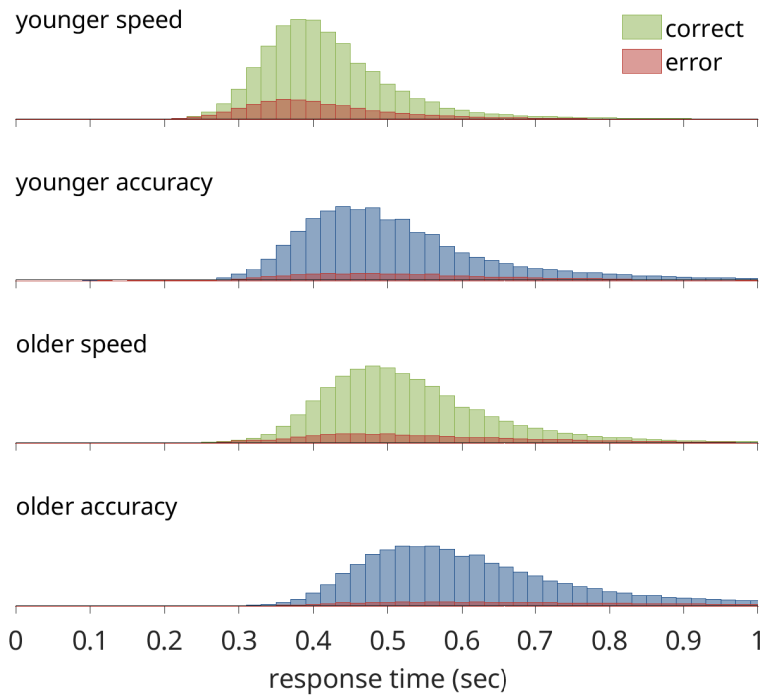

Figure 8.13: Aggregate response time and decision distributions for the four experimental groups in the numerosity data.

Figure 8.13 provides a summary of the numerosity data. Each panel corresponds to one of the experimental groups, showing the distribution of response time for correct and error responses aggregated over all participants. Because of the aggregation, this is a useful summary at the group level, but does not provide any information about the behavior of individuals in each of the groups.

Results

Group inferences

Figure 8.14 shows the marginal posterior distributions for the group means for younger and older participants in both speed and accuracy conditions. Each panel corresponds to a parameter of the EZ diffusion model, and the four experimental groups are color coded.

Another way of showing the same inferences is with a 2×2 ANOVA-style graph. In Figure 8.15, the x-axes correspond to speed versus accuracy instructions, and the posterior mean and 95% credible interval of the group means are shown in each panel.

Figure 8.14 has the advantage of representing fully the marginal posterior distributions. Figure 8.15 summarizes this uncertainty with means and credible intervals, but provides clearer intuitions in thinking about main effects and interactions on the model parameters between the four groups.

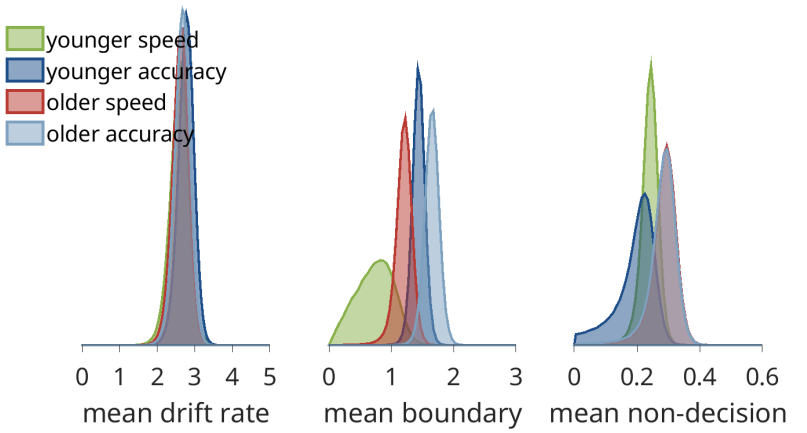

Figure 8.14: Posterior distributions for the group means for the numerosity data.

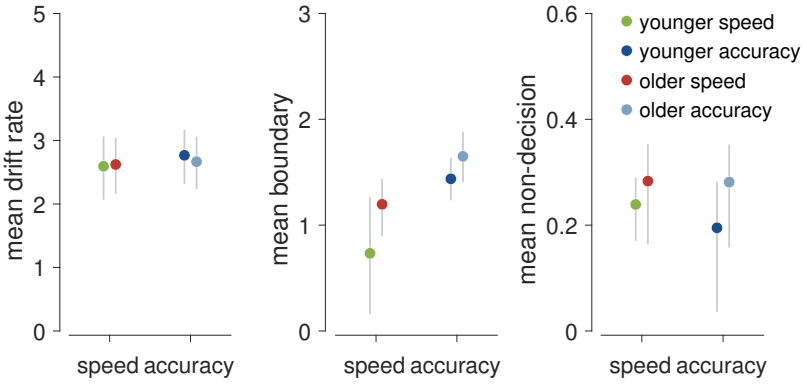

Figure 8.15: A 2×2 ANOVA-style summary of the posterior distributions for the group means for the numerosity data.

The clear result for drift rate seems to be that neither age nor instructions have an impact. This is consistent with the conception of drift rate as a property of the stimuli. There is evidence that accuracy instructions increase the boundary separation. This is consistent with its conception as reflecting the level of caution people use in making decisions. There is no clear difference in the non-decision time between the four groups. A reasonable theoretical expectation would be that non-decision time is greater for older people. While the group means for older participants are greater than those for young participants in both task conditions, the large uncertainties about all four means guard against this conclusion.

Representative individual inferences

Figure 8.16 shows the marginal posterior distributions for the representative individual in all four conditions. These distributions now represent the drift rate, boundary separation, and non-decision time, that could be expected for a random member of each of the four groups.

It is clear that there is large uncertainty about the parameter values a representative individual has for all the parameters in all four groups. Given the large numbers of trials individuals completed, this uncertainty is unlikely to result primarily from a lack of informative behavioral data. It is much more likely to result from large individual differences within the four groups among the 19 observed participants.

It is worth contrasting the inferences for group means in Figure 8.14 with those for the representative individual in Figure 8.16. In particular, while

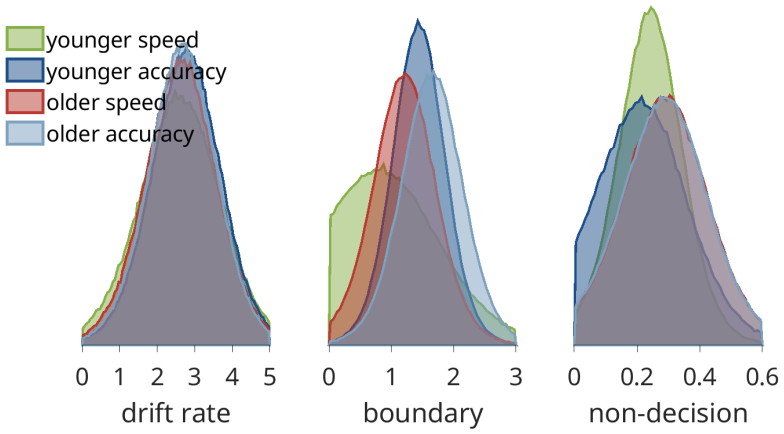

Figure 8.16: Posterior distributions for the representative individual for the numerosity data.

it seems clear the task instructions impacted boundary separation at the groups level, it is much less clear what that impact translates to at the individual level. Thus, a research question such as "is there an effect of speed versus accuracy instructions on response caution?" does not have a simple answer. If the question is asking about an average effect across many people, the answer is probably yes. If the question is whether there are effects for specific individuals, the answer is unclear. There is no inherent contradiction in these different answers. The nuanced answer is needed because of the lack of precision in the question.

Individual inferences

Figure 8.17 summarizes the parameter inferences for every individual in all four groups. The top row of panels is for the younger participants and the bottom row of panels is for the older participants. Within each panel, the posterior mean and 95% credible interval are shown. An advantage of this organization is that the within-participant change in model parameters across the speed and accuracy task instructions is represented.

Figure 8.17 shows the complexity of the evidence provided by the experimental data in relation to the question drift rate, boundary separation, and non-decision time vary with age and instructions. There are general regularities, such as the boundary separations being larger for most participants when given accuracy rather than speed instructions. This is visually clear because the red markers are generally above the blue markers, and the yellow markers are generally above the green markers, in the middle column relating to boundary separation. This regularity leads to the difference in group means. But, at the individual level, there are exceptions. For example, the second older participant clearly has a larger boundary separation in their speed condition.

In addition, there are clearly large individual differences in the level of evidence different participants need to make decisions. Many older participants have boundary separations around 1, but significant numbers have values as large as 2 and 3. This variability leads to the large uncertainty in the representative individual analysis. As anticipated earlier, the parameter inferences for the specific participants are relatively certain. But predicting the parameter value of the next participant (or a

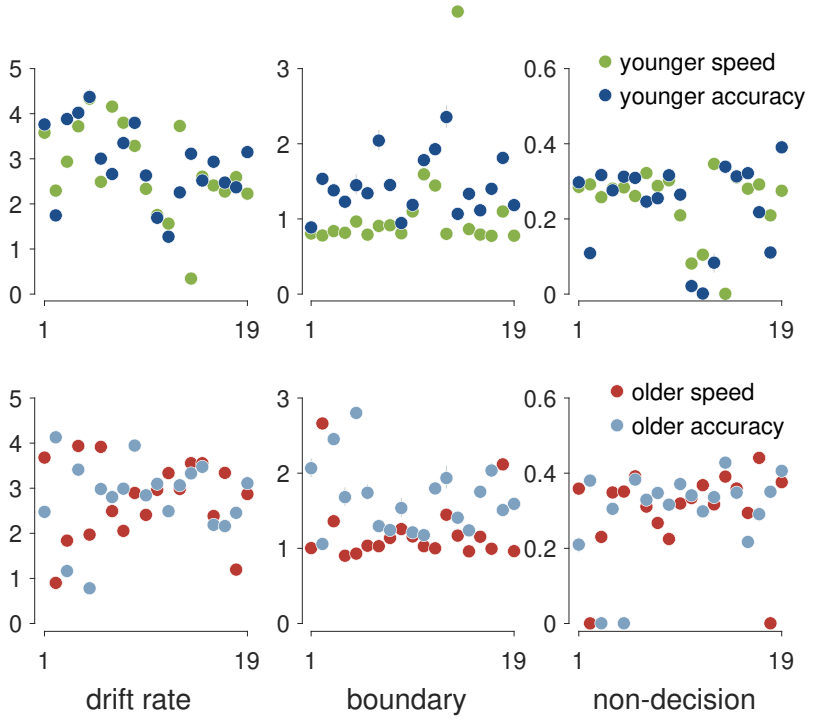

Figure 8.17: Posterior means and 95% credible intervals for each individual in the numerosity data.

random one among those observed) is much more uncertain, because of the variability caused by individual differences.

The results in Figure 8.17 for the drift rate and non-decision time are even more difficult to parse. The general point is that careful thinking about scientific claims at the individual and group level are often required when, as is very often the case, there are individual differences in psychological variables.

◇

8.7 Formal Details

8.7.1 EZ diffusion model

Likelihood

We follow the account given by Chávez De la Peña & Vandekerckhove (2025) in developing the probabilistic version of the EZ diffusion model, starting with the Wiener diffusion model. The Wiener diffusion model is described by the Wiener distribution. This is a bivariate distribution for both the binary decision y_{ij} individual i makes on trial j, and the response time t_{ij} for that decision.

$$(y_{ij}, t_{ij}) \sim \text{Wiener}(\delta_i, \alpha_i, \tau_i). \tag{8.1}$$

The EZ diffusion model is based on a system of equations that show how the parameters of the Wiener diffusion model predict the rate of accurate decisions $\tilde{R}_i$, and the mean $\tilde{M}_i$ and variance $\tilde{V}_i$ of the response time for

correct decisions. By setting $z_i = \exp(-\alpha_i \delta_i)$, these equations are given by

$$\tilde{R}_i = \frac{1}{z_i + 1} \tag{8.2}$$

$$\tilde{M}_i = \tau_i + \left(\frac{\alpha_i}{2\delta_i}\right)\left(\frac{z_i - 1}{z_i + 1}\right) \tag{8.3}$$

$$\tilde{V}_i = \left(\frac{\alpha_i}{2\delta_i^3}\right)\left(\frac{1 - 2\alpha_i\delta_i z_i - z_i^2}{(z_i + 1)^2}\right) \tag{8.4}$$

This system of equations can be inverted, providing point estimates $\hat{\delta}_i$, $\hat{\alpha}_i$. and $\hat{\tau}_i$ based on the accuracy rate, and mean and variance of correct response times.

The probabilistic EZ diffusion makes assumptions about the uncertainty inherent in $\tilde{R}_i$, $\tilde{M}_i$, and $\tilde{V}_i$. It assumes the number of correct responses k_i is binomially distributions with respect to the accuracy rate $\tilde{R}_i$ and the total number of trials n_i

$$k_i \sim \text{binomial}\big(\tilde{R}_i, n_i\big). \tag{8.5}$$

It assumes the observed mean of correct response times has a Gaussian distribution with mean $\tilde{M}_i$ and variance $\tilde{V}_i$ scaled by the number of trials n_i

$$m_i \sim \text{Gaussian}\big(\tilde{M}_i, \frac{1}{\tilde{V}_i / n_i}\big). \tag{8.6}$$

Finally, the probabilistic EZ diffusion assumes that $\tilde{V}_i$ follows the probability law

$$(n_i - 1)\, v_i / \tilde{V}_i \sim \text{Chi-squared}\big(n_i - 1\big). \tag{8.7}$$

After some simplification, and assuming the Gamma distribution is well-approximated by the Gaussian distribution for large enough n_i, this leads to

$$v_i \sim \text{Gaussian}\big(\tilde{V}_i, \frac{n_i - 1}{2\tilde{V}_i^2}\big). \tag{8.8}$$

Priors

Chávez De la Peña & Vandekerckhove (2025) specify the following priors for the EZ diffusion parameters:

$$\alpha_i \sim \text{Gamma}\big(2, 1\big) \tag{8.9}$$

$$\delta_i \sim \text{Gaussian}\big(0, 1\big) \tag{8.10}$$

$$\tau_i \sim \text{Exponential}\big(1\big). \tag{8.11}$$

8.7.2 JASP Implementation

```
# EZ Diffusion
data{
  nIndividuals = length(trials)
}
model {
 for (i in 1:nIndividuals){
    # Priors
    alpha[i] ~ dgamma(2, 1)
    delta[i] ~ dnorm(0, 1)
```

```
    tau[i] ~ dexp(1)
    # Forward equations from EZ Diffusion
    y[i]   = -alpha[i] * delta[i]
    Pc[i]  = 1/(1+exp(y[i]))
    PRT[i] = 2*pow(delta[i], 3)/alpha[i]*pow(exp(y[i])+1, 2)/(2*y[
    i]*exp(y[i])-exp(2*y[i])+1)
    MDT[i] = (alpha[i]/(2*delta[i]))*(1 - exp(y[i]))/(1 + exp(y[i
    ]))
    MRT[i] = MDT[i] + tau[i]
    # Loss functions using MRT, PRT, and Pc
    meanRT[i] ~ dnorm(MRT[i], PRT[i]*correct[i])
    varRT[i] ~ dnorm(1/PRT[i], 0.5*correct[i]*PRT[i]*PRT[i])
    correct[i] ~ dbin(Pc[i], trials[i])
  }
}
```

8.7.3 Hierarchical EZ diffusion model

$$
\begin{aligned}
\delta_i &\sim \text{Gaussian}\left(\mu^{\delta}_{g_i}, \frac{1}{\left(\sigma^{\delta}_{g_i}\right)^2}\right) \\
\alpha_i &\sim \text{Gaussian}_+\left(\mu^{\alpha}_{g_i}, \frac{1}{\left(\sigma^{\alpha}_{g_i}\right)^2}\right) \\
\tau_i &\sim \text{Gaussian}_+\left(\mu^{\tau}_{g_i}, \frac{1}{\left(\sigma^{\tau}_{g_i}\right)^2}\right)
\end{aligned}
$$

$$
\begin{aligned}
\tilde{\delta}_g &\sim \text{Gaussian}\left(\mu^{\delta}_{g_i}, \frac{1}{\left(\sigma^{\delta}_{g}\right)^2}\right) \\
\tilde{\alpha}_g &\sim \text{Gaussian}_+\left(\mu^{\alpha}_{g_i}, \frac{1}{\left(\sigma^{\alpha}_{g}\right)^2}\right) \\
\tilde{\tau}_g &\sim \text{Gaussian}_+\left(\mu^{\tau}_{g_i}, \frac{1}{\left(\sigma^{\tau}_{g}\right)^2}\right)
\end{aligned}
$$

8.7.4 JASP Implementation

```
# Hierarchical EZ Diffusion
data{
  nIndividuals = length(trials)
  nGroups = max(group)
}
model {
 for (i in 1:nIndividuals){
    # Individual parameters
    alpha[i] ~ dnorm(muAlpha[group[i]], 1/sigmaAlpha[group[i]]^2)T
    (0, )
    delta[i] ~ dnorm(muDelta[group[i]], 1/sigmaDelta[group[i]]^2)
    tau[i]   ~ dnorm(muTau[group[i]], 1/sigmaTau[group[i]]^2)T(0,
    )
    # Forward equations from EZ Diffusion
    y[i]   = -alpha[i]*delta[i]
    Pc[i]  = 1/(1 + exp(y[i]))
    PRT[i] = 2*pow(delta[i], 3)/alpha[i]*pow(exp(y[i])+1, 2)/(2*y[
    i]*exp(y[i])-exp(2*y[i])+1)
```

```
    MDT[i] = (alpha[i]/(2*delta[i]))*(1-exp(y[i]))/(1+exp(y[i]))
    MRT[i] = MDT[i] + tau[i]
    # Loss functions using MRT, PRT, and Pc
    meanRT[i] ~ dnorm(MRT[i], PRT[i]*correct[i])
    varRT[i] ~ dnorm(1/PRT[i], 0.5*correct[i]*PRT[i]*PRT[i])
    correct[i] ~ dbin(Pc[i], trials[i])
  }
  # Group priors
  for (g in 1:nGroups){
   muAlpha[g] ~ dgamma(2, 1)
   muDelta[g] ~ dnorm(0, 1)
   muTau[g] ~ dexp(1)
   sigmaAlpha[g] ~ dunif(0, 10)
   sigmaDelta[g] ~ dunif(0, 10)
   sigmaTau[g] ~ dunif(0, 10)
  }
  # Representative individuals
  for (g in 1:nGroups){
    alphaRep[g] ~ dnorm(muAlpha[g], 1/sigmaAlpha[g]^2)T(0, )
    deltaRep[g] ~ dnorm(muDelta[g], 1/sigmaDelta[g]^2)
    tauRep[g] ~ dnorm(muTau[g], 1/sigmaTau[g]^2)T(0, )
  }
}
```

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Exercises

Success Rate

Online sellers

This exercise is based on the *Online Sellers* example in Chapter 1.3. If you are not familiar with the example, read it first.

1. Open the file `onlineSellers.jasp` on the Open Science Framework (OSF). It shows the results from the JASP file, including tables and graphs for the success rate posteriors and success count posterior predictive distributions. Table 1.1 maps the statistical symbols (like θ_1) to the JASP variable names (like `theta[1]`). Make sure you can identify the information in the JASP output that corresponds the credible intervals and Figures 1.4 and 1.5. There is other information in the JASP file, some of which is useful, and some of which is not very useful. You can ignore that for now. Make sure you can see where the results used in Chapter 1.3 come from.
2. Download `onlineSellers.jasp` onto your computer and open it in JASP. The OSF interface may change over time, but the download button is currently at the top-right, and looks like an arrow pointing down to a hard disk. Familiarize yourself with the JASP interface with the JAGS module active. There are three main panes
   - the left pane has the data
   - the middle pane has the analysis tabs
     - success rate posterior
     - success count posterior prediction
     - success rate comparison
   - the right pane has the results, matching what was provided on the OSF

   The middle pane analysis tabs have JAGS code and other settings. You should generally not need to touch these. Doing your own analyses involves providing your own data on the left, and interpreting your results on the right.
3. Use `Save as -> Computer` to save the file under another name. Use `Export Data -> Computer` to download the data as a `.csv` (comma separated value) file. Open the `.csv` file in a text editor (not a spreadsheet application). It is worth making sure you have a reasonable text editor for coding and data management, such as sublime or vsCode.
4. Add a fourth online seller to the new data set, with $k = 1$ success from $n = 2$ sales, and continuing a posterior predictive count of $\hat{n} = 200$. The edited file should look like this:

   ```
   k,n,nPrd
   10,10,200
   48,50,200
   186,200,200
   1,2,200
   ```

5. Use `Sync Data -> Computer` to update the JASP file with the modified data. You should see the results update automatically in the right pane, and be able to see the modified data set in the left pane.

In your work document

- report the posterior mean and 95% credible interval for the fourth seller to two significant figures
- copy-and-paste the posterior marginal densities of the success rates for all four sellers

- report the probability that seller one has a higher success rate than seller four

6. Create a copy of the `.csv` file and insert a third column `group` with the entry `1` for all sellers. It should look like this:

```
k,n,group,nPrd
10,10,1,200
48,50,1,200
186,200,1,200
1,2,1,200
```

7. Download `hierarchicalGrades.jasp` from the OSF. Rename it to be called `hierarchicalOnlineSellers.jasp`. Sync this JASP file to the four-column modified online sellers data. You should see the results update automatically in the right pane, and be able to see the modified data set in the left pane.
8. These results provide an analysis of the four online sellers using the hierarchical success rate model.

In your work document

- report the posterior mean and 95% credible interval for the fourth seller to two significant figures
- copy-and-paste the posterior marginal densities of the success rates for all four sellers
- write a sentence or two comparing and explaning these results relative to those based on the non-hierarchical model

9. Sometimes working in a spreadsheet application is more convenient for curating data. If there is one you want to work with—Microsoft Excel, Google Sheets, Libre Office Calc, Apple Numbers, Grist—now is a good time to establish a workflow. Do this by creating a modified online sellers data set using the application you prefer, save the modified data as a `.csv` file, and Sync it to JASP. It is possible you will need to do some troubleshooting to make things work. These applications can use encodings, even when just asked to produce a `.csv` file, that trip up JASP.

Signal Detection Theory

Last Two Minutes

In 2015 the National Basketball Association (NBA) began issuing "Last Two Minutes" reports that detail the performance of officials at the ends of close games. An interesting feature of these reports is that they assess the accuracy of decisions not only in terms of whether calls made by officials were correct. They also consider whether officials missed calls, by considering every event in the short time period.

This approach means that the decisions made by officials can be analyzed in signal detection terms. There are events that should be "called" (i.e., officials should made the decision to intervene in play) and events that should not be called. For each of these events, officials either did or did not make the call. The 2×2 counts in Table 5 below represent this situation.

Table 5: Signal detection theory intepretation of last two minutes data.

	true call	true no call
call made	hit	false alarm
call not made	miss	correct rejection

A dataset compiling information from the two minute reports is available here. From that dataset, we consider a specific question:

> *Was decision making different between regular season games and playoff games for the 2023 season?*

It is possible that the higher stakes and more focused attention on playoff games makes it more or less difficult for officials to make correct decisions, or changes their bias towards misses or false alarms when they do make mistakes.

A tailored dataset to address this question is available in the file `lastTwoMinutes.csv`. It lists the signal detection counts for every game in 2023 season, indicating in the `group` column whether or not it was a playoff game. There are 474 games in total, of which 442 (93%) are regular season and 32 (7%) are playoff games. The games vary widely in the number of decisions they consider, ranging from 6 to 77, with a mean of 22 decisions. The data file sets the posterior predictive counts to 20 for both signal and noise trials.

This exercise asks you to use the data to address the research question based on the hierarchical signal detection theory model.

1. Download `grammaticalAcceptability.jasp` onto your computer and open it in JASP, and rename it to `lastTwoMinutes.jasp` or similar. Delete the first `individual discriminability, bias, criteria` analysis. We do not need this analysis to answer the research question, and it takes a long time to compute.
2. Download and sync the data file `lastTwoMinutes.csv`, wait for the results to update, and save the JASP file again.
3. The JASP analyses shows bivariate scatterplots for group means and posterior representatives. These could be improved by editing, but have the weakness that they do not display both regular season and playoff results on the same axes. The goal is to produce a figure that looks close to the one shown below. The key information needed for this figure is available in the JASP tables. It requires the means and 95% credible intervals for the posterior group means and representative individuals.

In your work document

- list the means and 95% credible intervals for the group means `muD` and `muC`
- list the means and 95% credible intervals for the posterior representatives `dRep` and `cRep`
- copy the numbers above into your preferred scientific programming language (e.g., Python, R, MATLAB, ...) and develop an approximation to the figure above. It really just needs to be an

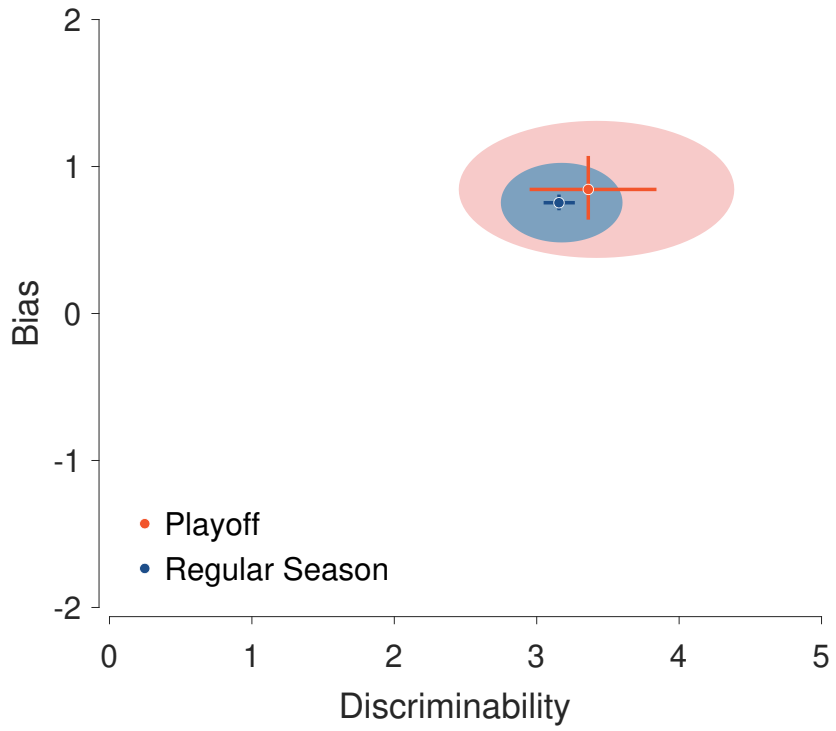

Figure 18: The circular markers show the posterior means of group discriminability and bias for the playoff and regular season officiating decisions. The crosses show the 95% credible intervals for the group means. The circles show the 95% posterior representative credible regions.

approximation. Any figure that displays all the relevant information in a visually accessible form is acceptable. Copy your figure into your notebook.

- interpret the results in terms of the motivating research question
- explain why there is more uncertainty about the playoff decision making

High Threshold Models

Continuous Recognition

This exercise involves recognition memory data from a *continuous recognition* experiment known as the continuous Mnemonic Similarity Task (Stark et al., 2019). In the task, participants were presented with a sequence of 256 images. As each image was presented, they were asked to decide if it was a "new" image that had not previously been presented, or if it was an "old" image previously presented earlier in the sequence. Over the 256 images, 64 had been presented before and the remaining 192 had not.

1. Download `amyloidMemoryTHT.jasp` onto your computer and open it in JASP, and rename it to `continuousRecognitionTHT.jasp` or similar.
2. Download and sync the data file `continuousRecognition.csv`, wait for the results to update, and save the JASP file again. This data file records the performance of 45 participants on the task.

> In your work document
> - copy the figure panels showing the joint posterior distribution of detection and bias for the group means and the representative individual
> - report the means and 95% credible intervals for both detection and bias for the group means and representative individuals
> - looking at the posteriors for detection and bias at the individual level, identify two participants who are outliers with much worse performance than the other participants

3. In the data pane, find a way to change the group membership of the two outlier *contaminant* participants from `1` to `2`. One way to do this is to export the data as a `.csv` file and edit it in a text editor. In theory it should also be possible to make the edits directly in JASP, although sometimes that seems to crash JASP.
4. However you manage to make the edits, you should end up with a new JASP file called `continuousRecognitionContaminantTHT.csv`. This will require file renaming and syncing if the editing is done externally to JASP, or just saving the file with the new name if the editing is done within JASP.
5. Once the results have updated, there are now two groups of participants: those who attended to the task and performed reasonably well, and two contaminants who most likely did not attend to the task.

> In your work document
> - again copy the figure panels showing the joint posterior distribution of detection and bias for the group means and the representative individual, this time for the two groups
> - again report the means and 95% credible intervals for both detection and bias for the group means and representative individuals, for the two groups
> - discuss how and why the removal of the contaminant participants affected the inference about the general performance of the remaining majority of participants

6. Create yet another JASP file `continuousRecognition4TrialsTHT.jasp` by syncing the data file `continuousRecognition.csv`. This data file has the performance of all 45 participants after their first four trials. Note that the posterior predictive signal and noise trial counts have been set to `nPrdSig=96` and `nPrdNoi=192`, consistent with the counts when the experiment is complete. This means the posterior predictive knows about four of the trials, but is making predictions about the other 252 trials.

> In your work document
> - copy the joint posterior predictive distribution of hit and false alarm counts for the representative individual

- discuss the strengths and weaknesses of the distribution as a prediction of the eventual final performance of the participants

LaTeX in Overleaf

This exercise provides an introduction to the LaTeX scientific document preparation system and the Overleaf interface for using it. You have free access to the professional edition of Overleaf via your UCI affiliation. There is information here.

1. The template for the class research report is here. Log in to Overleaf, go to this template, and *MAKE YOUR OWN COPY*. If you do not make a copy, you will be overwriting the original template (for everybody to see).
2. Read through the `.tex` document line by line, understanding as much as you can. Work through the exercises below, recompiling after each to see the changes they made to the typeset document.

In your copy of the template

- Add a sentence to the end of the abstract that says `This is the last sentence of the abstract.`
- Go to google scholar and search for "Bayesian benefits for the pragmatic researcher". Follow `cite` and `bibtex` to find the bibtex formatted reference. Cut and paste the reference into `references.bib` in Overleaf.
- After the second paragraph of the Introduction section, add `Another paper is \citeA{wagenmakers2016bayesian}, which could also have been referenced more tersely \shortciteA{wagenmakers2016bayesian}.`
- Download `wineTasting.jasp`
- Save the image showing the marginal posterior for `d`. Do this by clicking the downward triangle next to `d` (not the one next to `Marginal Density`) and using `Save Image As`. Call the file `dPrime.png` and save it anywhere on your hard drive.
- Upload the `dPrime.png` from your hard drive to Overleaf.
- The first LaTeX code after the `\section*{Results}` shows how a figure can be inserted. Copy and paste that block of LaTeX code, and go to the discriminability, criterion, and bias marginal posteriors analysis. Copy the figure panel showing the marginal posterior for `d` using `Save Image As`, using the figure filename `dPrime.png`.
- Change the caption to `The marginal posterior distribution of $d^\prime$.` and the `\label` to `\label{fig:dPrimeMarginal}`.
- Below the code that defines the two figures, add the sentence `Now there is both Figure~\ref{fig:sdtResults} and Figure~\ref{fig:dPrimeMarginal}.` The tilde ~ prevents a space between the word "Figure" and its figure number.
- Just before the Discussion section is an example of how a table is defined in LaTeX. This code can be copied from JASP. Go back to the analysis in `wineTasting.jasp` and copy the code for the *MCMC Summary* table above the figures. Use the downward triangle and `Copy LaTeX`.
- Paste the LaTeX table code into Overleaf immediately after the current table code, but before the Discussion. Immediately delete the first line that reads `\usepackage{booktabs}` and the comment above it. That is already installed and does not need to be added again. You can also delete the commented code that starts with `*`.
- Replace the parameter names in teh first column written `beta`, `c`, `d`, and `k` with `$\beta$`, `$c$`, `$d^\prime$`, and `$k$`.
- Remove all of the columns in the table *except* the posterior mean and lower and upper bounds of the 95% credible interval. Historically, this required learning how LaTeX defines tables. A modern approach is to get a genAI to do it for you. The prompt `remove all but the parameter, mean, lower, and upper columns from this` worked fine for me using Cursor. (It is good to learn some of the LaTeXcoding even as you interact with GenAI but, as with all of LaTeX, you can do this with experience.)
- Give the table a meaningful caption like `The posterior means and 95\% credible intervals for the four signal detection model parameters.`